

Krishna's

TEXT BOOK on
Vector Calculus



(For B.A./B.Sc. (Mathematics) IIIrd/IVth Semester Students of HNB Garhwal University, U.K.)

As per CHOICE BASED CREDIT SYSTEM (w.e.f. 2016-2017)

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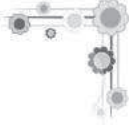
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Dedicated
to
Lord
Krishna

Authors & Publishers

Preface

This book on **VECTOR CALCULUS** has been specially written according to the latest **Syllabus** to meet the requirements of the **B.A. and B.Sc. IVth Semester Students** of all colleges affiliated to **Garhwal University (U.K.)**.

The subject matter has been discussed in such a simple way that the students will find no difficulty to understand it. The proofs of various theorems and examples have been given with minute details. Each chapter of this book contains complete theory and a fairly large number of solved examples. Sufficient problems have also been selected from various university examination papers. At the end of each chapter an exercise containing objective questions has been given.

We have tried our best to keep the book free from misprints. The authors shall be grateful to the readers who point out errors and omissions which, inspite of all care, might have been there.

The authors, in general, hope that the present book will be warmly received by the students and teachers. We shall indeed be very thankful to our colleagues for recommending this book to the students.

The authors wish to express their thanks to **Mr. S.K. Rastogi, M.D., Mr. Sugam Rastogi, Executive Director, Mrs. Kanupriya Rastogi, Director** and **entire team of KRISHNA Prakashan Media (P) Ltd., Meerut** for bringing out this book in the present nice form.

The authors will feel amply rewarded if the book serves the purpose for which it is meant. Suggestions for the improvement of the book are always welcome.

— *Authors*

Syllabus

Vector Calculus

H.N.B. Garhwal University, U.K.

Choice Based Credit System (*w.e.f.* 2016-2017)

B.A./B.Sc.—IVth Semester

Multiple products, Reciprocal vector, Vector differentiation and integration, differential vector operator Del, Gradient, Divergence, Curl, Vector identities. Vector integration along line, surface and volume, Simple applications of Green's theorem, Gauss's divergence and Stokes theorem (without proof)

Brief Contents

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Krishna's

VECTOR ANALYSIS

Chapters



1. Multiple Products
2. Differentiation of Vectors
3. Gradient, Divergence and Curl
4. Integration of Vectors
5. Line Integrals
6. Green's, Gauss's and Stoke's Theorems

Chapter

1



Multiple Products

1 Triple Products

We know that the vector product $\mathbf{a} \times \mathbf{b}$ of two vectors $\mathbf{a}$ and $\mathbf{b}$ is itself a vector quantity. Therefore we can multiply it by another vector $\mathbf{c}$ both scalarly and vectorially. The product $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ is called **scalar triple product**, which is a pure number. On the other hand the product $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ is called **vector triple product**, which is again a vector quantity.

Note. Since $\mathbf{a} \cdot \mathbf{b}$ is a scalar quantity, therefore the products $(\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{c}$ and $(\mathbf{a} \cdot \mathbf{b}) \times \mathbf{c}$ are meaningless. Moreover in the product $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ we can omit the parentheses and we can simply write it as $\mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$. Obviously the product $\mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$ has meaning only if we regard it as the product $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$.

2 Scalar Triple Product

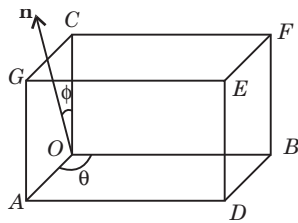
The scalar product of two vectors one of which is itself the vector product of two vectors is a scalar quantity called “Scalar Triple Product”. Thus if $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be three vectors, then $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ is called the scalar triple product of these three vectors.

Since the scalar triple product involves both the signs of ‘cross’ and ‘dot’ therefore it is sometimes also called the **mixed product**.

Geometrical Interpretation of Scalar Triple Product.

Let us consider a parallelopiped whose coterminous edges OA , OB , OC have the lengths and directions of the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ respectively. Let V be the volume of this parallelopiped. We shall regard V , as necessarily positive.

Let $\mathbf{a} \times \mathbf{b} = \mathbf{n}$. Then from our definition of vector product, the vector $\mathbf{n}$ is perpendicular to the face $OADB$, and its modulus n is the measure of the area of the parallelogram $OADB$. Also, by definition, the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{n}$ form a right handed triad.



Let ϕ denote the angle between the directions of the vectors $\vec{OC}$ and $\mathbf{n}$. Then the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ will form a right handed or a left handed triad according as ϕ is acute or obtuse.

$$\begin{aligned} \text{Now } (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} &= |\mathbf{a} \times \mathbf{b}| |\mathbf{c}| \cos \phi = |\mathbf{n}| |\mathbf{c}| \cos \phi \\ &= (\text{area of the parallelogram } OADB) \cdot (OC \cos \phi) \\ &\quad [\because |\mathbf{c}| = OC] \end{aligned}$$

Now $OC \cos \phi$ will be positive or negative according as ϕ is acute or obtuse. Its absolute value will give us the length of the perpendicular from C to the plane of the parallelogram $OADB$.

Now the volume V of the parallelopiped = (Area of the parallelogram $OADB$) $\times$ length of the perpendicular from C on this parallelogram. Therefore $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = +V$, if ϕ is acute i.e., if $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form a right handed triad and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = -V$, if ϕ is obtuse i.e., if $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form a left handed triad.

Now we know that if the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form a right handed triad, then the vector triads $\mathbf{b}$, $\mathbf{c}$, $\mathbf{a}$ and $\mathbf{c}$, $\mathbf{a}$, $\mathbf{b}$ are also right handed. Hence each of the products $(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}$ and $(\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b}$ will have the same value $+V$ or $-V$ according as $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form a right handed or a left handed triad. Thus we conclude that in all the cases

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} = (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b}.$$

$$\text{Now } \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} \text{ and } \mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}.$$

$$\begin{aligned} \therefore (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} &= \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} \\ &= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b} = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) \\ &= -(\mathbf{b} \times \mathbf{a}) \cdot \mathbf{c} = -\mathbf{c} \cdot (\mathbf{b} \times \mathbf{a}) = -(\mathbf{c} \times \mathbf{b}) \cdot \mathbf{a} \\ &= -\mathbf{a} \cdot (\mathbf{c} \times \mathbf{b}) = -(\mathbf{a} \times \mathbf{c}) \cdot \mathbf{b} = -\mathbf{b} \cdot (\mathbf{a} \times \mathbf{c}). \end{aligned}$$

From this we conclude that the value of a scalar triple product depends on the cyclic order of the factors and is independent of the position of the dot and cross. These may be interchanged at pleasure. However, an anticyclic permutation of the three factors changes the value of the product in sign but not in magnitude.

(Important)

Notation: In view of the properties discussed above, the scalar triple product is usually written as $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = [\mathbf{abc}]$ or $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$. This notation takes into consideration only the cyclic order of the three vectors and disregards the unimportant positions of dot and cross. Thus $[\mathbf{abc}] = [\mathbf{bca}] = [\mathbf{cab}] = -[\mathbf{cba}]$ etc.

The signs of dot and cross can be inserted at pleasure i.e.,

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \quad \text{or} \quad (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}.$$

Note 1: If $\mathbf{i}, \mathbf{j}, \mathbf{k}$ constitute an orthogonal right handed triad of unit vectors, then $[\mathbf{i}, \mathbf{j}, \mathbf{k}] = (\mathbf{i} \times \mathbf{j}) \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{k} = 1$.

Note 2: The scalar triple product $[\mathbf{abc}]$ is positive or negative according as $\mathbf{a}, \mathbf{b}, \mathbf{c}$ form a right handed or a left handed triad of vectors.

3 Distributive Law for Vector Product

To prove that $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$, where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are any three vectors.

$$\text{Let} \quad \mathbf{r} \equiv \mathbf{a} \times (\mathbf{b} + \mathbf{c}) - \mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c} \quad \dots(1)$$

Now forming the scalar product of both sides of (1) with an arbitrary vector $\mathbf{d}$, we get

$$\mathbf{d} \cdot \mathbf{r} = \mathbf{d} \cdot [\mathbf{a} \times (\mathbf{b} + \mathbf{c}) - \mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c}] \quad \dots(2)$$

$$\text{or} \quad \mathbf{d} \cdot \mathbf{r} = \mathbf{d} \cdot [\mathbf{a} \times (\mathbf{b} + \mathbf{c})] - \mathbf{d} \cdot (\mathbf{a} \times \mathbf{b}) - \mathbf{d} \cdot (\mathbf{a} \times \mathbf{c})$$

[Since scalar product is distributive]

Now in a scalar triple product the positions of dot and cross can be interchanged without affecting its value. Therefore from (2), we get

$$\begin{aligned} \mathbf{d} \cdot \mathbf{r} &= (\mathbf{d} \times \mathbf{a}) \cdot (\mathbf{b} + \mathbf{c}) - (\mathbf{d} \times \mathbf{a}) \cdot \mathbf{b} - (\mathbf{d} \times \mathbf{a}) \cdot \mathbf{c} \\ &= (\mathbf{d} \times \mathbf{a}) \cdot \mathbf{b} + (\mathbf{d} \times \mathbf{a}) \cdot \mathbf{c} - (\mathbf{d} \times \mathbf{a}) \cdot \mathbf{b} - (\mathbf{d} \times \mathbf{a}) \cdot \mathbf{c} \end{aligned}$$

[Since scalar product is distributive]

$$= 0.$$

Therefore either $\mathbf{d} = \mathbf{0}$, or $\mathbf{r} = \mathbf{0}$ or $\mathbf{d}$ is perpendicular to $\mathbf{r}$. But the vector $\mathbf{d}$ is arbitrary. Therefore we can take it to be non-zero and not perpendicular to $\mathbf{r}$.

$$\text{Hence} \quad \mathbf{r} = \mathbf{0} \quad \text{i.e.,} \quad \mathbf{a} \times (\mathbf{b} + \mathbf{c}) - \mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c} = \mathbf{0}$$

$$\text{i.e.,} \quad \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}.$$

4 Properties of Scalar Triple Product

(i) The value of a scalar triple product, if two of its vectors are equal, is zero.

$$\text{We have} \quad [\mathbf{aab}] = \mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}).$$

Now $\mathbf{a} \times \mathbf{b}$ is a vector perpendicular to the plane of $\mathbf{a}$ and $\mathbf{b}$.

$$\text{Therefore} \quad \mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0.$$

(ii) The value of a scalar triple product, if two of its vectors are parallel, is zero.

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be three vectors such that $\mathbf{a}$ and $\mathbf{b}$ are parallel i.e., $\mathbf{b} = t\mathbf{a}$, where t is some scalar.

$$\begin{aligned} \text{Now } [\mathbf{abc}] &= (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (\mathbf{a} \times t\mathbf{a}) \cdot \mathbf{c} = t (\mathbf{a} \times \mathbf{a}) \cdot \mathbf{c} = t (\mathbf{0} \cdot \mathbf{c}) \\ &= 0. \end{aligned} \quad [\because \mathbf{a} \times \mathbf{a} = \mathbf{0}]$$

- (iii) The necessary and sufficient condition that three non-parallel and non-zero vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be coplanar is that $[\mathbf{abc}] = 0$.

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be three coplanar vectors. Now $\mathbf{a} \times \mathbf{b}$ is a vector perpendicular to the plane of $\mathbf{a}$ and $\mathbf{b}$. Since $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar, therefore $\mathbf{a} \times \mathbf{b}$ is also perpendicular to $\mathbf{c}$. Now the dot product of two perpendicular vectors is equal to zero. Hence $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$ i.e., $[\mathbf{abc}] = 0$. Therefore the condition is necessary.

The condition is also sufficient: Because if $[\mathbf{abc}] = 0$ i.e., $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$, then $\mathbf{c}$ is perpendicular to $\mathbf{a} \times \mathbf{b}$. But $\mathbf{a} \times \mathbf{b}$ is a vector perpendicular to the plane of $\mathbf{a}$ and $\mathbf{b}$. Since $\mathbf{c}$ is perpendicular to $\mathbf{a} \times \mathbf{b}$, therefore $\mathbf{c}$ is parallel to the plane of $\mathbf{a}$ and $\mathbf{b}$.

Hence $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar.

- (iv) Since the distributive law holds for both scalar and vector products, it holds also for the scalar triple product.

Thus $[\mathbf{a}, \mathbf{b} + \mathbf{d}, \mathbf{c} + \mathbf{r}] = [\mathbf{abc}] + [\mathbf{abr}] + [\mathbf{adc}] + [\mathbf{adr}]$, the cyclic order of the factors being maintained in each term.

5 To Express The Value of The Scalar Triple Product $[\mathbf{abc}]$ In Terms of Rectangular Components of The Vectors

$$\begin{aligned} \text{Let } \mathbf{a} &= a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}, \mathbf{b} = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}, \\ \mathbf{c} &= c_1 \mathbf{i} + c_2 \mathbf{j} + c_3 \mathbf{k}. \end{aligned}$$

$$\text{Now } \mathbf{b} \times \mathbf{c} = (b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}) \times (c_1 \mathbf{i} + c_2 \mathbf{j} + c_3 \mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= (b_2 c_3 - b_3 c_2) \mathbf{i} - (b_1 c_3 - b_3 c_1) \mathbf{j} + (b_1 c_2 - b_2 c_1) \mathbf{k}.$$

$$\begin{aligned} \therefore \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) \cdot [(b_2 c_3 - b_3 c_2) \mathbf{i} \\ &\quad - (b_1 c_3 - b_3 c_1) \mathbf{j} + (b_1 c_2 - b_2 c_1) \mathbf{k}] \end{aligned}$$

$$= a_1 (b_2 c_3 - b_3 c_2) - a_2 (b_1 c_3 - b_3 c_1) + a_3 (b_1 c_2 - b_2 c_1)$$

$$[\because \mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1 \text{ and } \mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0]$$

$$\therefore [\mathbf{abc}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \quad \dots(1)$$

$$\text{Also } (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

showing that the value of a scalar triple product is independent of the positions of dot and cross.

Note: If OA, OB, OC be three concurrent edges of a parallelopiped and if $(a_1, a_2, a_3), (b_1, b_2, b_3), (c_1, c_2, c_3)$ be the rectangular coordinates of A, B, C referred to O as origin, then the determinant (1) gives the volume of that parallelopiped.

6 To Express the Scalar Triple Product $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$ in Terms of any Three non-Coplanar Vectors $\mathbf{l}, \mathbf{m}, \mathbf{n}$

Let $\mathbf{a} = a_1 \mathbf{l} + a_2 \mathbf{m} + a_3 \mathbf{n}, \mathbf{b} = b_1 \mathbf{l} + b_2 \mathbf{m} + b_3 \mathbf{n},$

and $\mathbf{c} = c_1 \mathbf{l} + c_2 \mathbf{m} + c_3 \mathbf{n}.$

$$\begin{aligned} \text{Now } \mathbf{b} \times \mathbf{c} &= (b_1 \mathbf{l} + b_2 \mathbf{m} + b_3 \mathbf{n}) \times (c_1 \mathbf{l} + c_2 \mathbf{m} + c_3 \mathbf{n}) \\ &= b_1 c_1 \mathbf{l} \times \mathbf{l} + b_1 c_2 \mathbf{l} \times \mathbf{m} + b_1 c_3 \mathbf{l} \times \mathbf{n} + b_2 c_1 \mathbf{m} \times \mathbf{l} \\ &\quad + b_2 c_2 \mathbf{m} \times \mathbf{m} + b_2 c_3 \mathbf{m} \times \mathbf{n} + b_3 c_1 \mathbf{n} \times \mathbf{l} \\ &\quad + b_3 c_2 \mathbf{n} \times \mathbf{m} + b_3 c_3 \mathbf{n} \times \mathbf{n} \\ &= (b_2 c_3 - b_3 c_2) \mathbf{m} \times \mathbf{n} - (b_1 c_3 - b_3 c_1) \mathbf{n} \times \mathbf{l} \\ &\quad + (b_1 c_2 - b_2 c_1) \mathbf{l} \times \mathbf{m} \\ &[\because \mathbf{l} \times \mathbf{l} = \mathbf{0} \text{ and } \mathbf{l} \times \mathbf{m} = -\mathbf{m} \times \mathbf{l} \text{ etc.}] \end{aligned}$$

$$\begin{aligned} \therefore \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= (a_1 \mathbf{l} + a_2 \mathbf{m} + a_3 \mathbf{n}) \cdot [(b_2 c_3 - b_3 c_2) \mathbf{m} \times \mathbf{n} \\ &\quad - (b_1 c_3 - b_3 c_1) \mathbf{n} \times \mathbf{l} + (b_1 c_2 - b_2 c_1) \mathbf{l} \times \mathbf{m}] \\ &= a_1 (b_2 c_3 - b_3 c_2) [\mathbf{lmn}] - a_2 (b_1 c_3 - b_3 c_1) [\mathbf{lmn}] \\ &\quad + a_3 (b_1 c_2 - b_2 c_1) [\mathbf{lmn}]. \end{aligned}$$

$[\because [\mathbf{lmn}] = [\mathbf{mnl}] = [\mathbf{nml}]$ and all the scalar triple products of the type $[\mathbf{lml}]$ in which two vectors are equal vanish]

$$\text{Hence, } [\mathbf{abc}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} [\mathbf{lmn}].$$

Note: Since $[\mathbf{i}, \mathbf{j}, \mathbf{k}] = 1$, therefore article 5 is particular case of article 6.

Illustrative Examples

Example 1: Find the volume of the parallelopiped whose edges are represented by

$$\mathbf{a} = 2\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}, \mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}, \mathbf{c} = 3\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}.$$

Solution: The required volume of the parallelopiped is equal to the absolute value of $[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$.

$$\begin{aligned} \text{Now} \quad [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] &= \begin{vmatrix} 2 & -4 & 5 \\ 1 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix} \\ &= 2(-2+5) + 4(2-3) + 5(-5+3) = 6 - 4 - 10 = -8. \end{aligned}$$

Neglecting the negative sign, we get the volume of the parallelopiped = 8 cubic units.

Example 2: Find the constant p such that the vectors $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{c} = 3\mathbf{i} + p\mathbf{j} + 5\mathbf{k}$ are coplanar. (Kumaun 2013)

Solution: If the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar, then we should have $[\mathbf{abc}] = 0$.

$$\begin{aligned} \text{Now} \quad [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] &= \begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & p & 5 \end{vmatrix} \\ &= 2(10 + 3p) + 1(5 + 9) + 1(p - 6) = 7p + 28. \end{aligned}$$

$\therefore [\mathbf{abc}]$ will be zero if $7p + 28 = 0$ or $p = -4$.

Hence for the given vectors to be coplanar, we should have $p = -4$.

Example 3: Prove that the four points $4\mathbf{i} + 5\mathbf{j} + \mathbf{k}$, $-(\mathbf{j} + \mathbf{k})$, $(3\mathbf{i} + 9\mathbf{j} + 4\mathbf{k})$ and $4(-\mathbf{i} + \mathbf{j} + \mathbf{k})$ are coplanar. (Kumaun 2012)

Solution: Let A, B, C, D be the four given points whose position vectors referred to some origin O are

$$4\mathbf{i} + 5\mathbf{j} + \mathbf{k}, -(\mathbf{j} + \mathbf{k}), (3\mathbf{i} + 9\mathbf{j} + 4\mathbf{k}) \text{ and } 4(-\mathbf{i} + \mathbf{j} + \mathbf{k}).$$

If the four points A, B, C, D are coplanar, then the vectors $\vec{AB}$, $\vec{AC}$ and $\vec{AD}$ should also be coplanar.

$$\begin{aligned} \text{We have} \quad \vec{AB} &= \text{position vector of } B - \text{position vector of } A \\ &= -(\mathbf{j} + \mathbf{k}) - (4\mathbf{i} + 5\mathbf{j} + \mathbf{k}) = -4\mathbf{i} - 6\mathbf{j} - 2\mathbf{k} = \mathbf{a} \quad (\text{say}). \end{aligned}$$

$$\text{Similarly} \quad \vec{AC} = (3\mathbf{i} + 9\mathbf{j} + 4\mathbf{k}) - (4\mathbf{i} + 5\mathbf{j} + \mathbf{k}) = -\mathbf{i} + 4\mathbf{j} + 3\mathbf{k} = \mathbf{b} \quad (\text{say}),$$

$$\text{and} \quad \vec{AD} = 4(-\mathbf{i} + \mathbf{j} + \mathbf{k}) - (4\mathbf{i} + 5\mathbf{j} + \mathbf{k}) = -8\mathbf{i} - \mathbf{j} + 3\mathbf{k} = \mathbf{c} \quad (\text{say}).$$

Now the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ will be coplanar if $[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 0$.

$$\begin{aligned}
 \text{Now} \quad [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] &= \begin{vmatrix} -4 & -6 & -2 \\ -1 & 4 & 3 \\ -8 & -1 & 3 \end{vmatrix} \\
 &= -4(12 + 3) + 6(-3 + 24) - 2(1 + 32) \\
 &= -60 + 126 - 66 = 0.
 \end{aligned}$$

$\therefore$ The points A, B, C, D are coplanar.

Example 4: Show that the four points $-\mathbf{a} + 4\mathbf{b} - 3\mathbf{c}$, $3\mathbf{a} + 2\mathbf{b} - 5\mathbf{c}$, $-3\mathbf{a} + 8\mathbf{b} - 5\mathbf{c}$ and $-3\mathbf{a} + 2\mathbf{b} + \mathbf{c}$ are coplanar.

Solution: Let A, B, C and D be the points whose position vectors are respectively $-\mathbf{a} + 4\mathbf{b} - 3\mathbf{c}$, $3\mathbf{a} + 2\mathbf{b} - 5\mathbf{c}$, $-3\mathbf{a} + 8\mathbf{b} - 5\mathbf{c}$ and $-3\mathbf{a} + 2\mathbf{b} + \mathbf{c}$.

$$\begin{aligned}
 \text{We have} \quad \vec{AB} &= \text{position vector of } B - \text{position vector of } A \\
 &= 3\mathbf{a} + 2\mathbf{b} - 5\mathbf{c} - (-\mathbf{a} + 4\mathbf{b} - 3\mathbf{c}) = 4\mathbf{a} - 2\mathbf{b} - 2\mathbf{c}, \\
 \vec{AC} &= \text{position vector of } C - \text{position vector of } A \\
 &= (-3\mathbf{a} + 8\mathbf{b} - 5\mathbf{c}) - (-\mathbf{a} + 4\mathbf{b} - 3\mathbf{c}) = -2\mathbf{a} + 4\mathbf{b} - 2\mathbf{c}, \\
 \text{and} \quad \vec{AD} &= (-3\mathbf{a} + 2\mathbf{b} + \mathbf{c}) - (-\mathbf{a} + 4\mathbf{b} - 3\mathbf{c}) = -2\mathbf{a} - 2\mathbf{b} + 4\mathbf{c}.
 \end{aligned}$$

Now the scalar triple product of the vectors $\vec{AB}$, $\vec{AC}$ and $\vec{AD}$

$$\begin{aligned}
 &= [\vec{AB}, \vec{AC}, \vec{AD}] = \begin{vmatrix} 4 & -2 & -2 \\ -2 & 4 & -2 \\ -2 & -2 & 4 \end{vmatrix} [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \quad [\text{Refer article 6}] \\
 &= \{4(16 - 4) + 2(-8 - 4) - 2(4 + 8)\} [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] \\
 &= (48 - 24 - 24) [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 0 \quad [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 0.
 \end{aligned}$$

Since the scalar triple product of the vectors $\vec{AB}$, $\vec{AC}$ and $\vec{AD}$ is zero, therefore these vectors are coplanar. Hence the points A, B, C and D are coplanar.

Example 5: If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are the position vectors of A, B, C prove that $\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}$ is a vector perpendicular to the plane of ABC .

Solution: We have $\vec{AB} = \mathbf{b} - \mathbf{a}$, $\vec{BC} = \mathbf{c} - \mathbf{b}$ and $\vec{CA} = \mathbf{a} - \mathbf{c}$.

$$\text{Let} \quad \mathbf{d} = \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}.$$

$$\begin{aligned}
 \text{Now} \quad \mathbf{d} \cdot \vec{AB} &= \mathbf{d} \cdot (\mathbf{b} - \mathbf{a}) = (\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a}) \\
 &= (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} - (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} + (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{b} - (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} \\
 &\quad + (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b} - (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{a} \\
 &= [\mathbf{abb}] - [\mathbf{aba}] + [\mathbf{bcb}] - [\mathbf{bca}] + [\mathbf{cab}] - [\mathbf{caa}] \\
 &= -[\mathbf{bca}] + [\mathbf{cab}], \text{ since } [\mathbf{abb}] = 0 \text{ etc.}
 \end{aligned}$$

$$= -[\mathbf{bca}] + [\mathbf{bca}], \text{ since } [\mathbf{cab}] = [\mathbf{bca}]$$

$$= 0.$$

Therefore vector $\mathbf{d}$ is perpendicular to $\overrightarrow{AB}$. Similarly, we can show that $\mathbf{d}$ is perpendicular to $\overrightarrow{BC}$.

Now since $\mathbf{d}$ is perpendicular to two lines in the plane ABC , hence it is perpendicular to the plane ABC .

Example 6: Prove that $[\mathbf{a} + \mathbf{b}, \mathbf{b} + \mathbf{c}, \mathbf{c} + \mathbf{a}] = 2[\mathbf{abc}]$. (Kumaun 2010)

Solution:

$$\begin{aligned} \text{L.H.S.} &= (\mathbf{a} + \mathbf{b}) \cdot [(\mathbf{b} + \mathbf{c}) \times (\mathbf{c} + \mathbf{a})] \\ &= (\mathbf{a} + \mathbf{b}) \cdot [\mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}] \\ &= (\mathbf{a} + \mathbf{b}) \cdot [\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{b} \times \mathbf{a}], \text{ since } \mathbf{c} \times \mathbf{c} = \mathbf{0} \\ &= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mathbf{a} \cdot (\mathbf{c} \times \mathbf{a}) + \mathbf{a} \cdot (\mathbf{b} \times \mathbf{a}) + \mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) \\ &\quad + \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) + \mathbf{b} \cdot (\mathbf{b} \times \mathbf{a}) \\ &= [\mathbf{abc}] + [\mathbf{aca}] + [\mathbf{aba}] + [\mathbf{bbc}] + [\mathbf{bca}] + [\mathbf{bba}] = [\mathbf{abc}] + [\mathbf{bca}], \end{aligned}$$

since all the scalar triple products in which two vectors are equal vanish.

But $[\mathbf{abc}] = [\mathbf{bca}]$.

Hence the L.H.S. = $2[\mathbf{abc}]$.

Example 7: Prove that $[\mathbf{lmn}][\mathbf{abc}] = \begin{vmatrix} \mathbf{l} \cdot \mathbf{a} & \mathbf{l} \cdot \mathbf{b} & \mathbf{l} \cdot \mathbf{c} \\ \mathbf{m} \cdot \mathbf{a} & \mathbf{m} \cdot \mathbf{b} & \mathbf{m} \cdot \mathbf{c} \\ \mathbf{n} \cdot \mathbf{a} & \mathbf{n} \cdot \mathbf{b} & \mathbf{n} \cdot \mathbf{c} \end{vmatrix}$.

Solution:

Let $\mathbf{l} = l_1 \mathbf{i} + l_2 \mathbf{j} + l_3 \mathbf{k}$, $\mathbf{m} = m_1 \mathbf{i} + m_2 \mathbf{j} + m_3 \mathbf{k}$,

$\mathbf{n} = n_1 \mathbf{i} + n_2 \mathbf{j} + n_3 \mathbf{k}$; $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$,

$\mathbf{b} = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}$, $\mathbf{c} = c_1 \mathbf{i} + c_2 \mathbf{j} + c_3 \mathbf{k}$.

Now

L.H.S. = $[\mathbf{lmn}][\mathbf{abc}]$

$$\begin{aligned} &= \begin{vmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \\ &= \begin{vmatrix} l_1 a_1 + l_2 a_2 + l_3 a_3 & l_1 b_1 + l_2 b_2 + l_3 b_3 & l_1 c_1 + l_2 c_2 + l_3 c_3 \\ m_1 a_1 + m_2 a_2 + m_3 a_3 & m_1 b_1 + m_2 b_2 + m_3 b_3 & m_1 c_1 + m_2 c_2 + m_3 c_3 \\ n_1 a_1 + n_2 a_2 + n_3 a_3 & n_1 b_1 + n_2 b_2 + n_3 b_3 & n_1 c_1 + n_2 c_2 + n_3 c_3 \end{vmatrix} \end{aligned}$$

by the rule for the multiplication of determinants of the same order.

$$\begin{aligned}\text{Now } \mathbf{l} \cdot \mathbf{a} &= (l_1 \mathbf{i} + l_2 \mathbf{j} + l_3 \mathbf{k}) \cdot (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) \\ &= l_1 a_1 + l_2 a_2 + l_3 a_3, \text{ etc.}\end{aligned}$$

$$\text{Hence the L.H.S.} = \begin{vmatrix} \mathbf{l} \cdot \mathbf{a} & \mathbf{l} \cdot \mathbf{b} & \mathbf{l} \cdot \mathbf{c} \\ \mathbf{m} \cdot \mathbf{a} & \mathbf{m} \cdot \mathbf{b} & \mathbf{m} \cdot \mathbf{c} \\ \mathbf{n} \cdot \mathbf{a} & \mathbf{n} \cdot \mathbf{b} & \mathbf{n} \cdot \mathbf{c} \end{vmatrix}.$$

Example 8: Prove that if $\mathbf{l}, \mathbf{m}, \mathbf{n}$ be three non-coplanar vectors, then

$$[\mathbf{lmn}](\mathbf{a} \times \mathbf{b}) = \begin{vmatrix} \mathbf{l} \cdot \mathbf{a} & \mathbf{l} \cdot \mathbf{b} & \mathbf{l} \\ \mathbf{m} \cdot \mathbf{a} & \mathbf{m} \cdot \mathbf{b} & \mathbf{m} \\ \mathbf{n} \cdot \mathbf{a} & \mathbf{n} \cdot \mathbf{b} & \mathbf{n} \end{vmatrix}.$$

Solution: Let

$$\mathbf{l} = l_1 \mathbf{i} + l_2 \mathbf{j} + l_3 \mathbf{k}, \quad \mathbf{m} = m_1 \mathbf{i} + m_2 \mathbf{j} + m_3 \mathbf{k}, \quad \mathbf{n} = n_1 \mathbf{i} + n_2 \mathbf{j} + n_3 \mathbf{k},$$

$$\text{and } \mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}, \quad \mathbf{b} = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}.$$

$$\text{Now } [\mathbf{lmn}] = \begin{vmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix} \quad \text{and } (\mathbf{a} \times \mathbf{b}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

$$\begin{aligned}\therefore [\mathbf{lmn}](\mathbf{a} \times \mathbf{b}) &= \begin{vmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= \begin{vmatrix} l_1 \mathbf{i} + l_2 \mathbf{j} + l_3 \mathbf{k} & l_1 a_1 + l_2 a_2 + l_3 a_3 & l_1 b_1 + l_2 b_2 + l_3 b_3 \\ m_1 \mathbf{i} + m_2 \mathbf{j} + m_3 \mathbf{k} & m_1 a_1 + m_2 a_2 + m_3 a_3 & m_1 b_1 + m_2 b_2 + m_3 b_3 \\ n_1 \mathbf{i} + n_2 \mathbf{j} + n_3 \mathbf{k} & n_1 a_1 + n_2 a_2 + n_3 a_3 & n_1 b_1 + n_2 b_2 + n_3 b_3 \end{vmatrix}.\end{aligned}$$

$$\begin{aligned}\text{Now } \mathbf{l} \cdot \mathbf{a} &= (l_1 \mathbf{i} + l_2 \mathbf{j} + l_3 \mathbf{k}) \cdot (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) \\ &= l_1 a_1 + l_2 a_2 + l_3 a_3 \text{ etc.}\end{aligned}$$

$$\therefore [\mathbf{lmn}](\mathbf{a} \times \mathbf{b}) = \begin{vmatrix} \mathbf{l} & \mathbf{l} \cdot \mathbf{a} & \mathbf{l} \cdot \mathbf{b} \\ \mathbf{m} & \mathbf{m} \cdot \mathbf{a} & \mathbf{m} \cdot \mathbf{b} \\ \mathbf{n} & \mathbf{n} \cdot \mathbf{a} & \mathbf{n} \cdot \mathbf{b} \end{vmatrix} = \begin{vmatrix} \mathbf{l} \cdot \mathbf{a} & \mathbf{l} \cdot \mathbf{b} & \mathbf{l} \\ \mathbf{m} \cdot \mathbf{a} & \mathbf{m} \cdot \mathbf{b} & \mathbf{m} \\ \mathbf{n} \cdot \mathbf{a} & \mathbf{n} \cdot \mathbf{b} & \mathbf{n} \end{vmatrix}.$$

Comprehensive Exercise 1

1. Define scalar triple product of three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and interpret the same geometrically.
2. Define scalar triple product of $\mathbf{u}, \mathbf{v}, \mathbf{w}$. Prove that the value of the scalar triple product of $\mathbf{u}, \mathbf{v}, \mathbf{w}$ remains unchanged if the cyclic order of the vectors is maintained.

3. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are any three vectors, prove that

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}).$$
4. Prove that $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$.
5. Show that $\mathbf{i} \cdot \mathbf{j} \times \mathbf{k} = 1$.
6. Show that $[\lambda \mathbf{a} + \mu \mathbf{b}, \mathbf{c}, \mathbf{d}] = \lambda [\mathbf{a}, \mathbf{c}, \mathbf{d}] + \mu [\mathbf{b}, \mathbf{c}, \mathbf{d}]$.
7. Prove that $[\mathbf{i} - \mathbf{j}, \mathbf{j} - \mathbf{k}, \mathbf{k} - \mathbf{i}] = 0$.
8. Find the volume of the parallelopiped whose edges are represented by
 (i) $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}, \mathbf{b} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}, \mathbf{c} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.
 (ii) $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}, \mathbf{b} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}, \mathbf{c} = \mathbf{j} + \mathbf{k}$.
9. Show that the vectors $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}, -2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}, \mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ are coplanar.
10. Show that the vectors $2\mathbf{a} - \mathbf{b} + 3\mathbf{c}, \mathbf{a} + \mathbf{b} - 2\mathbf{c}$ and $\mathbf{a} + \mathbf{b} - 3\mathbf{c}$ are non-coplanar where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are non-coplanar vectors.
 [Hint: Show that the scalar triple product of the three given vectors is not zero.]
11. Prove that the four points $6\mathbf{a} - 4\mathbf{b} + 10\mathbf{c}, -5\mathbf{a} + 3\mathbf{b} - 10\mathbf{c}, 4\mathbf{a} - 6\mathbf{b} - 10\mathbf{c}$ and $2\mathbf{b} + 10\mathbf{c}$ are coplanar.
12. Show that the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar if $\mathbf{b} + \mathbf{c}, \mathbf{c} + \mathbf{a}, \mathbf{a} + \mathbf{b}$ are coplanar.
13. Prove that four points with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ are coplanar if and only if $[\mathbf{b}, \mathbf{c}, \mathbf{d}] + [\mathbf{c}, \mathbf{a}, \mathbf{d}] + [\mathbf{a}, \mathbf{b}, \mathbf{d}] = [\mathbf{a}, \mathbf{b}, \mathbf{c}]$.

Answers 1

8. (i) 7 cubic units, (ii) 12 cubic units.

7 Vector Triple Product

The vector product of two vectors one of which is itself the vector product of two vectors is a vector quantity called a “Vector Triple Product”. Thus if $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ be three vectors, the products of the form $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ and $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ etc. are called “Vector Triple Products”.

Theorem: To prove that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$.

Let $\mathbf{r} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ and $\mathbf{b} \times \mathbf{c} = \mathbf{d}$.

Since $\mathbf{b} \times \mathbf{c} = \mathbf{d}$, therefore $\mathbf{d}$ is a vector perpendicular to the plane containing $\mathbf{b}$ and $\mathbf{c}$. Also $\mathbf{r} = \mathbf{a} \times \mathbf{d}$. Therefore $\mathbf{r}$ is a vector perpendicular to both $\mathbf{a}$ and $\mathbf{d}$. Now the vector $\mathbf{r}$ is perpendicular to the vector $\mathbf{d}$, whereas the vector $\mathbf{d}$ is perpendicular to the plane containing $\mathbf{b}$ and $\mathbf{c}$. Therefore the vector $\mathbf{r}$ must lie in the plane containing

b and **c**. Hence the vector **r** can be expressed linearly in terms of **b** and **c** in the form

$$\mathbf{r} = l\mathbf{b} + m\mathbf{c}, \quad \dots(1)$$

where l and m are scalars.

Since **r** is perpendicular to **a**, therefore $\mathbf{r} \cdot \mathbf{a} = 0$.

$$\therefore (l\mathbf{b} + m\mathbf{c}) \cdot \mathbf{a} = 0 \quad \text{or} \quad l(\mathbf{b} \cdot \mathbf{a}) + m(\mathbf{c} \cdot \mathbf{a}) = 0.$$

$$\therefore \frac{l}{\mathbf{c} \cdot \mathbf{a}} = \frac{-m}{\mathbf{b} \cdot \mathbf{a}} = \lambda \text{ (say).}$$

Putting the values of l and m in (1), we get

$$\mathbf{r} = \lambda(\mathbf{c} \cdot \mathbf{a})\mathbf{b} - \lambda(\mathbf{b} \cdot \mathbf{a})\mathbf{c} = \lambda[(\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\mathbf{c}] \quad \dots(2)$$

Now we are to find the value of λ .

Consider unit vectors **j** and **k**, the first parallel to **b** and the second perpendicular to it in the plane containing **b** and **c**. Then we may write $\mathbf{b} = b_2\mathbf{j}$ and $\mathbf{c} = c_2\mathbf{j} + c_3\mathbf{k}$.

In terms of **j** and **k** and the other unit vector **i** of the right handed system, the remaining vector **a** may be written as

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}.$$

$$\begin{aligned} \text{Now } \mathbf{b} \times \mathbf{c} &= b_2\mathbf{j} \times (c_2\mathbf{j} + c_3\mathbf{k}) = b_2c_2\mathbf{j} \times \mathbf{j} + b_2c_3\mathbf{j} \times \mathbf{k} = b_2c_3\mathbf{i} \\ &[\because \mathbf{j} \times \mathbf{j} = \mathbf{0} \text{ and } \mathbf{j} \times \mathbf{k} = \mathbf{i}] \end{aligned}$$

$$\begin{aligned} \therefore \mathbf{r} &= \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \times (b_2c_3\mathbf{i}) \\ &= a_1b_2c_3\mathbf{i} \times \mathbf{i} + a_2b_2c_3\mathbf{j} \times \mathbf{i} + a_3b_2c_3\mathbf{k} \times \mathbf{i} \\ &= a_3b_2c_3\mathbf{j} - a_2b_2c_3\mathbf{k} \quad \dots(3) \\ &[\because \mathbf{i} \times \mathbf{i} = \mathbf{0}, \mathbf{j} \times \mathbf{i} = -\mathbf{k} \text{ and } \mathbf{k} \times \mathbf{i} = \mathbf{j}] \end{aligned}$$

$$\begin{aligned} \text{Also } \mathbf{r} &= \lambda[(\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\mathbf{c}] \\ &= \lambda[(c_2\mathbf{j} + c_3\mathbf{k}) \cdot (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k})b_2\mathbf{j} \\ &\quad - (b_2\mathbf{j}) \cdot (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k})(c_2\mathbf{j} + c_3\mathbf{k})] \\ &= \lambda[c_2a_2b_2\mathbf{j} + c_3a_3b_2\mathbf{j} - b_2a_2c_2\mathbf{j} - b_2a_2c_3\mathbf{k}] \\ &[\because \mathbf{i} \cdot \mathbf{i} = 1, \mathbf{i} \cdot \mathbf{j} = 0 \text{ etc.}] \\ &= \lambda[a_3b_2c_3\mathbf{j} - a_2b_2c_3\mathbf{k}]. \quad \dots(4) \end{aligned}$$

Now from (3) and (4) we conclude that $\lambda = 1$.

$$\begin{aligned} \text{Hence } \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} \\ &[\because \mathbf{c} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{c}] \end{aligned}$$

$$\begin{aligned} \text{Corollary: } (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} &= -[\mathbf{c} \times (\mathbf{a} \times \mathbf{b})] \\ &= -[(\mathbf{c} \cdot \mathbf{b})\mathbf{a} - (\mathbf{c} \cdot \mathbf{a})\mathbf{b}] = (\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{c} \cdot \mathbf{b})\mathbf{a}. \end{aligned}$$

Rule to remember $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$. It is a vector to be expressed linearly in terms of **b** and **c** which are the vectors within the brackets. Also

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= [\text{Dot product of } \mathbf{a} \text{ and } \mathbf{c}]\mathbf{b} \\ &\quad - [\text{Dot product of } \mathbf{a} \text{ and } \mathbf{b}]\mathbf{c}. \end{aligned}$$

Similarly we may remember $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$.

8 Vector Triple Product is not Associative

If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be three vectors, then $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ gives a vector which lies in the plane of $\mathbf{b}$ and $\mathbf{c}$ and which is perpendicular to $\mathbf{a}$. Moreover $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ gives a vector which lies in the plane of $\mathbf{a}$ and $\mathbf{b}$ and which is perpendicular to $\mathbf{c}$. Hence, in general, $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \neq (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$. Thus in the case of vector triple product the position of brackets cannot be, in general, changed without altering the value of the product.

Illustrative Examples

Example 9: Prove that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$.

Solution: We have

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}, \quad \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) = (\mathbf{b} \cdot \mathbf{a}) \mathbf{c} - (\mathbf{b} \cdot \mathbf{c}) \mathbf{a},$$

and

$$\mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = (\mathbf{c} \cdot \mathbf{b}) \mathbf{a} - (\mathbf{c} \cdot \mathbf{a}) \mathbf{b}.$$

Adding these three expressions, we get

$$\begin{aligned} & \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) \\ &= (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} + (\mathbf{b} \cdot \mathbf{a}) \mathbf{c} - (\mathbf{b} \cdot \mathbf{c}) \mathbf{a} + (\mathbf{c} \cdot \mathbf{b}) \mathbf{a} - (\mathbf{c} \cdot \mathbf{a}) \mathbf{b} \\ &= \mathbf{0}. \quad [\because \mathbf{a} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a}, \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}, \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{b}] \end{aligned}$$

Example 10: Show that the vectors $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$, $\mathbf{b} \times (\mathbf{c} \times \mathbf{a})$, $\mathbf{c} \times (\mathbf{a} \times \mathbf{b})$ are coplanar.

Solution: Let $\mathbf{r}_1 = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$, $\mathbf{r}_2 = \mathbf{b} \times (\mathbf{c} \times \mathbf{a})$, $\mathbf{r}_3 = \mathbf{c} \times (\mathbf{a} \times \mathbf{b})$.

Now first prove that $\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 = \mathbf{0}$, as we have done in the previous exercise.

Since there exists a linear relation between the vectors $\mathbf{r}_1$, $\mathbf{r}_2$, $\mathbf{r}_3$ therefore any of these vectors can be expressed as a linear combination of the other two. Hence these three vectors are coplanar.

Example 11: If $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{c} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, find $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.

Solution: We have

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} \\ &= [(\mathbf{i} - 2\mathbf{j} + \mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j} - \mathbf{k})] (2\mathbf{i} + \mathbf{j} + \mathbf{k}) \\ &\quad - [(\mathbf{i} - 2\mathbf{j} + \mathbf{k}) \cdot (2\mathbf{i} + \mathbf{j} + \mathbf{k})] (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) \\ &= (1 - 4 - 1) (2\mathbf{i} + \mathbf{j} + \mathbf{k}) - (2 - 2 + 1) (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) \\ &= (-8\mathbf{i} - 4\mathbf{j} - 4\mathbf{k}) - (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = -9\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}. \end{aligned}$$

Example 12: Show that $\mathbf{i} \times (\mathbf{a} \times \mathbf{i}) + \mathbf{j} \times (\mathbf{a} \times \mathbf{j}) + \mathbf{k} \times (\mathbf{a} \times \mathbf{k}) = 2\mathbf{a}$.

(Kumaun 2009)

Solution: We have

$$\begin{aligned} \mathbf{i} \times (\mathbf{a} \times \mathbf{i}) &= (\mathbf{i} \cdot \mathbf{i}) \mathbf{a} - (\mathbf{i} \cdot \mathbf{a}) \mathbf{i} = \mathbf{a} - (\mathbf{i} \cdot \mathbf{a}) \mathbf{i} & [\because \mathbf{i} \cdot \mathbf{i} = 1] \\ \mathbf{j} \times (\mathbf{a} \times \mathbf{j}) &= (\mathbf{j} \cdot \mathbf{j}) \mathbf{a} - (\mathbf{j} \cdot \mathbf{a}) \mathbf{j} = \mathbf{a} - (\mathbf{j} \cdot \mathbf{a}) \mathbf{j} & [\because \mathbf{j} \cdot \mathbf{j} = 1] \end{aligned}$$

and $\mathbf{k} \times (\mathbf{a} \times \mathbf{k}) = (\mathbf{k} \cdot \mathbf{k}) \mathbf{a} - (\mathbf{k} \cdot \mathbf{a}) \mathbf{k} = \mathbf{a} - (\mathbf{k} \cdot \mathbf{a}) \mathbf{k}. \quad [\because \mathbf{k} \cdot \mathbf{k} = 1]$

Adding these three expressions, we get

$$\begin{aligned} & \mathbf{i} \times (\mathbf{a} \times \mathbf{i}) + \mathbf{j} \times (\mathbf{a} \times \mathbf{j}) + \mathbf{k} \times (\mathbf{a} \times \mathbf{k}) \\ &= 3\mathbf{a} - (\mathbf{i} \cdot \mathbf{a}) \mathbf{i} - (\mathbf{j} \cdot \mathbf{a}) \mathbf{j} - (\mathbf{k} \cdot \mathbf{a}) \mathbf{k} \\ &= 3\mathbf{a} - [(\mathbf{a} \cdot \mathbf{i}) \mathbf{i} + (\mathbf{a} \cdot \mathbf{j}) \mathbf{j} + (\mathbf{a} \cdot \mathbf{k}) \mathbf{k}]. \quad [\because \mathbf{a} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{a} \text{ etc.}] \end{aligned}$$

Now we shall show that

$$\mathbf{a} = (\mathbf{a} \cdot \mathbf{i}) \mathbf{i} + (\mathbf{a} \cdot \mathbf{j}) \mathbf{j} + (\mathbf{a} \cdot \mathbf{k}) \mathbf{k}.$$

Let $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$

Taking dot product of both sides with $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$ successively, we get

$$x = \mathbf{a} \cdot \mathbf{i}, \quad y = \mathbf{a} \cdot \mathbf{j}, \quad z = \mathbf{a} \cdot \mathbf{k}.$$

$$\therefore \mathbf{a} = (\mathbf{a} \cdot \mathbf{i}) \mathbf{i} + (\mathbf{a} \cdot \mathbf{j}) \mathbf{j} + (\mathbf{a} \cdot \mathbf{k}) \mathbf{k}.$$

$$\text{Hence } \mathbf{i} \times (\mathbf{a} \times \mathbf{i}) + \mathbf{j} \times (\mathbf{a} \times \mathbf{j}) + \mathbf{k} \times (\mathbf{a} \times \mathbf{k}) = 3\mathbf{a} - \mathbf{a} = 2\mathbf{a}.$$

Example 13: Show that $\mathbf{i} \times (\mathbf{j} \times \mathbf{k}) = \mathbf{0}.$

Solution: We have $\mathbf{i} \times (\mathbf{j} \times \mathbf{k}) = \mathbf{i} \times \mathbf{i} \quad [\because \mathbf{j} \times \mathbf{k} = \mathbf{i}]$
 $= \mathbf{0}. \quad [\because \mathbf{i} \times \mathbf{i} = \mathbf{0}]$

Example 14: Show that $[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = [\mathbf{abc}]^2$, and express the result by means of determinants. (Kumaun 2011)

Solution: We have

$$[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = (\mathbf{a} \times \mathbf{b}) \cdot [(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})].$$

Let us first find the value of $(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})$.

Let $\mathbf{b} \times \mathbf{c} = \mathbf{d}.$

$$\begin{aligned} \text{Then } (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) &= \mathbf{d} \times (\mathbf{c} \times \mathbf{a}) = (\mathbf{d} \cdot \mathbf{a}) \mathbf{c} - (\mathbf{d} \cdot \mathbf{c}) \mathbf{a} \\ &= [(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}] \mathbf{c} - [(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{c}] \mathbf{a} \\ &= [\mathbf{bca}] \mathbf{c} - [\mathbf{bcc}] \mathbf{a} = [\mathbf{abc}] \mathbf{c}, \\ &\quad \text{since } [\mathbf{bcc}] = 0 \text{ and } [\mathbf{bca}] = [\mathbf{abc}]. \end{aligned}$$

$$\begin{aligned} \therefore [\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] &= (\mathbf{a} \times \mathbf{b}) \cdot [\mathbf{abc}] \mathbf{c} \\ &= [\mathbf{abc}] (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = [\mathbf{abc}] [\mathbf{abc}] = [\mathbf{abc}]^2. \end{aligned}$$

Second part: Let

$$\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}, \quad \mathbf{b} = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k},$$

$$\mathbf{c} = c_1 \mathbf{i} + c_2 \mathbf{j} + c_3 \mathbf{k}.$$

$$\text{We have } [\mathbf{abc}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

$$\text{Again } \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2 b_3 - b_2 a_3) \mathbf{i} + (b_1 a_3 - a_1 b_3) \mathbf{j} + (a_1 b_2 - a_2 b_1) \mathbf{k}.$$

Similarly $\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

$$= (b_2 c_3 - b_3 c_2) \mathbf{i} + (c_1 b_3 - b_1 c_3) \mathbf{j} + (b_1 c_2 - c_1 b_2) \mathbf{k}$$

and

$$\mathbf{c} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$= (c_2 a_3 - a_2 c_3) \mathbf{i} + (a_1 c_3 - a_3 c_1) \mathbf{j} + (c_1 a_2 - a_1 c_2) \mathbf{k}.$$

$$\therefore [\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}]$$

$$= \begin{vmatrix} a_2 b_3 - b_2 a_3 & b_1 a_3 - a_1 b_3 & a_1 b_2 - a_2 b_1 \\ b_2 c_3 - b_3 c_2 & c_1 b_3 - b_1 c_3 & b_1 c_2 - b_2 c_1 \\ c_2 a_3 - c_3 a_2 & a_1 c_3 - a_3 c_1 & c_1 a_2 - c_2 a_1 \end{vmatrix}$$

$$= \begin{vmatrix} C_1 & C_2 & C_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix},$$

where the capital letters A_1, A_2, A_3 etc. denote the cofactors of the corresponding

small letters a_1, a_2, a_3 etc. in the determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$

Since $[\mathbf{abc}]^2 = [\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}],$

$$\therefore \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2 = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}.$$

Example 15: Prove that $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{a} \times \mathbf{c}) \cdot \mathbf{d} = (\mathbf{a} \cdot \mathbf{d}) [\mathbf{abc}].$

Solution: Let $\mathbf{a} \times \mathbf{b} = \mathbf{r}.$

Then $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{a} \times \mathbf{c}) = \mathbf{r} \times (\mathbf{a} \times \mathbf{c}) = (\mathbf{r} \cdot \mathbf{c}) \mathbf{a} - (\mathbf{r} \cdot \mathbf{a}) \mathbf{c}$

$$= [(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}] \mathbf{a} - [(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a}] \mathbf{c}$$

$$= [\mathbf{abc}] \mathbf{a} - [\mathbf{aba}] \mathbf{c} = [\mathbf{abc}] \mathbf{a}, \text{ since } [\mathbf{aba}] = 0.$$

Therefore $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{a} \times \mathbf{c}) \cdot \mathbf{d} = [\mathbf{abc}] \mathbf{a} \cdot \mathbf{d} = (\mathbf{a} \cdot \mathbf{d}) [\mathbf{abc}].$

Example 16: If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be three unit vectors such that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \frac{1}{2} \mathbf{b}$, find the angles which $\mathbf{a}$ makes with $\mathbf{b}$ and $\mathbf{c}$, $\mathbf{b}$ and $\mathbf{c}$ being non-parallel. (Kumaun 2009)

Solution: It is given that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \frac{1}{2} \mathbf{b}.$

$$\therefore (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} = \frac{1}{2} \mathbf{b},$$

$$\text{or} \quad \left(\mathbf{a} \cdot \mathbf{c} - \frac{1}{2} \right) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} = \mathbf{0}. \quad \dots(1)$$

Since $\mathbf{b}$ and $\mathbf{c}$ are non-parallel, therefore for the existence of the relation (1) the coefficients of $\mathbf{b}$ and $\mathbf{c}$ should vanish separately. Therefore, we get

$$\mathbf{a} \cdot \mathbf{c} - \frac{1}{2} = 0, \quad \text{i.e.,} \quad \mathbf{a} \cdot \mathbf{c} = \frac{1}{2} \quad \text{and} \quad \mathbf{a} \cdot \mathbf{b} = 0.$$

Let θ and ϕ be the angles which $\mathbf{a}$ makes with $\mathbf{b}$ and $\mathbf{c}$ respectively. Since $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors, we have

$$\mathbf{a} \cdot \mathbf{b} = \cos \theta = 0 \quad \Rightarrow \quad \theta = 90^\circ,$$

$$\text{and} \quad \mathbf{a} \cdot \mathbf{c} = \cos \phi = \frac{1}{2} \quad \Rightarrow \quad \phi = 60^\circ.$$

Example 17: Prove that $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$, if and only if $(\mathbf{c} \times \mathbf{a}) \times \mathbf{b} = \mathbf{0}$.

Solution: We have $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

$$\text{if and only if} \quad (\mathbf{c} \cdot \mathbf{a}) \mathbf{b} - (\mathbf{c} \cdot \mathbf{b}) \mathbf{a} = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$$

$$\text{i.e., if and only if} \quad -(\mathbf{c} \cdot \mathbf{b}) \mathbf{a} = -(\mathbf{a} \cdot \mathbf{b}) \mathbf{c}, \quad \text{since } \mathbf{c} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{c}$$

$$\text{i.e., if and only if} \quad (\mathbf{c} \cdot \mathbf{b}) \mathbf{a} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} = \mathbf{0}$$

$$\text{i.e., if and only if} \quad (\mathbf{b} \cdot \mathbf{c}) \mathbf{a} - (\mathbf{b} \cdot \mathbf{a}) \mathbf{c} = \mathbf{0}, \quad \text{since } \mathbf{c} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{c} \text{ and } \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$$

$$\text{i.e., if and only if} \quad (\mathbf{c} \times \mathbf{a}) \times \mathbf{b} = \mathbf{0}.$$

Note: $(\mathbf{c} \times \mathbf{a}) \times \mathbf{b} = \mathbf{0}$ is possible when (i) $\mathbf{a}$ and $\mathbf{c}$ are collinear because then $\mathbf{c} \times \mathbf{a} = \mathbf{0}$ or (ii) $\mathbf{b}$ is parallel to $\mathbf{c} \times \mathbf{a}$ i.e., $\mathbf{b}$ is perpendicular to both $\mathbf{c}$ and $\mathbf{a}$ or (iii) at least one of the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is a null vector.

Comprehensive Exercise 2

1. Evaluate $(\mathbf{b} \times \mathbf{c}) \times \mathbf{a}$,

where $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}$, $\mathbf{b} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$, and $\mathbf{c} = 4\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$.

2. (i) Verify the formula for vector triple product

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$$

by taking $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = -\mathbf{i} + 2\mathbf{k}$, $\mathbf{c} = \mathbf{j} + \mathbf{k}$.

- (ii) Verify $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$ for $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$

and $\mathbf{c} = 3\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$.

(Kumaun 2014)

3. Prove that $(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) = [\mathbf{abc}] \mathbf{c}$.

4. Prove that for any three vectors $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$,

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C}) \times (\mathbf{C} \times \mathbf{A}) = (\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})^2.$$

5. Prove that for any three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, $[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = [\mathbf{abc}]^2$.

Hence show that the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are non-coplanar if and only if the vectors $\mathbf{a} \times \mathbf{b}$, $\mathbf{b} \times \mathbf{c}$, $\mathbf{c} \times \mathbf{a}$ are non-coplanar.

6. Show that $[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = \begin{vmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c} \cdot \mathbf{c} \end{vmatrix} = [\mathbf{abc}]^2$.

[Hint: First prove that $[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = [\mathbf{abc}]^2$.

For proof see Ex. 6. For the next part proceed as in Ex. 7.]

7. Prove that $\mathbf{a} \times (\mathbf{b} \times \mathbf{a}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{a}$.

Answers 2

1. $8(-4\mathbf{i} + \mathbf{j} - \mathbf{k})$.

9 Scalar Product of Four Vectors

If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$ are four vectors, the products $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$, $(\mathbf{a} \times \mathbf{d}) \cdot (\mathbf{b} \times \mathbf{c})$ etc. are called scalar products of four vectors.

Theorem: To prove that

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = \begin{vmatrix} \mathbf{a} \cdot \mathbf{c} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{a} \cdot \mathbf{d} & \mathbf{b} \cdot \mathbf{d} \end{vmatrix}.$$

Let $\mathbf{a} \times \mathbf{b} = \mathbf{r}$.

Then $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = \mathbf{r} \cdot (\mathbf{c} \times \mathbf{d})$.

Now in a scalar triple product the position of dot and cross may be interchanged without altering the value of the product.

Therefore $\mathbf{r} \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{r} \times \mathbf{c}) \cdot \mathbf{d}$.

$$\begin{aligned} \therefore (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) &= [(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}] \cdot \mathbf{d} = [(\mathbf{c} \cdot \mathbf{a}) \mathbf{b} - (\mathbf{c} \cdot \mathbf{b}) \mathbf{a}] \cdot \mathbf{d} \\ &= (\mathbf{c} \cdot \mathbf{a}) (\mathbf{b} \cdot \mathbf{d}) - (\mathbf{c} \cdot \mathbf{b}) (\mathbf{a} \cdot \mathbf{d}) \\ &= (\mathbf{a} \cdot \mathbf{c}) (\mathbf{b} \cdot \mathbf{d}) - (\mathbf{b} \cdot \mathbf{c}) (\mathbf{a} \cdot \mathbf{d}) = \begin{vmatrix} \mathbf{a} \cdot \mathbf{c} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{a} \cdot \mathbf{d} & \mathbf{b} \cdot \mathbf{d} \end{vmatrix}. \end{aligned}$$

This relation is known as **Lagrange's Identity**.

10 Vector Product of Four Vectors

Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$ be four vectors. Consider the vector product of the vectors $\mathbf{a} \times \mathbf{b}$ and $\mathbf{c} \times \mathbf{d}$. This product can be written as $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d})$ and is called the vector

product of four vectors. It is a vector perpendicular to $\mathbf{a} \times \mathbf{b}$ and, therefore coplanar with $\mathbf{a}$ and $\mathbf{b}$. Similarly it is a vector coplanar with $\mathbf{c}$ and $\mathbf{d}$. Hence this vector must be parallel to the line of intersection of a plane parallel to $\mathbf{a}$ and $\mathbf{b}$ with another plane parallel to $\mathbf{c}$ and $\mathbf{d}$.

Theorem: *To prove that*

$$(i) \quad (\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = [\mathbf{abd}] \mathbf{c} - [\mathbf{abc}] \mathbf{d}$$

$$(ii) \quad (\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = [\mathbf{acd}] \mathbf{b} - [\mathbf{bcd}] \mathbf{a}.$$

Proof:

(i) $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d})$ is a vector which can be either expressed in terms of $\mathbf{c}$ and $\mathbf{d}$ or in terms of $\mathbf{a}$ and $\mathbf{b}$. To express it in terms of $\mathbf{c}$ and $\mathbf{d}$, let us put $\mathbf{a} \times \mathbf{b} = \mathbf{l}$. Then

$$\begin{aligned} (\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) &= \mathbf{l} \times (\mathbf{c} \times \mathbf{d}) = (\mathbf{l} \cdot \mathbf{d}) \mathbf{c} - (\mathbf{l} \cdot \mathbf{c}) \mathbf{d} \\ &= [(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{d}] \mathbf{c} - [(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}] \mathbf{d} \\ &= [\mathbf{abd}] \mathbf{c} - [\mathbf{abc}] \mathbf{d}. \end{aligned}$$

(ii) Again to express $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d})$ in terms of $\mathbf{a}$ and $\mathbf{b}$, let us put $\mathbf{c} \times \mathbf{d} = \mathbf{m}$.

$$\begin{aligned} \text{Then} \quad (\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) &= (\mathbf{a} \times \mathbf{b}) \times \mathbf{m} = -\mathbf{m} \times (\mathbf{a} \times \mathbf{b}) \\ &= -[(\mathbf{m} \cdot \mathbf{b}) \mathbf{a} - (\mathbf{m} \cdot \mathbf{a}) \mathbf{b}] = (\mathbf{m} \cdot \mathbf{a}) \mathbf{b} - (\mathbf{m} \cdot \mathbf{b}) \mathbf{a} \\ &= [(\mathbf{c} \times \mathbf{d}) \cdot \mathbf{a}] \mathbf{b} - [(\mathbf{c} \times \mathbf{d}) \cdot \mathbf{b}] \mathbf{a} \\ &= [\mathbf{cda}] \mathbf{b} - [\mathbf{cdb}] \mathbf{a} = [\mathbf{acd}] \mathbf{b} - [\mathbf{bcd}] \mathbf{a}. \end{aligned}$$

Linear Relation connecting four vectors: Equating the above two expressions for the value of $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d})$, we get

$$[\mathbf{abd}] \mathbf{c} - [\mathbf{abc}] \mathbf{d} = [\mathbf{acd}] \mathbf{b} - [\mathbf{bcd}] \mathbf{a}$$

$$\text{or} \quad [\mathbf{bcd}] \mathbf{a} - [\mathbf{acd}] \mathbf{b} + [\mathbf{abd}] \mathbf{c} - [\mathbf{abc}] \mathbf{d} = \mathbf{0}, \quad \dots(1)$$

which is the required linear relation connecting the four vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$.

To find an expression for any vector $\mathbf{r}$, in space, as a linear combination of three non-coplanar vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$.

Replacing $\mathbf{d}$ by $\mathbf{r}$ in the relation (1) just established, we get

$$[\mathbf{bcr}] \mathbf{a} - [\mathbf{acr}] \mathbf{b} + [\mathbf{abr}] \mathbf{c} - [\mathbf{abc}] \mathbf{r} = \mathbf{0}$$

$$\text{or} \quad [\mathbf{abc}] \mathbf{r} = [\mathbf{bcr}] \mathbf{a} - [\mathbf{acr}] \mathbf{b} + [\mathbf{abr}] \mathbf{c}. \quad \dots(2)$$

Since $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are non-coplanar, therefore $[\mathbf{abc}] \neq 0$.

Therefore dividing both sides of (2) by $[\mathbf{abc}]$, we get

$$\mathbf{r} = \frac{[\mathbf{bcr}] \mathbf{a} - [\mathbf{acr}] \mathbf{b} + [\mathbf{abr}] \mathbf{c}}{[\mathbf{abc}]},$$

$$\text{or} \quad \mathbf{r} = \frac{[\mathbf{bcr}] \mathbf{a} + [\mathbf{car}] \mathbf{b} + [\mathbf{abr}] \mathbf{c}}{[\mathbf{abc}]}, \text{ since } [\mathbf{acr}] = -[\mathbf{car}]$$

$$\text{or} \quad \mathbf{r} = \frac{[\mathbf{rbc}] \mathbf{a} + [\mathbf{rca}] \mathbf{b} + [\mathbf{rab}] \mathbf{c}}{[\mathbf{abc}]}, \quad \dots(3)$$

which is the required expression for $\mathbf{r}$.

11 Reciprocal System of Vectors

If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be any three non-coplanar vectors so that $[\mathbf{abc}] \neq 0$, then the three vectors $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ defined by the equations

$$\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{[\mathbf{abc}]}, \quad \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{[\mathbf{abc}]}, \quad \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{[\mathbf{abc}]}$$

are called reciprocal system of vectors to the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

(i) To show that $\mathbf{a} \cdot \mathbf{a}' = \mathbf{b} \cdot \mathbf{b}' = \mathbf{c} \cdot \mathbf{c}' = 1$.

We have
$$\mathbf{a} \cdot \mathbf{a}' = \mathbf{a} \cdot \frac{\mathbf{b} \times \mathbf{c}}{[\mathbf{abc}]} = \frac{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}{[\mathbf{abc}]} = \frac{[\mathbf{abc}]}{[\mathbf{abc}]} = 1.$$

Similarly
$$\mathbf{b} \cdot \mathbf{b}' = \mathbf{b} \cdot \frac{\mathbf{c} \times \mathbf{a}}{[\mathbf{abc}]} = \frac{\mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})}{[\mathbf{abc}]} = \frac{[\mathbf{bca}]}{[\mathbf{abc}]} = \frac{[\mathbf{abc}]}{[\mathbf{abc}]} = 1$$

and
$$\mathbf{c} \cdot \mathbf{c}' = \mathbf{c} \cdot \frac{\mathbf{a} \times \mathbf{b}}{[\mathbf{abc}]} = \frac{\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})}{[\mathbf{abc}]} = \frac{[\mathbf{cab}]}{[\mathbf{abc}]} = \frac{[\mathbf{abc}]}{[\mathbf{abc}]} = 1.$$

Note: The reason for the name reciprocal lies in the relations

$$\mathbf{a} \cdot \mathbf{a}' = \mathbf{b} \cdot \mathbf{b}' = \mathbf{c} \cdot \mathbf{c}' = 1.$$

(ii) The scalar product of any other pair of vectors, one from each system, is zero i.e.,

$$\mathbf{a} \cdot \mathbf{b}' = \mathbf{a} \cdot \mathbf{c}' = \mathbf{b} \cdot \mathbf{a}' = \mathbf{b} \cdot \mathbf{c}' = \mathbf{c} \cdot \mathbf{a}' = \mathbf{c} \cdot \mathbf{b}' = 0.$$

We have

$$\mathbf{a} \cdot \mathbf{b}' = \mathbf{a} \cdot \frac{\mathbf{c} \times \mathbf{a}}{[\mathbf{abc}]} = \frac{\mathbf{a} \cdot (\mathbf{c} \times \mathbf{a})}{[\mathbf{abc}]} = \frac{[\mathbf{aca}]}{[\mathbf{abc}]} = 0, \text{ since } [\mathbf{aca}] = 0.$$

Similarly we can prove the other results.

(iii) The scalar triple product $[\mathbf{abc}]$ formed from three non-coplanar vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is the reciprocal of the scalar triple product $[\mathbf{a}' \mathbf{b}' \mathbf{c}']$ formed from the reciprocal system $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ i.e., $[\mathbf{abc}][\mathbf{a}' \mathbf{b}' \mathbf{c}'] = 1$.

$$\begin{aligned} \text{We have } [\mathbf{a}' \mathbf{b}' \mathbf{c}'] &= \mathbf{a}' \cdot (\mathbf{b}' \times \mathbf{c}') = \frac{\mathbf{b} \times \mathbf{c}}{[\mathbf{abc}]} \cdot \left\{ \frac{\mathbf{c} \times \mathbf{a}}{[\mathbf{abc}]} \times \frac{(\mathbf{a} \times \mathbf{b})}{[\mathbf{abc}]} \right\} \\ &= \frac{(\mathbf{b} \times \mathbf{c}) \cdot [(\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b})]}{[\mathbf{abc}]^3}. \end{aligned}$$

Now expanding $(\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b})$ by vector triple product treating $\mathbf{c} \times \mathbf{a}$ as one vector, we get

$$\begin{aligned} (\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b}) &= [(\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b}] \mathbf{a} - [(\mathbf{c} \times \mathbf{a}) \cdot \mathbf{a}] \mathbf{b} = [\mathbf{cab}] \mathbf{a} - [\mathbf{caa}] \mathbf{b} \\ &= [\mathbf{abc}] \mathbf{a}, \text{ since } [\mathbf{caa}] = 0 \end{aligned}$$

and
$$[\mathbf{cab}] = [\mathbf{abc}].$$

$$\therefore [\mathbf{a}' \mathbf{b}' \mathbf{c}'] = \frac{(\mathbf{b} \times \mathbf{c}) \cdot [\mathbf{abc}] \mathbf{a}}{[\mathbf{abc}]^3} = \frac{[(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}] [\mathbf{abc}]}{[\mathbf{abc}]^3}$$

$$= \frac{[\mathbf{bca}][\mathbf{abc}]}{[\mathbf{abc}]^3} = \frac{[\mathbf{abc}]^2}{[\mathbf{abc}]^3} = \frac{1}{[\mathbf{abc}]}.$$

$$\therefore [\mathbf{a}' \mathbf{b}' \mathbf{c}'] [\mathbf{abc}] = 1.$$

Note 1: Since $[\mathbf{abc}] \neq 0$, therefore from the relation, $[\mathbf{a}' \mathbf{b}' \mathbf{c}'] [\mathbf{abc}] = 1$, we conclude that $[\mathbf{a}' \mathbf{b}' \mathbf{c}'] \neq 0$.

Hence the vectors $\mathbf{a}'$, $\mathbf{b}'$, $\mathbf{c}'$ are also non-coplanar.

Note 2: The symmetry of results proved in properties (i), (ii) and (iii) suggest that if $\mathbf{a}'$, $\mathbf{b}'$, $\mathbf{c}'$ is the reciprocal system to $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ then $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ is also the reciprocal system to $\mathbf{a}'$, $\mathbf{b}'$, $\mathbf{c}'$.

Note 3: The relation $[\mathbf{abc}][\mathbf{a}' \mathbf{b}' \mathbf{c}'] = 1$ shows that the scalar triple products $[\mathbf{abc}]$ and $[\mathbf{a}' \mathbf{b}' \mathbf{c}']$ are either both positive or both negative. Hence the two systems of vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{a}'$, $\mathbf{b}'$, $\mathbf{c}'$ are either both right handed or both left handed.

(iv) The orthonormal vector triads $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ form a self reciprocal system.

Let $\mathbf{i}'$, $\mathbf{j}'$, $\mathbf{k}'$ be the system of vectors reciprocal to the system $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$.

$$\text{Then by definition } \mathbf{i}' = \frac{\mathbf{j} \times \mathbf{k}}{[\mathbf{i} \mathbf{j} \mathbf{k}]} = \frac{\mathbf{i}}{1} = \mathbf{i}.$$

$$\text{Similarly, } \mathbf{j}' = \mathbf{j} \text{ and } \mathbf{k}' = \mathbf{k}.$$

Hence the result.

Theorem: If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be three non-coplanar vectors and $\mathbf{a}'$, $\mathbf{b}'$, $\mathbf{c}'$ constitute the reciprocal system of vectors, then any vector $\mathbf{r}$ can be expressed as

$$\mathbf{r} = (\mathbf{r} \cdot \mathbf{a}') \mathbf{a} + (\mathbf{r} \cdot \mathbf{b}') \mathbf{b} + (\mathbf{r} \cdot \mathbf{c}') \mathbf{c}.$$

Proof: Let $\mathbf{r}$ be expressed as a linear combination of the non-coplanar vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ in the form

$$\mathbf{r} = x\mathbf{a} + y\mathbf{b} + z\mathbf{c} \quad \dots(1)$$

where x , y , z are some scalars.

Multiplying both sides of (1) scalarly with $\mathbf{b} \times \mathbf{c}$, we get

$$\begin{aligned} \mathbf{r} \cdot (\mathbf{b} \times \mathbf{c}) &= x\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + y\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) + z\mathbf{c} \cdot (\mathbf{b} \times \mathbf{c}) \\ &= x[\mathbf{abc}] + y[\mathbf{bbc}] + z[\mathbf{cbc}] = x[\mathbf{abc}], \end{aligned}$$

$$\text{since } [\mathbf{bbc}] = 0 = [\mathbf{cbc}].$$

$$\therefore x = \frac{\mathbf{r} \cdot (\mathbf{b} \times \mathbf{c})}{[\mathbf{abc}]} = \mathbf{r} \cdot \frac{(\mathbf{b} \times \mathbf{c})}{[\mathbf{abc}]} = \mathbf{r} \cdot \mathbf{a}', \text{ since } \mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{[\mathbf{abc}]}.$$

Similarly multiplying both sides of (1) scalarly with $\mathbf{c} \times \mathbf{a}$ and $\mathbf{a} \times \mathbf{b}$, we can show that

$$y = \mathbf{r} \cdot \mathbf{b}' \text{ and } z = \mathbf{r} \cdot \mathbf{c}'.$$

Putting the values of x , y and z in (1), we get

$$\mathbf{r} = (\mathbf{r} \cdot \mathbf{a}') \mathbf{a} + (\mathbf{r} \cdot \mathbf{b}') \mathbf{b} + (\mathbf{r} \cdot \mathbf{c}') \mathbf{c}. \quad \dots(2)$$

Note 1: In a similar manner, we can prove that

$$\mathbf{r} = (\mathbf{r} \cdot \mathbf{a}) \mathbf{a}' + (\mathbf{r} \cdot \mathbf{b}) \mathbf{b}' + (\mathbf{r} \cdot \mathbf{c}) \mathbf{c}'.$$

Note 2: Since the system of vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ is self-reciprocal, therefore from (2) we conclude that $\mathbf{r} = (\mathbf{r} \cdot \mathbf{i}) \mathbf{i} + (\mathbf{r} \cdot \mathbf{j}) \mathbf{j} + (\mathbf{r} \cdot \mathbf{k}) \mathbf{k}$.

Illustrative Examples

Example 16: Find a set of vectors reciprocal to the set $2\mathbf{i} + 3\mathbf{j} - \mathbf{k}, \mathbf{i} - \mathbf{j} - 2\mathbf{k}, -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

Solution: Let $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}, \mathbf{b} = \mathbf{i} - \mathbf{j} - 2\mathbf{k}, \mathbf{c} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

Let $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ be the set of vectors reciprocal to the set $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Then by definition

$$\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{[\mathbf{abc}]}, \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{[\mathbf{abc}]}, \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{[\mathbf{abc}]}.$$

Now $[\mathbf{abc}] = \begin{vmatrix} 2 & 3 & -1 \\ 1 & -1 & -2 \\ -1 & 2 & 2 \end{vmatrix} = 2(2) - 3(0) - 1(1) = 3$

and $\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & -2 \\ -1 & 2 & 2 \end{vmatrix} = 2\mathbf{i} + 0\mathbf{j} + \mathbf{k} = 2\mathbf{i} + \mathbf{k}.$

$\therefore \mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{[\mathbf{abc}]} = \frac{2\mathbf{i} + \mathbf{k}}{3} = \left(\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{k}\right).$

Similarly $\mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{[\mathbf{abc}]} = \frac{1}{3} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 2 & 2 \\ 2 & 3 & -1 \end{vmatrix} = \frac{-8\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}}{3}$

and $\mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{[\mathbf{abc}]} = \frac{1}{3} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & -1 \\ 1 & -1 & -2 \end{vmatrix} = \frac{-7\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}}{3}.$

Example 17: Prove that $\mathbf{a} \times \{\mathbf{b} \times (\mathbf{c} \times \mathbf{d})\} = (\mathbf{b} \cdot \mathbf{d})(\mathbf{a} \times \mathbf{c}) - (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \times \mathbf{d}).$

Hence expand $\mathbf{a} \times [\mathbf{b} \times \{\mathbf{c} \times (\mathbf{d} \times \mathbf{e})\}].$

Solution: First part: We have

$$\begin{aligned} \mathbf{a} \times \{\mathbf{b} \times (\mathbf{c} \times \mathbf{d})\} &= \mathbf{a} \times \{(\mathbf{b} \cdot \mathbf{d}) \mathbf{c} - (\mathbf{b} \cdot \mathbf{c}) \mathbf{d}\} \\ &= (\mathbf{b} \cdot \mathbf{d})(\mathbf{a} \times \mathbf{c}) - (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \times \mathbf{d}). \end{aligned}$$

Second part: We have

$$\begin{aligned} \mathbf{b} \times \{\mathbf{c} \times (\mathbf{d} \times \mathbf{e})\} &= \mathbf{b} \times \{(\mathbf{c} \cdot \mathbf{e}) \mathbf{d} - (\mathbf{c} \cdot \mathbf{d}) \mathbf{e}\} \\ &= (\mathbf{c} \cdot \mathbf{e})(\mathbf{b} \times \mathbf{d}) - (\mathbf{c} \cdot \mathbf{d})(\mathbf{b} \times \mathbf{e}). \end{aligned}$$

Also $(\mathbf{a} \times \mathbf{d}) \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{d}) \times \mathbf{n}$, where $\mathbf{n} = \mathbf{b} \times \mathbf{c}$

$$\begin{aligned}
 &= (\mathbf{n} \cdot \mathbf{a}) \mathbf{d} - (\mathbf{n} \cdot \mathbf{d}) \mathbf{a} \\
 &= [(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}] \mathbf{d} - [(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{d}] \mathbf{a} \\
 &= [\mathbf{bca}] \mathbf{d} - [\mathbf{bcd}] \mathbf{a} = [\mathbf{abc}] \mathbf{d} - [\mathbf{bcd}] \mathbf{a}. \quad \dots(3)
 \end{aligned}$$

Adding (1), (2) and (3), we get

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) + (\mathbf{a} \times \mathbf{c}) \times (\mathbf{d} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{d}) \times (\mathbf{b} \times \mathbf{c}) = -2 [\mathbf{bcd}] \mathbf{a}.$$

Example 21: Prove that

$$[\mathbf{a} \times \mathbf{p}, \mathbf{b} \times \mathbf{q}, \mathbf{c} \times \mathbf{r}] + [\mathbf{a} \times \mathbf{q}, \mathbf{b} \times \mathbf{r}, \mathbf{c} \times \mathbf{p}] + [\mathbf{a} \times \mathbf{r}, \mathbf{b} \times \mathbf{p}, \mathbf{c} \times \mathbf{q}] = 0.$$

(Kumaun 2012)

Solution: We have

$$\begin{aligned} [\mathbf{a} \times \mathbf{p}, \mathbf{b} \times \mathbf{q}, \mathbf{c} \times \mathbf{r}] &= (\mathbf{a} \times \mathbf{p}) \cdot [(\mathbf{b} \times \mathbf{q}) \times (\mathbf{c} \times \mathbf{r})] \\ &= (\mathbf{a} \times \mathbf{p}) \cdot \{[(\mathbf{b} \times \mathbf{q}) \cdot \mathbf{r}] \mathbf{c} - [(\mathbf{b} \times \mathbf{q}) \cdot \mathbf{c}] \mathbf{r}\} \\ &= (\mathbf{a} \times \mathbf{p}) \cdot \{[\mathbf{bqr}] \mathbf{c} - [\mathbf{bqc}] \mathbf{r}\} \\ &= [\mathbf{apc}] [\mathbf{bqr}] - [\mathbf{apr}] [\mathbf{bqc}]. \end{aligned} \quad \dots(1)$$

Again

$$\begin{aligned} [\mathbf{a} \times \mathbf{q}, \mathbf{b} \times \mathbf{r}, \mathbf{c} \times \mathbf{p}] &= [\mathbf{b} \times \mathbf{r}, \mathbf{c} \times \mathbf{p}, \mathbf{a} \times \mathbf{q}] \\ &= (\mathbf{b} \times \mathbf{r}) \cdot [(\mathbf{c} \times \mathbf{p}) \times (\mathbf{a} \times \mathbf{q})] \\ &= (\mathbf{b} \times \mathbf{r}) \cdot \{[(\mathbf{c} \times \mathbf{p}) \cdot \mathbf{q}] \mathbf{a} - [(\mathbf{c} \times \mathbf{p}) \cdot \mathbf{a}] \mathbf{q}\} \\ &= [\mathbf{bra}] [\mathbf{cpq}] - [\mathbf{brq}] [\mathbf{cpa}]. \end{aligned} \quad \dots(2)$$

and

$$\begin{aligned} [\mathbf{a} \times \mathbf{r}, \mathbf{b} \times \mathbf{p}, \mathbf{c} \times \mathbf{q}] &= [\mathbf{c} \times \mathbf{q}, \mathbf{a} \times \mathbf{r}, \mathbf{b} \times \mathbf{p}] \\ &= (\mathbf{c} \times \mathbf{q}) \cdot [(\mathbf{a} \times \mathbf{r}) \times (\mathbf{b} \times \mathbf{p})] \\ &= (\mathbf{c} \times \mathbf{q}) \cdot \{[(\mathbf{a} \times \mathbf{r}) \cdot \mathbf{p}] \mathbf{b} - [(\mathbf{a} \times \mathbf{r}) \cdot \mathbf{b}] \mathbf{p}\} \\ &= [\mathbf{cqb}] [\mathbf{arp}] - [\mathbf{cqp}] [\mathbf{arb}]. \end{aligned} \quad \dots(3)$$

Adding (1), (2) and (3) we get

$$\begin{aligned} &[\mathbf{a} \times \mathbf{p}, \mathbf{b} \times \mathbf{q}, \mathbf{c} \times \mathbf{r}] + [\mathbf{a} \times \mathbf{q}, \mathbf{b} \times \mathbf{r}, \mathbf{c} \times \mathbf{p}] + [\mathbf{a} \times \mathbf{r}, \mathbf{b} \times \mathbf{p}, \mathbf{c} \times \mathbf{q}] \\ &= [\mathbf{apc}] [\mathbf{bqr}] - [\mathbf{apr}] [\mathbf{bqc}] + [\mathbf{bra}] [\mathbf{cpq}] \\ &\quad - [\mathbf{brq}] [\mathbf{cpa}] + [\mathbf{cqb}] [\mathbf{arp}] - [\mathbf{cqp}] [\mathbf{arb}] \\ &= [\mathbf{apc}] [\mathbf{bqr}] - [\mathbf{apr}] [\mathbf{bqc}] + [\mathbf{bra}] [\mathbf{cpq}] \\ &\quad - [\mathbf{bqr}] [\mathbf{apc}] + [\mathbf{bqc}] [\mathbf{apr}] - [\mathbf{cpq}] [\mathbf{bra}] \\ &= 0, \text{ since } [\mathbf{brq}] = -[\mathbf{bqr}], [\mathbf{cpa}] = -[\mathbf{apc}] \text{ etc.} \end{aligned}$$

Example 22: Prove that $[\mathbf{a} \times \mathbf{b}, \mathbf{c} \times \mathbf{d}, \mathbf{e} \times \mathbf{f}] = [\mathbf{abd}] [\mathbf{cef}] - [\mathbf{abc}] [\mathbf{def}]$

$$= [\mathbf{abe}] [\mathbf{fcd}] - [\mathbf{abf}] [\mathbf{ecd}] = [\mathbf{cda}] [\mathbf{bef}] - [\mathbf{cdb}] [\mathbf{aef}].$$

(Kumaun 2012, 15)

Solution: We have $[\mathbf{a} \times \mathbf{b}, \mathbf{c} \times \mathbf{d}, \mathbf{e} \times \mathbf{f}] = (\mathbf{a} \times \mathbf{b}) \cdot [(\mathbf{c} \times \mathbf{d}) \times (\mathbf{e} \times \mathbf{f})]$

$$\begin{aligned} &= (\mathbf{a} \times \mathbf{b}) \cdot [\mathbf{l} \times (\mathbf{e} \times \mathbf{f})], \text{ where } \mathbf{l} = \mathbf{c} \times \mathbf{d} \\ &= (\mathbf{a} \times \mathbf{b}) \cdot [(\mathbf{l} \cdot \mathbf{f}) \mathbf{e} - (\mathbf{l} \cdot \mathbf{e}) \mathbf{f}] \\ &= (\mathbf{a} \times \mathbf{b}) \cdot \{[(\mathbf{c} \times \mathbf{d}) \cdot \mathbf{f}] \mathbf{e} - [(\mathbf{c} \times \mathbf{d}) \cdot \mathbf{e}] \mathbf{f}\} \\ &= [\mathbf{cdf}] [\mathbf{abe}] - [\mathbf{cde}] [\mathbf{abf}] \\ &= [\mathbf{abe}] [\mathbf{fcd}] - [\mathbf{abf}] [\mathbf{ecd}], \text{ since } [\mathbf{cdf}] = [\mathbf{fcd}] \text{ etc.} \end{aligned}$$

Again $[a \times b, c \times d, e \times f] = [c \times d, e \times f, a \times b]$

$$= (c \times d) \cdot [(e \times f) \times (a \times b)]$$

$$= (c \times d) \cdot \{[(e \times f) \cdot b] a - [(e \times f) \cdot a] b\}$$

$$= [cda][efb] - [cdb][efa]$$

$$= [cda][bef] - [cdb][aef].$$

and $[a \times b, c \times d, e \times f] = [e \times f, a \times b, c \times d]$

$$= (e \times f) \cdot [(a \times b) \times (c \times d)]$$

$$= (e \times f) \cdot \{[(a \times b) \cdot d] c - [(a \times b) \cdot c] d\}$$

$$= [efc][abd] - [efd][abc]$$

$$= [abd][cef] - [abc][def].$$

Example 23: Prove that $(b \times c) \cdot (a \times d) + (c \times a) \cdot (b \times d) + (a \times b) \cdot (c \times d) = 0$.

Solution: We have

$$(b \times c) \cdot (a \times d) = \begin{vmatrix} b \cdot a & b \cdot d \\ c \cdot a & c \cdot d \end{vmatrix}$$

$$= (b \cdot a)(c \cdot d) - (c \cdot a)(b \cdot d) \quad \dots(1)$$

Similarly, $(c \times a) \cdot (b \times d) = \begin{vmatrix} c \cdot b & c \cdot d \\ a \cdot b & a \cdot d \end{vmatrix}$

$$= (c \cdot b)(a \cdot d) - (a \cdot b)(c \cdot d) \quad \dots(2)$$

and $(a \times b) \cdot (c \times d) = \begin{vmatrix} a \cdot c & a \cdot d \\ b \cdot c & b \cdot d \end{vmatrix}$

$$= (a \cdot c)(b \cdot d) - (b \cdot c)(a \cdot d) \quad \dots(3)$$

Adding (1), (2) and (3), we get

$$(b \times c) \cdot (a \times d) + (c \times a) \cdot (b \times d) + (a \times b) \cdot (c \times d) = 0,$$

since $a \cdot b = b \cdot a$ etc.

Comprehensive Exercise 3

1. Prove the identity $a \times [a \times (a \times b)] = (a \cdot a)(b \times a)$. (Kumaun 2011)
2. Establish the identity $[a \ b \ c] d = [b \ c \ d] a + [c \ a \ d] b + [a \ b \ d] c$ for any four vectors a, b, c, d . Hence show that any vector r can always be expressed as a linear combination of three given non-coplanar vectors.
3. Obtain a set of vectors reciprocal to the three vectors $-i + j + k, i - j + k, i + j + k$.

4. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be a set of non-coplanar vectors and

$$\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{[\mathbf{abc}]}, \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{[\mathbf{abc}]}, \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{[\mathbf{abc}]},$$

then prove that

$$\mathbf{a} = \frac{\mathbf{b}' \times \mathbf{c}'}{[\mathbf{a}' \mathbf{b}' \mathbf{c}']}, \mathbf{b} = \frac{\mathbf{c}' \times \mathbf{a}'}{[\mathbf{a}' \mathbf{b}' \mathbf{c}']}, \text{ and } \mathbf{c} = \frac{\mathbf{a}' \times \mathbf{b}'}{[\mathbf{a}' \mathbf{b}' \mathbf{c}']}.$$

5. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are reciprocal system of vectors, prove that

(i) $\mathbf{a} \times \mathbf{a}' + \mathbf{b} \times \mathbf{b}' + \mathbf{c} \times \mathbf{c}' = \mathbf{0}$

(ii) $\mathbf{a}' \times \mathbf{b}' + \mathbf{b}' \times \mathbf{c}' + \mathbf{c}' \times \mathbf{a}' = \frac{\mathbf{a} + \mathbf{b} + \mathbf{c}}{[\mathbf{a} \mathbf{b} \mathbf{c}]}$

(iii) $\mathbf{a} \cdot \mathbf{a}' + \mathbf{b} \cdot \mathbf{b}' + \mathbf{c} \cdot \mathbf{c}' = 3$.

Answers 3

3. $-\frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{k}, -\frac{1}{2}\mathbf{j} + \frac{1}{2}\mathbf{k}, \frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}.$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The value of $\mathbf{i} \cdot (\mathbf{j} \times \mathbf{k}) + \mathbf{j} \cdot (\mathbf{k} \times \mathbf{i}) + \mathbf{k} \cdot (\mathbf{i} \times \mathbf{j})$ is
 (a) 0 (b) 1
 (c) 2 (d) 3
- The volume of the parallelepiped whose edges are given by
 $\vec{OA} = 2\mathbf{i} - 3\mathbf{j}, \vec{OB} = \mathbf{i} + \mathbf{j} - \mathbf{k}, \vec{OC} = 3\mathbf{i} - \mathbf{k}$ is
 (a) 1 (b) 4
 (c) 2/7 (d) None of these
- If $[\mathbf{a} \mathbf{b} \mathbf{c}]$ is the scalar triple product of three vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$, then $[\mathbf{a} \mathbf{b} \mathbf{c}]$ is equal to
 (a) $[\mathbf{b} \mathbf{a} \mathbf{c}]$ (b) $[\mathbf{c} \mathbf{b} \mathbf{a}]$
 (c) $[\mathbf{b} \mathbf{c} \mathbf{a}]$ (d) $[\mathbf{a} \mathbf{c} \mathbf{b}]$

4. If $\mathbf{i}, \mathbf{j}, \mathbf{k}$ constitute an orthogonal right handed triad of unit vectors and $\mathbf{a}$ is any vector, then $\mathbf{i} \times (\mathbf{a} \times \mathbf{i}) + \mathbf{j} \times (\mathbf{a} \times \mathbf{j}) + \mathbf{k} \times (\mathbf{a} \times \mathbf{k})$ is equal to
 (a) $\mathbf{a}$ (b) $2\mathbf{a}$
 (c) $3\mathbf{a}$ (d) $\mathbf{0}$
5. $[\bar{a}', \bar{b}', \bar{c}']$ is equal to
 (a) $[\bar{a}, \bar{b}, \bar{c}]$ (b) $[\bar{a}, \bar{b}, \bar{c}]^2$
 (c) $\frac{1}{[\bar{a}, \bar{b}, \bar{c}]}$ (d) None of these
 (Kumaun 2009)

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. If $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ be a set of orthonormal unit vectors, then $[\mathbf{i} \ \mathbf{j} \ \mathbf{k}] = \dots\dots$
2. If $\vec{A}, \vec{B}, \vec{C}$ be three non-coplanar vectors, then

$$\frac{\vec{A} \cdot \vec{B} \times \vec{C}}{\vec{C} \times \vec{A} \cdot \vec{B}} + \frac{\vec{B} \cdot \vec{A} \times \vec{C}}{\vec{C} \cdot \vec{A} \times \vec{B}} = \dots\dots$$
3. For any three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \dots$
4. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are reciprocal system of vectors, then

$$\mathbf{a} \cdot \mathbf{a}' + \mathbf{b} \cdot \mathbf{b}' + \mathbf{c} \cdot \mathbf{c}' = \dots\dots$$
5. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are any three coplanar vectors, then $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \dots\dots$
 (Kumaun 2008)
6. $(\bar{a} \times \bar{b}) \times \bar{c} = \dots\dots\dots$
 (Kumaun 2015)
7. If $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are unit vectors, then $(\mathbf{i} \times \mathbf{j}) \cdot \mathbf{k} = \dots\dots\dots$
 (Kumaun 2009)
8. Value of vector $[(\mathbf{b} + \mathbf{c}) \times (\mathbf{c} + \mathbf{a})] = \dots\dots\dots$
 (Kumaun 2013)

True or False

Write ‘T’ for true and ‘F’ for false statement.

1. If $\mathbf{x} \cdot \mathbf{a} = \mathbf{x} \cdot \mathbf{b} = \mathbf{x} \cdot \mathbf{c} = 0$, for some non-zero vector $\mathbf{x}$, then $[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 0$.
2. If $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ be a set of orthonormal unit vectors, then $\mathbf{i} \times (\mathbf{j} \times \mathbf{k}) \neq \mathbf{0}$.
3. The orthonormal vector triads $\mathbf{i}, \mathbf{j}, \mathbf{k}$ form a self reciprocal system.

Answers

Multiple Choice Questions

1. (d) 2. (b) 3. (c) 4. (b) 5. (c)

Fill in the Blank(s)

1. 1 2. 0 3. 0 4. 3 5. 0
6. $(\mathbf{c} \cdot \mathbf{a}) \mathbf{b} - (\mathbf{c} \cdot \mathbf{b}) \mathbf{a}$ 7. 1 8. $\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{b} \times \mathbf{a}$

True or False

1. T 2. F 3. T



Chapter

2



Differentiation of Vectors

1 Vector Function

We know that a scalar quantity possesses only magnitude and has no concern with direction. A single real number gives us a complete representation of a scalar quantity. Thus a scalar quantity is nothing but a real number.

Let D be any subset of the set of all real numbers. If to each element t of D , we associate by some rule a unique real number $f(t)$, then this rule defines a **scalar function** of the scalar variable t . Here $f(t)$ is a scalar quantity and thus f is a scalar function.

In a similar manner we define a vector function.

*Let D be any subset of the set of all real numbers. If to each element t of D , we associate by some rule a unique vector $\mathbf{f}(t)$, then this rule defines a **vector function** of the scalar variable t .*

Here $\mathbf{f}(t)$ is a vector quantity and thus $\mathbf{f}$ is a vector function.

We know that every vector can be uniquely expressed as a linear combination of three fixed non-coplanar vectors. Therefore we may write

$$\mathbf{f}(t) = f_1(t) \mathbf{i} + f_2(t) \mathbf{j} + f_3(t) \mathbf{k}$$

where $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ denote a fixed right handed triad of three mutually perpendicular non-coplanar unit vectors.

2 Scalar Fields and Vector Fields

If to each point $P(x, y, z)$ of a region R in space there corresponds a unique scalar $f(P)$, then f is called a **scalar point function** and we say that a **scalar field** f has been defined in R .

Examples: (1) The temperature at any point within or on the surface of earth at a certain time defines a scalar field.

(2) $f(x, y, z) = x^2 - y^3 - 3z^2$ defines a scalar field.

If to each point $P(x, y, z)$ of a region R in space there corresponds a unique vector $\mathbf{f}(P)$, then $\mathbf{f}$ is called a **vector point function** and we say that a **vector field** $\mathbf{f}$ has been defined in R .

Examples: (1) If the velocity at any point (x, y, z) of a particle moving in a curve is known at a certain time, then a vector field is defined.

(2) $\mathbf{f}(x, y, z) = xy^2\mathbf{i} + 3yz^3\mathbf{j} - 2x^2z\mathbf{k}$ defines a vector field.

3 Limit and Continuity of a Vector Function

Definition 1: A vector function $\mathbf{f}(t)$ is said to tend to a limit $\mathbf{l}$, when t tends to t_0 , if for any given positive number ϵ , however small, there corresponds a positive number δ such that

$$|\mathbf{f}(t) - \mathbf{l}| < \epsilon \quad \text{whenever} \quad 0 < |t - t_0| < \delta.$$

If $\mathbf{f}(t)$ tends to a limit $\mathbf{l}$ as t tends to t_0 , we write $\lim_{t \rightarrow t_0} \mathbf{f}(t) = \mathbf{l}$.

Definition 2: A vector function $\mathbf{f}(t)$ is said to be continuous for a value t_0 of t if

(i) $\mathbf{f}(t_0)$ is defined and

(ii) for any given positive number ϵ , however small, there corresponds a positive number δ such that $|\mathbf{f}(t) - \mathbf{f}(t_0)| < \epsilon$, whenever $|t - t_0| < \delta$.

Further a vector function $\mathbf{f}(t)$ is said to be continuous if it is continuous for every value of t for which it has been defined.

We shall give here (without proof) some important results about the limits and continuity of a vector function.

Theorem 1: The necessary and sufficient condition for a vector function $\mathbf{f}(t)$ to be

continuous at $t = t_0$ is that $\lim_{t \rightarrow t_0} \mathbf{f}(t) = \mathbf{f}(t_0)$.

(Purvanchal 2014)

Theorem 2: If $\mathbf{f}(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}$, then $\mathbf{f}(t)$ is continuous if and only if $f_1(t)$, $f_2(t)$, $f_3(t)$ are continuous.

Theorem 3: Let $\mathbf{f}(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}$ and $\mathbf{l} = l_1\mathbf{i} + l_2\mathbf{j} + l_3\mathbf{k}$.

Then the necessary and sufficient conditions that $\lim_{t \rightarrow t_0} \mathbf{f}(t) = \mathbf{l}$ are

$$\lim_{t \rightarrow t_0} f_1(t) = l_1, \quad \lim_{t \rightarrow t_0} f_2(t) = l_2 \quad \text{and} \quad \lim_{t \rightarrow t_0} f_3(t) = l_3.$$

Theorem 4: If $\mathbf{f}(t)$, $\mathbf{g}(t)$ are vector functions of scalar variable t and $\phi(t)$ is a scalar function of scalar variable t , then

$$(i) \quad \lim_{t \rightarrow t_0} [\mathbf{f}(t) \pm \mathbf{g}(t)] = \lim_{t \rightarrow t_0} \mathbf{f}(t) \pm \lim_{t \rightarrow t_0} \mathbf{g}(t)$$

$$(ii) \quad \lim_{t \rightarrow t_0} [\mathbf{f}(t) \bullet \mathbf{g}(t)] = \left[\lim_{t \rightarrow t_0} \mathbf{f}(t) \right] \bullet \left[\lim_{t \rightarrow t_0} \mathbf{g}(t) \right]$$

$$(iii) \quad \lim_{t \rightarrow t_0} [\mathbf{f}(t) \times \mathbf{g}(t)] = \left[\lim_{t \rightarrow t_0} \mathbf{f}(t) \right] \times \left[\lim_{t \rightarrow t_0} \mathbf{g}(t) \right]$$

$$(iv) \quad \lim_{t \rightarrow t_0} [\phi(t) \mathbf{f}(t)] = \left[\lim_{t \rightarrow t_0} \phi(t) \right] \left[\lim_{t \rightarrow t_0} \mathbf{f}(t) \right]$$

$$(v) \quad \lim_{t \rightarrow t_0} |\mathbf{f}(t)| = \left| \lim_{t \rightarrow t_0} \mathbf{f}(t) \right|.$$

4 Derivative of a Vector Function with Respect to a Scalar

Definition: Let $\mathbf{r} = \mathbf{f}(t)$ be a vector function of the scalar variable t . We define

$$\mathbf{r} + \delta\mathbf{r} = \mathbf{f}(t + \delta t).$$

$$\therefore \quad \delta\mathbf{r} = \mathbf{f}(t + \delta t) - \mathbf{f}(t).$$

$$\text{Consider the vector } \frac{\delta\mathbf{r}}{\delta t} = \frac{\mathbf{f}(t + \delta t) - \mathbf{f}(t)}{\delta t}.$$

If $\lim_{\delta t \rightarrow 0} \frac{\delta\mathbf{r}}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{\mathbf{f}(t + \delta t) - \mathbf{f}(t)}{\delta t}$ exists, then the value of this limit, which we shall denote by $\frac{d\mathbf{r}}{dt}$, is called the derivative of the vector function $\mathbf{r}$ with respect to the scalar t .

Symbolically

$$\frac{d\mathbf{r}}{dt} = \lim_{\delta t \rightarrow 0} \frac{(\mathbf{r} + \delta\mathbf{r}) - \mathbf{r}}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{\mathbf{f}(t + \delta t) - \mathbf{f}(t)}{\delta t}.$$

If $\frac{d\mathbf{r}}{dt}$ exists, then $\mathbf{r}$ is said to be differentiable. Since $\frac{\delta\mathbf{r}}{\delta t}$ is a vector quantity, therefore $\frac{d\mathbf{r}}{dt}$ is also a vector quantity.

Successive Derivatives: If we differentiate $\frac{d\mathbf{r}}{dt}$ again, we get $\frac{d^2\mathbf{r}}{dt^2}$ which is called the second derivative of $\mathbf{r}$ w.r.t. t , and so on.

Thus, we continue differentiating successively upto n times and get

$$\frac{d\mathbf{r}}{dt}, \frac{d^2\mathbf{r}}{dt^2}, \frac{d^3\mathbf{r}}{dt^3}, \dots, \frac{d^n\mathbf{r}}{dt^n},$$

where $\frac{d^n\mathbf{r}}{dt^n}$ is called the n^{th} diff. coeff. (or derivative) of $\mathbf{r}$ w.r.t. t . (Here diff. coeff. means differential coefficient.)

We often represent $\frac{d\mathbf{r}}{dt}, \frac{d^2\mathbf{r}}{dt^2}, \dots$ by $\dot{\mathbf{r}}, \ddot{\mathbf{r}}, \dots$ respectively.

A scalar or vector function of t is called differentiable of order n if its n^{th} order derivative exists.

5 Differentiation Formulae

Theorem: If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are differentiable vector functions of a scalar t and ϕ is a differentiable scalar function of the same variable t , then

1. $\frac{d}{dt}(\mathbf{a} + \mathbf{b}) = \frac{d\mathbf{a}}{dt} + \frac{d\mathbf{b}}{dt}$
2. $\frac{d}{dt}(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \cdot \mathbf{b}$
3. $\frac{d}{dt}(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \times \mathbf{b}$
4. $\frac{d}{dt}(\phi\mathbf{a}) = \phi \frac{d\mathbf{a}}{dt} + \frac{d\phi}{dt} \mathbf{a}$
5. $\frac{d}{dt}[\mathbf{a} \mathbf{b} \mathbf{c}] = \left[\frac{d\mathbf{a}}{dt} \mathbf{b} \mathbf{c} \right] + \left[\mathbf{a} \frac{d\mathbf{b}}{dt} \mathbf{c} \right] + \left[\mathbf{a} \mathbf{b} \frac{d\mathbf{c}}{dt} \right]$ (Meerut 2013B; Purvanchal 08)
6. $\frac{d}{dt} \{ \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \} = \frac{d\mathbf{a}}{dt} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{a} \times \left(\frac{d\mathbf{b}}{dt} \times \mathbf{c} \right) + \mathbf{a} \times \left(\mathbf{b} \times \frac{d\mathbf{c}}{dt} \right).$
(Meerut 2000; Purvanchal 08)

Proof: 1. $\frac{d}{dt}(\mathbf{a} + \mathbf{b}) = \lim_{\delta t \rightarrow 0} \frac{\{(\mathbf{a} + \delta\mathbf{a}) + (\mathbf{b} + \delta\mathbf{b})\} - (\mathbf{a} + \mathbf{b})}{\delta t}$

$$= \lim_{\delta t \rightarrow 0} \frac{\delta\mathbf{a} + \delta\mathbf{b}}{\delta t} = \lim_{\delta t \rightarrow 0} \left(\frac{\delta\mathbf{a}}{\delta t} + \frac{\delta\mathbf{b}}{\delta t} \right)$$

$$= \lim_{\delta t \rightarrow 0} \frac{\delta\mathbf{a}}{\delta t} + \lim_{\delta t \rightarrow 0} \frac{\delta\mathbf{b}}{\delta t} = \frac{d\mathbf{a}}{dt} + \frac{d\mathbf{b}}{dt}.$$

Thus the derivative of the sum of two vectors is equal to the sum of their derivatives, as it is also in Scalar Calculus.

Similarly we can prove that $\frac{d}{dt} (\mathbf{a} - \mathbf{b}) = \frac{d\mathbf{a}}{dt} - \frac{d\mathbf{b}}{dt}$.

In general if $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$ are vector functions of a scalar t , then

$$\frac{d}{dt} (\mathbf{r}_1 + \mathbf{r}_2 + \dots + \mathbf{r}_n) = \frac{d\mathbf{r}_1}{dt} + \frac{d\mathbf{r}_2}{dt} + \dots + \frac{d\mathbf{r}_n}{dt}.$$

$$\begin{aligned} 2. \quad \frac{d}{dt} (\mathbf{a} \cdot \mathbf{b}) &= \lim_{\delta t \rightarrow 0} \frac{(\mathbf{a} + \delta \mathbf{a}) \cdot (\mathbf{b} + \delta \mathbf{b}) - \mathbf{a} \cdot \mathbf{b}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \frac{\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \delta \mathbf{b} + \delta \mathbf{a} \cdot \mathbf{b} + \delta \mathbf{a} \cdot \delta \mathbf{b} - \mathbf{a} \cdot \mathbf{b}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \frac{\mathbf{a} \cdot \delta \mathbf{b} + \delta \mathbf{a} \cdot \mathbf{b} + \delta \mathbf{a} \cdot \delta \mathbf{b}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \left\{ \mathbf{a} \cdot \frac{\delta \mathbf{b}}{\delta t} + \frac{\delta \mathbf{a}}{\delta t} \cdot \mathbf{b} + \frac{\delta \mathbf{a}}{\delta t} \cdot \delta \mathbf{b} \right\} \\ &= \lim_{\delta t \rightarrow 0} \mathbf{a} \cdot \frac{\delta \mathbf{b}}{\delta t} + \lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{a}}{\delta t} \cdot \mathbf{b} + \lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{a}}{\delta t} \cdot \delta \mathbf{b} \\ &= \mathbf{a} \cdot \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \cdot \mathbf{b} + \frac{d\mathbf{a}}{dt} \cdot \mathbf{0}, \quad \text{since } \delta \mathbf{b} \rightarrow \text{zero vector as } \delta t \rightarrow 0 \\ &= \mathbf{a} \cdot \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \cdot \mathbf{b} \end{aligned}$$

Note: We know that $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$. Therefore while evaluating $\frac{d}{dt} (\mathbf{a} \cdot \mathbf{b})$, we should not bother about the order of the factors.

$$\begin{aligned} 3. \quad \frac{d}{dt} (\mathbf{a} \times \mathbf{b}) &= \lim_{\delta t \rightarrow 0} \frac{(\mathbf{a} + \delta \mathbf{a}) \times (\mathbf{b} + \delta \mathbf{b}) - \mathbf{a} \times \mathbf{b}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \frac{\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \delta \mathbf{b} + \delta \mathbf{a} \times \mathbf{b} + \delta \mathbf{a} \times \delta \mathbf{b} - \mathbf{a} \times \mathbf{b}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \frac{\mathbf{a} \times \delta \mathbf{b} + \delta \mathbf{a} \times \mathbf{b} + \delta \mathbf{a} \times \delta \mathbf{b}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \left\{ \mathbf{a} \times \frac{\delta \mathbf{b}}{\delta t} + \frac{\delta \mathbf{a}}{\delta t} \times \mathbf{b} + \frac{\delta \mathbf{a}}{\delta t} \times \delta \mathbf{b} \right\} \\ &= \lim_{\delta t \rightarrow 0} \mathbf{a} \times \frac{\delta \mathbf{b}}{\delta t} + \lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{a}}{\delta t} \times \mathbf{b} + \lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{a}}{\delta t} \times \delta \mathbf{b} \\ &= \mathbf{a} \times \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \times \mathbf{b} + \frac{d\mathbf{a}}{dt} \times \mathbf{0}, \quad \text{since } \delta \mathbf{b} \rightarrow \text{zero vector as } \delta t \rightarrow 0 \\ &= \mathbf{a} \times \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \times \mathbf{b} + \mathbf{0} = \mathbf{a} \times \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \times \mathbf{b}. \end{aligned}$$

Note: We know that cross product of two vectors is not commutative because $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$. Therefore while evaluating $\frac{d}{dt}(\mathbf{a} \times \mathbf{b})$, we must maintain the order of the factors $\mathbf{a}$ and $\mathbf{b}$.

$$\begin{aligned}
 4. \quad \frac{d}{dt}(\phi \mathbf{a}) &= \lim_{\delta t \rightarrow 0} \frac{(\phi + \delta\phi)(\mathbf{a} + \delta\mathbf{a}) - \phi \mathbf{a}}{\delta t} \\
 &= \lim_{\delta t \rightarrow 0} \frac{\phi \mathbf{a} + \phi \delta\mathbf{a} + \delta\phi \mathbf{a} + \delta\phi \delta\mathbf{a} - \phi \mathbf{a}}{\delta t} \\
 &= \lim_{\delta t \rightarrow 0} \frac{\phi \delta\mathbf{a} + \delta\phi \mathbf{a} + \delta\phi \delta\mathbf{a}}{\delta t} \\
 &= \lim_{\delta t \rightarrow 0} \left\{ \phi \frac{\delta\mathbf{a}}{\delta t} + \frac{\delta\phi}{\delta t} \mathbf{a} + \frac{\delta\phi}{\delta t} \delta\mathbf{a} \right\} \\
 &= \lim_{\delta t \rightarrow 0} \phi \frac{\delta\mathbf{a}}{\delta t} + \lim_{\delta t \rightarrow 0} \frac{\delta\phi}{\delta t} \mathbf{a} + \lim_{\delta t \rightarrow 0} \frac{\delta\phi}{\delta t} \delta\mathbf{a} \\
 &= \phi \frac{d\mathbf{a}}{dt} + \frac{d\phi}{dt} \mathbf{a} + \frac{d\phi}{dt} \mathbf{0}, \text{ since } \delta\mathbf{a} \rightarrow \text{zero vector as } \delta t \rightarrow 0 \\
 &= \phi \frac{d\mathbf{a}}{dt} + \frac{d\phi}{dt} \mathbf{a} + \mathbf{0} = \phi \frac{d\mathbf{a}}{dt} + \frac{d\phi}{dt} \mathbf{a}.
 \end{aligned}$$

Note: $\phi \mathbf{a}$ is the multiplication of a vector by a scalar. In the case of such multiplication we usually write the scalar in the first position and the vector in the second position.

$$\begin{aligned}
 5. \quad \frac{d}{dt}[\mathbf{a} \mathbf{b} \mathbf{c}] &= \frac{d}{dt} \{ \mathbf{a} \bullet (\mathbf{b} \times \mathbf{c}) \} = \mathbf{a} \bullet \frac{d}{dt}(\mathbf{b} \times \mathbf{c}) + \frac{d\mathbf{a}}{dt} \bullet (\mathbf{b} \times \mathbf{c}) \quad [\text{by rule (2)}] \\
 &= \mathbf{a} \bullet \left(\mathbf{b} \times \frac{d\mathbf{c}}{dt} + \frac{d\mathbf{b}}{dt} \times \mathbf{c} \right) + \frac{d\mathbf{a}}{dt} \bullet (\mathbf{b} \times \mathbf{c}) \quad [\text{by rule (3)}] \\
 &= \mathbf{a} \bullet \left(\mathbf{b} \times \frac{d\mathbf{c}}{dt} \right) + \mathbf{a} \bullet \left(\frac{d\mathbf{b}}{dt} \times \mathbf{c} \right) + \frac{d\mathbf{a}}{dt} \bullet (\mathbf{b} \times \mathbf{c}) \\
 &= \left[\mathbf{a} \mathbf{b} \frac{d\mathbf{c}}{dt} \right] + \left[\mathbf{a} \frac{d\mathbf{b}}{dt} \mathbf{c} \right] + \left[\frac{d\mathbf{a}}{dt} \mathbf{b} \mathbf{c} \right] \\
 &= \left[\frac{d\mathbf{a}}{dt} \mathbf{b} \mathbf{c} \right] + \left[\mathbf{a} \frac{d\mathbf{b}}{dt} \mathbf{c} \right] + \left[\mathbf{a} \mathbf{b} \frac{d\mathbf{c}}{dt} \right].
 \end{aligned}$$

Note: Here $[\mathbf{abc}]$ is the scalar triple product of three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. Therefore while evaluating $\frac{d}{dt}[\mathbf{abc}]$ we must maintain the cyclic order of each factor.

$$\begin{aligned}
 6. \quad \frac{d}{dt} \{ \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \} &= \mathbf{a} \times \frac{d}{dt}(\mathbf{b} \times \mathbf{c}) + \frac{d\mathbf{a}}{dt} \times (\mathbf{b} \times \mathbf{c}) \quad [\text{by rule (3)}] \\
 &= \mathbf{a} \times \left(\frac{d\mathbf{b}}{dt} \times \mathbf{c} + \mathbf{b} \times \frac{d\mathbf{c}}{dt} \right) + \frac{d\mathbf{a}}{dt} \times (\mathbf{b} \times \mathbf{c})
 \end{aligned}$$

$$\begin{aligned}
&= \mathbf{a} \times \left(\frac{d\mathbf{b}}{dt} \times \mathbf{c} \right) + \mathbf{a} \times \left(\mathbf{b} \times \frac{d\mathbf{c}}{dt} \right) + \frac{d\mathbf{a}}{dt} \times (\mathbf{b} \times \mathbf{c}) \\
&= \frac{d\mathbf{a}}{dt} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{a} \times \left(\frac{d\mathbf{b}}{dt} \times \mathbf{c} \right) + \mathbf{a} \times \left(\mathbf{b} \times \frac{d\mathbf{c}}{dt} \right).
\end{aligned}$$

6 Derivative of a Function of a Function

Suppose $\mathbf{r}$ is a differentiable vector function of a scalar variable s and s is a differentiable scalar function of another scalar variable t . Then $\mathbf{r}$ is a function of t .

An increment δt in t produces an increment $\delta \mathbf{r}$ in $\mathbf{r}$ and an increment δs in s . When $\delta t \rightarrow 0$, $\delta \mathbf{r} \rightarrow \mathbf{0}$ and $\delta s \rightarrow 0$.

We have

$$\begin{aligned}
\frac{d\mathbf{r}}{dt} &= \lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{r}}{\delta t} = \lim_{\delta t \rightarrow 0} \left(\frac{\delta s}{\delta t} \frac{\delta \mathbf{r}}{\delta s} \right) \\
&= \left(\lim_{\delta t \rightarrow 0} \frac{\delta s}{\delta t} \right) \left(\lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{r}}{\delta s} \right) = \frac{ds}{dt} \frac{d\mathbf{r}}{ds}.
\end{aligned}$$

7 Derivative of a Constant Vector

A vector is said to be constant only if both its magnitude and direction are fixed. If either of these changes then the vector will change and thus it will not be constant.

Let $\mathbf{r}$ be a constant vector function of the scalar variable t . Let $\mathbf{r} = \mathbf{c}$, where $\mathbf{c}$ is a constant vector. Then $\mathbf{r} + \delta \mathbf{r} = \mathbf{c}$.

$$\begin{aligned}
\therefore \quad \delta \mathbf{r} &= \mathbf{0} \text{ (zero vector).} & \therefore \quad \frac{\delta \mathbf{r}}{\delta t} &= \frac{\mathbf{0}}{\delta t} = \mathbf{0}. \\
\therefore \quad \lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{r}}{\delta t} &= \lim_{\delta t \rightarrow 0} \mathbf{0} = \mathbf{0}. & \therefore \quad \frac{d\mathbf{r}}{dt} &= \mathbf{0} \text{ (zero vector).}
\end{aligned}$$

Thus the derivative of a constant vector is equal to the null vector.

8 Derivative of a Vector Function in Terms of its Components

Let $\mathbf{r}$ be a vector function of the scalar variable t . (Avadh 2014)

Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ where the components x, y, z are scalar functions of the scalar variable t and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are fixed unit vectors.

We have $\mathbf{r} + \delta \mathbf{r} = (x + \delta x)\mathbf{i} + (y + \delta y)\mathbf{j} + (z + \delta z)\mathbf{k}$.

$$\begin{aligned}
\therefore \quad \delta \mathbf{r} &= (\mathbf{r} + \delta \mathbf{r}) - \mathbf{r} = \delta x \mathbf{i} + \delta y \mathbf{j} + \delta z \mathbf{k}. \\
\therefore \quad \frac{\delta \mathbf{r}}{\delta t} &= \frac{\delta x}{\delta t} \mathbf{i} + \frac{\delta y}{\delta t} \mathbf{j} + \frac{\delta z}{\delta t} \mathbf{k}. \\
\therefore \quad \lim_{\delta t \rightarrow 0} \frac{\delta \mathbf{r}}{\delta t} &= \lim_{\delta t \rightarrow 0} \left\{ \frac{\delta x}{\delta t} \mathbf{i} + \frac{\delta y}{\delta t} \mathbf{j} + \frac{\delta z}{\delta t} \mathbf{k} \right\}.
\end{aligned}$$

$$\therefore \frac{d\mathbf{r}}{dt} = \frac{dx}{dt} \mathbf{i} + \frac{dy}{dt} \mathbf{j} + \frac{dz}{dt} \mathbf{k}.$$

Thus in order to differentiate a vector we should differentiate its components.

Note: If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then sometimes we also write it as $\mathbf{r} = (x, y, z)$. In this

notation $\frac{d\mathbf{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$, $\frac{d^2\mathbf{r}}{dt^2} = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2} \right)$, and so on.

Alternative Method:

We have $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are constant vectors and so their derivatives will be zero.

$$\begin{aligned} \text{Now, } \frac{d\mathbf{r}}{dt} &= \frac{d}{dt} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \frac{d}{dt} (x\mathbf{i}) + \frac{d}{dt} (y\mathbf{j}) + \frac{d}{dt} (z\mathbf{k}) \\ &= \frac{dx}{dt} \mathbf{i} + x \frac{d\mathbf{i}}{dt} + \frac{dy}{dt} \mathbf{j} + y \frac{d\mathbf{j}}{dt} + \frac{dz}{dt} \mathbf{k} + z \frac{d\mathbf{k}}{dt} \\ &= \frac{dx}{dt} \mathbf{i} + \frac{dy}{dt} \mathbf{j} + \frac{dz}{dt} \mathbf{k}, \text{ since } \frac{d\mathbf{i}}{dt}, \text{ etc. vanish.} \end{aligned}$$

9 Some Important Results

Theorem 1: The necessary and sufficient condition for the vector function $\mathbf{a}(t)$ to be constant is that $\frac{d\mathbf{a}}{dt} = \mathbf{0}$. (Purvanchal 2014)

Proof: **The condition is necessary.** Let $\mathbf{a}(t)$ be a constant vector function of the scalar variable t . Then $\mathbf{a}(t + \delta t) = \mathbf{a}(t)$. We have

$$\frac{d\mathbf{a}}{dt} = \lim_{\delta t \rightarrow 0} \frac{\mathbf{a}(t + \delta t) - \mathbf{a}(t)}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{\mathbf{0}}{\delta t} = \mathbf{0}.$$

Therefore the condition is necessary.

The condition is sufficient. Let $\frac{d\mathbf{a}}{dt} = \mathbf{0}$. Then to prove that $\mathbf{a}$ is a constant vector. Let

$$\mathbf{a}(t) = a_1(t) \mathbf{i} + a_2(t) \mathbf{j} + a_3(t) \mathbf{k}.$$

$$\text{Then } \frac{d\mathbf{a}}{dt} = \frac{da_1}{dt} \mathbf{i} + \frac{da_2}{dt} \mathbf{j} + \frac{da_3}{dt} \mathbf{k}.$$

$$\text{Therefore } \frac{d\mathbf{a}}{dt} = \mathbf{0} \text{ gives, } \frac{da_1}{dt} \mathbf{i} + \frac{da_2}{dt} \mathbf{j} + \frac{da_3}{dt} \mathbf{k} = \mathbf{0}.$$

Equating to zero the coefficients of $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$, we get

$$\frac{da_1}{dt} = 0, \frac{da_2}{dt} = 0, \frac{da_3}{dt} = 0.$$

Hence a_1, a_2, a_3 are constant scalars i.e., they are independent of t . Therefore $\mathbf{a}(t)$ is a constant vector function.

Theorem 2: If $\mathbf{a}$ is a differentiable vector function of the scalar variable t and if $|\mathbf{a}| = a$, then

$$(i) \quad \frac{d}{dt} (\mathbf{a}^2) = 2a \frac{da}{dt}; \quad (ii) \quad \mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = a \frac{da}{dt}.$$

Proof: (i) We have $\mathbf{a}^2 = \mathbf{a} \bullet \mathbf{a} = (a)(a) \cos 0 = a^2$.

Therefore $\frac{d}{dt} (\mathbf{a}^2) = \frac{d}{dt} (a^2) = 2a \frac{da}{dt}$.

(ii) We have $\frac{d}{dt} (\mathbf{a}^2) = \frac{d}{dt} (\mathbf{a} \bullet \mathbf{a}) = \frac{d\mathbf{a}}{dt} \bullet \mathbf{a} + \mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 2\mathbf{a} \bullet \frac{d\mathbf{a}}{dt}$.

Also $\frac{d}{dt} (\mathbf{a}^2) = \frac{d}{dt} (a^2) = 2a \frac{da}{dt}$.

$\therefore 2\mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 2a \frac{da}{dt} \quad \text{or} \quad \mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = a \frac{da}{dt}.$

Theorem 3: If $\mathbf{a}$ has constant length (fixed magnitude), then $\mathbf{a}$ and $\frac{d\mathbf{a}}{dt}$ are perpendicular provided $\left| \frac{d\mathbf{a}}{dt} \right| \neq 0$.

Proof: Let $|\mathbf{a}| = a = \text{constant}$. Then $\mathbf{a} \bullet \mathbf{a} = a^2 = \text{constant}$.

$\therefore \frac{d}{dt} (\mathbf{a} \bullet \mathbf{a}) = 0 \quad \text{or} \quad \frac{d\mathbf{a}}{dt} \bullet \mathbf{a} + \mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 0$

or $2\mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 0 \quad \text{or} \quad \mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 0.$

Thus the scalar product of two vectors $\mathbf{a}$ and $\frac{d\mathbf{a}}{dt}$ is zero.

Therefore $\mathbf{a}$ is perpendicular to $\frac{d\mathbf{a}}{dt}$ provided $\frac{d\mathbf{a}}{dt}$ is not null vector i.e., provided

$$\left| \frac{d\mathbf{a}}{dt} \right| \neq 0.$$

Thus the derivative of a vector of constant length is perpendicular to the vector provided the vector itself is not constant.

Theorem 4: The necessary and sufficient condition for the vector $\mathbf{a}(t)$ to have constant magnitude is $\mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 0$. (Meerut 2001, 04B, 06, 07, 11, 13; Avadh 13;

Kashi 10, 12; Purvanchal 11)

Proof: Let $\mathbf{a}$ be a vector function of the scalar variable t . Let $|\mathbf{a}| = a = \text{constant}$. Then $\mathbf{a} \bullet \mathbf{a} = a^2 = \text{constant}$.

$\therefore \frac{d}{dt} (\mathbf{a} \bullet \mathbf{a}) = 0 \quad \text{or} \quad \mathbf{a} \bullet \frac{d\mathbf{a}}{dt} + \frac{d\mathbf{a}}{dt} \bullet \mathbf{a} = 0$

or $2\mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 0 \quad \text{or} \quad \mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 0.$

Therefore the condition is necessary.

Condition is sufficient. If $\mathbf{a} \bullet \frac{d\mathbf{a}}{dt} = 0$, then

$$\mathbf{a} \bullet \frac{d\mathbf{a}}{dt} + \frac{d\mathbf{a}}{dt} \bullet \mathbf{a} = 0 \quad \text{or} \quad \frac{d}{dt} (\mathbf{a} \bullet \mathbf{a}) = 0$$

$$\text{or} \quad \mathbf{a} \bullet \mathbf{a} = \text{constant} \quad \text{or} \quad a^2 = \text{constant}$$

$$\text{or} \quad a^2 = \text{constant} \quad \text{or} \quad |\mathbf{a}| = \text{constant}$$

Theorem 5: If $\mathbf{a}$ is a differentiable vector function of the scalar variable t , then

$$\frac{d}{dt} \left(\mathbf{a} \times \frac{d\mathbf{a}}{dt} \right) = \mathbf{a} \times \frac{d^2\mathbf{a}}{dt^2}.$$

Proof: We have

$$\frac{d}{dt} \left(\mathbf{a} \times \frac{d\mathbf{a}}{dt} \right) = \frac{d\mathbf{a}}{dt} \times \frac{d\mathbf{a}}{dt} + \mathbf{a} \times \frac{d^2\mathbf{a}}{dt^2} = \mathbf{0} + \mathbf{a} \times \frac{d^2\mathbf{a}}{dt^2},$$

since the cross product of two equal vectors $\frac{d\mathbf{a}}{dt}$ is zero $= \mathbf{a} \times \frac{d^2\mathbf{a}}{dt^2}$.

Theorem 6: The necessary and sufficient condition for the vector $\mathbf{a}(t)$ to have constant direction is $\mathbf{a} \times \frac{d\mathbf{a}}{dt} = \mathbf{0}$. (Meerut 2004B, 06, 07, 10, 11, 13B; Avadh 14)

Proof: Let $\mathbf{a}$ be a vector function of the scalar variable t . Let $\mathbf{A}$ be a unit vector in the direction of $\mathbf{a}$. If a be the magnitude of $\mathbf{a}$, then $\mathbf{a} = a\mathbf{A}$.

$$\therefore \frac{d\mathbf{a}}{dt} = a \frac{d\mathbf{A}}{dt} + \frac{da}{dt} \mathbf{A}.$$

$$\begin{aligned} \text{Hence} \quad \mathbf{a} \times \frac{d\mathbf{a}}{dt} &= (a \mathbf{A}) \times \left(a \frac{d\mathbf{A}}{dt} + \frac{da}{dt} \mathbf{A} \right) \\ &= a^2 \mathbf{A} \times \frac{d\mathbf{A}}{dt} + a \frac{da}{dt} \mathbf{A} \times \mathbf{A} \\ &= a^2 \mathbf{A} \times \frac{d\mathbf{A}}{dt} \quad [\because \mathbf{A} \times \mathbf{A} = \mathbf{0}] \quad \dots(1) \end{aligned}$$

The condition is necessary. Suppose $\mathbf{a}$ has a constant direction. Then $\mathbf{A}$ is a constant vector because it has constant direction as well as constant magnitude. Therefore $\frac{d\mathbf{A}}{dt} = \mathbf{0}$.

$$\therefore \text{From (1), we get } \mathbf{a} \times \frac{d\mathbf{a}}{dt} = a^2 \mathbf{A} \times \mathbf{0} = \mathbf{0}.$$

Therefore the condition is necessary.

The condition is sufficient. Suppose that $\mathbf{a} \times \frac{d\mathbf{a}}{dt} = \mathbf{0}$.

Then from (1), we get

$$a^2 \mathbf{A} \times \frac{d\mathbf{A}}{dt} = \mathbf{0} \quad \text{or} \quad \mathbf{A} \times \frac{d\mathbf{A}}{dt} = \mathbf{0}. \quad \dots(2)$$

Since $\mathbf{A}$ is of constant length, therefore $\mathbf{A} \cdot \frac{d\mathbf{A}}{dt} = 0$ (3)

From (2) and (3), we get $\frac{d\mathbf{A}}{dt} = \mathbf{0}$.

Hence $\mathbf{A}$ is a constant vector i.e., the direction of $\mathbf{a}$ is constant.

10 Curves in Space

A curve in a three dimensional Euclidean space may be regarded as the intersection of two surfaces represented by two equations of the form

$$F_1(x, y, z) = 0, F_2(x, y, z) = 0.$$

It can be easily seen that the parametric equations of the form

$$x = f_1(t), y = f_2(t), z = f_3(t)$$

where x, y, z are scalar functions of the scalar t , also represent a curve in three dimensional space. Here (x, y, z) are the coordinates of a current point on the curve. The scalar variable t may range over a set of values $a \leq t \leq b$.

In vector notation an equation of the form $\mathbf{r} = \mathbf{f}(t)$, represents a curve in three dimensional space if $\mathbf{r}$ is the position vector of a current point on the curve. As t changes, $\mathbf{r}$ will give position vectors of different points on the curve. The vector $\mathbf{f}(t)$ can be expressed as

$$f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}.$$

Also if (x, y, z) are the coordinates of a current point on the curve whose position vector is $\mathbf{r}$, then $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

Therefore the single vector equation $\mathbf{r} = \mathbf{f}(t)$

i.e., $x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}$

is equivalent to the three parametric equations

$$x = f_1(t), y = f_2(t), z = f_3(t).$$

Thus a curve in a space may be defined as the locus of a point whose coordinates may be expressed as a function of a **single** parameter.

For example, the two equations

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad x = a \cosh \frac{z}{a}$$

specify a curve in three dimensional space. The parametric equations of this curve are

$$x = a \cosh u, \quad y = b \sinh u, \quad z = au.$$

And its vectorial equation is

$$\mathbf{r} = a \cosh u \mathbf{i} + b \sinh u \mathbf{j} + au \mathbf{k}.$$

The vector equation $\mathbf{r} = a \cos t \mathbf{i} + b \sin t \mathbf{j} + 0 \mathbf{k}$ represents an ellipse, as for different values of t , the end point of $\mathbf{r}$ describes an ellipse.

Similarly $\mathbf{r} = at^2 \mathbf{i} + 2at \mathbf{j} + 0 \mathbf{k}$ is the vector equation of a parabola.

The terms **skew**, **twisted** or **tortuous** are often used for curves in a space.

11 Geometrical Significance of $d\mathbf{r}/dt$ and Unit Tangent Vector to a Curve

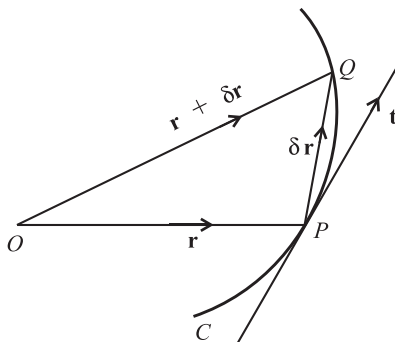
Geometrical Significance of $d\mathbf{r}/dt$:

Let $\mathbf{r} = \mathbf{f}(t)$ be the vector equation of a curve in space. Let $\mathbf{r}$ and $\mathbf{r} + \delta\mathbf{r}$ be the position vectors of two neighbouring points P and Q on this curve. Thus we have

$$\vec{OP} = \mathbf{r} = \mathbf{f}(t)$$

and $\vec{OQ} = \mathbf{r} + \delta\mathbf{r} = \mathbf{f}(t + \delta t).$

$$\begin{aligned} \therefore \vec{PQ} &= \vec{OQ} - \vec{OP} \\ &= (\mathbf{r} + \delta\mathbf{r}) - \mathbf{r} = \delta\mathbf{r}. \end{aligned}$$



Thus $\frac{\delta\mathbf{r}}{\delta t}$ is a vector parallel to the chord PQ .

As $Q \rightarrow P$ i.e., as $\delta t \rightarrow 0$, chord $PQ \rightarrow$ tangent at P to the curve.

$\therefore \lim_{\delta t \rightarrow 0} \frac{\delta\mathbf{r}}{\delta t} = \frac{d\mathbf{r}}{dt}$ is a vector parallel to the tangent at P to the curve $\mathbf{r} = \mathbf{f}(t)$.

Unit tangent vector to a curve:

Suppose in place of the scalar parameter t , we take the parameter as s where s denotes the arc length measured along the curve from any convenient fixed point C on the curve. Thus arc $CP = s$ and arc $CQ = s + \delta s$.

In this case $\frac{d\mathbf{r}}{ds}$ will be a vector along the tangent at P to the curve and in the direction of s increasing. Also we have

$$\left| \frac{d\mathbf{r}}{ds} \right| = \lim_{\delta s \rightarrow 0} \left| \frac{\delta\mathbf{r}}{\delta s} \right| = \lim_{\delta s \rightarrow 0} \frac{|\delta\mathbf{r}|}{\text{arc } PQ} = \lim_{Q \rightarrow P} \frac{\text{chord } PQ}{\text{arc } PQ} = 1.$$

Thus $\frac{d\mathbf{r}}{ds}$ is a unit vector along the tangent at P in the direction of s increasing. We denote it by $\mathbf{t}$. Note that $\mathbf{t}$ always points in the direction of motion along the curve.

12 Application to Velocity and Acceleration

Velocity: If the scalar variable t be the time and $\mathbf{r}$ be the position vector of a moving particle P with respect to the origin O , then $\delta\mathbf{r}$ is the displacement of the particle in time δt .

The vector $\frac{\delta\mathbf{r}}{\delta t}$ is the average velocity of the particle during the interval δt . If $\mathbf{v}$

represents the velocity vector of the particle at P , then $\mathbf{v} = \lim_{\delta t \rightarrow 0} \frac{\delta\mathbf{r}}{\delta t} = \frac{d\mathbf{r}}{dt}$.

Since $\frac{d\mathbf{r}}{dt}$ is a vector along the tangent at P to the curve in which the particle is moving, therefore the direction of velocity is along the tangent.

Acceleration: If $\delta\mathbf{v}$ be the change in the velocity $\mathbf{v}$ during the time δt , then $\frac{\delta\mathbf{v}}{\delta t}$ is the average acceleration during that interval. If $\mathbf{a}$ represents the acceleration of the particle at time t , then

$$\mathbf{a} = \lim_{\delta t \rightarrow 0} \frac{\delta\mathbf{v}}{\delta t} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt} \left(\frac{d\mathbf{r}}{dt} \right) = \frac{d^2\mathbf{r}}{dt^2}.$$

Illustrative Examples

Example 1: If $\mathbf{r} = \sin t \mathbf{i} + \cos t \mathbf{j} + t\mathbf{k}$, find

$$(i) \frac{d\mathbf{r}}{dt}, \quad (\text{Kashi 2013}) \quad (ii) \frac{d^2\mathbf{r}}{dt^2}, \quad (\text{Meerut 2010B})$$

$$(iii) \left| \frac{d\mathbf{r}}{dt} \right|, \quad (\text{Kashi 2013}) \quad (iv) \left| \frac{d^2\mathbf{r}}{dt^2} \right|. \quad (\text{Meerut 2009})$$

Solution: Since $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are constant vectors, therefore $\frac{d\mathbf{i}}{dt} = \mathbf{0}$ etc. Therefore

$$(i) \frac{d\mathbf{r}}{dt} = \frac{d}{dt} (\sin t) \mathbf{i} + \frac{d}{dt} (\cos t) \mathbf{j} + \frac{d}{dt} (t) \mathbf{k} = \cos t \mathbf{i} - \sin t \mathbf{j} + \mathbf{k}.$$

$$(ii) \frac{d^2\mathbf{r}}{dt^2} = \frac{d}{dt} \left(\frac{d\mathbf{r}}{dt} \right) = \frac{d}{dt} (\cos t) \mathbf{i} - \frac{d}{dt} (\sin t) \mathbf{j} + \frac{d\mathbf{k}}{dt} \\ = -\sin t \mathbf{i} - \cos t \mathbf{j} + \mathbf{0} = -\sin t \mathbf{i} - \cos t \mathbf{j}.$$

$$(iii) \left| \frac{d\mathbf{r}}{dt} \right| = \sqrt{(\cos t)^2 + (-\sin t)^2 + (1)^2} = \sqrt{2}.$$

$$(iv) \left| \frac{d^2\mathbf{r}}{dt^2} \right| = \sqrt{(-\sin t)^2 + (-\cos t)^2} = 1.$$

Example 2: If $\mathbf{a}$, $\mathbf{b}$ are constant vectors, ω is a constant, and $\mathbf{r}$ is a vector function of the scalar variable t given by $\mathbf{r} = \cos \omega t \mathbf{a} + \sin \omega t \mathbf{b}$, show that

$$(i) \frac{d^2\mathbf{r}}{dt^2} + \omega^2 \mathbf{r} = \mathbf{0} \quad (\text{Rohilkhand 2006})$$

$$(ii) \mathbf{r} \times \frac{d\mathbf{r}}{dt} = \omega \mathbf{a} \times \mathbf{b}. \quad (\text{Bundelkhand 2004; Kanpur 05})$$

Solution: Since $\mathbf{a}$, $\mathbf{b}$ are constant vectors, therefore

$$\frac{d\mathbf{a}}{dt} = \mathbf{0}, \quad \frac{d\mathbf{b}}{dt} = \mathbf{0}.$$

$$(i) \quad \frac{d\mathbf{r}}{dt} = \frac{d}{dt} (\cos \omega t) \mathbf{a} + \frac{d}{dt} (\sin \omega t) \mathbf{b}$$

$$= -\omega \sin \omega t \mathbf{a} + \omega \cos \omega t \mathbf{b}.$$

$$\therefore \quad \frac{d^2 \mathbf{r}}{dt^2} = -\omega^2 \cos \omega t \mathbf{a} - \omega^2 \sin \omega t \mathbf{b}$$

$$= -\omega^2 (\cos \omega t \mathbf{a} + \sin \omega t \mathbf{b}) = -\omega^2 \mathbf{r}.$$

$$\therefore \quad \frac{d^2 \mathbf{r}}{dt^2} + \omega^2 \mathbf{r} = \mathbf{0}.$$

$$(ii) \quad \mathbf{r} \times \frac{d\mathbf{r}}{dt} = (\cos \omega t \mathbf{a} + \sin \omega t \mathbf{b}) \times (-\omega \sin \omega t \mathbf{a} + \omega \cos \omega t \mathbf{b})$$

$$= \omega \cos^2 \omega t \mathbf{a} \times \mathbf{b} - \omega \sin^2 \omega t \mathbf{b} \times \mathbf{a} \quad [\because \mathbf{a} \times \mathbf{a} = \mathbf{0}, \mathbf{b} \times \mathbf{b} = \mathbf{0}]$$

$$= \omega \cos^2 \omega t \mathbf{a} \times \mathbf{b} + \omega \sin^2 \omega t \mathbf{a} \times \mathbf{b}$$

$$= \omega (\cos^2 \omega t + \sin^2 \omega t) \mathbf{a} \times \mathbf{b} = \omega \mathbf{a} \times \mathbf{b}.$$

Example 3: If $\mathbf{r} = a \cos t \mathbf{i} + a \sin t \mathbf{j} + a \tan \alpha \mathbf{k}$, find

$$\left| \frac{d\mathbf{r}}{dt} \times \frac{d^2 \mathbf{r}}{dt^2} \right| \text{ and } \left[\frac{d\mathbf{r}}{dt}, \frac{d^2 \mathbf{r}}{dt^2}, \frac{d^3 \mathbf{r}}{dt^3} \right]. \quad (\text{Purvanchal 07, 10; Agra 14})$$

Solution: We have $\frac{d\mathbf{r}}{dt} = -a \sin t \mathbf{i} + a \cos t \mathbf{j} + a \tan \alpha \mathbf{k}$

$$\frac{d^2 \mathbf{r}}{dt^2} = -a \cos t \mathbf{i} - a \sin t \mathbf{j}, \quad \left[\because \frac{d\mathbf{k}}{dt} = \mathbf{0} \right]$$

$$\frac{d^3 \mathbf{r}}{dt^3} = a \sin t \mathbf{i} - a \cos t \mathbf{j}.$$

$$\therefore \quad \frac{d\mathbf{r}}{dt} \times \frac{d^2 \mathbf{r}}{dt^2} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -a \sin t & a \cos t & a \tan \alpha \\ -a \cos t & -a \sin t & 0 \end{vmatrix}$$

$$= a^2 \sin t \tan \alpha \mathbf{i} - a^2 \cos t \tan \alpha \mathbf{j} + a^2 \mathbf{k}.$$

$$\therefore \quad \left| \frac{d\mathbf{r}}{dt} \times \frac{d^2 \mathbf{r}}{dt^2} \right| = \sqrt{(a^4 \sin^2 t \tan^2 \alpha + a^4 \cos^2 t \tan^2 \alpha + a^4)}$$

$$= a^2 \sec \alpha.$$

Also

$$\left[\frac{d\mathbf{r}}{dt}, \frac{d^2 \mathbf{r}}{dt^2}, \frac{d^3 \mathbf{r}}{dt^3} \right] = \left(\frac{d\mathbf{r}}{dt} \times \frac{d^2 \mathbf{r}}{dt^2} \right) \cdot \frac{d^3 \mathbf{r}}{dt^3}$$

$$= (a^2 \sin t \tan \alpha \mathbf{i} - a^2 \cos t \tan \alpha \mathbf{j} + a^2 \mathbf{k}) \cdot (a \sin t \mathbf{i} - a \cos t \mathbf{j})$$

$$= a^3 \sin^2 t \tan \alpha \mathbf{i} \cdot \mathbf{i} + a^3 \cos^2 t \tan \alpha \mathbf{j} \cdot \mathbf{j} \quad [\because \mathbf{i} \cdot \mathbf{j} = 0 \text{ etc.}]$$

$$= a^3 \tan \alpha (\sin^2 t + \cos^2 t) \quad [\because \mathbf{i} \cdot \mathbf{i} = 1 = \mathbf{j} \cdot \mathbf{j}]$$

$$= a^3 \tan \alpha.$$

Example 4: If $\frac{d\mathbf{u}}{dt} = \mathbf{w} \times \mathbf{u}$, $\frac{d\mathbf{v}}{dt} = \mathbf{w} \times \mathbf{v}$, show that $\frac{d}{dt} (\mathbf{u} \times \mathbf{v}) = \mathbf{w} \times (\mathbf{u} \times \mathbf{v})$.

(Garhwal 2003; Bundelkhand 09)

Solution: We have

$$\begin{aligned} \frac{d}{dt} (\mathbf{u} \times \mathbf{v}) &= \frac{d\mathbf{u}}{dt} \times \mathbf{v} + \mathbf{u} \times \frac{d\mathbf{v}}{dt} = (\mathbf{w} \times \mathbf{u}) \times \mathbf{v} + \mathbf{u} \times (\mathbf{w} \times \mathbf{v}) \\ &= (\mathbf{v} \bullet \mathbf{w}) \mathbf{u} - (\mathbf{v} \bullet \mathbf{u}) \mathbf{w} + (\mathbf{u} \bullet \mathbf{v}) \mathbf{w} - (\mathbf{u} \bullet \mathbf{w}) \mathbf{v} \\ &= (\mathbf{v} \bullet \mathbf{w}) \mathbf{u} - (\mathbf{u} \bullet \mathbf{w}) \mathbf{v} \quad [\because \mathbf{u} \bullet \mathbf{v} = \mathbf{v} \bullet \mathbf{u}] \\ &= (\mathbf{w} \bullet \mathbf{v}) \mathbf{u} - (\mathbf{w} \bullet \mathbf{u}) \mathbf{v} = \mathbf{w} \times (\mathbf{u} \times \mathbf{v}). \end{aligned}$$

Example 5: If $\mathbf{R}$ be a unit vector in the direction of $\mathbf{r}$, prove that

$$\mathbf{R} \times \frac{d\mathbf{R}}{dt} = \frac{1}{r^2} \mathbf{r} \times \frac{d\mathbf{r}}{dt}, \text{ where } r = |\mathbf{r}|. \quad (\text{Bundelkhand 2008, 11})$$

Solution: We have $\mathbf{r} = r\mathbf{R}$; so that $\mathbf{R} = \frac{1}{r} \mathbf{r}$.

$$\therefore \frac{d\mathbf{R}}{dt} = \frac{1}{r} \frac{d\mathbf{r}}{dt} - \frac{1}{r^2} \frac{dr}{dt} \mathbf{r}.$$

$$\begin{aligned} \text{Hence } \mathbf{R} \times \frac{d\mathbf{R}}{dt} &= \frac{1}{r} \mathbf{r} \times \left(\frac{1}{r} \frac{d\mathbf{r}}{dt} - \frac{1}{r^2} \frac{dr}{dt} \mathbf{r} \right) \\ &= \frac{1}{r^2} \mathbf{r} \times \frac{d\mathbf{r}}{dt} - \frac{1}{r^3} \frac{dr}{dt} \mathbf{r} \times \mathbf{r} \\ &= \frac{1}{r^2} \mathbf{r} \times \frac{d\mathbf{r}}{dt}. \quad [\because \mathbf{r} \times \mathbf{r} = \mathbf{0}] \end{aligned}$$

Example 6: If $\mathbf{r}$ is a vector function of a scalar t and $\mathbf{a}$ is a constant vector, m a constant, differentiate the following with respect to t :

- | | |
|---|---|
| (i) $\mathbf{r} \bullet \mathbf{a}$, | (ii) $\mathbf{r} \times \mathbf{a}$, |
| (iii) $\mathbf{r} \times \frac{d\mathbf{r}}{dt}$, | (iv) $\mathbf{r} \bullet \frac{d\mathbf{r}}{dt}$, |
| (v) $\mathbf{r}^2 + \frac{1}{r^2}$, | (vi) $m \left(\frac{d\mathbf{r}}{dt} \right)^2$, |
| (vii) $\frac{\mathbf{r} + \mathbf{a}}{r^2 + a^2}$, | (viii) $\frac{\mathbf{r} \times \mathbf{a}}{\mathbf{r} \bullet \mathbf{a}}$. |

Solution: (i) Let $R = \mathbf{r} \bullet \mathbf{a}$.

[Note that $\mathbf{r} \bullet \mathbf{a}$ is a scalar]

$$\begin{aligned} \text{Then } \frac{dR}{dt} &= \frac{d\mathbf{r}}{dt} \bullet \mathbf{a} + \mathbf{r} \bullet \frac{d\mathbf{a}}{dt} \\ &= \frac{d\mathbf{r}}{dt} \bullet \mathbf{a} + \mathbf{r} \bullet \mathbf{0} \\ &= \frac{d\mathbf{r}}{dt} \bullet \mathbf{a} + \mathbf{0} \\ &= \frac{d\mathbf{r}}{dt} \bullet \mathbf{a}. \end{aligned} \quad \left[\because \frac{d\mathbf{a}}{dt} = \mathbf{0}, \text{ as } \mathbf{a} \text{ is constant} \right]$$

(ii) Let $\mathbf{R} = \mathbf{r} \times \mathbf{a}$.

$$\begin{aligned}
 \text{Then } \frac{d\mathbf{R}}{dt} &= \frac{d\mathbf{r}}{dt} \times \mathbf{a} + \mathbf{r} \times \frac{d\mathbf{a}}{dt} \\
 &= \frac{d\mathbf{r}}{dt} \times \mathbf{a} + \mathbf{r} \times \mathbf{0} \quad \left[\because \frac{d\mathbf{a}}{dt} = \mathbf{0} \right] \\
 &= \frac{d\mathbf{r}}{dt} \times \mathbf{a} + \mathbf{0} = \frac{d\mathbf{r}}{dt} \times \mathbf{a}.
 \end{aligned}$$

(iii) Let $\mathbf{R} = \mathbf{r} \times \frac{d\mathbf{r}}{dt}$.

$$\begin{aligned}
 \text{Then } \frac{d\mathbf{R}}{dt} &= \frac{d\mathbf{r}}{dt} \times \frac{d\mathbf{r}}{dt} + \mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2} \\
 &= \mathbf{0} + \mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2} \quad \left[\because \frac{d\mathbf{r}}{dt} \times \frac{d\mathbf{r}}{dt} = \mathbf{0} \right] \\
 &= \mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2}.
 \end{aligned}$$

(iv) Let $R = \mathbf{r} \bullet \frac{d\mathbf{r}}{dt}$.

$$\begin{aligned}
 \text{Then } \frac{dR}{dt} &= \frac{d\mathbf{r}}{dt} \bullet \frac{d\mathbf{r}}{dt} + \mathbf{r} \bullet \frac{d^2\mathbf{r}}{dt^2} \\
 &= \left(\frac{d\mathbf{r}}{dt} \right)^2 + \mathbf{r} \bullet \frac{d^2\mathbf{r}}{dt^2}.
 \end{aligned}$$

(v) Let $R = \mathbf{r}^2 + \frac{1}{\mathbf{r}^2}$.

$$\begin{aligned}
 \text{Then } \frac{dR}{dt} &= \frac{d}{dt} (\mathbf{r}^2) + \frac{d}{dt} \left(\frac{1}{\mathbf{r}^2} \right) \\
 &= \frac{d}{dt} (r^2) + \frac{d}{dt} \left(\frac{1}{r^2} \right), \text{ where } r = |\mathbf{r}| \\
 &= 2r \frac{dr}{dt} - \frac{2}{r^3} \frac{dr}{dt}.
 \end{aligned}$$

(vi) Let $R = m \left(\frac{d\mathbf{r}}{dt} \right)^2$.

$$\begin{aligned}
 \text{Then } \frac{dR}{dt} &= m \frac{d}{dt} \left(\frac{d\mathbf{r}}{dt} \right)^2 \\
 &= 2m \frac{d\mathbf{r}}{dt} \bullet \frac{d^2\mathbf{r}}{dt^2} \quad \left[\text{Note: } \frac{d\mathbf{r}^2}{dt} = 2 \mathbf{r} \bullet \frac{d\mathbf{r}}{dt} \right] \\
 &= 2m \frac{d\mathbf{r}}{dt} \bullet \frac{d^2\mathbf{r}}{dt^2}.
 \end{aligned}$$

(vii) Let $\mathbf{R} = \frac{\mathbf{r} + \mathbf{a}}{\mathbf{r}^2 + \mathbf{a}^2}$.

Then
$$\begin{aligned} \frac{d\mathbf{R}}{dt} &= \frac{1}{(\mathbf{r}^2 + \mathbf{a}^2)} \frac{d}{dt} (\mathbf{r} + \mathbf{a}) + \left\{ \frac{d}{dt} \left(\frac{1}{\mathbf{r}^2 + \mathbf{a}^2} \right) \right\} (\mathbf{r} + \mathbf{a}) \\ &\quad \text{[Note that } \mathbf{r}^2 + \mathbf{a}^2 \text{ is a scalar]} \\ &= \frac{1}{\mathbf{r}^2 + \mathbf{a}^2} \left(\frac{d\mathbf{r}}{dt} + \frac{d\mathbf{a}}{dt} \right) - \left\{ \frac{1}{(\mathbf{r}^2 + \mathbf{a}^2)^2} \frac{d}{dt} (\mathbf{r}^2 + \mathbf{a}^2) \right\} (\mathbf{r} + \mathbf{a}) \\ &= \frac{1}{\mathbf{r}^2 + \mathbf{a}^2} \frac{d\mathbf{r}}{dt} - \frac{2 \mathbf{r} \bullet \frac{d\mathbf{r}}{dt}}{(\mathbf{r}^2 + \mathbf{a}^2)^2} (\mathbf{r} + \mathbf{a}) \\ &\quad \left[\because \frac{d\mathbf{a}}{dt} = \mathbf{0}, \frac{d}{dt} \mathbf{r}^2 = 2\mathbf{r} \bullet \frac{d\mathbf{r}}{dt}, \frac{d}{dt} \mathbf{a}^2 = 0 \right] \end{aligned}$$

(viii) Let $\mathbf{R} = \frac{\mathbf{r} \times \mathbf{a}}{\mathbf{r} \bullet \mathbf{a}}$.

Then
$$\begin{aligned} \frac{d\mathbf{R}}{dt} &= \frac{1}{\mathbf{r} \bullet \mathbf{a}} \frac{d}{dt} (\mathbf{r} \times \mathbf{a}) + \left\{ \frac{d}{dt} \left(\frac{1}{\mathbf{r} \bullet \mathbf{a}} \right) \right\} (\mathbf{r} \times \mathbf{a}) \\ &\quad \text{[Note that } \mathbf{r} \bullet \mathbf{a} \text{ is a scalar quantity]} \\ &= \frac{1}{\mathbf{r} \bullet \mathbf{a}} \left(\frac{d\mathbf{r}}{dt} \times \mathbf{a} + \mathbf{r} \times \frac{d\mathbf{a}}{dt} \right) - \left\{ \frac{1}{(\mathbf{r} \bullet \mathbf{a})^2} \frac{d}{dt} (\mathbf{r} \bullet \mathbf{a}) \right\} (\mathbf{r} \times \mathbf{a}) \\ &= \frac{\frac{d\mathbf{r}}{dt} \times \mathbf{a}}{\mathbf{r} \bullet \mathbf{a}} - \left\{ \frac{1}{(\mathbf{r} \bullet \mathbf{a})^2} \left(\frac{d\mathbf{r}}{dt} \bullet \mathbf{a} + \mathbf{r} \bullet \frac{d\mathbf{a}}{dt} \right) \right\} (\mathbf{r} \times \mathbf{a}) \\ &= \frac{\frac{d\mathbf{r}}{dt} \times \mathbf{a}}{\mathbf{r} \bullet \mathbf{a}} - \frac{\frac{d\mathbf{r}}{dt} \bullet \mathbf{a}}{(\mathbf{r} \bullet \mathbf{a})^2} (\mathbf{r} \times \mathbf{a}). \quad \left[\because \frac{d\mathbf{a}}{dt} = \mathbf{0} \right] \end{aligned}$$

Example 7: Find

(i) $\frac{d}{dt} \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^2\mathbf{r}}{dt^2} \right];$

(ii) $\frac{d^2}{dt^2} \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^2\mathbf{r}}{dt^2} \right];$

(iii) $\frac{d}{dt} \left[\mathbf{r} \times \left(\frac{d\mathbf{r}}{dt} \times \frac{d^2\mathbf{r}}{dt^2} \right) \right].$

(Kumaun 2010)

Solution: (i) Let $R = \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^2\mathbf{r}}{dt^2} \right]$. Then R is the scalar triple product of three vectors $\mathbf{r}$, $\frac{d\mathbf{r}}{dt}$ and $\frac{d^2\mathbf{r}}{dt^2}$. Therefore using the rule for finding the derivative of a scalar triple product, we have

$$\begin{aligned}\frac{dR}{dt} &= \left[\frac{d\mathbf{r}}{dt}, \frac{d\mathbf{r}}{dt}, \frac{d^2\mathbf{r}}{dt^2} \right] + \left[\mathbf{r}, \frac{d^2\mathbf{r}}{dt^2}, \frac{d^2\mathbf{r}}{dt^2} \right] + \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^3\mathbf{r}}{dt^3} \right] \\ &= \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^3\mathbf{r}}{dt^3} \right],\end{aligned}$$

since scalar triple products having two equal vectors vanish.

(ii) Let $R = \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^2\mathbf{r}}{dt^2} \right]$. Then as in part (i)

$$\frac{dR}{dt} = \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^3\mathbf{r}}{dt^3} \right].$$

Differentiating again, we get

$$\begin{aligned}\frac{d^2R}{dt^2} &= \left[\frac{d\mathbf{r}}{dt}, \frac{d\mathbf{r}}{dt}, \frac{d^3\mathbf{r}}{dt^3} \right] + \left[\mathbf{r}, \frac{d^2\mathbf{r}}{dt^2}, \frac{d^3\mathbf{r}}{dt^3} \right] + \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^4\mathbf{r}}{dt^4} \right] \\ &= \left[\mathbf{r}, \frac{d^2\mathbf{r}}{dt^2}, \frac{d^3\mathbf{r}}{dt^3} \right] + \left[\mathbf{r}, \frac{d\mathbf{r}}{dt}, \frac{d^4\mathbf{r}}{dt^4} \right].\end{aligned}$$

(iii) Let $\mathbf{R} = \mathbf{r} \times \left(\frac{d\mathbf{r}}{dt} \times \frac{d^2\mathbf{r}}{dt^2} \right)$.

Then $\mathbf{R}$ is the vector triple product of three vectors. Therefore using the rule for finding the derivative of a vector triple product, we have

$$\begin{aligned}\frac{d\mathbf{R}}{dt} &= \frac{d\mathbf{r}}{dt} \times \left(\frac{d\mathbf{r}}{dt} \times \frac{d^2\mathbf{r}}{dt^2} \right) + \mathbf{r} \times \left(\frac{d^2\mathbf{r}}{dt^2} \times \frac{d^2\mathbf{r}}{dt^2} \right) + \mathbf{r} \times \left(\frac{d\mathbf{r}}{dt} \times \frac{d^3\mathbf{r}}{dt^3} \right) \\ &= \frac{d\mathbf{r}}{dt} \times \left(\frac{d\mathbf{r}}{dt} \times \frac{d^2\mathbf{r}}{dt^2} \right) + \mathbf{r} \times \left(\frac{d\mathbf{r}}{dt} \times \frac{d^3\mathbf{r}}{dt^3} \right),\end{aligned}$$

since $\frac{d^2\mathbf{r}}{dt^2} \times \frac{d^2\mathbf{r}}{dt^2} = \mathbf{0}$, being vector product of two equal vectors.

Example 8: If $\mathbf{a} = \sin \theta \mathbf{i} + \cos \theta \mathbf{j} + \theta \mathbf{k}$, $\mathbf{b} = \cos \theta \mathbf{i} - \sin \theta \mathbf{j} - 3\mathbf{k}$, and

$\mathbf{c} = 2\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$, find $\frac{d}{d\theta} \{ \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \}$ at $\theta = \frac{\pi}{2}$. (Kumaun 2010)

Solution: We have $\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos \theta & -\sin \theta & -3 \\ 2 & 3 & -3 \end{vmatrix}$

$$= (3 \sin \theta + 9) \mathbf{i} + (3 \cos \theta - 6) \mathbf{j} + (3 \cos \theta + 2 \sin \theta) \mathbf{k}.$$

$\therefore \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \sin \theta & \cos \theta & \theta \\ 3 \sin \theta + 9 & 3 \cos \theta - 6 & 3 \cos \theta + 2 \sin \theta \end{vmatrix}$

$$\begin{aligned}
 &= (3 \cos^2 \theta + 2 \sin \theta \cos \theta - 3 \theta \cos \theta + 6 \theta) \mathbf{i} \\
 &\quad + (3 \theta \sin \theta + 9 \theta - 3 \sin \theta \cos \theta - 2 \sin^2 \theta) \mathbf{j} \\
 &\quad + (-6 \sin \theta - 9 \cos \theta) \mathbf{k} . \\
 \therefore \quad \frac{d}{d\theta} \{\mathbf{a} \times (\mathbf{b} \times \mathbf{c})\} &= (-6 \cos \theta \sin \theta + 2 \cos^2 \theta - 2 \sin^2 \theta \\
 &\quad - 3 \cos \theta + 3 \theta \sin \theta + 6) \mathbf{i} + (3 \sin \theta + 3 \theta \cos \theta + 9 \\
 &\quad - 3 \cos^2 \theta + 3 \sin^2 \theta - 4 \sin \theta \cos \theta) \mathbf{j} \\
 &\quad + (-6 \cos \theta + 9 \sin \theta) \mathbf{k} .
 \end{aligned}$$

Putting $\theta = \pi / 2$, we get the required derivative

$$= \left(4 + \frac{3}{2} \pi\right) \mathbf{i} + 15 \mathbf{j} + 9 \mathbf{k} .$$

Example 9: A particle moves along the curve $x = 4 \cos t$, $y = 4 \sin t$, $z = 6t$. Find the velocity and acceleration at time $t = 0$ and $t = \frac{1}{2} \pi$. Find also the magnitudes of the velocity and acceleration at any time t . (Bundelkhand 2007, 14)

Solution: Let $\mathbf{r}$ be the position vector of the particle at time t .

Then $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = 4 \cos t \mathbf{i} + 4 \sin t \mathbf{j} + 6t \mathbf{k}$. If $\mathbf{v}$ is the velocity of the particle at time t and $\mathbf{a}$ its acceleration at that time, then

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = -4 \sin t \mathbf{i} + 4 \cos t \mathbf{j} + 6 \mathbf{k}$$

$$\text{and} \quad \mathbf{a} = \frac{d^2 \mathbf{r}}{dt^2} = -4 \cos t \mathbf{i} - 4 \sin t \mathbf{j} .$$

Magnitude of the velocity at time $t = |\mathbf{v}|$

$$= \sqrt{(16 \sin^2 t + 16 \cos^2 t + 36)} = \sqrt{(52)} = 2 \sqrt{(13)} .$$

Magnitude of the acceleration $= |\mathbf{a}| = \sqrt{(16 \cos^2 t + 16 \sin^2 t)} = 4$.

At $t = 0$, $\mathbf{v} = 4 \mathbf{j} + 6 \mathbf{k}$, $\mathbf{a} = -4 \mathbf{i}$.

At $t = \frac{1}{2} \pi$, $\mathbf{v} = -4 \mathbf{i} + 6 \mathbf{k}$, $\mathbf{a} = -4 \mathbf{j}$.

Example 10: A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 5$, where t is the time. Find the components of its velocity and acceleration at $t = 1$ in the direction $\mathbf{i} + \mathbf{j} + 3\mathbf{k}$. (Purvanchal 2012)

Solution: If $\mathbf{r}$ is the position vector of any point (x, y, z) on the given curve, then

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (t^3 + 1) \mathbf{i} + t^2 \mathbf{j} + (2t + 5) \mathbf{k} .$$

$$\text{Velocity} = \mathbf{v} = \frac{d\mathbf{r}}{dt} = 3t^2 \mathbf{i} + 2t \mathbf{j} + 2 \mathbf{k} = 3 \mathbf{i} + 2 \mathbf{j} + 2 \mathbf{k} \text{ at } t = 1 .$$

$$\text{Acceleration} = \mathbf{a} = \frac{d^2 \mathbf{r}}{dt^2} = \frac{d}{dt} \left(\frac{d\mathbf{r}}{dt} \right) = 6t \mathbf{i} + 2 \mathbf{j} = 6 \mathbf{i} + 2 \mathbf{j} \text{ at } t = 1 .$$

Now the unit vector in the given direction $\mathbf{i} + \mathbf{j} + 3\mathbf{k}$

$$= \frac{\mathbf{i} + \mathbf{j} + 3\mathbf{k}}{|\mathbf{i} + \mathbf{j} + 3\mathbf{k}|} = \frac{\mathbf{i} + \mathbf{j} + 3\mathbf{k}}{\sqrt{(11)}} = \mathbf{b}, \text{ say.}$$

$\therefore$ the component of velocity in the given direction

$$= \mathbf{v} \cdot \mathbf{b} = \frac{(3\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) \cdot (\mathbf{i} + \mathbf{j} + 3\mathbf{k})}{\sqrt{(11)}} = \frac{11}{\sqrt{(11)}} = \sqrt{(11)};$$

and the component of acceleration in the given direction

$$= \mathbf{a} \cdot \mathbf{b} = \frac{(6\mathbf{i} + 2\mathbf{j}) \cdot (\mathbf{i} + \mathbf{j} + 3\mathbf{k})}{\sqrt{(11)}} = \frac{8}{\sqrt{(11)}}.$$

Comprehensive Exercise 1

1. (i) If $\mathbf{r}$ is the position vector of a moving point and r is the modulus of $\mathbf{r}$, show that $\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = r \frac{dr}{dt}$. Interpret the relations $\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0$ and $\mathbf{r} \times \frac{d\mathbf{r}}{dt} = \mathbf{0}$.
 (ii) If $\mathbf{r} \times d\mathbf{r} = \mathbf{0}$, show that $\hat{\mathbf{r}} = \text{constant}$. (Garhwal 2001; Meerut 05B; Purvanchal 12; Bundelkhand 12; Avadh 13)
2. (i) If $\mathbf{r} = (t+1)\mathbf{i} + (t^2+t+1)\mathbf{j} + (t^3+t^2+t+1)\mathbf{k}$, find $\frac{d\mathbf{r}}{dt}$ and $\frac{d^2\mathbf{r}}{dt^2}$.
 (ii) If $\mathbf{r} = t^2\mathbf{i} - t\mathbf{j} + (2t+1)\mathbf{k}$, find at $t=0$, the values of $\frac{d\mathbf{r}}{dt}$, $\frac{d^2\mathbf{r}}{dt^2}$, $\left| \frac{d\mathbf{r}}{dt} \right|$, $\left| \frac{d^2\mathbf{r}}{dt^2} \right|$. (Kanpur 2002; Bundelkhand 2010, 13)
3. Show that $\hat{\mathbf{r}} \times d\hat{\mathbf{r}} = (\mathbf{r} \times d\mathbf{r}) / r^2$, where $\mathbf{r} = r\hat{\mathbf{r}}$.
4. (i) The position vector of a moving particle at time t is given by $\mathbf{r} = (3t-4)\mathbf{i} + (t^2-2)\mathbf{j} + 4t^3\mathbf{k}$. Find its velocity and acceleration at time $t=2$. (Kanpur 2005; Bundelkhand 13)
 (ii) If $\mathbf{r} = t^3\mathbf{i} + \left(2t^3 - \frac{1}{5t^2}\right)\mathbf{j}$, show that $\mathbf{r} \times \frac{d\mathbf{r}}{dt} = \mathbf{k}$. (Agra 2007)
 (iii) If $\mathbf{r} = (\cos nt)\mathbf{i} + (\sin nt)\mathbf{j}$, where n is a constant and t varies, show that $\mathbf{r} \times \frac{d\mathbf{r}}{dt} = n\mathbf{k}$. (Garhwal 2002)
5. (i) If $\mathbf{r} = (\sinh t)\mathbf{a} + (\cosh t)\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors, then show that $\frac{d^2\mathbf{r}}{dt^2} = \mathbf{r}$.
 (ii) If $\mathbf{u} = t^2\mathbf{i} - t\mathbf{j} + (2t+1)\mathbf{k}$ and $\mathbf{v} = (2t-3)\mathbf{i} + \mathbf{j} - t\mathbf{k}$, find $\frac{d}{dt}(\mathbf{u} \cdot \mathbf{v})$, when $t=1$. (Bundelkhand 2007)

6. (i) If $\mathbf{r} = e^{nt} \mathbf{a} + e^{-nt} \mathbf{b}$, where $\mathbf{a}, \mathbf{b}$ are constant vectors, show that

$$\frac{d^2 \mathbf{r}}{dt^2} - n^2 \mathbf{r} = \mathbf{0}. \quad (\text{Kumaun 2000; Rohilkhand 07; Agra 07; Kashi 14})$$

- (ii) Show that $\mathbf{r} = \mathbf{a} e^{mt} + \mathbf{b} e^{nt}$, where $\mathbf{a}$ and $\mathbf{b}$ are the constant vectors, is the solution of the differential equation $\frac{d^2 \mathbf{r}}{dt^2} - (m+n) \frac{d\mathbf{r}}{dt} + mn \mathbf{r} = \mathbf{0}$.

Hence solve the equation

$$\frac{d^2 \mathbf{r}}{dt^2} - \frac{d\mathbf{r}}{dt} - 2\mathbf{r} = \mathbf{0}, \quad \text{where } \mathbf{r} = \mathbf{i} \text{ and } \frac{d\mathbf{r}}{dt} = \mathbf{j} \text{ for } t = 0.$$

- (iii) If $\mathbf{r} = \mathbf{a} \sin \omega t + \mathbf{b} \cos \omega t + \frac{\mathbf{c}}{\omega^2} \sin \omega t$, prove that

$$\frac{d^2 \mathbf{r}}{dt^2} + \omega^2 \mathbf{r} = \frac{2\mathbf{c}}{\omega} \cos \omega t,$$

where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are constant vectors and ω is a constant scalar.

7. (i) A particle moves along the curve $x = e^{-t}, y = 2 \cos 3t, z = 2 \sin 3t$.

Determine the velocity and acceleration at any time t and their magnitudes at $t = 0$. (Bundelkhand 2005; Rohilkhand 08)

- (ii) Show that if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are constant vectors, then $\mathbf{r} = \mathbf{a}t^2 + \mathbf{b}t + \mathbf{c}$ is the path of a particle moving with constant acceleration.

8. A particle moves so that its position vector is given by $\mathbf{r} = \cos \omega t \mathbf{i} + \sin \omega t \mathbf{j}$ where ω is a constant; show that

- (i) the velocity of the particle is perpendicular to $\mathbf{r}$,
 (ii) the acceleration is directed towards the origin and has magnitude proportional to the distance from the origin,
 (iii) $\mathbf{r} \times \frac{d\mathbf{r}}{dt}$ is a constant vector.

9. If $\mathbf{A} = 5t^2 \mathbf{i} + t \mathbf{j} - t^3 \mathbf{k}$ and $\mathbf{B} = \sin t \mathbf{i} - \cos t \mathbf{j}$, find

$$(i) \frac{d}{dt} (\mathbf{A} \cdot \mathbf{B}); \quad (ii) \frac{d}{dt} (\mathbf{A} \times \mathbf{B}); \quad (iii) \frac{d}{dt} (\mathbf{A} \cdot \mathbf{A}). \quad (\text{Agra 2005})$$

10. Prove the following :

$$(i) \frac{d}{dt} \left[\mathbf{a} \cdot \frac{d\mathbf{b}}{dt} - \frac{d\mathbf{a}}{dt} \cdot \mathbf{b} \right] = \mathbf{a} \cdot \frac{d^2 \mathbf{b}}{dt^2} - \frac{d^2 \mathbf{a}}{dt^2} \cdot \mathbf{b}$$

$$(ii) \frac{d}{dt} \left[\mathbf{a} \times \frac{d\mathbf{b}}{dt} - \frac{d\mathbf{a}}{dt} \times \mathbf{b} \right] = \mathbf{a} \times \frac{d^2 \mathbf{b}}{dt^2} - \frac{d^2 \mathbf{a}}{dt^2} \times \mathbf{b}$$

11. If $\mathbf{r}$ is a unit vector, then prove that

$$\left| \mathbf{r} \times \frac{d\mathbf{r}}{dt} \right| = \left| \frac{d\mathbf{r}}{dt} \right|.$$

(Purvanchal 2006)

12. If $\mathbf{r}$ is a vector function of a scalar t , r its module, and $\mathbf{a}$, $\mathbf{b}$ are constant vectors, differentiate the following with respect to t :
- (i) $r^3 \mathbf{r} + \mathbf{a} \times \frac{d\mathbf{r}}{dt}$, (ii) $r^2 \mathbf{r} + (\mathbf{a} \cdot \mathbf{r}) \mathbf{b}$,
 (iii) $r^n \mathbf{r}$, (iv) $(a\mathbf{r} + r \mathbf{b})^2$.
13. Find the unit tangent vector to any point on the curve
 $x = a \cos t$, $y = a \sin t$, $z = bt$.
14. If the direction of a differentiable vector function $\mathbf{r}(t)$ is constant, show that
 $\mathbf{r} \times \left(\frac{d\mathbf{r}}{dt} \right) = \mathbf{0}$. Or

If $\mathbf{r}(t)$ is a vector of constant direction, show that its derivative is collinear with it.

15. If $\mathbf{e}$ is the unit vector making an angle θ with x -axis, show that $d\mathbf{e}/d\theta$ is a unit vector obtained by rotating $\mathbf{e}$ through a right angle in the direction of θ increasing.

Answers 1

2. (i) $\mathbf{i} + (2t + 1) \mathbf{j} + (3t^2 + 2t + 1) \mathbf{k}$; $2\mathbf{j} + (6t + 2) \mathbf{k}$
 (ii) $-\mathbf{j} + 2\mathbf{k}$; $2i$; $\sqrt{5}$; 2
4. (i) $\mathbf{V} = 3\mathbf{i} + 4\mathbf{j} + 48\mathbf{k}$, $\mathbf{a} = 2\mathbf{j} + 48\mathbf{k}$
5. (i) $\mathbf{r}$ (ii) $6t^2 - 10t - 2$; -6
7. (i) $\sqrt{37}$; $\sqrt{325}$
9. (i) $(5t^2 - 1) \cos t + 11t \sin t$
 (ii) $t^2 (t \sin t - 3 \cos t) \mathbf{i} - t^2 (t \cos t + 3 \sin t) \mathbf{j}$
 $- (11t \cos t - 5t^2 \sin t + \sin t) \mathbf{k}$
 (iii) $100 t^3 + 2t + 6t^5$
12. (i) $3r^2 \frac{d\mathbf{r}}{dt} + r^3 \frac{d^2\mathbf{r}}{dt^2} + \mathbf{a} \times \frac{d^2\mathbf{r}}{dt^2}$
 (ii) $2r \frac{dr}{dt} \mathbf{r} + r^2 \frac{d\mathbf{r}}{dt} + \left(\mathbf{a} \cdot \frac{d\mathbf{r}}{dt} \right) \mathbf{b}$
 (iii) $\left(nr^{n-1} \frac{dr}{dt} \right) \mathbf{r} + r^n \frac{d\mathbf{r}}{dt}$
 (iv) $2(a\mathbf{r} + r\mathbf{b}) \cdot \left(a \frac{d\mathbf{r}}{dt} + \frac{dr}{dt} \mathbf{b} \right)$
13. $\frac{1}{\sqrt{(a^2 + b^2)}} (-a \sin t \mathbf{i} + a \cos t \mathbf{j} + b \mathbf{k})$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- If $\mathbf{r} = \mathbf{a} e^{\omega t} + \mathbf{b} e^{-\omega t}$, where $\mathbf{a}$, $\mathbf{b}$ are constant vectors then $\frac{d^2 \mathbf{r}}{dt^2} - \omega^2 \mathbf{r}$ is equal to
 (a) 1 (b) 0
 (c) 2 (d) none of these
 (Bundelkhand 2001; Agra 06)
- A particle moves along the curve $\mathbf{r} = e^{-t} \cos t \mathbf{i} + e^{-t} \sin t \mathbf{j} + e^{-t} \mathbf{k}$. The magnitude of its velocity at $t = 0$ is
 (a) $2\sqrt{3}$ (b) $\sqrt{3}/2$
 (c) $\sqrt{3}$ (d) none of these
- If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then $\frac{d^2 \mathbf{r}}{dt^2} = \dots\dots$
 (a) $\mathbf{i}x + \mathbf{j}y + \mathbf{k}z$ (b) $\mathbf{i}\frac{dx}{dt} + \mathbf{j}\frac{dy}{dt} + \mathbf{k}\frac{dz}{dt}$
 (c) $\mathbf{i}\frac{d^2 x}{dt^2} + \mathbf{j}\frac{d^2 y}{dt^2} + \mathbf{k}\frac{d^2 z}{dt^2}$ (d) none of these
 (Kumaun 2011)

Fill in the Blank(s)

Fill in the blanks “.....”, so that the following statements are complete and correct.

- If $\mathbf{r} = 3\mathbf{i} - 6t^2 \mathbf{j} + 4t \mathbf{k}$, then $\frac{d\mathbf{r}}{dt} = \dots\dots$; $\frac{d^2 \mathbf{r}}{dt^2} = \dots\dots$ (Bundelkhand 2008)
- If $\mathbf{u} = t^2 \mathbf{i} - t \mathbf{j} + (2t + 1) \mathbf{k}$, $\mathbf{v} = (2t - 3) \mathbf{i} + \mathbf{j} - t \mathbf{k}$, then $\frac{d}{dt} (\mathbf{u} \cdot \mathbf{v}) = \dots\dots$
- If $\mathbf{r} = (\cos \omega t) \mathbf{i} + (\sin \omega t) \mathbf{j}$, then $\mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} = \dots\dots$
- The necessary and sufficient condition for the vector $\mathbf{a}(t)$ to have constant direction is
- If $\mathbf{r} = 5\mathbf{i} + 3t^2 \mathbf{j} + 2t \mathbf{k}$, then $\frac{d\mathbf{r}}{dt} = \dots\dots$; $\frac{d^2 \mathbf{r}}{dt^2} = \dots\dots$ (Bundelkhand 2010)
- $\frac{d}{dt} (\mathbf{a} \cdot \mathbf{b}) = \dots\dots$
 (Kumaun 2009)

True or False

Write 'T' for true and 'F' for false statement.

1. A vector is said to be constant only if its magnitude is fixed and direction changes.
2. The necessary and sufficient condition for the vector $\mathbf{a}(t)$ to have constant magnitude is $\mathbf{a} \cdot \frac{d\mathbf{a}}{dt} = 0$.

Answers

Multiple Choice Questions

1. (b)
2. (c)
3. (c)

Fill in the Blank(s)

1. $-12t\mathbf{j} + 4\mathbf{k}; -12\mathbf{j}$
2. $6t^2 - 10t - 2$
3. 0
4. $\mathbf{a} \times \frac{d\mathbf{a}}{dt} = \mathbf{0}$
5. $6t\mathbf{j} + 2\mathbf{k}; 6\mathbf{j}$
6. $\mathbf{a} \cdot \frac{d\mathbf{b}}{dt} + \frac{d\mathbf{a}}{dt} \cdot \mathbf{b}$

True or False

1. F
2. T



Chapter

3



Gradient, Divergence and Curl

1 Partial Derivatives of Vectors

Suppose $\mathbf{r}$ is a vector depending on more than one scalar variable. Let $\mathbf{r} = \mathbf{f}(x, y, z)$ i.e., let $\mathbf{r}$ be a function of three scalar variables x , y and z . The partial derivative of $\mathbf{r}$ with respect to x is defined as

$$\frac{\partial \mathbf{r}}{\partial x} = \lim_{\delta x \rightarrow 0} \frac{\mathbf{f}(x + \delta x, y, z) - \mathbf{f}(x, y, z)}{\delta x}$$

if this limit exists. Thus $\partial \mathbf{r} / \partial x$ is nothing but the ordinary derivative of $\mathbf{r}$ with respect to x provided the other variables y and z are regarded as constants.

Similarly we may define the partial derivatives $\frac{\partial \mathbf{r}}{\partial y}$ and $\frac{\partial \mathbf{r}}{\partial z}$.

Higher partial derivatives can also be defined as in Scalar Calculus. Thus, for example,

$$\begin{aligned} \frac{\partial^2 \mathbf{r}}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial \mathbf{r}}{\partial x} \right), \quad \frac{\partial^2 \mathbf{r}}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial \mathbf{r}}{\partial y} \right), \quad \frac{\partial^2 \mathbf{r}}{\partial z^2} = \frac{\partial}{\partial z} \left(\frac{\partial \mathbf{r}}{\partial z} \right), \\ \frac{\partial^2 \mathbf{r}}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial \mathbf{r}}{\partial y} \right), \quad \frac{\partial^2 \mathbf{r}}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial \mathbf{r}}{\partial x} \right). \end{aligned}$$

If $\mathbf{r}$ has continuous partial derivatives of the second order at least, then, $\frac{\partial^2 \mathbf{r}}{\partial x \partial y} = \frac{\partial^2 \mathbf{r}}{\partial y \partial x}$ i.e., the order of differentiation is immaterial. If $\mathbf{r} = \mathbf{f}(x, y, z)$, the

total differential $d\mathbf{r}$ of $\mathbf{r}$ is given by $d\mathbf{r} = \frac{\partial \mathbf{r}}{\partial x} dx + \frac{\partial \mathbf{r}}{\partial y} dy + \frac{\partial \mathbf{r}}{\partial z} dz$.

2 The Vector Differential Operator Del (∇)

The vector differential operator ∇ (read as *del* or *nabla*) is defined as

$$\nabla \equiv \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \equiv \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

and operates distributively.

The vector operator ∇ can generally be treated to behave as an ordinary vector. It possesses properties like ordinary vectors. The symbols $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$, $\frac{\partial}{\partial z}$ can be treated as its components along $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$.

3 Gradient of a Scalar Field

(Agra 2005)

Definition: Let $f(x, y, z)$ be defined and differentiable at each point (x, y, z) in a certain region of space (i.e., defines a differentiable scalar field). Then the gradient of f , written as ∇f or $\text{grad } f$, is defined as

$$\nabla f = \left(\frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}.$$

It should be noted that ∇f is a vector whose three successive components are $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$

and $\frac{\partial f}{\partial z}$. Thus the gradient of a scalar field defines a vector field. If f is a scalar point function, then ∇f is a vector point function.

4 Formulas Involving Gradient

Theorem 1: Gradient of the sum of two scalar point functions. If f and g are two scalar point functions, then

$$\text{grad}(f + g) = \text{grad } f + \text{grad } g \quad \text{or} \quad \nabla(f + g) = \nabla f + \nabla g.$$

Proof: We have

$$\text{grad}(f + g) = \nabla(f + g) = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) (f + g)$$

$$\begin{aligned}
 &= \mathbf{i} \frac{\partial}{\partial x} (f + g) + \mathbf{j} \frac{\partial}{\partial y} (f + g) + \mathbf{k} \frac{\partial}{\partial z} (f + g) \\
 &= \mathbf{i} \frac{\partial f}{\partial x} + \mathbf{i} \frac{\partial g}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{j} \frac{\partial g}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} + \mathbf{k} \frac{\partial g}{\partial z} \\
 &= \left(\mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} \right) + \left(\mathbf{i} \frac{\partial g}{\partial x} + \mathbf{j} \frac{\partial g}{\partial y} + \mathbf{k} \frac{\partial g}{\partial z} \right) \\
 &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) f + \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) g \\
 &= \nabla f + \nabla g = \text{grad } f + \text{grad } g .
 \end{aligned}$$

Similarly, we can prove that $\nabla (f - g) = \nabla f - \nabla g$.

Theorem 2: Gradient of a constant. The necessary and sufficient condition for a scalar point function to be constant is that $\nabla f = \mathbf{0}$.

Proof: If $f(x, y, z)$ is constant, then $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0, \frac{\partial f}{\partial z} = 0$.

Therefore, $\text{grad } f = \mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}$.

Hence the condition is necessary.

Conversely, let $\text{grad } f = \mathbf{0}$. Then $\mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} = \mathbf{0}$.

Therefore, $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0, \frac{\partial f}{\partial z} = 0$.

$\therefore f$ must be independent of x, y and z .

$\therefore f$ must be a constant. Hence the condition is sufficient.

Theorem 3: Gradient of the product of two scalar point functions.

If f and g are scalar point functions, then $\text{grad}(fg) = f \text{grad } g + g \text{grad } f$

or $\nabla (fg) = f \nabla g + g \nabla f$. (Kumaun 2014)

Proof: We have $\nabla (fg) = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) (fg)$

$$\begin{aligned}
 &= \mathbf{i} \frac{\partial}{\partial x} (fg) + \mathbf{j} \frac{\partial}{\partial y} (fg) + \mathbf{k} \frac{\partial}{\partial z} (fg) \\
 &= \mathbf{i} \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right) + \mathbf{j} \left(f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} \right) + \mathbf{k} \left(f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right) \\
 &= f \left(\mathbf{i} \frac{\partial g}{\partial x} + \mathbf{j} \frac{\partial g}{\partial y} + \mathbf{k} \frac{\partial g}{\partial z} \right) + g \left(\mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} \right) \\
 &= f \nabla g + g \nabla f = f \text{grad } g + g \text{grad } f .
 \end{aligned}$$

In particular, if c is a constant, then

$$\nabla (cf) = c \nabla f + f \nabla c = c \nabla f + \mathbf{0} = c \nabla f .$$

Theorem 4: Gradient of the quotient of two scalar functions. If f and g are two scalar point functions, then $\nabla \left(\frac{f}{g} \right) = \frac{g \nabla f - f \nabla g}{g^2}$.

(Kumaun 2013)

Proof: We have

$$\begin{aligned}\nabla \left(\frac{f}{g} \right) &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \left(\frac{f}{g} \right) \\ &= \mathbf{i} \frac{\partial}{\partial x} \left(\frac{f}{g} \right) + \mathbf{j} \frac{\partial}{\partial y} \left(\frac{f}{g} \right) + \mathbf{k} \frac{\partial}{\partial z} \left(\frac{f}{g} \right).\end{aligned}$$

But $\frac{\partial}{\partial x} \left(\frac{f}{g} \right) = \frac{g \frac{\partial f}{\partial x} - f \frac{\partial g}{\partial x}}{g^2}, \quad \frac{\partial}{\partial y} \left(\frac{f}{g} \right) = \frac{g \frac{\partial f}{\partial y} - f \frac{\partial g}{\partial y}}{g^2},$

and $\frac{\partial}{\partial z} \left(\frac{f}{g} \right) = \frac{g \frac{\partial f}{\partial z} - f \frac{\partial g}{\partial z}}{g^2}.$

$$\begin{aligned}\therefore \nabla \left(\frac{f}{g} \right) &= \frac{1}{g^2} \left\{ \mathbf{i} \left(g \frac{\partial f}{\partial x} - f \frac{\partial g}{\partial x} \right) + \mathbf{j} \left(g \frac{\partial f}{\partial y} - f \frac{\partial g}{\partial y} \right) + \mathbf{k} \left(g \frac{\partial f}{\partial z} - f \frac{\partial g}{\partial z} \right) \right\} \\ &= \frac{1}{g^2} \left\{ g \left(\mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} \right) - f \left(\mathbf{i} \frac{\partial g}{\partial x} + \mathbf{j} \frac{\partial g}{\partial y} + \mathbf{k} \frac{\partial g}{\partial z} \right) \right\} \\ &= \frac{1}{g^2} \{ g \nabla f - f \nabla g \}.\end{aligned}$$

Illustrative Examples

Example 1: If $\mathbf{A} = x^2 yz \mathbf{i} - 2xz^3 \mathbf{j} + xz^2 \mathbf{k}$, $\mathbf{B} = 2z\mathbf{i} + y\mathbf{j} - x^2 \mathbf{k}$, find the value of $\frac{\partial^2}{\partial x \partial y} (\mathbf{A} \times \mathbf{B})$ at $(1, 0, -2)$.

Solution: We have $\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x^2 yz & -2xz^3 & xz^2 \\ 2z & y & -x^2 \end{vmatrix}$

$$\begin{aligned}&= (2x^3 z^3 - xyz^2) \mathbf{i} + (2xz^3 + x^4 yz) \mathbf{j} \\ &\quad + (x^2 y^2 z + 4xz^4) \mathbf{k}.\end{aligned}$$

$\therefore \frac{\partial}{\partial y} (\mathbf{A} \times \mathbf{B}) = -xz^2 \mathbf{i} + x^4 z \mathbf{j} + 2x^2 yz \mathbf{k}.$

Again $\frac{\partial^2}{\partial x \partial y} (\mathbf{A} \times \mathbf{B}) = \frac{\partial}{\partial x} \left\{ \frac{\partial}{\partial y} (\mathbf{A} \times \mathbf{B}) \right\} = -z^2 \mathbf{i} + 4x^3 z \mathbf{j} + 4xyz \mathbf{k} \dots (1)$

Putting $x = 1, y = 0$ and $z = -2$ in (1), we get the required derivative at the point $(1, 0, -2) = -4\mathbf{i} - 8\mathbf{j}$.

Example 2: If $f(x, y, z) = 3x^2y - y^3z^2$, find $\text{grad } f$ at the point $(1, -2, -1)$.

(Rohilkhand 2009)

Solution: We have

$$\begin{aligned}\text{grad } f &= \nabla f = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) (3x^2y - y^3z^2) \\ &= \mathbf{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \mathbf{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) \\ &\quad + \mathbf{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2) \\ &= \mathbf{i} (6xy) + \mathbf{j} (3x^2 - 3y^2z^2) + \mathbf{k} (-2y^3z) \\ &= 6xy \mathbf{i} + (3x^2 - 3y^2z^2) \mathbf{j} - 2y^3z \mathbf{k}.\end{aligned}$$

Putting $x = 1, y = -2, z = -1$, we get

$$\begin{aligned}\nabla f &= 6(1)(-2) \mathbf{i} + \{3(1)^2 - 3(-2)^2(-1)^2\} \mathbf{j} - 2(-2)^3(-1) \mathbf{k} \\ &= -12 \mathbf{i} - 9\mathbf{j} - 16\mathbf{k}.\end{aligned}$$

Example 3: If $r = |\mathbf{r}|$ where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, prove that

$$(i) \quad \nabla f(r) = f'(r) \nabla r, \quad (\text{Rohilkhand 2008; Meerut 11})$$

$$(ii) \quad \nabla r = \frac{1}{r} \mathbf{r},$$

$$(iii) \quad \nabla f(r) \times \mathbf{r} = \mathbf{0},$$

$$(iv) \quad \nabla \left(\frac{1}{r} \right) = -\frac{\mathbf{r}}{r^3}, \quad (\text{Rohilkhand 2008, 09B})$$

$$(v) \quad \nabla \log |\mathbf{r}| = \frac{\mathbf{r}}{r^2}, \quad (\text{Garhwal 2000; Kumaun 07; Rohilkhand 08; Bundelkhand 08; Meerut 11})$$

$$(vi) \quad \nabla r^n = nr^{n-2} \mathbf{r}. \quad (\text{Rohilkhand 2008, 11; Bundelkhand 11})$$

Solution: If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$.

$$\therefore r^2 = x^2 + y^2 + z^2.$$

$$\begin{aligned}(i) \quad \nabla f(r) &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) f(r) = \mathbf{i} \frac{\partial}{\partial x} f(r) + \mathbf{j} \frac{\partial}{\partial y} f(r) + \mathbf{k} \frac{\partial}{\partial z} f(r) \\ &= \mathbf{i} f'(r) \frac{\partial r}{\partial x} + \mathbf{j} f'(r) \frac{\partial r}{\partial y} + \mathbf{k} f'(r) \frac{\partial r}{\partial z} = f'(r) \left(\mathbf{i} \frac{\partial r}{\partial x} + \mathbf{j} \frac{\partial r}{\partial y} + \mathbf{k} \frac{\partial r}{\partial z} \right) \\ &= f'(r) \nabla r.\end{aligned}$$

$$(ii) \quad \text{We have } \nabla r = \mathbf{i} \frac{\partial r}{\partial x} + \mathbf{j} \frac{\partial r}{\partial y} + \mathbf{k} \frac{\partial r}{\partial z}.$$

Now $r^2 = x^2 + y^2 + z^2$.

$$\therefore 2r \frac{\partial r}{\partial x} = 2x \quad \text{i.e.,} \quad \frac{\partial r}{\partial x} = \frac{x}{r}.$$

Similarly, $\frac{\partial r}{\partial y} = \frac{y}{r}$ and $\frac{\partial r}{\partial z} = \frac{z}{r}$.

$$\therefore \nabla r = \frac{x}{r} \mathbf{i} + \frac{y}{r} \mathbf{j} + \frac{z}{r} \mathbf{k} = \frac{1}{r} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \frac{1}{r} \mathbf{r} = \hat{\mathbf{r}}.$$

(iii) We have as in part (i), $\nabla f(r) = f'(r) \nabla r$.

But as in part (ii), $\nabla r = \frac{1}{r} \mathbf{r}$.

$$\therefore \nabla f(r) = f'(r) \frac{1}{r} \mathbf{r}.$$

$$\therefore \nabla f(r) \times \mathbf{r} = \left\{ f'(r) \frac{1}{r} \mathbf{r} \right\} \times \mathbf{r} = \left\{ \frac{1}{r} f'(r) \right\} (\mathbf{r} \times \mathbf{r}) = \mathbf{0}.$$

[$\because \mathbf{r} \times \mathbf{r} = \mathbf{0}$]

$$\begin{aligned} \text{(iv) We have } \nabla \left(\frac{1}{r} \right) &= \mathbf{i} \frac{\partial}{\partial x} \left(\frac{1}{r} \right) + \mathbf{j} \frac{\partial}{\partial y} \left(\frac{1}{r} \right) + \mathbf{k} \frac{\partial}{\partial z} \left(\frac{1}{r} \right) \\ &= \mathbf{i} \left(-\frac{1}{r^2} \frac{\partial r}{\partial x} \right) + \mathbf{j} \left(-\frac{1}{r^2} \frac{\partial r}{\partial y} \right) + \mathbf{k} \left(-\frac{1}{r^2} \frac{\partial r}{\partial z} \right) \\ &= -\frac{1}{r^2} \left(\frac{\partial r}{\partial x} \mathbf{i} + \frac{\partial r}{\partial y} \mathbf{j} + \frac{\partial r}{\partial z} \mathbf{k} \right) \\ &= -\frac{1}{r^2} \left(\frac{x}{r} \mathbf{i} + \frac{y}{r} \mathbf{j} + \frac{z}{r} \mathbf{k} \right) = -\frac{1}{r^3} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = -\frac{1}{r^3} \mathbf{r}. \end{aligned}$$

[see part (ii)]

(v) We have $\nabla \log |\mathbf{r}| = \nabla \log r$

$$\begin{aligned} &= \mathbf{i} \frac{\partial}{\partial x} \log r + \mathbf{j} \frac{\partial}{\partial y} \log r + \mathbf{k} \frac{\partial}{\partial z} \log r \\ &= \frac{1}{r} \frac{\partial r}{\partial x} \mathbf{i} + \frac{1}{r} \frac{\partial r}{\partial y} \mathbf{j} + \frac{1}{r} \frac{\partial r}{\partial z} \mathbf{k} = \frac{1}{r} \left(\frac{x}{r} \mathbf{i} + \frac{y}{r} \mathbf{j} + \frac{z}{r} \mathbf{k} \right) \\ &= \frac{1}{r^2} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \frac{1}{r^2} \mathbf{r}. \end{aligned}$$

(vi) We have $\nabla r^n = \mathbf{i} \frac{\partial}{\partial x} r^n + \mathbf{j} \frac{\partial}{\partial y} r^n + \mathbf{k} \frac{\partial}{\partial z} r^n$

$$\begin{aligned} &= \mathbf{i} n r^{n-1} \frac{\partial r}{\partial x} + \mathbf{j} n r^{n-1} \frac{\partial r}{\partial y} + \mathbf{k} n r^{n-1} \frac{\partial r}{\partial z} \\ &= n r^{n-1} \left(\mathbf{i} \frac{\partial r}{\partial x} + \mathbf{j} \frac{\partial r}{\partial y} + \mathbf{k} \frac{\partial r}{\partial z} \right) = n r^{n-1} \nabla r \end{aligned}$$

$$\begin{aligned}
 &= nr^{n-1} \frac{1}{r} \mathbf{r} \\
 &= nr^{n-2} \mathbf{r}.
 \end{aligned}
 \quad \left[\because \nabla r = \frac{\mathbf{r}}{r} \text{ as in part (ii)} \right]$$

Example 4: (i) Interpret the symbol $\mathbf{a} \cdot \nabla$ (ii) Show that $(\mathbf{a} \cdot \nabla) \phi = \mathbf{a} \cdot \nabla \phi$

(iii) Show that $(\mathbf{a} \cdot \nabla) \mathbf{r} = \mathbf{a}$. (Kumaun 2008; Purvanchal 14)

Solution: (i) Let $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$. Then

$$\begin{aligned}
 \mathbf{a} \cdot \nabla &= (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) \cdot \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \\
 &= a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z}.
 \end{aligned}$$

$$(ii) \quad (\mathbf{a} \cdot \nabla) \phi = \left(a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z} \right) \phi.$$

$$\begin{aligned}
 \text{Also} \quad \mathbf{a} \cdot \nabla \phi &= (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) \cdot \left(\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \\
 &= a_1 \frac{\partial \phi}{\partial x} + a_2 \frac{\partial \phi}{\partial y} + a_3 \frac{\partial \phi}{\partial z}.
 \end{aligned}$$

Hence $(\mathbf{a} \cdot \nabla) \phi = \mathbf{a} \cdot \nabla \phi$.

$$(iii) \quad (\mathbf{a} \cdot \nabla) \mathbf{r} = \left(a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z} \right) \mathbf{r} = a_1 \frac{\partial \mathbf{r}}{\partial x} + a_2 \frac{\partial \mathbf{r}}{\partial y} + a_3 \frac{\partial \mathbf{r}}{\partial z}.$$

But $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

$$\therefore \frac{\partial \mathbf{r}}{\partial x} = \mathbf{i}, \frac{\partial \mathbf{r}}{\partial y} = \mathbf{j}, \frac{\partial \mathbf{r}}{\partial z} = \mathbf{k}.$$

$$\therefore (\mathbf{a} \cdot \nabla) \mathbf{r} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k} = \mathbf{a}.$$

Comprehensive Exercise 1

1. If $\mathbf{F} = e^{xy} \mathbf{i} + (x - 2y) \mathbf{j} + x \sin y \mathbf{k}$, calculate

$$\begin{aligned}
 (i) \quad & \frac{\partial \mathbf{F}}{\partial x}, & (ii) \quad & \frac{\partial \mathbf{F}}{\partial y}, & (iii) \quad & \frac{\partial^2 \mathbf{F}}{\partial x^2}, \\
 (iv) \quad & \frac{\partial^2 \mathbf{F}}{\partial x \partial y}, & (v) \quad & \frac{\partial^2 \mathbf{F}}{\partial y^2}.
 \end{aligned}$$

2. If $\mathbf{f} = (2x^2y - x^4) \mathbf{i} + (e^{xy} - y \sin x) \mathbf{j} + x^2 \cos y \mathbf{k}$, verify that

$$\frac{\partial^2 \mathbf{f}}{\partial y \partial x} = \frac{\partial^2 \mathbf{f}}{\partial x \partial y}.$$

3. If $\mathbf{u} = x y z \mathbf{i} + x z^2 \mathbf{j} - y^3 \mathbf{k}$ and $\mathbf{v} = x^3 \mathbf{i} - x y z \mathbf{j} + x^2 z \mathbf{k}$, calculate $\frac{\partial^2 \mathbf{u}}{\partial y^2} \times \frac{\partial^2 \mathbf{v}}{\partial x^2}$ at the point (1, 1, 0).
4. If $\phi(x, y, z) = x^2 y + y^2 x + z^2$, find $\nabla \phi$ at the point (1, 1, 1).
5. Find $\text{grad } f$, where f is given by $f = x^3 - y^3 + xz^2$, at the point (1, -1, 2).
6. If $\phi(x, y, z) = xy^2 z$ and $\mathbf{f} = xz\mathbf{i} - xy\mathbf{j} + yz^2\mathbf{k}$, show that $\frac{\partial^3}{\partial x^2 \partial z} (\phi \mathbf{f})$ at (2, -1, 1) is $4\mathbf{i} + 2\mathbf{j}$.
7. If $u = x + y + z$, $v = x^2 + y^2 + z^2$, $w = yz + zx + xy$, prove that $(\text{grad } u) \bullet [(\text{grad } v) \times (\text{grad } w)] = 0$. (Meerut 2007B)
8. If $\mathbf{F} = \left(y \frac{\partial f}{\partial z} - z \frac{\partial f}{\partial y} \right) \mathbf{i} + \left(z \frac{\partial f}{\partial x} - x \frac{\partial f}{\partial z} \right) \mathbf{j} + \left(x \frac{\partial f}{\partial y} - y \frac{\partial f}{\partial x} \right) \mathbf{k}$, prove that
 (i) $\mathbf{F} = \mathbf{r} \times \nabla f$, (ii) $\mathbf{F} \bullet \mathbf{r} = 0$, (iii) $\mathbf{F} \bullet \nabla f = 0$.
9. If $\phi = (3r^2 - 4r^{1/2} + 6r^{-1/3})$, show that $\nabla \phi = 2(3 - r^{-3/2} - r^{-7/3}) \mathbf{r}$.
10. Prove that $\nabla \phi \bullet d\mathbf{r} = d\phi$. (Meerut 2005, 06, 09B; Kumaun 08)
11. Prove that $f(u) \nabla u = \nabla \int f(u) du$. (Kumaun 2012, 13)
12. ρ and p are two scalar point functions such that ρ is a function of p , show that $\nabla \rho = \frac{d\rho}{dp} \nabla p$.
13. Show that $\frac{d\phi}{ds} = \nabla \phi \bullet \frac{d\mathbf{r}}{ds}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and ϕ is a function of x, y and z .
14. Prove that $\mathbf{A} \bullet \left(\nabla \frac{1}{r} \right) = -\frac{\mathbf{A} \bullet \mathbf{r}}{r^3}$. (Meerut 2010)
15. Prove that $\nabla r^{-3} = -3r^{-5} \mathbf{r}$. (Meerut 2009, 12)
16. Show that
 (i) $\text{grad}(\mathbf{r} \bullet \mathbf{a}) = \mathbf{a}$, (Avadh 2010)
 (ii) $\text{grad}[\mathbf{r}, \mathbf{a}, \mathbf{b}] = \mathbf{a} \times \mathbf{b}$,
 where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors.

Answers 1

1. (i) $ye^{xy} \mathbf{i} + \mathbf{j} + \sin y \mathbf{k}$ (ii) $xe^{xy} \mathbf{i} - 2\mathbf{j} + x \cos y \mathbf{k}$
 (iii) $y^2 e^{xy} \mathbf{i}$ (iv) $e^{xy} (xy + 1) \mathbf{i} + \cos y \mathbf{k}$

$$(v) \quad x^2 e^{xy} \mathbf{i} - x \sin y \mathbf{k}$$

$$3. \quad -36 \mathbf{j}$$

$$4. \quad 3\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$$

$$5. \quad 7\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$$

5 Equipotential Surfaces or Level Surfaces

Let $f(x, y, z)$ be a scalar field over a region R . The points satisfying an equation of the type

$$f(x, y, z) = c, \text{ (arbitrary constant)}$$

constitute a family of surfaces in three dimensional space. The surfaces of this family are called **level surfaces**. Any surface of this family is such that the value of the function f at any point of it is the same. Therefore these surfaces are also called **iso- f -surfaces**.

Theorem 1: Let $f(x, y, z)$ be a scalar field over a region R . Then through any point of R there passes one and only one level surface.

Proof: Let (x_1, y_1, z_1) be any point of the region R .

Then the level surface $f(x, y, z) = f(x_1, y_1, z_1)$ passes through this point.

Now suppose the level surfaces $f(x, y, z) = c_1$ and $f(x, y, z) = c_2$ pass through the point (x_1, y_1, z_1) . Then

$$f(x_1, y_1, z_1) = c_1 \quad \text{and} \quad f(x_1, y_1, z_1) = c_2.$$

Since $f(x, y, z)$ has a unique value at (x_1, y_1, z_1) , therefore we have $c_1 = c_2$.

Hence only one level surface passes through the point (x_1, y_1, z_1) .

Theorem 2: ∇f is a vector normal to the surface $f(x, y, z) = c$ where c is a constant.

Proof: Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ be the position vector of any point $P(x, y, z)$ on the level surface $f(x, y, z) = c$. Let $Q(x + \delta x, y + \delta y, z + \delta z)$ be a neighbouring point on this surface.

Then the position vector of $Q = \mathbf{r} + \delta \mathbf{r} = (x + \delta x)\mathbf{i} + (y + \delta y)\mathbf{j} + (z + \delta z)\mathbf{k}$.

$$\therefore \quad \vec{PQ} = (\mathbf{r} + \delta \mathbf{r}) - \mathbf{r} = \delta \mathbf{r} = \delta x \mathbf{i} + \delta y \mathbf{j} + \delta z \mathbf{k}.$$

As $Q \rightarrow P$, the line PQ tends to tangent at P to the level surface.

Therefore $d\mathbf{r} = dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}$ lies in the tangent plane to the surface at P .

From the differential calculus, we have

$$\begin{aligned} df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \\ &= \left(\mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} \right) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) = \nabla f \cdot d\mathbf{r}. \end{aligned}$$

Since $f(x, y, z) = \text{constant}$, therefore $df = 0$.

$\therefore \nabla f \bullet d\mathbf{r} = 0$ so that ∇f is a vector perpendicular to $d\mathbf{r}$ and therefore to the tangent plane at P to the surface $f(x, y, z) = c$.

Hence ∇f is a vector normal to the surface $f(x, y, z) = c$.

Thus if $f(x, y, z)$ is a scalar field defined over a region R , then ∇f at any point (x, y, z) is a vector in the direction of normal at that point to the level surface $f(x, y, z) = c$ passing through that point.

6 Directional Derivative of a Scalar Point Function

Definition: Let $f(x, y, z)$ define a scalar field in a region R and let P be any point in this region. Suppose Q is a point in this region in the neighbourhood of P in the direction of a given unit vector $\hat{\mathbf{a}}$.

Then $\lim_{Q \rightarrow P} \frac{f(Q) - f(P)}{PQ}$, if it exists, is called the **directional derivative** of f at P in the direction of $\hat{\mathbf{a}}$.

Interpretation of Directional Derivative: Let P be the point (x, y, z) and let Q be the point $(x + \delta x, y + \delta y, z + \delta z)$. Suppose $PQ = \delta s$. Then δs is a small element at P in the direction of $\hat{\mathbf{a}}$. If

$$\delta f = f(x + \delta x, y + \delta y, z + \delta z) - f(x, y, z) = f(Q) - f(P),$$

then $\frac{\delta f}{\delta s}$ represents the average rate of change of f per unit distance in the direction

of $\hat{\mathbf{a}}$. Now the directional derivative of f at P in the direction of $\hat{\mathbf{a}}$ is

$$\lim_{Q \rightarrow P} \frac{f(Q) - f(P)}{PQ} = \lim_{\delta s \rightarrow 0} \frac{\delta f}{\delta s} = \frac{df}{ds}. \text{ It represents the rate of change of } f \text{ with}$$

respect to distance s at the point P in the direction of unit vector $\hat{\mathbf{a}}$.

Theorem 1: The directional derivative of a scalar field f at a point $P(x, y, z)$ in the direction of a unit vector $\hat{\mathbf{a}}$ is given by $\frac{df}{ds} = \nabla f \bullet \hat{\mathbf{a}}$.

Proof: Let $f(x, y, z)$ define a scalar field in the region R . Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ denote the position vector of any point $P(x, y, z)$ in this region. If s denotes the distance of P from some fixed point A in the direction of $\hat{\mathbf{a}}$, then δs denotes small element at P in the direction of $\hat{\mathbf{a}}$. Therefore $\frac{d\mathbf{r}}{ds}$ is a unit vector at P in this

direction i.e., $\frac{d\mathbf{r}}{ds} = \hat{\mathbf{a}}$.

But $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

$$\therefore \frac{d \mathbf{r}}{ds} = \frac{dx}{ds} \mathbf{i} + \frac{dy}{ds} \mathbf{j} + \frac{dz}{ds} \mathbf{k} = \hat{\mathbf{a}}.$$

$$\begin{aligned} \text{Now } \nabla f \cdot \hat{\mathbf{a}} &= \left(\mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} \right) \cdot \left(\frac{dx}{ds} \mathbf{i} + \frac{dy}{ds} \mathbf{j} + \frac{dz}{ds} \mathbf{k} \right) \\ &= \frac{\partial f}{\partial x} \frac{dx}{ds} + \frac{\partial f}{\partial y} \frac{dy}{ds} + \frac{\partial f}{\partial z} \frac{dz}{ds} = \frac{d f}{ds} \\ &= \text{directional derivative of } f \text{ at } P \text{ in the direction of } \hat{\mathbf{a}}. \end{aligned}$$

Alternative Proof: Let Q be a point in the neighbourhood of P in the direction of the given unit vector $\hat{\mathbf{a}}$. If l, m, n are the direction cosines of the line PQ , then $l \mathbf{i} + m \mathbf{j} + n \mathbf{k}$ = the unit vector in the direction of $PQ = \hat{\mathbf{a}}$. Further if $PQ = \delta s$, then the co-ordinates of Q are $(x + l\delta s, y + m\delta s, z + n\delta s)$. Now the directional derivative of f at P in the direction of $\hat{\mathbf{a}}$ is

$$\begin{aligned} &= \lim_{Q \rightarrow P} \frac{f(Q) - f(P)}{PQ} \\ &= \lim_{\delta s \rightarrow 0} \frac{f(x + l\delta s, y + m\delta s, z + n\delta s) - f(x, y, z)}{\delta s} \\ &= \lim_{\delta s \rightarrow 0} \frac{f(x, y, z) + \left(l\delta s \frac{\partial f}{\partial x} + m\delta s \frac{\partial f}{\partial y} + n\delta s \frac{\partial f}{\partial z} \right) + \dots - f(x, y, z)}{\delta s}, \\ &\quad \text{on expanding by Taylor's theorem} \\ &= l \frac{\partial f}{\partial x} + m \frac{\partial f}{\partial y} + n \frac{\partial f}{\partial z} \\ &= \left(\frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \right) \cdot (l \mathbf{i} + m \mathbf{j} + n \mathbf{k}) = \nabla f \cdot \hat{\mathbf{a}}. \end{aligned}$$

Theorem 2: If $\hat{\mathbf{n}}$ be a unit vector normal to the level surface $f(x, y, z) = c$ at a point $P(x, y, z)$ and n be the distance of P from some fixed point A in the direction of $\hat{\mathbf{n}}$ so that δn represents element of normal at P in the direction of $\hat{\mathbf{n}}$, then $\text{grad } f = \frac{df}{dn} \hat{\mathbf{n}}$.

Proof: We have $\text{grad } f = \nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$.

Also $\text{grad } f$ is a vector normal to the surface $f(x, y, z) = c$. Since $\hat{\mathbf{n}}$ is a unit vector normal to the surface $f(x, y, z) = c$, therefore let $\text{grad } f = A \hat{\mathbf{n}}$, where A is some scalar to be determined.

Now $\frac{df}{dn}$ = directional derivative of f in the direction of $\hat{\mathbf{n}}$

$$= \nabla f \cdot \hat{\mathbf{n}} = A \hat{\mathbf{n}} \cdot \hat{\mathbf{n}} = A. \quad [\because \nabla f = \text{grad } f = A \hat{\mathbf{n}}]$$

$$\therefore \text{grad } f = \nabla f = \frac{df}{dn} \hat{\mathbf{n}}.$$

Note: If the vector $\hat{\mathbf{n}}$ is in the direction of f increasing, then $\frac{df}{dn}$ is positive.

Therefore ∇f is a vector normal to the surface $f(x, y, z) = c$ in the direction of f increasing.

Theorem 3: *Grad f is a vector in the direction of which the maximum value of the directional derivative of f i.e., $\frac{df}{ds}$ occurs.*

Proof: The directional derivative of f in the direction of $\hat{\mathbf{a}}$ is given by

$$\begin{aligned} \frac{df}{ds} &= \nabla f \cdot \hat{\mathbf{a}} = \left(\frac{df}{dn} \hat{\mathbf{n}} \right) \cdot \hat{\mathbf{a}} \quad \left[\because \nabla f = \frac{df}{dn} \hat{\mathbf{n}} \right] \\ &= \frac{df}{dn} (\hat{\mathbf{n}} \cdot \hat{\mathbf{a}}) = \frac{df}{dn} \cos \theta, \text{ where } \theta \text{ is the angle between } \hat{\mathbf{a}} \text{ and } \hat{\mathbf{n}}. \end{aligned}$$

Now $\frac{df}{dn}$ is fixed. Therefore $\frac{df}{dn} \cos \theta$ is maximum when $\cos \theta$ is maximum i.e., when $\cos \theta = 1$. But $\cos \theta$ will be 1 when the angle between $\hat{\mathbf{a}}$ and $\hat{\mathbf{n}}$ is 0 i.e., when $\hat{\mathbf{a}}$ is along the unit normal vector $\hat{\mathbf{n}}$.

Therefore the directional derivative is maximum along the normal to the surface.

Its maximum value is $= \frac{df}{dn} = |\text{grad } f|$.

7 Tangent Plane and Normal to a Level Surface

Tangent plane to a surface: The tangent to any curve drawn on a surface is called a *tangent line* to the surface. All the tangent lines to a surface at the point P lie in a plane. This plane is called the *tangent plane* to the surface at the point P on it.

Normal to a surface: The normal to a surface at the point P is a straight line passing through P and perpendicular to the tangent plane at P .

Angle between two surfaces: The angle between the two surfaces at a point P is the angle between the normals to the two surfaces at that point.

Equations of the Tangent Plane and Normal to a Surface:

To find the equations of the tangent plane and normal to the surface $f(x, y, z) = c$.

Let $f(x, y, z) = c$ be the equation of a level surface. Let $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$ be the position vector of any point $P(x, y, z)$ on this surface.

Then $\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$ is a vector along the normal to the surface at P i.e., ∇f is perpendicular to the tangent plane at P .

Tangent plane at P : Let $\mathbf{R} = X \mathbf{i} + Y \mathbf{j} + Z \mathbf{k}$ be the position vector of any current point $Q(X, Y, Z)$ on the tangent plane at P to the surface. The vector

$$\vec{PQ} = \mathbf{R} - \mathbf{r} = (X - x) \mathbf{i} + (Y - y) \mathbf{j} + (Z - z) \mathbf{k}$$

lies in the tangent plane at P . Therefore it is perpendicular to the vector ∇f .

$$\therefore (\mathbf{R} - \mathbf{r}) \cdot \nabla f = 0$$

$$\text{or } [(X - x) \mathbf{i} + (Y - y) \mathbf{j} + (Z - z) \mathbf{k}] \cdot \left(\frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \right) = 0$$

$$\text{or } (X - x) \frac{\partial f}{\partial x} + (Y - y) \frac{\partial f}{\partial y} + (Z - z) \frac{\partial f}{\partial z} = 0, \quad \dots(1)$$

is the equation of the tangent plane at P .

Normal at P : Let $\mathbf{R} = X \mathbf{i} + Y \mathbf{j} + Z \mathbf{k}$

be the position vector of any current point $Q(X, Y, Z)$ on the normal at P to the surface. The vector $\vec{PQ} = \mathbf{R} - \mathbf{r} = (X - x) \mathbf{i} + (Y - y) \mathbf{j} + (Z - z) \mathbf{k}$ lies along the normal at P to the surface. Therefore it is parallel to the vector ∇f .

$$\therefore (\mathbf{R} - \mathbf{r}) \times \nabla f = \mathbf{0} \quad \dots(2)$$

is the vector equation of the normal at P to the given surface.

Cartesian form: The vectors

$$(X - x) \mathbf{i} + (Y - y) \mathbf{j} + (Z - z) \mathbf{k} \text{ and } \nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k},$$

will be parallel if

$$(X - x) \mathbf{i} + (Y - y) \mathbf{j} + (Z - z) \mathbf{k} = p \left(\frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \right),$$

where p is some scalar.

Equating the coefficients of $\mathbf{i}, \mathbf{j}, \mathbf{k}$, we get

$$X - x = p \frac{\partial f}{\partial x}, Y - y = p \frac{\partial f}{\partial y}, Z - z = p \frac{\partial f}{\partial z}$$

$$\text{or } \frac{X - x}{\frac{\partial f}{\partial x}} = \frac{Y - y}{\frac{\partial f}{\partial y}} = \frac{Z - z}{\frac{\partial f}{\partial z}} \quad \dots(3)$$

which are the equations of the normal at P .

Note: The vector $\left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right)$ is along the normal to the surface

$F(x, y, z) = 0$ at the point (x, y, z) .

Illustrative Examples

Example 5: Find a unit normal vector to the level surface $x^2 y + 2xz = 4$ at the point $(2, -2, 3)$. (Kashi 2014)

Solution: The equation of the level surface is $f(x, y, z) \equiv x^2 y + 2xz = 4$.

The vector $\text{grad } f$ is along the normal to the surface at the point (x, y, z) .

We have $\text{grad } f = \nabla (x^2 y + 2xz) = (2xy + 2z) \mathbf{i} + x^2 \mathbf{j} + 2x \mathbf{k}$.

$\therefore$ at the point $(2, -2, 3)$, $\text{grad } f = -2 \mathbf{i} + 4 \mathbf{j} + 4 \mathbf{k}$.

$\therefore -2 \mathbf{i} + 4 \mathbf{j} + 4 \mathbf{k}$ is a vector along the normal to the given surface at the point $(2, -2, 3)$.

Hence a unit normal vector to the surface at this point

$$= \frac{-2 \mathbf{i} + 4 \mathbf{j} + 4 \mathbf{k}}{|-2 \mathbf{i} + 4 \mathbf{j} + 4 \mathbf{k}|} = \frac{-2 \mathbf{i} + 4 \mathbf{j} + 4 \mathbf{k}}{\sqrt{4 + 16 + 16}} = -\frac{1}{3} \mathbf{i} + \frac{2}{3} \mathbf{j} + \frac{2}{3} \mathbf{k}.$$

The vector $-\left(-\frac{1}{3} \mathbf{i} + \frac{2}{3} \mathbf{j} + \frac{2}{3} \mathbf{k}\right)$ i.e., $\frac{1}{3} \mathbf{i} - \frac{2}{3} \mathbf{j} - \frac{2}{3} \mathbf{k}$ is also a unit normal vector to the given surface at the point $(2, -2, 3)$.

Alternate Solution: The given surface is

$$\phi(x, y, z) \equiv x^2 y + 2xz - 4 = 0.$$

We have $\frac{\partial \phi}{\partial x} = 2xy + 2z$, $\frac{\partial \phi}{\partial y} = x^2$, $\frac{\partial \phi}{\partial z} = 2x$.

$\therefore$ At the point $(2, -2, 3)$, we have

$$\frac{\partial \phi}{\partial x} = -2, \frac{\partial \phi}{\partial y} = 4, \frac{\partial \phi}{\partial z} = 4.$$

$\therefore$ A vector along the normal to the given surface at the point $(2, -2, 3)$

$$= \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z} \right) = (-2, 4, 4) \quad \text{or} \quad (-1, 2, 2).$$

We have $|(-1, 2, 2)| = \sqrt{(-1)^2 + 2^2 + 2^2} = 3$.

Hence, a unit normal vector to the given surface at the point $(2, -2, 3)$

$$= \frac{1}{3} (-1, 2, 2) = \left(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right).$$

Example 6: Find the directional derivative of $f(x, y, z) = x^2 yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction of the vector $2 \mathbf{i} - \mathbf{j} - 2 \mathbf{k}$.

(Bundelkhand 2007; Rohilkhand 11; Kashi 13; Agra 14)

Solution: We have $f(x, y, z) = x^2 yz + 4xz^2$.

$\therefore \text{grad } f = (2xyz + 4z^2) \mathbf{i} + x^2 z \mathbf{j} + (x^2 y + 8xz) \mathbf{k}$
 $= 8 \mathbf{i} - \mathbf{j} - 10 \mathbf{k}$ at the point $(1, -2, -1)$.

If $\hat{\mathbf{a}}$ be the unit vector in the direction of the vector $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$, then

$$\hat{\mathbf{a}} = \frac{2\mathbf{i} - \mathbf{j} - 2\mathbf{k}}{\sqrt{4 + 1 + 4}} = \frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}.$$

Therefore the required directional derivative is

$$\begin{aligned} \frac{df}{ds} &= \text{grad } f \cdot \hat{\mathbf{a}} = (8\mathbf{i} - \mathbf{j} - 10\mathbf{k}) \cdot \left(\frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k} \right) \\ &= \frac{16}{3} + \frac{1}{3} + \frac{20}{3} = \frac{37}{3}. \end{aligned}$$

Since this is positive, f is increasing in this direction.

Example 7: Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ where Q is the point $(5, 0, 4)$.

(Purvanchal 2007; Kashi 14)

Solution: Here

$$\begin{aligned} \text{grad } f &= \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k} \\ &= 2x\mathbf{i} - 2y\mathbf{j} + 4z\mathbf{k} = 2\mathbf{i} - 4\mathbf{j} + 12\mathbf{k} \text{ at the point } (1, 2, 3). \end{aligned}$$

Also $\vec{PQ}$ = position vector of Q - position vector of P

$$= (5\mathbf{i} + 0\mathbf{j} + 4\mathbf{k}) - (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) = 4\mathbf{i} - 2\mathbf{j} + \mathbf{k}.$$

If $\hat{\mathbf{a}}$ be the unit vector in the direction of the vector $\vec{PQ}$, then

$$\hat{\mathbf{a}} = \frac{4\mathbf{i} - 2\mathbf{j} + \mathbf{k}}{\sqrt{16 + 4 + 1}} = \frac{4\mathbf{i} - 2\mathbf{j} + \mathbf{k}}{\sqrt{21}}.$$

$\therefore$ the required directional derivative

$$\begin{aligned} &= (\text{grad } f) \cdot \hat{\mathbf{a}} = (2\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}) \cdot \left\{ \frac{4\mathbf{i} - 2\mathbf{j} + \mathbf{k}}{\sqrt{21}} \right\} \\ &= \frac{28}{\sqrt{21}} = \frac{28}{21} \sqrt{21} = \frac{4}{3} \sqrt{21}. \end{aligned}$$

Example 8: In what direction from the point $(1, 1, -1)$ is the directional derivative of $f = x^2 - 2y^2 + 4z^2$ a maximum? Also find the value of this maximum directional derivative.

(Kanpur 2008)

Solution: We have

$$\begin{aligned} \text{grad } f &= 2x\mathbf{i} - 4y\mathbf{j} + 8z\mathbf{k} \\ &= 2\mathbf{i} - 4\mathbf{j} - 8\mathbf{k} \text{ at the point } (1, 1, -1). \end{aligned}$$

The directional derivative of f is maximum in the direction of $\text{grad } f$

$$= 2\mathbf{i} - 4\mathbf{j} - 8\mathbf{k}.$$

The maximum value of this directional derivative

$$= |\text{grad } f| = |2\mathbf{i} - 4\mathbf{j} - 8\mathbf{k}| = \sqrt{4 + 16 + 64} = \sqrt{84} = 2\sqrt{21}.$$

Example 9: What is the greatest rate of increase of $u = xyz^2$ at the point $(1, 0, 3)$?

Solution: We have $\nabla u = yz^2 \mathbf{i} + xz^2 \mathbf{j} + 2xyz \mathbf{k}$.

$\therefore$ at the point $(1, 0, 3)$, we have $\nabla u = 0 \mathbf{i} + 9 \mathbf{j} + 0 \mathbf{k} = 9 \mathbf{j}$.

The greatest rate of increase of u at the point $(1, 0, 3)$

= the maximum value of $\frac{du}{ds}$ at the point $(1, 0, 3)$

$= |\nabla u|$, at the point $(1, 0, 3) = |9\mathbf{j}| = 9$.

Example 10: Find the equations of the tangent plane and normal to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$.

Solution: The equation of the surface is

$$f(x, y, z) \equiv 2xz^2 - 3xy - 4x = 7.$$

We have

$$\text{grad } f = (2z^2 - 3y - 4) \mathbf{i} - 3x \mathbf{j} + 4xz \mathbf{k}$$

$$= 7\mathbf{i} - 3\mathbf{j} + 8\mathbf{k}, \text{ at the point } (1, -1, 2).$$

$\therefore 7\mathbf{i} - 3\mathbf{j} + 8\mathbf{k}$ is a vector along the normal to the surface at the point $(1, -1, 2)$.

The position vector of the point $(1, -1, 2)$ is $\mathbf{r} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

If $\mathbf{R} = X \mathbf{i} + Y \mathbf{j} + Z \mathbf{k}$ is the position vector of any current point (X, Y, Z) on the tangent plane at $(1, -1, 2)$, then the vector $\mathbf{R} - \mathbf{r}$ is perpendicular to the vector $\text{grad } f$.

$\therefore$ the equation of the tangent plane is $(\mathbf{R} - \mathbf{r}) \cdot \text{grad } f = 0$

$$\text{i.e., } \{(X \mathbf{i} + Y \mathbf{j} + Z \mathbf{k}) - (\mathbf{i} - \mathbf{j} + 2\mathbf{k})\} \cdot (7\mathbf{i} - 3\mathbf{j} + 8\mathbf{k}) = 0$$

$$\text{i.e., } \{(X - 1) \mathbf{i} + (Y + 1) \mathbf{j} + (Z - 2) \mathbf{k}\} \cdot (7\mathbf{i} - 3\mathbf{j} + 8\mathbf{k}) = 0$$

$$\text{i.e., } 7(X - 1) - 3(Y + 1) + 8(Z - 2) = 0.$$

The equations of the normal to the surface at the point $(1, -1, 2)$ are

$$\frac{X-1}{\frac{\partial f}{\partial x}} = \frac{Y+1}{\frac{\partial f}{\partial y}} = \frac{Z-2}{\frac{\partial f}{\partial z}} \quad \text{i.e.,} \quad \frac{X-1}{7} = \frac{Y+1}{-3} = \frac{Z-2}{8}.$$

Example 11: Find the equations of the tangent plane and normal to the surface $xyz = 4$ at the point $(1, 2, 2)$. (Meerut 2000)

Solution: The equation of the surface is $f(x, y, z) \equiv xyz - 4 = 0$.

We have $\text{grad } f = yz \mathbf{i} + xz \mathbf{j} + xy \mathbf{k} = 4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, at the point $(1, 2, 2)$.

$\therefore 4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ is a vector along the normal to the surface at the point $(1, 2, 2)$.

The position vector of the point $(1, 2, 2)$ is $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

If $\mathbf{R} = X \mathbf{i} + Y \mathbf{j} + Z \mathbf{k}$ is the position vector of any current point (X, Y, Z) on the tangent plane at $(1, 2, 2)$, the equation of the tangent plane is

$$(\mathbf{R} - \mathbf{r}) \cdot \text{grad } f = 0,$$

$$\begin{aligned} \text{i.e.,} & \quad \{(X \mathbf{i} + Y \mathbf{j} + Z \mathbf{k}) - (\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})\} \cdot (4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) = 0 \\ \text{i.e.,} & \quad \{(X - 1) \mathbf{i} + (Y - 2) \mathbf{j} + (Z - 2) \mathbf{k}\} \cdot (4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) = 0 \\ \text{i.e.,} & \quad 4(X - 1) + 2(Y - 2) + 2(Z - 2) = 0 \\ \text{i.e.,} & \quad 4X + 2Y + 2Z = 12, \text{ i.e., } 2X + Y + Z = 6. \end{aligned}$$

The equations of the normal to the surface at the point $(1, 2, 2)$ are

$$\frac{X - 1}{\frac{\partial f}{\partial x}} = \frac{Y - 2}{\frac{\partial f}{\partial y}} = \frac{Z - 2}{\frac{\partial f}{\partial z}}$$

$$\text{i.e.,} \quad \frac{X - 1}{4} = \frac{Y - 2}{2} = \frac{Z - 2}{2}, \quad \text{i.e.,} \quad \frac{X - 1}{2} = \frac{Y - 2}{1} = \frac{Z - 2}{1}.$$

Example 12: Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$, and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$. (Meerut 2001)

Solution: Angle between two surfaces at a point is the angle between the normals to the surfaces at that point. Let $f_1 = x^2 + y^2 + z^2$ and $f_2 = x^2 + y^2 - z$.

Then $\text{grad } f_1 = 2x \mathbf{i} + 2y \mathbf{j} + 2z \mathbf{k}$ and $\text{grad } f_2 = 2x \mathbf{i} + 2y \mathbf{j} - \mathbf{k}$.

Let $\mathbf{n}_1 = \text{grad } f_1$ at the point $(2, -1, 2)$ and $\mathbf{n}_2 = \text{grad } f_2$ at the point $(2, -1, 2)$.

Then $\mathbf{n}_1 = 4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\mathbf{n}_2 = 4\mathbf{i} - 2\mathbf{j} - \mathbf{k}$.

The vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ are along normals to the two surfaces at the point $(2, -1, 2)$. If θ is the angle between these vectors, then

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = |\mathbf{n}_1| |\mathbf{n}_2| \cos \theta$$

$$\text{or} \quad 16 + 4 - 4 = \sqrt{(16 + 4 + 16)} \sqrt{(16 + 4 + 1)} \cos \theta.$$

$$\therefore \quad \cos \theta = \frac{16}{6\sqrt{(21)}} \quad \text{or} \quad \theta = \cos^{-1} \frac{8}{3\sqrt{(21)}}.$$

Comprehensive Exercise 2

- Find the gradient and the unit normal to the level surface $x^2 + y - z = 4$ at the point $(2, 0, 0)$.
 - Find the unit normal to the surface $z = x^2 + y^2$ at the point $(-1, -2, 5)$.
- Find the unit vector normal to the surface $x^2 - y^2 + z = 2$ at the point $(1, -1, 2)$.
 - Find the unit normal to the surface $x^4 - 3xyz + z^2 + 1 = 0$ at the point $(1, 1, 1)$.
 - Find a unit normal vector to the surface $x^2 y + 2xz = 4$ at the point $(2, -2, 3)$.

3. (i) Find the directional derivatives of a scalar point function f in the direction of coordinate axes.
 (ii) Find the directional derivative of $\phi = xy + yz + zx$ in the direction of the vector $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ at $(1, 2, 0)$. (Kumaun 2015)
 (iii) Find the directional derivative of $\phi(x, y, z) = x^2yz + 4xz^2$ at the point $(1, -2, 1)$ in the direction of $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$.
 (iv) Find the directional derivative of $f(x, y, z) = x^2 - 2y^2 + 4z^2$ at the point $(1, 1, -1)$ in the direction of $2\mathbf{i} + \mathbf{j} - \mathbf{k}$.
 (v) Find the directional derivative of the function $f = xy + yz + zx$ in the direction of the vector $2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ at the point $(3, 1, 2)$.
 (vi) Find the directional derivative of the function $xy^2 + yz^2 + zx^2$ along the tangent to the curve $x = t, y = t^2, z = t^3$ at the point $(1, 1, 1)$.
4. Find the directional derivatives of $\phi = xyz$ at the point $(2, 2, 2)$, in the directions (i) $\mathbf{i}$, (ii) $\mathbf{j}$, (iii) $\mathbf{i} + \mathbf{j} + \mathbf{k}$.
5. For the function $f = y / (x^2 + y^2)$, find the value of the directional derivative making an angle 30° with the positive x -axis at the point $(0, 1)$.
6. (i) Find the greatest value of the directional derivative of the function $2x^2 - y - z^4$ at the point $(2, -1, 1)$.
 (ii) In what direction the directional derivative of $\phi = x^2y^2z$ from $(1, 1, 2)$ will be maximum and what is its magnitude? Also find a unit normal vector to the surface $x^2y^2z = 2$ at the point $(1, 1, 2)$.
 (iii) Find the maximum value of the directional derivative of $\phi = x^2yz$ at the point $(1, 4, 1)$.
 (iv) Calculate the maximum rate of change and the corresponding direction for the function $\phi = x^2y^3z^4$ at the point $2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.
 (v) Find the values of the constants a, b, c so that the directional derivative of $\phi = ax^2 + by^2 + cz^2$ at $(1, 1, 2)$ has a maximum magnitude 4 in the direction parallel to y -axis.
7. Find the equation of the tangent plane to the surface $yz - zx + xy + 5 = 0$, at the point $(1, -1, 2)$.
8. (i) Find the equations of the tangent plane and normal to the surface $x^2 + y^2 + z^2 = 25$ at the point $(4, 0, 3)$.
 (ii) Given the curve $x^2 + y^2 + z^2 = 1, x + y + z = 1$ (intersection of two surfaces), find the equations of the tangent line at the point $(1, 0, 0)$.
 (iii) Find the equations of the tangent plane and normal to the surface $z = x^2 + y^2$ at the point $(2, -1, 5)$.

9. (i) Find the equations of the tangent plane and the normal to the surface $x^2 + 2y^2 + 3z^2 = 12$ at the point $(1, 2, -1)$.
 (ii) Find the equations of the tangent plane and the normal to the surface $xy + yz + zx = 1$, at the point $(2, 3, -1)$.
 (iii) Find the equations of the tangent plane and the normal to the surface $z = x^2 - 2xy - y^2$ at the point $(1, 2, -7)$.
 (iv) Find the equation of the tangent plane to the surface $x^2 + y^2 + z^2 = 9$ at $(2, -1, 2)$.
 (v) Find the equation of the tangent plane to the surface $z = x^2 + y^2$ at the point $(1, -1, 2)$.
10. Show that the directional derivative of a scalar point function at any point along any tangent line to the level surface at the point is zero.
11. If $\mathbf{F}$ and f are point functions, show that the components of the former, tangential and normal to the level surface $f = 0$ are
$$\frac{\nabla f \times (\mathbf{F} \times \nabla f)}{(\nabla f)^2} \text{ and } \frac{(\mathbf{F} \cdot \nabla f) \nabla f}{(\nabla f)^2}.$$
12. Find the angle of intersection at $(4, -3, 2)$ of spheres $x^2 + y^2 + z^2 = 29$ and $x^2 + y^2 + z^2 + 4x - 6y - 8z - 47 = 0$.
13. Find the constants a and b so that the surface $ax^2 - byz = (a + 2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$.
14. Show that the sum of the squares of the intercepts on the coordinate axes made by the tangent plane to the surface $x^{2/3} + y^{2/3} + z^{2/3} = a^{2/3}$ is constant.

Answers 2

1. (i) $4\mathbf{i} + \mathbf{j} - \mathbf{k}$; $\frac{1}{3\sqrt{2}}(4\mathbf{i} + \mathbf{j} - \mathbf{k})$ (ii) $\frac{-(2\mathbf{i} + 4\mathbf{j} + \mathbf{k})}{\sqrt{21}}$
2. (i) $\frac{1}{3}(2\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ (ii) $\frac{\mathbf{i} - 3\mathbf{j} - \mathbf{k}}{\sqrt{11}}$ (iii) $\left(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$
3. (i) $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$ in the directions of $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$
 (ii) $\frac{10}{3}$ (iii) $-\frac{13}{3}$ (iv) $\frac{8}{\sqrt{6}}$
 (v) $\frac{45}{7}$ (vi) $\frac{18}{\sqrt{14}}$

4. (i) 4 (ii) 4 (iii) $4\sqrt{3}$
5. $-1/2$
6. (i) 9
 (ii) $\sqrt{(33)}$ in the direction of the vector $4\mathbf{i} + 4\mathbf{j} + \mathbf{k}$; $\frac{4\mathbf{i} + 4\mathbf{j} + \mathbf{k}}{\sqrt{(33)}}$
 (iii) 9
 (iv) $324\sqrt{2}$ in the direction of the vector $108(\mathbf{i} + \mathbf{j} - 4\mathbf{k})$
 (v) $a = 0, b = 2, c = 0$
7. $3X - 3Y + 2Z = 10$
8. (i) $4x + 3z = 25$; $\frac{x-4}{4} = \frac{y}{0} = \frac{z-3}{3}$
 (ii) $X = 1, \frac{Y}{-1} = \frac{Z}{1}$ (iii) $4x - 2y - z = 5$; $\frac{x-2}{4} = \frac{y+1}{-2} = \frac{z-5}{-1}$
9. (i) $x + 4y - 3z = 12$; $\frac{x-1}{1} = \frac{y-2}{4} = \frac{z+1}{-3}$
 (ii) $2x + y + 5z = 2$; $\frac{x-2}{2} = \frac{y-3}{1} = \frac{z+1}{5}$
 (iii) $2x + 6y + z = 7$; $\frac{x-1}{2} = \frac{y-2}{6} = \frac{z+7}{1}$
 (iv) $2x - y + 2z = 9$ (v) $2x - 2y - z = 2$
12. $\cos^{-1} \sqrt{(19/29)}$ 13. $a = 5/2, b = 1$

8 Divergence of a Vector Point Function

(Agra 2005)

Definition: Let $\mathbf{V}$ be any given differentiable vector point function. Then the divergence of $\mathbf{V}$, written as, $\nabla \cdot \mathbf{V}$ or $\text{div } \mathbf{V}$,

is defined as
$$\text{div } \mathbf{V} = \nabla \cdot \mathbf{V} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot \mathbf{V}$$

$$= \mathbf{i} \cdot \frac{\partial \mathbf{V}}{\partial x} + \mathbf{j} \cdot \frac{\partial \mathbf{V}}{\partial y} + \mathbf{k} \cdot \frac{\partial \mathbf{V}}{\partial z} = \sum \mathbf{i} \cdot \frac{\partial \mathbf{V}}{\partial x}.$$

It should be noted that $\text{div } \mathbf{V}$ is a scalar quantity. Thus the divergence of a vector point function is a scalar point function.

Theorem: If $\mathbf{V} = V_1 \mathbf{i} + V_2 \mathbf{j} + V_3 \mathbf{k}$ is a differentiable vector point function, then

$$\text{div } \mathbf{V} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}.$$

Proof: We have by definition

$$\operatorname{div} \mathbf{V} = \nabla \cdot \mathbf{V} = \mathbf{i} \cdot \frac{\partial \mathbf{V}}{\partial x} + \mathbf{j} \cdot \frac{\partial \mathbf{V}}{\partial y} + \mathbf{k} \cdot \frac{\partial \mathbf{V}}{\partial z}.$$

Now

$$\mathbf{V} = V_1 \mathbf{i} + V_2 \mathbf{j} + V_3 \mathbf{k}.$$

$$\therefore \frac{\partial \mathbf{V}}{\partial x} = \frac{\partial V_1}{\partial x} \mathbf{i} + \frac{\partial V_2}{\partial x} \mathbf{j} + \frac{\partial V_3}{\partial x} \mathbf{k}.$$

$$\therefore \mathbf{i} \cdot \frac{\partial \mathbf{V}}{\partial x} = \mathbf{i} \cdot \left(\frac{\partial V_1}{\partial x} \mathbf{i} + \frac{\partial V_2}{\partial x} \mathbf{j} + \frac{\partial V_3}{\partial x} \mathbf{k} \right) = \frac{\partial V_1}{\partial x}.$$

Similarly, $\mathbf{j} \cdot \frac{\partial \mathbf{V}}{\partial y} = \frac{\partial V_2}{\partial y} \quad \text{and} \quad \mathbf{k} \cdot \frac{\partial \mathbf{V}}{\partial z} = \frac{\partial V_3}{\partial z}.$

Hence, $\operatorname{div} \mathbf{V} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}.$

Solenoidal Vector: A vector $\mathbf{V}$ is said to be solenoidal if $\operatorname{div} \mathbf{V} = 0$.

9 Curl of a Vector Point Function

(Agra 2005)

Definition: Let $\mathbf{f}$ be any given differentiable vector point function. Then the curl or rotation of $\mathbf{f}$, written as $\nabla \times \mathbf{f}$, curl $\mathbf{f}$ or rot $\mathbf{f}$ is defined as

$$\begin{aligned} \operatorname{curl} \mathbf{f} = \nabla \times \mathbf{f} &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \times \mathbf{f} \\ &= \mathbf{i} \times \frac{\partial \mathbf{f}}{\partial x} + \mathbf{j} \times \frac{\partial \mathbf{f}}{\partial y} + \mathbf{k} \times \frac{\partial \mathbf{f}}{\partial z} = \sum \mathbf{i} \times \frac{\partial \mathbf{f}}{\partial x}. \end{aligned}$$

It should be noted that curl $\mathbf{f}$ is a vector quantity. Thus the curl of a vector point function is a vector point function.

Theorem: If $\mathbf{f} = f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}$ is a differentiable vector point function, then

$$\operatorname{curl} \mathbf{f} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \mathbf{k}.$$

Proof: We have by definition

$$\begin{aligned} \operatorname{curl} \mathbf{f} = \nabla \times \mathbf{f} &= \mathbf{i} \times \frac{\partial \mathbf{f}}{\partial x} + \mathbf{j} \times \frac{\partial \mathbf{f}}{\partial y} + \mathbf{k} \times \frac{\partial \mathbf{f}}{\partial z} \\ &= \mathbf{i} \times \frac{\partial}{\partial x} (f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}) + \mathbf{j} \times \frac{\partial}{\partial y} (f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}) \\ &\quad + \mathbf{k} \times \frac{\partial}{\partial z} (f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}) \\ &= \mathbf{i} \times \left(\frac{\partial f_1}{\partial x} \mathbf{i} + \frac{\partial f_2}{\partial x} \mathbf{j} + \frac{\partial f_3}{\partial x} \mathbf{k} \right) + \mathbf{j} \times \left(\frac{\partial f_1}{\partial y} \mathbf{i} + \frac{\partial f_2}{\partial y} \mathbf{j} + \frac{\partial f_3}{\partial y} \mathbf{k} \right) \\ &\quad + \mathbf{k} \times \left(\frac{\partial f_1}{\partial z} \mathbf{i} + \frac{\partial f_2}{\partial z} \mathbf{j} + \frac{\partial f_3}{\partial z} \mathbf{k} \right) \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{\partial f_2}{\partial x} \mathbf{k} - \frac{\partial f_3}{\partial x} \mathbf{j} \right) + \left(-\frac{\partial f_1}{\partial y} \mathbf{k} + \frac{\partial f_3}{\partial y} \mathbf{i} \right) + \left(\frac{\partial f_1}{\partial z} \mathbf{j} - \frac{\partial f_2}{\partial z} \mathbf{i} \right) \\
&= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \mathbf{k}.
\end{aligned}$$

Note: It should be noted that the expression for curl $\mathbf{f}$ can be written immediately if we treat the operator ∇ as a vector quantity. Thus

$$\begin{aligned}
\text{Curl } \mathbf{f} &= \nabla \times \mathbf{f} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \times (f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}) \\
&= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} \\
&= \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_2 & f_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ f_1 & f_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ f_1 & f_2 \end{vmatrix} \mathbf{k} \\
&= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \mathbf{k}.
\end{aligned}$$

But we must take care that in the expansion of the determinant the operators $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$ must *precede* the functions f_1, f_2, f_3 .

Irrotational vector: A vector $\mathbf{f}$ is said to be irrotational if $\nabla \times \mathbf{f} = \mathbf{0}$.

10 The Laplacian Operator ∇^2

The Laplacian operator ∇^2 is defined as $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$.

If f is a scalar point function, then $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$.

It should be noted that $\nabla^2 f$ is also a scalar quantity.

If $\mathbf{f}$ is a vector point function, then $\nabla^2 \mathbf{f} = \frac{\partial^2 \mathbf{f}}{\partial x^2} + \frac{\partial^2 \mathbf{f}}{\partial y^2} + \frac{\partial^2 \mathbf{f}}{\partial z^2}$.

It should be noted that $\nabla^2 \mathbf{f}$ is also a vector quantity.

Laplace's equation: The equation $\nabla^2 f = 0$ is called Laplace's equation. A function which satisfies Laplace's equation is called a **harmonic function**.

Illustrative Examples

Example 13: Prove that $\text{div } \mathbf{r} = 3$. (Kumaun 2000; Garhwal 01; Bundelkhand 01, 04; Meerut 03, 04, 07B, 08; Agra 08; Lucknow 05; Kanpur 13; Avadh 09; Purvanchal 11)

Solution: We have $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$.

$$\begin{aligned} \text{By definition, } \text{div } \mathbf{r} &= \nabla \cdot \mathbf{r} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot \mathbf{r} = \mathbf{i} \cdot \frac{\partial \mathbf{r}}{\partial x} + \mathbf{j} \cdot \frac{\partial \mathbf{r}}{\partial y} + \mathbf{k} \cdot \frac{\partial \mathbf{r}}{\partial z} \\ &= \mathbf{i} \cdot \mathbf{i} + \mathbf{j} \cdot \mathbf{j} + \mathbf{k} \cdot \mathbf{k} \quad \left[\because \frac{\partial \mathbf{r}}{\partial x} = \mathbf{i}, \frac{\partial \mathbf{r}}{\partial y} = \mathbf{j}, \frac{\partial \mathbf{r}}{\partial z} = \mathbf{k} \right] \\ &= 1 + 1 + 1 = 3. \end{aligned}$$

Example 14: Prove that $\text{curl } \mathbf{r} = \mathbf{0}$. (Kumaun 2000; Garhwal 02; Meerut 11; Bundelkhand 06; Agra 08; Avadh 09; Kashi 13; Purvanchal 11; Rohilkhand 14)

Solution: We have by definition

$$\begin{aligned} \text{curl } \mathbf{r} &= \nabla \times \mathbf{r} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \times \mathbf{r} \\ &= \mathbf{i} \times \frac{\partial \mathbf{r}}{\partial x} + \mathbf{j} \times \frac{\partial \mathbf{r}}{\partial y} + \mathbf{k} \times \frac{\partial \mathbf{r}}{\partial z}. \end{aligned}$$

Now $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$.

$$\therefore \frac{\partial \mathbf{r}}{\partial x} = \mathbf{i}, \frac{\partial \mathbf{r}}{\partial y} = \mathbf{j}, \frac{\partial \mathbf{r}}{\partial z} = \mathbf{k}.$$

$$\begin{aligned} \therefore \text{curl } \mathbf{r} &= \mathbf{i} \times \mathbf{i} + \mathbf{j} \times \mathbf{j} + \mathbf{k} \times \mathbf{k} \\ &= \mathbf{0} + \mathbf{0} + \mathbf{0} = \mathbf{0}. \end{aligned}$$

Example 15: If $\mathbf{f} = x^2 y \mathbf{i} - 2xz \mathbf{j} + 2yz \mathbf{k}$, find

- (i) $\text{div } \mathbf{f}$, (Kumaun 2008)
 (ii) $\text{curl } \mathbf{f}$, (Meerut 2012)
 (iii) $\text{curl curl } \mathbf{f}$. (Meerut 2001; Kanpur 14)

Solution: (i) We have

$$\begin{aligned} \text{div } \mathbf{f} &= \nabla \cdot \mathbf{f} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (x^2 y \mathbf{i} - 2xz \mathbf{j} + 2yz \mathbf{k}) \\ &= \frac{\partial}{\partial x} (x^2 y) + \frac{\partial}{\partial y} (-2xz) + \frac{\partial}{\partial z} (2yz) \\ &= 2xy + 0 + 2y \\ &= 2y (x + 1). \end{aligned}$$

$$\begin{aligned}
 \text{(ii) We have } \operatorname{curl} \mathbf{f} &= \nabla \times \mathbf{f} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y & -2xz & 2yz \end{vmatrix} \\
 &= \left[\frac{\partial}{\partial y} (2yz) - \frac{\partial}{\partial z} (-2xz) \right] \mathbf{i} - \left[\frac{\partial}{\partial x} (2yz) - \frac{\partial}{\partial z} (x^2 y) \right] \mathbf{j} \\
 &\quad + \left[\frac{\partial}{\partial x} (-2xz) - \frac{\partial}{\partial y} (x^2 y) \right] \mathbf{k} \\
 &= (2z + 2x) \mathbf{i} - 0 \mathbf{j} + (-2z - x^2) \mathbf{k} \\
 &= (2x + 2z) \mathbf{i} - (x^2 + 2z) \mathbf{k}.
 \end{aligned}$$

$$\text{(iii) We have } \operatorname{curl} \operatorname{curl} \mathbf{f} = \nabla \times (\nabla \times \mathbf{f}) = \nabla \times [(2x + 2z) \mathbf{i} - (x^2 + 2z) \mathbf{k}]$$

$$\begin{aligned}
 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x + 2z & 0 & -x^2 - 2z \end{vmatrix} \\
 &= \left[\frac{\partial}{\partial y} (-x^2 - 2z) \right] \mathbf{i} - \left[\frac{\partial}{\partial x} (-x^2 - 2z) - \frac{\partial}{\partial z} (2x + 2z) \right] \mathbf{j} \\
 &\quad + \left[0 - \frac{\partial}{\partial y} (2x + 2z) \right] \mathbf{k} \\
 &= 0 \mathbf{i} - (-2x - 2) \mathbf{j} + (0 - 0) \mathbf{k} = (2x + 2) \mathbf{j}.
 \end{aligned}$$

Example 16: Determine the constant a so that the vector

$$\mathbf{V} = (x + 3y) \mathbf{i} + (y - 2z) \mathbf{j} + (x + az) \mathbf{k} \text{ is solenoidal.}$$

(Rohilkhand 2009B)

Solution: A vector $\mathbf{V}$ is said to be solenoidal if $\operatorname{div} \mathbf{V} = 0$.

$$\begin{aligned}
 \text{We have } \operatorname{div} \mathbf{V} &= \nabla \cdot \mathbf{V} = \frac{\partial}{\partial x} (x + 3y) + \frac{\partial}{\partial y} (y - 2z) + \frac{\partial}{\partial z} (x + az) \\
 &= 1 + 1 + a = 2 + a.
 \end{aligned}$$

$$\text{Now } \operatorname{div} \mathbf{V} = 0 \text{ if } 2 + a = 0 \text{ i.e., if } a = -2.$$

Example 17: Show that the vector $\mathbf{V} = (\sin y + z) \mathbf{i} + (x \cos y - z) \mathbf{j} + (x - y) \mathbf{k}$ is irrotational.

Solution: A vector $\mathbf{V}$ is said to be irrotational if $\operatorname{curl} \mathbf{V} = \mathbf{0}$. We have

$$\begin{aligned}
 \operatorname{curl} \mathbf{V} &= \nabla \times \mathbf{V} \\
 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin y + z & x \cos y - z & x - y \end{vmatrix}
 \end{aligned}$$

$$= \left[\frac{\partial}{\partial y} (x - y) - \frac{\partial}{\partial z} (x \cos y - z) \right] \mathbf{i} - \left[\frac{\partial}{\partial x} (x - y) - \frac{\partial}{\partial z} (\sin y + z) \right] \mathbf{j} \\ + \left[\frac{\partial}{\partial x} (x \cos y - z) - \frac{\partial}{\partial y} (\sin y + z) \right] \mathbf{k}$$

$$= (-1 + 1) \mathbf{i} - (1 - 1) \mathbf{j} + (\cos y - \cos y) \mathbf{k} = \mathbf{0}.$$

$\therefore \mathbf{V}$ is irrotational.

Example 18: Prove that $\nabla \cdot (r^3 \mathbf{r}) = 6r^3$. (Purvanchal 2006, 10)

Solution: We have $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$.

$$\therefore r^3 \mathbf{r} = r^3 (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) = r^3 x \mathbf{i} + r^3 y \mathbf{j} + r^3 z \mathbf{k}.$$

$$\therefore \nabla \cdot (r^3 \mathbf{r}) = \text{div} (r^3 \mathbf{r}) \\ = \frac{\partial}{\partial x} (r^3 x) + \frac{\partial}{\partial y} (r^3 y) + \frac{\partial}{\partial z} (r^3 z) \\ = r^3 + 3r^2 x \frac{\partial r}{\partial x} + r^3 + 3r^2 y \frac{\partial r}{\partial y} + r^3 + 3r^2 z \frac{\partial r}{\partial z} \\ = 3r^3 + 3r^2 \left(x \frac{\partial r}{\partial x} + y \frac{\partial r}{\partial y} + z \frac{\partial r}{\partial z} \right) \quad \dots(1)$$

$$\text{Now } r^2 = x^2 + y^2 + z^2.$$

$$\therefore 2r \frac{\partial r}{\partial x} = 2x \quad \text{or} \quad \frac{\partial r}{\partial x} = \frac{x}{r}.$$

$$\text{Similarly } \frac{\partial r}{\partial y} = \frac{y}{r} \quad \text{and} \quad \frac{\partial r}{\partial z} = \frac{z}{r}.$$

$$\therefore \text{from (1), } \nabla \cdot (r^3 \mathbf{r}) = 3r^3 + 3r^2 \left(x \cdot \frac{x}{r} + y \cdot \frac{y}{r} + z \cdot \frac{z}{r} \right) \\ = 3r^3 + 3r^2 \left(\frac{x^2 + y^2 + z^2}{r} \right) \\ = 3r^3 + 3r^2 \cdot \frac{r^2}{r} = 3r^3 + 3r^3 = 6r^3.$$

Example 19: If $\mathbf{V}$ is a constant vector, show that

$$(i) \quad \text{div } \mathbf{V} = 0, \quad (\text{Rohilkhand 2005; Purvanchal 14})$$

$$(ii) \quad \text{curl } \mathbf{V} = \mathbf{0}. \quad (\text{Purvanchal 2014; Kumaun 15})$$

Solution: (i) We have

$$\text{div } \mathbf{V} = \mathbf{i} \cdot \frac{\partial \mathbf{V}}{\partial x} + \mathbf{j} \cdot \frac{\partial \mathbf{V}}{\partial y} + \mathbf{k} \cdot \frac{\partial \mathbf{V}}{\partial z} \\ = \mathbf{i} \cdot \mathbf{0} + \mathbf{j} \cdot \mathbf{0} + \mathbf{k} \cdot \mathbf{0} = 0.$$

(ii) We have

$$\begin{aligned}\operatorname{curl} \mathbf{V} &= \mathbf{i} \times \frac{\partial \mathbf{V}}{\partial x} + \mathbf{j} \times \frac{\partial \mathbf{V}}{\partial y} + \mathbf{k} \times \frac{\partial \mathbf{V}}{\partial z} \\ &= \mathbf{i} \times \mathbf{0} + \mathbf{j} \times \mathbf{0} + \mathbf{k} \times \mathbf{0} = \mathbf{0}.\end{aligned}$$

Example 20: If $\mathbf{a}$ is a constant vector, find

- (i) $\operatorname{div} (\mathbf{r} \times \mathbf{a})$, (Bundelkhand 2006; Kumaun 07, 11)
(ii) $\operatorname{curl} (\mathbf{r} \times \mathbf{a})$. (Bundelkhand 2006; Kanpur 06; Kumaun 07; Agra 08; Rohilkhand 14)

Solution: We have $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$.

Let $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$. Then the scalars a_1, a_2, a_3 are all constants.

We have

$$\begin{aligned}\mathbf{r} \times \mathbf{a} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ a_1 & a_2 & a_3 \end{vmatrix} \\ &= (a_3 y - a_2 z) \mathbf{i} + (a_1 z - a_3 x) \mathbf{j} + (a_2 x - a_1 y) \mathbf{k}.\end{aligned}$$

$$\begin{aligned}\text{(i)} \quad \operatorname{div} (\mathbf{r} \times \mathbf{a}) &= \frac{\partial}{\partial x} (a_3 y - a_2 z) + \frac{\partial}{\partial y} (a_1 z - a_3 x) + \frac{\partial}{\partial z} (a_2 x - a_1 y) \\ &= 0 + 0 + 0 = 0.\end{aligned}$$

$$\text{(ii)} \quad \operatorname{curl} (\mathbf{r} \times \mathbf{a}) = \nabla \times (\mathbf{r} \times \mathbf{a})$$

$$\begin{aligned}&= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_3 y - a_2 z & a_1 z - a_3 x & a_2 x - a_1 y \end{vmatrix} \\ &= \left[\frac{\partial}{\partial y} (a_2 x - a_1 y) - \frac{\partial}{\partial z} (a_1 z - a_3 x) \right] \mathbf{i} \\ &\quad - \left[\frac{\partial}{\partial x} (a_2 x - a_1 y) - \frac{\partial}{\partial z} (a_3 y - a_2 z) \right] \mathbf{j} \\ &\quad + \left[\frac{\partial}{\partial x} (a_1 z - a_3 x) - \frac{\partial}{\partial y} (a_3 y - a_2 z) \right] \mathbf{k} \\ &= -2a_1 \mathbf{i} - 2a_2 \mathbf{j} - 2a_3 \mathbf{k} = -2 (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) = -2\mathbf{a}.\end{aligned}$$

Example 21: If $\mathbf{V} = e^{xyz} (\mathbf{i} + \mathbf{j} + \mathbf{k})$, find $\operatorname{curl} \mathbf{V}$.

$$\text{Solution: We have } \operatorname{curl} \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{xyz} & e^{xyz} & e^{xyz} \end{vmatrix}$$

$$\begin{aligned}
 &= \left[\frac{\partial}{\partial y} (e^{xyz}) - \frac{\partial}{\partial z} (e^{xyz}) \right] \mathbf{i} - \left[\frac{\partial}{\partial x} (e^{xyz}) - \frac{\partial}{\partial z} (e^{xyz}) \right] \mathbf{j} \\
 &\quad + \left[\frac{\partial}{\partial x} (e^{xyz}) - \frac{\partial}{\partial y} (e^{xyz}) \right] \mathbf{k} \\
 &= e^{xyz} (xz - xy) \mathbf{i} + e^{xyz} (xy - yz) \mathbf{j} + e^{xyz} (yz - xz) \mathbf{k}.
 \end{aligned}$$

Example 22: Evaluate $\text{div } \mathbf{f}$ where $\mathbf{f} = 2x^2z \mathbf{i} - xy^2z \mathbf{j} + 3y^2x \mathbf{k}$.

Solution: We have

$$\begin{aligned}
 \text{div } \mathbf{f} &= \nabla \cdot \mathbf{f} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (2x^2z \mathbf{i} - xy^2z \mathbf{j} + 3y^2x \mathbf{k}) \\
 &= \frac{\partial}{\partial x} (2x^2z) + \frac{\partial}{\partial y} (-xy^2z) + \frac{\partial}{\partial z} (3y^2x) \\
 &= 4xz - 2xyz + 0 = 2xz (2 - y).
 \end{aligned}$$

Example 23: Show that $\nabla^2 (x/r^3) = 0$.

(Kumaun 2008)

Solution:
$$\nabla^2 \left(\frac{x}{r^3} \right) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left(\frac{x}{r^3} \right).$$

Now
$$\begin{aligned}
 \frac{\partial^2}{\partial x^2} \left(\frac{x}{r^3} \right) &= \frac{\partial}{\partial x} \left\{ \frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) \right\} = \frac{\partial}{\partial x} \left\{ \frac{1}{r^3} - \frac{3x}{r^4} \frac{\partial r}{\partial x} \right\} \\
 &= \frac{\partial}{\partial x} \left\{ \frac{1}{r^3} - \frac{3x}{r^4} \frac{x}{r} \right\} \quad \left[\because r^2 = x^2 + y^2 + z^2 \text{ gives } \frac{\partial r}{\partial x} = \frac{x}{r} \right] \\
 &= \frac{\partial}{\partial x} \left\{ \frac{1}{r^3} - \frac{3x^2}{r^5} \right\} = -\frac{3}{r^4} \frac{\partial r}{\partial x} - \frac{6x}{r^5} + \frac{15x^2}{r^6} \frac{\partial r}{\partial x} \\
 &= -\frac{3}{r^4} \frac{x}{r} - \frac{6x}{r^5} + \frac{15x^2}{r^6} \frac{x}{r} = -\frac{9x}{r^5} + \frac{15x^3}{r^7}.
 \end{aligned}$$

Again
$$\begin{aligned}
 \frac{\partial^2}{\partial y^2} \left(\frac{x}{r^3} \right) &= \frac{\partial}{\partial y} \left\{ \frac{\partial}{\partial y} \left(\frac{x}{r^3} \right) \right\} = \frac{\partial}{\partial y} \left\{ -\frac{3x}{r^4} \frac{\partial r}{\partial y} \right\} \\
 &= \frac{\partial}{\partial y} \left\{ -\frac{3x}{r^4} \frac{y}{r} \right\} \quad \left[\because \frac{\partial r}{\partial y} = \frac{y}{r} \right] \\
 &= \frac{\partial}{\partial y} \left(-\frac{3xy}{r^5} \right) = -\frac{3x}{r^5} + \frac{15xy}{r^6} \frac{\partial r}{\partial y} = -\frac{3x}{r^5} + \frac{15x y^2}{r^7}.
 \end{aligned}$$

Similarly
$$\frac{\partial^2}{\partial z^2} \left(\frac{x}{r^3} \right) = -\frac{3x}{r^5} + \frac{15xz^2}{r^7}.$$

Therefore, adding we get

$$\nabla^2 \left(\frac{x}{r^3} \right) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left(\frac{x}{r^3} \right)$$

$$\begin{aligned}
 &= -\frac{9x}{r^5} + \frac{15x^3}{r^7} - \frac{3x}{r^5} + \frac{15xy^2}{r^7} - \frac{3x}{r^5} + \frac{15xz^2}{r^7} \\
 &= -\frac{15x}{r^5} + \frac{15x}{r^7} (x^2 + y^2 + z^2) = -\frac{15x}{r^5} + \frac{15x}{r^7} r^2 = 0.
 \end{aligned}$$

Comprehensive Exercise 3

1. (i) If $\mathbf{F} = x^2z \mathbf{i} - 2y^3z^2 \mathbf{j} + xy^2z \mathbf{k}$, find $\text{div } \mathbf{F}$, $\text{curl } \mathbf{F}$ at $(1, -1, 1)$.
 (ii) If $\mathbf{f} = (y^2 + z^2 - x^2) \mathbf{i} + (z^2 + x^2 - y^2) \mathbf{j} + (x^2 + y^2 - z^2) \mathbf{k}$, find $\text{div } \mathbf{f}$ and $\text{curl } \mathbf{f}$.
 (iii) If $\mathbf{f} = xy^2 \mathbf{i} + 2x^2yz \mathbf{j} - 3yz^2 \mathbf{k}$, find $\text{div } \mathbf{f}$ and $\text{curl } \mathbf{f}$. What are their values at the point $(1, -1, 1)$? (Rohilkhand 2005)
2. Find $\text{div } \mathbf{F}$ and $\text{curl } \mathbf{F}$ where $\mathbf{F} = \text{grad } (x^3 + y^3 + z^3 - 3xyz)$. (Rohilkhand 2007)
3. Find the divergence and curl of the vector $\mathbf{f} = (x^2 - y^2) \mathbf{i} + 2xy \mathbf{j} + (y^2 - xy) \mathbf{k}$. (Bundelkhand 2004)
4. Given $\phi = 2x^3y^2z^4$, find $\text{div } (\text{grad } \phi)$.
5. If $u = x^2 - y^2 + 4z$, show that $\nabla^2 u = 0$.
6. If $u = 3x^2y$ and $v = xz^2 - 2y$, then find $\text{grad } [(\text{grad } u) \cdot (\text{grad } v)]$.
7. If $\mathbf{f} = (x + y + 1) \mathbf{i} + \mathbf{j} + (-x - y) \mathbf{k}$, prove that $\mathbf{f} \cdot \text{curl } \mathbf{f} = 0$.
8. (i) If $\mathbf{f} = f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}$, show that $\nabla \cdot \mathbf{f} = \nabla f_1 \cdot \mathbf{i} + \nabla f_2 \cdot \mathbf{j} + \nabla f_3 \cdot \mathbf{k}$. (Bundelkhand 2001)
 (ii) Prove that $\nabla \cdot (r^3 \mathbf{r}) = 6r^3$.
9. Find the constants a, b, c so that the vector $\mathbf{F} = (x + 2y + az) \mathbf{i} + (bx - 3y - z) \mathbf{j} + (4x + cy + 2z) \mathbf{k}$ is irrotational. (Bundelkhand 2005; Rohilkhand 08B)
10. Show that the vector $\mathbf{F} = 3y^4z^2 \mathbf{i} + 4x^3z^2 \mathbf{j} - 3x^2y^2 \mathbf{k}$ is solenoid but not irrotational. (Kashi 2014)

Answers 3

1. (i) $-3; -6i$
 (ii) $-2(x + y + z); 2(y - z) \mathbf{i} + 2(z - x) \mathbf{j} + 2(x - y) \mathbf{k}$

$$(iii) y^2 + 2x^2z - 6yz; 9; -(3z^2 + 2x^2y) \mathbf{i} + (4xyz - 2xy) \mathbf{k}; -\mathbf{i} - 2\mathbf{k}$$

$$2. \quad 6(x + y + z); 0 \qquad 3. \quad 4x; (2y - x) \mathbf{i} + y \mathbf{j} + 4y \mathbf{k}$$

$$4. \quad 12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2$$

$$6. \quad (6yz^2 - 12x) \mathbf{i} + (6xz^2) \mathbf{j} + (12xyz) \mathbf{k}$$

$$9. \quad a = 4, b = 2, c = -1$$

11 Vector Identities

1. Prove that $\text{div} (\mathbf{A} + \mathbf{B}) = \text{div} \mathbf{A} + \text{div} \mathbf{B}$ or $\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}$.

Proof: We have

$$\begin{aligned} \text{div} (\mathbf{A} + \mathbf{B}) &= \nabla \cdot (\mathbf{A} + \mathbf{B}) = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (\mathbf{A} + \mathbf{B}) \\ &= \mathbf{i} \cdot \frac{\partial}{\partial x} (\mathbf{A} + \mathbf{B}) + \mathbf{j} \cdot \frac{\partial}{\partial y} (\mathbf{A} + \mathbf{B}) + \mathbf{k} \cdot \frac{\partial}{\partial z} (\mathbf{A} + \mathbf{B}) \\ &= \mathbf{i} \cdot \left(\frac{\partial \mathbf{A}}{\partial x} + \frac{\partial \mathbf{B}}{\partial x} \right) + \mathbf{j} \cdot \left(\frac{\partial \mathbf{A}}{\partial y} + \frac{\partial \mathbf{B}}{\partial y} \right) + \mathbf{k} \cdot \left(\frac{\partial \mathbf{A}}{\partial z} + \frac{\partial \mathbf{B}}{\partial z} \right) \\ &= \left(\mathbf{i} \cdot \frac{\partial \mathbf{A}}{\partial x} + \mathbf{j} \cdot \frac{\partial \mathbf{A}}{\partial y} + \mathbf{k} \cdot \frac{\partial \mathbf{A}}{\partial z} \right) + \left(\mathbf{i} \cdot \frac{\partial \mathbf{B}}{\partial x} + \mathbf{j} \cdot \frac{\partial \mathbf{B}}{\partial y} + \mathbf{k} \cdot \frac{\partial \mathbf{B}}{\partial z} \right) \\ &= \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B} = \text{div} \mathbf{A} + \text{div} \mathbf{B}. \end{aligned}$$

2. Prove that $\text{curl} (\mathbf{A} + \mathbf{B}) = \text{curl} \mathbf{A} + \text{curl} \mathbf{B}$ or $\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$.

(Rohilkhand 2008)

Proof: We have $\text{curl} (\mathbf{A} + \mathbf{B}) = \nabla \times (\mathbf{A} + \mathbf{B})$

$$\begin{aligned} &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \times (\mathbf{A} + \mathbf{B}) \\ &= \Sigma \mathbf{i} \times \frac{\partial}{\partial x} (\mathbf{A} + \mathbf{B}) = \Sigma \mathbf{i} \times \left(\frac{\partial \mathbf{A}}{\partial x} + \frac{\partial \mathbf{B}}{\partial x} \right) \\ &= \Sigma \mathbf{i} \times \frac{\partial \mathbf{A}}{\partial x} + \Sigma \mathbf{i} \times \frac{\partial \mathbf{B}}{\partial x} = \text{curl} \mathbf{A} + \text{curl} \mathbf{B}. \end{aligned}$$

3. If $\mathbf{A}$ is a differentiable vector function and ϕ is a differentiable scalar function, then

$$\text{div} (\phi \mathbf{A}) = (\text{grad } \phi) \cdot \mathbf{A} + \phi \text{div} \mathbf{A} \quad \text{or} \quad \nabla \cdot (\phi \mathbf{A}) = (\nabla \phi) \cdot \mathbf{A} + \phi (\nabla \cdot \mathbf{A}).$$

(Garhwal 2001; Agra 05; Kashi 14)

Proof: We have

$$\text{div} (\phi \mathbf{A}) = \nabla \cdot (\phi \mathbf{A}) = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (\phi \mathbf{A})$$

$$\begin{aligned}
&= \mathbf{i} \cdot \frac{\partial}{\partial x} (\phi \mathbf{A}) + \mathbf{j} \cdot \frac{\partial}{\partial y} (\phi \mathbf{A}) + \mathbf{k} \cdot \frac{\partial}{\partial z} (\phi \mathbf{A}) \\
&= \Sigma \left\{ \mathbf{i} \cdot \left(\frac{\partial}{\partial x} (\phi \mathbf{A}) \right) \right\} = \Sigma \left\{ \mathbf{i} \cdot \left(\frac{\partial \phi}{\partial x} \mathbf{A} + \phi \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \\
&= \Sigma \left\{ \mathbf{i} \cdot \left(\frac{\partial \phi}{\partial x} \mathbf{A} \right) \right\} + \Sigma \left\{ \mathbf{i} \cdot \left(\phi \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \\
&= \Sigma \left\{ \left(\frac{\partial \phi}{\partial x} \mathbf{i} \right) \cdot \mathbf{A} \right\} + \Sigma \left\{ \phi \left(\mathbf{i} \cdot \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \\
&\quad [\because \mathbf{a} \cdot (m\mathbf{b}) = (m\mathbf{a}) \cdot \mathbf{b} = m(\mathbf{a} \cdot \mathbf{b})] \\
&= \left\{ \Sigma \frac{\partial \phi}{\partial x} \mathbf{i} \right\} \cdot \mathbf{A} + \phi \Sigma \left(\mathbf{i} \cdot \frac{\partial \mathbf{A}}{\partial x} \right) = (\nabla \phi) \cdot \mathbf{A} + \phi (\nabla \cdot \mathbf{A}).
\end{aligned}$$

4. Prove that $\text{curl} (\phi \mathbf{A}) = (\text{grad } \phi) \times \mathbf{A} + \phi \text{curl } \mathbf{A}$

or $\nabla \times (\phi \mathbf{A}) = (\nabla \phi) \times \mathbf{A} + \phi (\nabla \times \mathbf{A}).$

(Garhwal 2002, 03; Bundelkhand 06)

Proof: We have $\text{curl} (\phi \mathbf{A}) = \nabla \times (\phi \mathbf{A}) = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \times (\phi \mathbf{A})$

$$\begin{aligned}
&= \Sigma \left\{ \mathbf{i} \times \frac{\partial}{\partial x} (\phi \mathbf{A}) \right\} = \Sigma \left\{ \mathbf{i} \times \left(\frac{\partial \phi}{\partial x} \mathbf{A} + \phi \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \\
&= \Sigma \left\{ \mathbf{i} \times \left(\frac{\partial \phi}{\partial x} \mathbf{A} \right) \right\} + \Sigma \left\{ \mathbf{i} \times \left(\phi \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \\
&= \Sigma \left\{ \left(\frac{\partial \phi}{\partial x} \mathbf{i} \right) \times \mathbf{A} \right\} + \Sigma \left\{ \phi \left(\mathbf{i} \times \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \\
&\quad [\because \mathbf{a} \times (m\mathbf{b}) = (m\mathbf{a}) \times \mathbf{b} = m(\mathbf{a} \times \mathbf{b})] \\
&= \left\{ \Sigma \left(\frac{\partial \phi}{\partial x} \mathbf{i} \right) \right\} \times \mathbf{A} + \phi \Sigma \left(\mathbf{i} \times \frac{\partial \mathbf{A}}{\partial x} \right) = (\nabla \phi) \times \mathbf{A} + \phi (\nabla \times \mathbf{A}).
\end{aligned}$$

5. Prove that $\text{div} (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \text{curl } \mathbf{A} - \mathbf{A} \cdot \text{curl } \mathbf{B}$

or $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}).$

(Garhwal 2003; Agra 05; Meerut 04, 05B, 08, 09;
Bundelkhand 05, 07; Kashi 13; Avadh 09)

Proof: We have

$$\begin{aligned}
\text{div} (\mathbf{A} \times \mathbf{B}) &= \Sigma \left\{ \mathbf{i} \cdot \frac{\partial}{\partial x} (\mathbf{A} \times \mathbf{B}) \right\} = \Sigma \left\{ \mathbf{i} \cdot \left(\frac{\partial \mathbf{A}}{\partial x} \times \mathbf{B} + \mathbf{A} \times \frac{\partial \mathbf{B}}{\partial x} \right) \right\} \\
&= \Sigma \left\{ \mathbf{i} \cdot \left(\frac{\partial \mathbf{A}}{\partial x} \times \mathbf{B} \right) \right\} + \Sigma \left\{ \mathbf{i} \cdot \left(\mathbf{A} \times \frac{\partial \mathbf{B}}{\partial x} \right) \right\} \\
&= \Sigma \left\{ \left(\mathbf{i} \times \frac{\partial \mathbf{A}}{\partial x} \right) \cdot \mathbf{B} \right\} - \Sigma \left\{ \mathbf{i} \cdot \left(\frac{\partial \mathbf{B}}{\partial x} \times \mathbf{A} \right) \right\}
\end{aligned}$$

$$[\because \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} \text{ and } \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = -\mathbf{a} \cdot (\mathbf{c} \times \mathbf{b})]$$

$$\begin{aligned}
 &= \left\{ \Sigma \left(\mathbf{i} \times \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \cdot \mathbf{B} - \Sigma \left\{ \left(\mathbf{i} \times \frac{\partial \mathbf{B}}{\partial x} \right) \cdot \mathbf{A} \right\} \\
 &= (\text{curl } \mathbf{A}) \cdot \mathbf{B} - \left\{ \Sigma \left(\mathbf{i} \times \frac{\partial \mathbf{B}}{\partial x} \right) \right\} \cdot \mathbf{A} \\
 &= (\text{curl } \mathbf{A}) \cdot \mathbf{B} - (\text{curl } \mathbf{B}) \cdot \mathbf{A} = \mathbf{B} \cdot \text{curl } \mathbf{A} - \mathbf{A} \cdot \text{curl } \mathbf{B}.
 \end{aligned}$$

6. Prove that $\text{curl } (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} - \mathbf{B} \text{ div } \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A} \text{ div } \mathbf{B}$.

(Garhwal 2001; Meerut 06B; Kumaun 09, 11, 12)

Proof: We have $\text{curl } (\mathbf{A} \times \mathbf{B}) = \nabla \times (\mathbf{A} \times \mathbf{B})$

$$\begin{aligned}
 &= \Sigma \left\{ \mathbf{i} \times \frac{\partial}{\partial x} (\mathbf{A} \times \mathbf{B}) \right\} = \Sigma \left\{ \mathbf{i} \times \left(\mathbf{A} \times \frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x} \times \mathbf{B} \right) \right\} \\
 &= \Sigma \left\{ \mathbf{i} \times \left(\mathbf{A} \times \frac{\partial \mathbf{B}}{\partial x} \right) \right\} + \Sigma \left\{ \mathbf{i} \times \left(\frac{\partial \mathbf{A}}{\partial x} \times \mathbf{B} \right) \right\} \\
 &= \Sigma \left\{ \left(\mathbf{i} \cdot \frac{\partial \mathbf{B}}{\partial x} \right) \mathbf{A} - (\mathbf{i} \cdot \mathbf{A}) \frac{\partial \mathbf{B}}{\partial x} \right\} + \Sigma \left\{ (\mathbf{i} \cdot \mathbf{B}) \frac{\partial \mathbf{A}}{\partial x} - \left(\mathbf{i} \cdot \frac{\partial \mathbf{A}}{\partial x} \right) \mathbf{B} \right\} \\
 &= \Sigma \left\{ \left(\mathbf{i} \cdot \frac{\partial \mathbf{B}}{\partial x} \right) \mathbf{A} \right\} - \Sigma \left\{ (\mathbf{A} \cdot \mathbf{i}) \frac{\partial \mathbf{B}}{\partial x} \right\} \\
 &\quad + \Sigma \left\{ (\mathbf{B} \cdot \mathbf{i}) \frac{\partial \mathbf{A}}{\partial x} \right\} - \Sigma \left\{ \left(\mathbf{i} \cdot \frac{\partial \mathbf{A}}{\partial x} \right) \mathbf{B} \right\} \\
 &= \left\{ \Sigma \left(\mathbf{i} \cdot \frac{\partial \mathbf{B}}{\partial x} \right) \right\} \mathbf{A} - \left\{ \mathbf{A} \cdot \Sigma \mathbf{i} \frac{\partial}{\partial x} \right\} \mathbf{B} + \left\{ \mathbf{B} \cdot \Sigma \mathbf{i} \frac{\partial}{\partial x} \right\} \mathbf{A} \\
 &\quad - \left\{ \Sigma \left(\mathbf{i} \cdot \frac{\partial \mathbf{A}}{\partial x} \right) \right\} \mathbf{B} \\
 &= (\text{div } \mathbf{B}) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A} - (\text{div } \mathbf{A}) \mathbf{B}.
 \end{aligned}$$

7. Prove that $\text{grad } (\mathbf{A} \cdot \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} + (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{B} \times \text{curl } \mathbf{A} + \mathbf{A} \times \text{curl } \mathbf{B}$.

(Agra 2001; Kumaun 15)

Proof: We have

$$\begin{aligned}
 \text{grad } (\mathbf{A} \cdot \mathbf{B}) &= \nabla (\mathbf{A} \cdot \mathbf{B}) = \Sigma \mathbf{i} \frac{\partial}{\partial x} (\mathbf{A} \cdot \mathbf{B}) = \Sigma \mathbf{i} \left(\mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x} \cdot \mathbf{B} \right) \\
 &= \Sigma \left\{ \left(\mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial x} \right) \mathbf{i} \right\} + \Sigma \left\{ \left(\mathbf{B} \cdot \frac{\partial \mathbf{A}}{\partial x} \right) \mathbf{i} \right\}. \quad \dots(1)
 \end{aligned}$$

Now we know that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$.

$$\therefore (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - \mathbf{a} \times (\mathbf{b} \times \mathbf{c}).$$

$$\begin{aligned}
 \therefore \left(\mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial x} \right) \mathbf{i} &= (\mathbf{A} \cdot \mathbf{i}) \frac{\partial \mathbf{B}}{\partial x} - \mathbf{A} \times \left(\frac{\partial \mathbf{B}}{\partial x} \times \mathbf{i} \right) \\
 &= (\mathbf{A} \cdot \mathbf{i}) \frac{\partial \mathbf{B}}{\partial x} + \mathbf{A} \times \left(\mathbf{i} \times \frac{\partial \mathbf{B}}{\partial x} \right).
 \end{aligned}$$

Thus
$$\begin{aligned}\Sigma \left\{ \left(\mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial x} \right) \mathbf{i} \right\} &= \Sigma \left\{ (\mathbf{A} \cdot \mathbf{i}) \frac{\partial \mathbf{B}}{\partial x} \right\} + \Sigma \left\{ \mathbf{A} \times \left(\mathbf{i} \times \frac{\partial \mathbf{B}}{\partial x} \right) \right\} \\ &= \left\{ \mathbf{A} \cdot \Sigma \mathbf{i} \frac{\partial}{\partial x} \right\} \mathbf{B} + \mathbf{A} \times \Sigma \left(\mathbf{i} \times \frac{\partial \mathbf{B}}{\partial x} \right) \\ &= (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A} \times (\nabla \times \mathbf{B}).\end{aligned}\quad \dots(2)$$

Similarly
$$\Sigma \left\{ \left(\mathbf{B} \cdot \frac{\partial \mathbf{A}}{\partial x} \right) \mathbf{i} \right\} = (\mathbf{B} \cdot \nabla) \mathbf{A} + \mathbf{B} \times (\nabla \times \mathbf{A}). \quad \dots(3)$$

Putting the values from (2) and (3) in (1), we get

$$\text{grad } (\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A} \times (\nabla \times \mathbf{B}) + (\mathbf{B} \cdot \nabla) \mathbf{A} + \mathbf{B} \times (\nabla \times \mathbf{A}).$$

Note: If we put $\mathbf{A}$ in place of $\mathbf{B}$, then

$$\text{grad } (\mathbf{A} \cdot \mathbf{A}) = 2 (\mathbf{A} \cdot \nabla) \mathbf{A} + 2 \mathbf{A} \times (\nabla \times \mathbf{A})$$

or
$$\frac{1}{2} \text{grad } \mathbf{A}^2 = (\mathbf{A} \cdot \nabla) \mathbf{A} + \mathbf{A} \times \text{curl } \mathbf{A}.$$

8. Prove that $\text{div grad } \phi = \nabla^2 \phi$ i.e., $\nabla \cdot (\nabla \phi) = \nabla^2 \phi$.

Proof: We have

$$\begin{aligned}\nabla \cdot (\nabla \phi) &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot \left(\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} \right) \\ &= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi = \nabla^2 \phi.\end{aligned}$$

9. Prove that $\text{curl of the gradient of } \phi \text{ is zero i.e., } \nabla \times (\nabla \phi) = \mathbf{0}$, i.e., $\text{curl grad } \phi = \mathbf{0}$.

(Bundelkhand 2014)

Proof: We have $\text{grad } \phi = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}$.

$\therefore$

$$\begin{aligned}\text{curl grad } \phi &= \nabla \times \text{grad } \phi \\ &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \times \left(\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right) \mathbf{i} + \left(\frac{\partial^2 \phi}{\partial z \partial x} - \frac{\partial^2 \phi}{\partial x \partial z} \right) \mathbf{j} + \left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right) \mathbf{k}\end{aligned}$$

$$= 0 \mathbf{i} + 0 \mathbf{j} + 0 \mathbf{k} = \mathbf{0},$$

provided we suppose that ϕ has continuous second partial derivatives so that the order of differentiation is immaterial.

10. Prove that $\text{div curl } \mathbf{A} = 0$, i.e., $\nabla \cdot (\nabla \times \mathbf{A}) = 0$.

(Garhwal 2000; Meerut 02, 06B, 09; Agra 2000; Rohilkhand 05; Bundelkhand 08; Kumaun 14)

Proof: Let $\mathbf{A} = A_1 \mathbf{i} + A_2 \mathbf{j} + A_3 \mathbf{k}$. Then

$$\begin{aligned} \text{curl } \mathbf{A} = \nabla \times \mathbf{A} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\ &= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \mathbf{k}. \end{aligned}$$

Now

$$\begin{aligned} \text{div curl } \mathbf{A} &= \nabla \cdot (\nabla \times \mathbf{A}) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \\ &= \frac{\partial^2 A_3}{\partial x \partial y} - \frac{\partial^2 A_2}{\partial x \partial z} + \frac{\partial^2 A_1}{\partial y \partial z} - \frac{\partial^2 A_3}{\partial y \partial x} + \frac{\partial^2 A_2}{\partial z \partial x} - \frac{\partial^2 A_1}{\partial z \partial y} \\ &= 0, \text{ assuming that } \mathbf{A} \text{ has continuous second partial derivatives.} \end{aligned}$$

11. Prove that $\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$.

(Meerut 2000, 07B, 10B; Rohilkhand 14; Kumaun 10)

Proof: Let $\mathbf{A} = A_1 \mathbf{i} + A_2 \mathbf{j} + A_3 \mathbf{k}$.

$$\begin{aligned} \text{Then } \nabla \times \mathbf{A} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\ &= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \mathbf{k}. \end{aligned}$$

$$\therefore \nabla \times (\nabla \times \mathbf{A}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} & \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} & \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \end{vmatrix}$$

$$\begin{aligned}
&= \Sigma \left[\left\{ \frac{\partial}{\partial y} \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \right\} \mathbf{i} \right] \\
&= \Sigma \left[\left\{ \left(\frac{\partial^2 A_2}{\partial y \partial x} + \frac{\partial^2 A_3}{\partial z \partial x} \right) - \left(\frac{\partial^2 A_1}{\partial y^2} + \frac{\partial^2 A_1}{\partial z^2} \right) \right\} \mathbf{i} \right] \\
&= \Sigma \left[\left\{ \frac{\partial}{\partial x} \left(\frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) - \left(\frac{\partial^2 A_1}{\partial y^2} + \frac{\partial^2 A_1}{\partial z^2} \right) \right\} \mathbf{i} \right] \\
&= \Sigma \left[\left\{ \frac{\partial}{\partial x} \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) - \left(\frac{\partial^2 A_1}{\partial x^2} + \frac{\partial^2 A_1}{\partial y^2} + \frac{\partial^2 A_1}{\partial z^2} \right) \right\} \mathbf{i} \right] \\
&= \Sigma \left[\left\{ \frac{\partial}{\partial x} (\nabla \cdot \mathbf{A}) - (\nabla^2 A_1) \right\} \mathbf{i} \right] \\
&= \Sigma \left[\left\{ \frac{\partial}{\partial x} (\nabla \cdot \mathbf{A}) \right\} \mathbf{i} \right] - \nabla^2 \Sigma A_1 \mathbf{i} = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.
\end{aligned}$$

Illustrative Examples

Example 24: Taking $\mathbf{F} = x^2 y \mathbf{i} + xz \mathbf{j} + 2yz \mathbf{k}$, verify that $\text{div curl } \mathbf{F} = 0$.

(Garhwal 2002; Bundelkhand 04; Kanpur 10)

Solution: We have $\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y & xz & 2yz \end{vmatrix}$

$$\begin{aligned}
&= \left[\frac{\partial}{\partial y} (2yz) - \frac{\partial}{\partial z} (xz) \right] \mathbf{i} - \left[\frac{\partial}{\partial x} (2yz) - \frac{\partial}{\partial z} (x^2 y) \right] \mathbf{j} \\
&\quad + \left[\frac{\partial}{\partial x} (xz) - \frac{\partial}{\partial y} (x^2 y) \right] \mathbf{k} \\
&= (2z - x) \mathbf{i} - 0 \mathbf{j} + (z - x^2) \mathbf{k} = (2z - x) \mathbf{i} + (z - x^2) \mathbf{k}.
\end{aligned}$$

Now $\text{div curl } \mathbf{F} = \text{div} [(2z - x) \mathbf{i} + (z - x^2) \mathbf{k}]$

$$= \frac{\partial}{\partial x} (2z - x) + \frac{\partial}{\partial z} (z - x^2) = -1 + 1 = 0.$$

Example 25: Prove that $\text{div} (r^n \mathbf{r}) = (n + 3) r^n$.

(Bundelkhand 2006)

Solution: We have

$$\text{div} (\phi \mathbf{A}) = \phi (\text{div } \mathbf{A}) + \mathbf{A} \cdot \text{grad } \phi.$$

Putting $\mathbf{A} = \mathbf{r}$ and $\phi = r^n$ in this identity, we get

$$\begin{aligned}
 \operatorname{div} (r^n \mathbf{r}) &= r^n \operatorname{div} \mathbf{r} + \mathbf{r} \bullet \operatorname{grad} r^n \\
 &= 3r^n + \mathbf{r} \bullet (n r^{n-1} \operatorname{grad} r) \\
 &\quad [\because \operatorname{div} \mathbf{r} = 3 \text{ and } \operatorname{grad} f(u) = f'(u) \operatorname{grad} u] \\
 &= 3r^n + \mathbf{r} \bullet \left[n r^{n-1} \frac{1}{r} \mathbf{r} \right] \quad \left[\because \operatorname{grad} r = \hat{\mathbf{r}} = \frac{1}{r} \mathbf{r} \right] \\
 &= 3r^n + n r^{n-2} (\mathbf{r} \bullet \mathbf{r}) \\
 &= 3r^n + n r^{n-2} r^2 \\
 &= (n+3) r^n.
 \end{aligned}$$

Example 26: Prove that $\operatorname{div} \left(\frac{\mathbf{r}}{r^3} \right) = 0$.

(Bundelkhand 2010)

Solution: We have

$$\begin{aligned}
 \operatorname{div} \left(\frac{1}{r^3} \mathbf{r} \right) &= \operatorname{div} (r^{-3} \mathbf{r}) = r^{-3} \operatorname{div} \mathbf{r} + \mathbf{r} \bullet \operatorname{grad} r^{-3} \\
 &= 3r^{-3} + \mathbf{r} \bullet (-3r^{-4} \operatorname{grad} r) = 3r^{-3} + \mathbf{r} \bullet \left(-3r^{-4} \frac{1}{r} \mathbf{r} \right) \\
 &= 3r^{-3} - 3r^{-5} (\mathbf{r} \bullet \mathbf{r}) = 3r^{-3} - 3r^{-5} r^2 \\
 &= 3r^{-3} - 3r^{-3} = 0.
 \end{aligned}$$

$\therefore$ the vector $r^{-3} \mathbf{r}$ is solenoidal.

Example 27: Prove that $\operatorname{div} \hat{\mathbf{r}} = 2/r$.

(Bundelkhand 2007; Kanpur 05)

Solution: $\operatorname{div} (\hat{\mathbf{r}}) = \operatorname{div} \left(\frac{1}{r} \mathbf{r} \right)$. Now proceed as in Example 25.

Alternative Method:

$$\begin{aligned}
 \operatorname{div} \hat{\mathbf{r}} &= \operatorname{div} \left(\frac{1}{r} \mathbf{r} \right) = \operatorname{div} \left[\frac{1}{r} (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) \right] \\
 &= \operatorname{div} \left(\frac{x}{r} \mathbf{i} + \frac{y}{r} \mathbf{j} + \frac{z}{r} \mathbf{k} \right) = \frac{\partial}{\partial x} \left(\frac{x}{r} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r} \right) \\
 &= \left(\frac{1}{r} - \frac{x}{r^2} \frac{\partial r}{\partial x} \right) + \left(\frac{1}{r} - \frac{y}{r^2} \frac{\partial r}{\partial y} \right) + \left(\frac{1}{r} - \frac{z}{r^2} \frac{\partial r}{\partial z} \right).
 \end{aligned}$$

Now $r^2 = x^2 + y^2 + z^2$.

$\therefore 2r \frac{\partial r}{\partial x} = 2x$ i.e., $\frac{\partial r}{\partial x} = \frac{x}{r}$.

Similarly $\frac{\partial r}{\partial y} = \frac{y}{r}$ and $\frac{\partial r}{\partial z} = \frac{z}{r}$.

$\therefore \operatorname{div} \hat{\mathbf{r}} = \frac{3}{r} - \left(\frac{x}{r^2} \frac{x}{r} + \frac{y}{r^2} \frac{y}{r} + \frac{z}{r^2} \frac{z}{r} \right)$

$$= \frac{3}{r} - \frac{x^2 + y^2 + z^2}{r^3} = \frac{3}{r} - \frac{r^2}{r^3} = \frac{3}{r} - \frac{1}{r} = \frac{2}{r}.$$

Example 28: Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$. (Meerut 2003, 05, 06B)

Solution: We know that if ϕ is a scalar function, then $\nabla^2 \phi = \nabla \cdot (\nabla \phi)$.

$$\begin{aligned} \therefore \nabla^2 f(r) &= \nabla \cdot \{\nabla f(r)\} = \text{div} \{\text{grad } f(r)\} \\ &= \text{div} \{f'(r) \text{ grad } r\} = \text{div} \left\{ \frac{1}{r} f'(r) \mathbf{r} \right\} \\ &= \frac{1}{r} f'(r) \text{div } \mathbf{r} + \mathbf{r} \cdot \text{grad} \left\{ \frac{1}{r} f'(r) \right\} \\ &= \frac{3}{r} f'(r) + \mathbf{r} \cdot \left[\frac{d}{dr} \left\{ \frac{1}{r} f'(r) \right\} \text{grad } r \right] \\ &= \frac{3}{r} f'(r) + \mathbf{r} \cdot \left[\left\{ -\frac{1}{r^2} f'(r) + \frac{1}{r} f''(r) \right\} \frac{1}{r} \mathbf{r} \right] \\ &= \frac{3}{r} f'(r) + \left[\frac{1}{r} \left\{ -\frac{1}{r^2} f'(r) + \frac{1}{r} f''(r) \right\} \right] (\mathbf{r} \cdot \mathbf{r}) \\ &= \frac{3}{r} f'(r) + \left[\frac{1}{r} \left\{ -\frac{1}{r^2} f'(r) + \frac{1}{r} f''(r) \right\} \right] r^2 \\ &= \frac{3}{r} f'(r) - \frac{1}{r} f'(r) + f''(r) = f''(r) + \frac{2}{r} f'(r). \end{aligned}$$

Example 29: Prove that $\nabla^2 \left(\frac{1}{r} \right) = 0$ or $\text{div} \left(\text{grad} \frac{1}{r} \right) = 0$. (Agra 2002; Meerut 07, 13)

Solution: We have

$$\begin{aligned} \nabla^2 \left(\frac{1}{r} \right) &= \nabla \cdot \left(\nabla \frac{1}{r} \right) = \text{div} \left(\text{grad} \frac{1}{r} \right) \\ &= \text{div} \left(-\frac{1}{r^2} \text{grad } r \right) = \text{div} \left(-\frac{1}{r^2} \frac{1}{r} \mathbf{r} \right) = \text{div} \left(-\frac{1}{r^3} \mathbf{r} \right) \\ &= \left(-\frac{1}{r^3} \right) \text{div } \mathbf{r} + \mathbf{r} \cdot \text{grad} \left(-\frac{1}{r^3} \right) = -\frac{3}{r^3} + \mathbf{r} \cdot \left[\frac{d}{dr} \left(-\frac{1}{r^3} \right) \text{grad } r \right] \\ &= -\frac{3}{r^3} + \mathbf{r} \cdot \left(\frac{3}{r^4} \frac{1}{r} \mathbf{r} \right) = -\frac{3}{r^3} + \frac{3}{r^5} (\mathbf{r} \cdot \mathbf{r}) = -\frac{3}{r^3} + \frac{3}{r^5} r^2 = 0. \end{aligned}$$

$\therefore 1/r$ is a solution of Laplace's equation.

Example 30: Prove that $\text{div grad } r^n = n(n+1)r^{n-2}$, i.e., $\nabla^2 r^n = n(n+1)r^{n-2}$.

(Bundelkhand 2005, 10; Meerut 2000, 08; Avadh 10; Agra 14)

Solution: We have $\nabla^2 r^n = \nabla \cdot (\nabla r^n) = \text{div} (\text{grad } r^n)$

$$\begin{aligned}
 &= \operatorname{div} (nr^{n-1} \operatorname{grad} r) = \operatorname{div} \left(nr^{n-1} \frac{1}{r} \mathbf{r} \right) = \operatorname{div} (nr^{n-2} \mathbf{r}) \\
 &= (nr^{n-2}) \operatorname{div} \mathbf{r} + \mathbf{r} \bullet (\operatorname{grad} nr^{n-2}) \\
 &= 3nr^{n-2} + \mathbf{r} \bullet [n(n-2)r^{n-3} \operatorname{grad} r] \\
 &= 3nr^{n-2} + \mathbf{r} \bullet \left[n(n-2)r^{n-3} \frac{1}{r} \mathbf{r} \right] \\
 &= 3nr^{n-2} + \mathbf{r} \bullet [n(n-2)r^{n-4} \mathbf{r}] = 3nr^{n-2} + n(n-2)r^{n-4} (\mathbf{r} \bullet \mathbf{r}) \\
 &= 3nr^{n-2} + n(n-2)r^{n-4} r^2 = nr^{n-2} (3+n-2) = n(n+1)r^{n-2}.
 \end{aligned}$$

Note: If $n = -1$, then $\nabla^2 (r^{-1}) = (-1)(-1+1)r^{-3} = 0$.

Example 31: Prove that $\nabla \bullet \left\{ r \nabla \left(\frac{1}{r^3} \right) \right\} = \frac{3}{r^4}$ or, $\operatorname{div} [r \operatorname{grad} r^{-3}] = 3r^{-4}$.
(Meerut 2009B)

Solution: We have $\nabla \left(\frac{1}{r^3} \right) = \operatorname{grad} r^{-3}$

$$= \frac{\partial}{\partial x} (r^{-3}) \mathbf{i} + \frac{\partial}{\partial y} (r^{-3}) \mathbf{j} + \frac{\partial}{\partial z} (r^{-3}) \mathbf{k}.$$

Now $\frac{\partial}{\partial x} (r^{-3}) = -3r^{-4} \frac{\partial r}{\partial x}.$

But $r^2 = x^2 + y^2 + z^2.$

Therefore $2r \frac{\partial r}{\partial x} = 2x$ or $\frac{\partial r}{\partial x} = \frac{x}{r}.$

So $\frac{\partial}{\partial x} (r^{-3}) = -3r^{-4} \frac{x}{r} = -3r^{-5} x.$

Similarly $\frac{\partial}{\partial y} (r^{-3}) = -3r^{-5} y$ and $\frac{\partial}{\partial z} (r^{-3}) = -3r^{-5} z.$

Therefore $\nabla \left(\frac{1}{r^3} \right) = -3r^{-5} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$

$\therefore r \nabla \left(\frac{1}{r^3} \right) = -3r^{-4} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$

$\therefore \nabla \bullet \left(r \nabla \frac{1}{r^3} \right) = \frac{\partial}{\partial x} (-3r^{-4} x) + \frac{\partial}{\partial y} (-3r^{-4} y) + \frac{\partial}{\partial z} (-3r^{-4} z).$

Now $\frac{\partial}{\partial x} (-3r^{-4} x) = 12r^{-5} \frac{\partial r}{\partial x} x - 3r^{-4}$
 $= 12r^{-5} \frac{x}{r} x - 3r^{-4} = 12r^{-6} x^2 - 3r^{-4}.$

Similarly $\frac{\partial}{\partial y} (-3r^{-4} y) = 12r^{-6} y^2 - 3r^{-4}$

and $\frac{\partial}{\partial z} (-3r^{-4} z) = 12r^{-6} z^2 - 3r^{-4}.$

Hence
$$\nabla \cdot \left(r \nabla \frac{1}{r^3} \right) = 12 r^{-6} (x^2 + y^2 + z^2) - 9r^{-4}$$

$$= 12 r^{-6} r^2 - 9r^{-4} = 12 r^{-4} - 9r^{-4} = 3r^{-4}.$$

Example 32: If $\mathbf{a}$ is a constant vector, prove that $\text{div} \{r^n (\mathbf{a} \times \mathbf{r})\} = 0$ (Kanpur 2008)

Solution: We have $\text{div} (\phi \mathbf{A}) = \phi \text{div} \mathbf{A} + \mathbf{A} \cdot \text{grad} \phi$.

$$\begin{aligned} \therefore \text{div} \{r^n (\mathbf{a} \times \mathbf{r})\} &= r^n \text{div} (\mathbf{a} \times \mathbf{r}) + (\mathbf{a} \times \mathbf{r}) \cdot \text{grad} r^n \\ &= r^n \text{div} (\mathbf{a} \times \mathbf{r}) + (\mathbf{a} \times \mathbf{r}) \cdot (nr^{n-1} \text{grad} r) \\ &= r^n (\mathbf{r} \cdot \text{curl} \mathbf{a} - \mathbf{a} \cdot \text{curl} \mathbf{r}) + (\mathbf{a} \times \mathbf{r}) \cdot \left(nr^{n-1} \frac{1}{r} \mathbf{r} \right) \\ &= r^n (\mathbf{r} \cdot \mathbf{0} - \mathbf{a} \cdot \mathbf{0}) + nr^{n-2} (\mathbf{a} \times \mathbf{r}) \cdot \mathbf{r} \\ &\quad [\because \text{curl of a constant vector is zero and } \text{curl} \mathbf{r} = \mathbf{0}] \\ &= nr^{n-2} [\mathbf{a}, \mathbf{r}, \mathbf{r}] \\ &= 0, \text{ since a scalar triple product having two equal vectors is zero.} \end{aligned}$$

Example 33: If $\mathbf{a}$ and $\mathbf{b}$ are constant vectors, prove that

(i) $\text{div} [(\mathbf{r} \times \mathbf{a}) \times \mathbf{b}] = -2 \mathbf{b} \cdot \mathbf{a}$. (Bundelkhand 2010)

(ii) $\text{curl} [(\mathbf{r} \times \mathbf{a}) \times \mathbf{b}] = \mathbf{b} \times \mathbf{a}$. (Bundelkhand 2010)

Solution: (i) We have $(\mathbf{r} \times \mathbf{a}) \times \mathbf{b} = (\mathbf{b} \cdot \mathbf{r}) \mathbf{a} - (\mathbf{b} \cdot \mathbf{a}) \mathbf{r}$.

$$\begin{aligned} \therefore \text{div} [(\mathbf{r} \times \mathbf{a}) \times \mathbf{b}] &= \text{div} [(\mathbf{b} \cdot \mathbf{r}) \mathbf{a} - (\mathbf{b} \cdot \mathbf{a}) \mathbf{r}] \\ &= \text{div} [(\mathbf{b} \cdot \mathbf{r}) \mathbf{a}] - \text{div} [(\mathbf{b} \cdot \mathbf{a}) \mathbf{r}] \end{aligned} \quad \dots(1)$$

But $\text{div} (\phi \mathbf{A}) = \phi \text{div} \mathbf{A} + \mathbf{A} \cdot \text{grad} \phi$.

Taking $\phi = \mathbf{b} \cdot \mathbf{r}$ and $\mathbf{A} = \mathbf{a}$, we get

$$\text{div} [(\mathbf{b} \cdot \mathbf{r}) \mathbf{a}] = (\mathbf{b} \cdot \mathbf{r}) \text{div} \mathbf{a} + \mathbf{a} \cdot \text{grad} (\mathbf{b} \cdot \mathbf{r}).$$

Since $\mathbf{a}$ is a constant vector, therefore $\text{div} \mathbf{a} = 0$.

Also let $\mathbf{b} = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}$.

Then $\mathbf{b} \cdot \mathbf{r} = (b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}) \cdot (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$
 $= b_1 x + b_2 y + b_3 z$, where b_1, b_2, b_3 are constants.

$$\therefore \text{grad} (\mathbf{b} \cdot \mathbf{r}) = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k} = \mathbf{b}.$$

$$\therefore \text{div} [(\mathbf{b} \cdot \mathbf{r}) \mathbf{a}] = \mathbf{a} \cdot \mathbf{b}. \quad \dots(2)$$

Again $\text{div} [(\mathbf{b} \cdot \mathbf{a}) \mathbf{r}] = (\mathbf{b} \cdot \mathbf{a}) \text{div} \mathbf{r} + \mathbf{r} \cdot \text{grad} (\mathbf{b} \cdot \mathbf{a}).$

But $\text{div} \mathbf{r} = 3$.

Also $\text{grad} (\mathbf{b} \cdot \mathbf{a}) = \mathbf{0}$ because $\mathbf{b} \cdot \mathbf{a}$ is constant.

$$\therefore \text{div} [(\mathbf{b} \cdot \mathbf{a}) \mathbf{r}] = 3 (\mathbf{b} \cdot \mathbf{a}). \quad \dots(3)$$

Substituting the values from (2) and (3) in (1), we get

$$\text{div} [(\mathbf{r} \times \mathbf{a}) \times \mathbf{b}] = (\mathbf{a} \cdot \mathbf{b}) - 3 (\mathbf{b} \cdot \mathbf{a}) = -2 \mathbf{b} \cdot \mathbf{a}.$$

(ii) $\text{curl} [(\mathbf{r} \times \mathbf{a}) \times \mathbf{b}] = \text{curl} [(\mathbf{b} \cdot \mathbf{r}) \mathbf{a} - (\mathbf{b} \cdot \mathbf{a}) \mathbf{r}]$
 $= \text{curl} [(\mathbf{b} \cdot \mathbf{r}) \mathbf{a}] - \text{curl} [(\mathbf{b} \cdot \mathbf{a}) \mathbf{r}].$

But $\text{curl} (\phi \mathbf{A}) = \text{grad } \phi \times \mathbf{A} + \phi \text{curl } \mathbf{A}$.

$$\therefore \text{curl} [(\mathbf{b} \bullet \mathbf{r}) \mathbf{a}] = [\text{grad } (\mathbf{b} \bullet \mathbf{r})] \times \mathbf{a} + (\mathbf{b} \bullet \mathbf{r}) \text{curl } \mathbf{a} = \mathbf{b} \times \mathbf{a}$$

$$[\because \text{curl } \mathbf{a} = \mathbf{0} \text{ and } \text{grad } (\mathbf{b} \bullet \mathbf{r}) = \mathbf{b}]$$

Also $\text{curl} [(\mathbf{b} \bullet \mathbf{a}) \mathbf{r}] = [\text{grad } (\mathbf{b} \bullet \mathbf{a})] \times \mathbf{r} + (\mathbf{b} \bullet \mathbf{a}) \text{curl } \mathbf{r} = \mathbf{0}$

$$[\because \text{grad } (\mathbf{b} \bullet \mathbf{a}) = \mathbf{0} \text{ and } \text{curl } \mathbf{r} = \mathbf{0}]$$

$$\therefore \text{curl} [(\mathbf{r} \times \mathbf{a}) \times \mathbf{b}] = \mathbf{b} \times \mathbf{a} - \mathbf{0} = \mathbf{b} \times \mathbf{a}.$$

Example 34: Prove that $\text{curl} [\mathbf{r} \times (\mathbf{a} \times \mathbf{r})] = 3\mathbf{r} \times \mathbf{a}$, where $\mathbf{a}$ is a constant vector.

(Purvanchal 2009)

Solution:

$$\begin{aligned} \text{curl} [\mathbf{r} \times (\mathbf{a} \times \mathbf{r})] &= \nabla \times [(\mathbf{r} \bullet \mathbf{r}) \mathbf{a} - (\mathbf{r} \bullet \mathbf{a}) \mathbf{r}] \quad [\because \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \bullet \mathbf{c}) \mathbf{b} - (\mathbf{a} \bullet \mathbf{b}) \mathbf{c}] \\ &= \nabla \times [r^2 \mathbf{a} - (\mathbf{r} \bullet \mathbf{a}) \mathbf{r}] \quad [\because \mathbf{r} \bullet \mathbf{r} = r^2 = r^2] \\ &= \nabla \times (r^2 \mathbf{a}) - \nabla \times [(\mathbf{r} \bullet \mathbf{a}) \mathbf{r}] \quad [\because \nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}] \\ &= (\nabla r^2) \times \mathbf{a} + r^2 (\nabla \times \mathbf{a}) - [\nabla (\mathbf{r} \bullet \mathbf{a})] \times \mathbf{r} - (\mathbf{r} \bullet \mathbf{a}) (\nabla \times \mathbf{r}) \\ &\quad [\because \nabla \times (\phi \mathbf{A}) = (\nabla \phi) \times \mathbf{A} + \phi (\nabla \times \mathbf{A})] \\ &= (2r \nabla r) \times \mathbf{a} + r^2 \mathbf{0} - [\nabla (\mathbf{r} \bullet \mathbf{a})] \times \mathbf{r} - (\mathbf{r} \bullet \mathbf{a}) \mathbf{0} \\ &\quad [\because \nabla f(r) = f'(r) \nabla r; \nabla \times \mathbf{a} = \mathbf{0}, \mathbf{a} \text{ being a constant vector; and } \nabla \times \mathbf{r} = \mathbf{0}] \\ &= \left(2r \frac{1}{r} \mathbf{r}\right) \times \mathbf{a} - [\nabla (\mathbf{r} \bullet \mathbf{a})] \times \mathbf{r} \\ &= 2\mathbf{r} \times \mathbf{a} - \mathbf{a} \times \mathbf{r} \quad [\because \nabla (\mathbf{r} \bullet \mathbf{a}) = \mathbf{a}, \text{ if } \mathbf{a} \text{ is a constant vector}] \\ &= 2\mathbf{r} \times \mathbf{a} + \mathbf{r} \times \mathbf{a} = 3\mathbf{r} \times \mathbf{a}. \end{aligned}$$

Example 35: If $\mathbf{a}$ is a constant vector, prove that

$$\text{curl} \frac{\mathbf{a} \times \mathbf{r}}{r^3} = -\frac{\mathbf{a}}{r^3} + \frac{3\mathbf{r}}{r^5} (\mathbf{a} \bullet \mathbf{r}).$$

(Meerut 2009B; Bundelkhand 09, 11)

Solution: We have $\text{curl} \frac{\mathbf{a} \times \mathbf{r}}{r^3} = \nabla \times \left(\frac{\mathbf{a} \times \mathbf{r}}{r^3} \right) = \Sigma \left\{ \mathbf{i} \times \frac{\partial}{\partial x} \left(\frac{\mathbf{a} \times \mathbf{r}}{r^3} \right) \right\}.$

Now $\frac{\partial}{\partial x} \left(\frac{\mathbf{a} \times \mathbf{r}}{r^3} \right) = -\frac{3}{r^4} \frac{\partial r}{\partial x} (\mathbf{a} \times \mathbf{r}) + \frac{1}{r^3} \left(\mathbf{a} \times \frac{\partial \mathbf{r}}{\partial x} \right) + \frac{1}{r^3} \left(\frac{\partial \mathbf{a}}{\partial x} \times \mathbf{r} \right) \dots (1)$

Now $\frac{\partial \mathbf{a}}{\partial x} = \mathbf{0}$ because $\mathbf{a}$ is a constant vector.

Also $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k} \therefore \frac{\partial \mathbf{r}}{\partial x} = \mathbf{i}.$

Further $\frac{\partial r}{\partial x} = \frac{x}{r}.$

$\therefore$ (1) becomes $\frac{\partial}{\partial x} \left(\frac{\mathbf{a} \times \mathbf{r}}{r^3} \right) = -\frac{3}{r^4} \frac{x}{r} (\mathbf{a} \times \mathbf{r}) + \frac{1}{r^3} (\mathbf{a} \times \mathbf{i})$

$$= -\frac{3x}{r^5} (\mathbf{a} \times \mathbf{r}) + \frac{1}{r^3} (\mathbf{a} \times \mathbf{i}).$$

$$\begin{aligned} \therefore \mathbf{i} \times \frac{\partial}{\partial x} \left(\frac{\mathbf{a} \times \mathbf{r}}{r^3} \right) &= -\frac{3x}{r^5} \mathbf{i} \times (\mathbf{a} \times \mathbf{r}) + \frac{1}{r^3} \mathbf{i} \times (\mathbf{a} \times \mathbf{i}) \\ &= -\frac{3x}{r^5} [(\mathbf{i} \bullet \mathbf{r}) \mathbf{a} - (\mathbf{i} \bullet \mathbf{a}) \mathbf{r}] + \frac{1}{r^3} [(\mathbf{i} \bullet \mathbf{i}) \mathbf{a} - (\mathbf{i} \bullet \mathbf{a}) \mathbf{i}] \\ &= -\frac{3x}{r^5} x\mathbf{a} + \frac{3x}{r^5} a_1 \mathbf{r} + \frac{1}{r^3} \mathbf{a} - \frac{1}{r^3} a_1 \mathbf{i} \end{aligned}$$

$$[\because \mathbf{i} \bullet \mathbf{r} = x \text{ and } \mathbf{i} \bullet \mathbf{a} = a_1 \text{ if } \mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}]$$

$$= -\frac{3x^2}{r^5} \mathbf{a} + \frac{3}{r^5} a_1 x \mathbf{r} + \frac{1}{r^3} \mathbf{a} - \frac{1}{r^3} a_1 \mathbf{i}.$$

$$\begin{aligned} \therefore \Sigma \left\{ \mathbf{i} \times \frac{\partial}{\partial x} \left(\frac{\mathbf{a} \times \mathbf{r}}{r^3} \right) \right\} &= \left\{ -\frac{3}{r^5} \Sigma x^2 \right\} \mathbf{a} + \left\{ \frac{3}{r^5} \Sigma a_1 x \right\} \mathbf{r} + \frac{3}{r^3} \mathbf{a} - \frac{1}{r^3} \Sigma a_1 \mathbf{i} \\ &= -\frac{3}{r^5} r^2 \mathbf{a} + \frac{3}{r^5} (\mathbf{r} \bullet \mathbf{a}) \mathbf{r} + \frac{3}{r^3} \mathbf{a} - \frac{1}{r^3} \mathbf{a} \\ &\quad [\because \Sigma x^2 = r^2, \Sigma a_1 x = \mathbf{r} \bullet \mathbf{a}, \Sigma a_1 \mathbf{i} = \mathbf{a}] \\ &= -\frac{\mathbf{a}}{r^3} + \frac{3}{r^5} (\mathbf{a} \bullet \mathbf{r}) \mathbf{r}. \end{aligned}$$

Comprehensive Exercise 4

1. (i) Verify that $\text{curl grad } f = \mathbf{0}$, where $f = x^2 y + 2xy + z^2$.
 (ii) Prove that $\text{grad } f(u) = f'(u) \text{ grad } u$.
 (iii) Find $\nabla \phi$ and $|\nabla \phi|$ when $\phi = (x^2 + y^2 + z^2) e^{-(x^2 + y^2 + z^2)^{1/2}}$.
(Kumaun 2012)
2. (i) Prove that $\text{curl}(\psi \nabla \phi) = \nabla \psi \times \nabla \phi = -\text{curl}(\phi \nabla \psi)$.
 (ii) Prove that $\nabla^2(\phi \psi) = \phi \nabla^2 \psi + 2\nabla \phi \bullet \nabla \psi + \psi \nabla^2 \phi$.
(Kumaun 2014)
 (iii) Prove that $\text{div}(\nabla \phi \times \nabla \psi) = 0$.
 (iv) If $\mathbf{A}$ and $\mathbf{B}$ are irrotational, prove that $\mathbf{A} \times \mathbf{B}$ is solenoidal.
 (v) Prove that $\text{curl}(\phi \text{ grad } \phi) = \mathbf{0}$.
3. (i) Prove that $\mathbf{a} \bullet \{\nabla(\mathbf{v} \bullet \mathbf{a}) - \nabla \times (\mathbf{v} \times \mathbf{a})\} = \text{div } \mathbf{v}$, where $\mathbf{a}$ is a constant unit vector.
 (ii) Prove that vector $f(r) \mathbf{r}$ is irrotational.
 (iii) If f and g are two scalar point functions, prove that

$$\text{div}(f \nabla g) = f \nabla^2 g + \nabla f \bullet \nabla g.$$

(iv) Show that $\text{curl} (\mathbf{a} \bullet \mathbf{r}) \mathbf{a} = \mathbf{0}$, where $\mathbf{a}$ is a constant vector.

(Kumaun 2007, 09)

[**Hint:** Use identity 4. Note that $\nabla (\mathbf{a} \bullet \mathbf{r}) = \mathbf{a}$, if $\mathbf{a}$ is a constant vector.]

4. If $\mathbf{a}$ is a constant vector, then prove that
 - (i) $\nabla (\mathbf{a} \bullet \mathbf{u}) = (\mathbf{a} \bullet \nabla) \mathbf{u} + \mathbf{a} \times \text{curl } \mathbf{u}$,
 - (ii) $\nabla \bullet (\mathbf{a} \times \mathbf{u}) = -\mathbf{a} \bullet \text{curl } \mathbf{u}$,
 - (iii) $\nabla \times (\mathbf{a} \times \mathbf{u}) = \mathbf{a} \text{ div } \mathbf{u} - (\mathbf{a} \bullet \nabla) \mathbf{u}$,
5. (i) Given that $\rho \mathbf{F} = \nabla p$, where $\rho, p, \mathbf{F}$ are point functions, prove that $\mathbf{F} \bullet \text{curl } \mathbf{F} = 0$.
- (ii) A vector function $\mathbf{f}$ is the product of a scalar function and the gradient of a scalar function. Show that $\mathbf{f} \bullet \text{curl } \mathbf{f} = 0$.
- (iii) Show that $\text{curl } \mathbf{a} \phi(r) = \frac{1}{r} \phi'(r) \mathbf{r} \times \mathbf{a}$, where $\mathbf{a}$ is a constant vector.
6. (i) Prove that $\text{curl} [r^n (\mathbf{a} \times \mathbf{r})] = (n+2) r^n \mathbf{a} - n r^{n-2} (\mathbf{r} \bullet \mathbf{a}) \mathbf{r}$, where $\mathbf{a}$ is a constant vector. (Kumaun 2008)
- (ii) Prove that $\nabla^2 (r^n \mathbf{r}) = n(n+3) r^{n-2} \mathbf{r}$.
7. (i) Prove that $\text{curl grad } r^n = \mathbf{0}$. (Avadh 2010; Kanpur 11)
- (ii) If $\nabla^2 f(r) = 0$, show that $f(r) = \frac{c_1}{r} + c_2$, where $r^2 = x^2 + y^2 + z^2$ and c_1, c_2 are arbitrary constants.
8. If $\mathbf{r}$ is the position vector of the point (x, y, z) show that $\text{curl}(r^n \mathbf{r}) = \mathbf{0}$, where r is the module of $\mathbf{r}$.
9. Prove that $r^n \mathbf{r}$ is an irrotational vector for any value of n but is solenoidal only if $n+3=0$.
10. (i) If $\mathbf{u} = (1/r) \mathbf{r}$, show that $\nabla \times \mathbf{u} = \mathbf{0}$. (Avadh 2010)
- (ii) Prove that $\text{div} (\mathbf{A} \times \mathbf{r}) = \mathbf{r} \bullet \text{curl } \mathbf{A}$.
11. (i) If $\nabla^2 f(r) = 0$ show that $f(r) = c_1 \log r + c_2$ where $r^2 = x^2 + y^2$ and c_1, c_2 are arbitrary constants.
- (ii) If $\mathbf{a}$ and $\mathbf{b}$ are constant vectors, then show that $\nabla \bullet (\mathbf{a} \bullet \mathbf{b} \mathbf{r}) = 3\mathbf{a} \bullet \mathbf{b}$.
12. If $\mathbf{u} = (1/r) \mathbf{r}$ find $\text{grad} (\text{div } \mathbf{u})$.
13. (i) Prove that $\nabla^2 \left[\nabla \bullet \left(\frac{\mathbf{r}}{r^2} \right) \right] = 2r^{-4}$. (Kumaun 2011, 13)
- (ii) Prove that $\mathbf{a} \bullet \left(\nabla \frac{1}{r} \right) = -\frac{\mathbf{a} \bullet \mathbf{r}}{r^3}$.
- (iii) Prove that $\nabla \bullet (U \nabla V - V \nabla U) = U \nabla^2 V - V \nabla^2 U$.

(iv) Prove that $\mathbf{b} \cdot \nabla \left(\mathbf{a} \cdot \nabla \frac{1}{r} \right) = \frac{3(\mathbf{a} \cdot \mathbf{r})(\mathbf{b} \cdot \mathbf{r})}{r^5} - \frac{\mathbf{a} \cdot \mathbf{b}}{r^3},$

where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors.

(Kumaun 2015)

14. Evaluate $\text{div} \{ \mathbf{a} \times (\mathbf{r} \times \mathbf{a}) \}$, where $\mathbf{a}$ is a constant vector. (Kanpur 2007)

15. (i) Prove that $\text{div} \left\{ \frac{f(r) \mathbf{r}}{r} \right\} = \frac{1}{r^2} \frac{d}{dr} (r^2 f).$ (Kumaun 2007, 14)

(ii) Prove that $\frac{1}{2} \nabla \mathbf{a}^2 = (\mathbf{a} \cdot \nabla) \mathbf{a} + \mathbf{a} \times \text{curl } \mathbf{a}.$

16. Prove that $\text{curl} [\mathbf{r} \times (\mathbf{a} \times \mathbf{r})] = 3\mathbf{r} \times \mathbf{a}$, where $\mathbf{a}$ is a constant vector.

17. Prove that $\nabla \times (\mathbf{F} \times \mathbf{r}) = 2\mathbf{F} - (\nabla \cdot \mathbf{F}) \mathbf{r} + (\mathbf{r} \cdot \nabla) \mathbf{F}.$ (Kumaun 2015)

18. If $\mathbf{a}$ and $\mathbf{b}$ are constant vectors, prove that

$$\text{grad} [(\mathbf{r} \times \mathbf{a}) \cdot (\mathbf{r} \times \mathbf{b})] = (\mathbf{b} \times \mathbf{r}) \times \mathbf{a} + (\mathbf{a} \times \mathbf{r}) \times \mathbf{b}. \quad (\text{Kumaun 2015})$$

Answers 4

1. (iii) $(2-r)e^{-r} \mathbf{r}; (2-r)e^{-r} r$

12. $-\frac{2}{r^3} \mathbf{r}$

14. $2\mathbf{a}^2$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If the vector $\mathbf{V} = (x+3y) \mathbf{i} + (y-2z) \mathbf{j} + (x+az) \mathbf{k}$ is solenoidal, then the constant a is

(a) 0

(b) 1

(c) -2

(d) 2

2. The directional derivative of $\phi(x, y, z) = x^2 yz + 4xz^2$ at $(1, -2, -1)$ in the direction of the vector $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ is

(a) $37/3$

(b) $3/37$

(c) 3

(d) none of these

3. If $r = |\mathbf{r}|$ where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then $\nabla^2 r^n =$

(a) $n(n+1)r^n$

(b) $n(n+1)r^{n-1}$

(c) $n(n+1)r^{n-2}$

(d) none of these

(Agra 2007)

4. $\nabla^2 \left(\frac{1}{r} \right) =$
 - (a) $-2 / r^3$
 - (b) 0
 - (c) $2 / r^3$
 - (d) none of these

(Kumaun 2014)
5. If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{a}$ is a constant vector, then $\text{curl}(\mathbf{r} \times \mathbf{a})$ is
 - (a) $-\mathbf{a}$
 - (b) $-3\mathbf{a}$
 - (c) $-2\mathbf{a}$
 - (d) none of these
6. The value of $\text{div } \hat{\mathbf{r}}$ is
 - (a) $2 / r$
 - (b) 0
 - (c) $1 / r$
 - (d) none of these

(Agra 2014)
7. The $\mathbf{V}$ is a constant vector, then $\text{div } \mathbf{V}$ is :
 - (a) 3
 - (b) $3\mathbf{V}$
 - (c) 0
 - (d) none of these

(Kumaun 2007, 11)
8. If $\mathbf{f} = x^2 y\mathbf{i} - 2xz\mathbf{j} + 2yz\mathbf{k}$, then $\text{div } \mathbf{f}$ is equal to
 - (a) $2x(x+1)$
 - (b) $2y(x+1)$
 - (c) $y(y+1)$
 - (d) none of these

(Kumaun 2008)
9. $\nabla \times (\nabla f)$ is equal to
 - (a) $\nabla^2 f$
 - (b) 0
 - (c) 0
 - (d) none of these

(Kumaun 2010)

Fill in the Blank(s)

Fill in the blanks “.....”, so that the following statements are complete and correct.

1. If $\mathbf{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$, then $\mathbf{F} \cdot d\mathbf{r} = \dots\dots\dots$
2. If $\mathbf{P} = e^{xy}\mathbf{i} + (x - 2y)\mathbf{j} + (x \sin y)\mathbf{k}$, then $\frac{\partial \mathbf{P}}{\partial x} = \dots\dots\dots$
3. If $\mathbf{a}$ is a constant vector then $\text{grad}(\mathbf{a} \cdot \mathbf{r}) = \dots\dots\dots$

(Rohilkhand 2007; Kumaun 12)
4. If $\mathbf{a}$ is a constant vector, then $\nabla \cdot (\mathbf{a} \times \mathbf{r}) = \dots\dots\dots$
5. If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then the value of $\text{div } \mathbf{r} = \dots\dots\dots$

(Agra 2008; Kumaun 15)
6. If $\mathbf{A} = x^2 z\mathbf{i} - 2y^3 z^2 \mathbf{j} + xy^2 z\mathbf{k}$, then $\text{div } \mathbf{A}$ at $(1, -1, 1) = \dots\dots\dots$
7. If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then the value of $\text{curl } \mathbf{r} = \dots\dots\dots$

(Agra 2008; Rohilkhand 14)

8. For any vector $\mathbf{A}$, $\text{div curl } \mathbf{A} = \dots\dots$
9. A vector $\mathbf{V}$ is said to be solenoidal if $\dots\dots$ (Rohilkhand 2005; Agra 06)
10. A vector $\mathbf{F}$ is said to be irrotational if $\dots\dots$
11. If $\phi = x^2 y + 2xy + z^2$, then $\text{curl grad } \phi = \dots\dots$ (Bundelkhand 2007)
12. If $\mathbf{f} = x^2 y\mathbf{i} + 2xz\mathbf{j} + 2yz\mathbf{k}$, then $\text{div (curl } \mathbf{f}) = \dots\dots$ (Kumaun 2010, 12)
13. Value of $\text{div grad } \phi$ is $\dots\dots$ (Kumaun 2011)
14. If $\mathbf{a}$ is a constant vector, then $\text{curl } (\mathbf{r} \times \mathbf{a}) = \dots\dots$ (Kumaun 2013)

True or False

Write 'T' for true and 'F' for false statement.

1. If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then $\mathbf{r}$ is solenoidal.
2. If $\mathbf{V}$ is a constant vector, then $\text{div } \mathbf{V} = 0$. (Rohilkhand 2005)
3. If $\mathbf{F} = 2xyz \mathbf{i} + y^2 z\mathbf{j} - 2yz^2 \mathbf{k}$, then $\mathbf{F}$ is irrotational.
4. If ϕ is a differentiable scalar function, then $\text{curl grad } \phi = \mathbf{0}$.
5. If ϕ is a differentiable scalar function then $\text{div grad } \phi = \nabla^2 \phi$.
6. $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{A} \cdot (\nabla \times \mathbf{B}) - \mathbf{B} \cdot (\nabla \times \mathbf{A})$.
7. A function which satisfies Laplace's equation is called a harmonic function.

Answers

Multiple Choice Questions

- | | | |
|--------|--------|--------|
| 1. (c) | 2. (a) | 3. (c) |
| 4. (b) | 5. (c) | 6. (a) |
| 7. (c) | 8. (b) | 9. (b) |

Fill in the Blank(s)

- | | | |
|--|---|---------------------------------|
| 1. $(x^2 + y^2) dx - 2xydy$ | 2. $y e^{xy} \mathbf{i} + \mathbf{j} + \sin y \mathbf{k}$ | 3. $\mathbf{a}$ |
| 4. 0 | 5. 3 | 6. -3 |
| 7. 0 | 8. 0 | 9. $\text{div } \mathbf{V} = 0$ |
| 10. $\text{curl } \mathbf{F} = \mathbf{0}$ | 11. 0 | 12. 0 |
| 13. $\nabla^2 \phi$ | 14. $-2\mathbf{a}$ | |

True or False

- | | | |
|------|------|------|
| 1. F | 2. T | 3. F |
| 4. T | 5. T | 6. F |
| 7. T | | |



Chapter

4



Integration of Vectors

1 Integration of Vector Functions

We shall define *integration as the reverse process of differentiation*. Let $\mathbf{f}(t)$ and $\mathbf{F}(t)$ be two vector functions of the scalar t such that

$$\frac{d}{dt} \mathbf{F}(t) = \mathbf{f}(t).$$

Then $\mathbf{F}(t)$ is called the *indefinite integral* of $\mathbf{f}(t)$ with respect to t and symbolically we write

$$\int \mathbf{f}(t) dt = \mathbf{F}(t). \quad \dots(1)$$

The function $\mathbf{f}(t)$ to be integrated is called the *integrand*.

If $\mathbf{c}$ is any *arbitrary constant vector* independent of t , then

$$\frac{d}{dt} \{\mathbf{F}(t) + \mathbf{c}\} = \mathbf{f}(t).$$

This is equivalent to

$$\int \mathbf{f}(t) dt = \mathbf{F}(t) + \mathbf{c}. \quad \dots(2)$$

From (2) it is obvious that the integral $\mathbf{F}(t)$ of $\mathbf{f}(t)$ is indefinite to the extent of an additive arbitrary constant $\mathbf{c}$. Therefore $\mathbf{F}(t)$ is called the *indefinite integral* of $\mathbf{f}(t)$.

The constant vector $\mathbf{c}$ is called the *constant of integration*. It can be determined if we are given some initial conditions.

If $\frac{d}{dt} \mathbf{F}(t) = \mathbf{f}(t)$ for all t in the interval $[a, b]$, then the *definite integral* between the limits $t = a$ and $t = b$ can, in such case, be written as

$$\int_a^b \mathbf{f}(t) dt = \int_a^b \left\{ \frac{d}{dt} \mathbf{F}(t) \right\} dt = [\mathbf{F}(t) + \mathbf{c}]_a^b = \mathbf{F}(b) - \mathbf{F}(a).$$

Theorem: If $\mathbf{f}(t) = f_1(t) \mathbf{i} + f_2(t) \mathbf{j} + f_3(t) \mathbf{k}$, then

$$\int \mathbf{f}(t) dt = \mathbf{i} \int f_1(t) dt + \mathbf{j} \int f_2(t) dt + \mathbf{k} \int f_3(t) dt.$$

Proof: Let $\frac{d}{dt} \mathbf{F}(t) = \mathbf{f}(t)$ (1)

Then $\int \mathbf{f}(t) dt = \mathbf{F}(t)$ (2)

Let $\mathbf{F}(t) = F_1(t) \mathbf{i} + F_2(t) \mathbf{j} + F_3(t) \mathbf{k}$.

Then from (1), we have

$$\frac{d}{dt} \{ F_1(t) \mathbf{i} + F_2(t) \mathbf{j} + F_3(t) \mathbf{k} \} = \mathbf{f}(t)$$

$$\begin{aligned} \text{or} \quad \left\{ \frac{d}{dt} F_1(t) \right\} \mathbf{i} + \left\{ \frac{d}{dt} F_2(t) \right\} \mathbf{j} + \left\{ \frac{d}{dt} F_3(t) \right\} \mathbf{k} \\ = f_1(t) \mathbf{i} + f_2(t) \mathbf{j} + f_3(t) \mathbf{k}. \end{aligned}$$

Equating the coefficients of $\mathbf{i}, \mathbf{j}, \mathbf{k}$, we get

$$\frac{d}{dt} F_1(t) = f_1(t), \frac{d}{dt} F_2(t) = f_2(t), \frac{d}{dt} F_3(t) = f_3(t).$$

$$\therefore F_1(t) = \int f_1(t) dt, F_2(t) = \int f_2(t) dt, F_3(t) = \int f_3(t) dt.$$

$$\therefore \mathbf{F}(t) = \left\{ \int f_1(t) dt \right\} \mathbf{i} + \left\{ \int f_2(t) dt \right\} \mathbf{j} + \left\{ \int f_3(t) dt \right\} \mathbf{k}.$$

So from (2), we get

$$\int \mathbf{f}(t) dt = \mathbf{i} \int f_1(t) dt + \mathbf{j} \int f_2(t) dt + \mathbf{k} \int f_3(t) dt.$$

Note: From this theorem we conclude that the definition of the integral of a vector function implies the definition of integrals of three scalar functions which are the components of that vector function. **Thus in order to integrate a vector function we should integrate its components.**

2 Some Standard Results

We have already obtained some standard results for differentiation. With the help of these results we can obtain some standard results for integration.

$$1. \quad \text{We have } \frac{d}{dt} (\mathbf{r} \cdot \mathbf{s}) = \frac{d\mathbf{r}}{dt} \cdot \mathbf{s} + \mathbf{r} \cdot \frac{d\mathbf{s}}{dt}.$$

Therefore $\int \left(\frac{d\mathbf{r}}{dt} \bullet \mathbf{s} + \mathbf{r} \bullet \frac{d\mathbf{s}}{dt} \right) dt = \mathbf{r} \bullet \mathbf{s} + c,$

where c is the constant of integration. It should be noted that c is here a scalar quantity since the integrand is also scalar.

2. We have $\frac{d}{dt} (\mathbf{r}^2) = 2\mathbf{r} \bullet \frac{d\mathbf{r}}{dt}$. Therefore $\int \left(2\mathbf{r} \bullet \frac{d\mathbf{r}}{dt} \right) dt = \mathbf{r}^2 + c.$

Here the constant of integration c is a scalar quantity.

3. We have $\frac{d}{dt} \left(\frac{d\mathbf{r}}{dt} \right)^2 = 2 \frac{d\mathbf{r}}{dt} \bullet \frac{d^2\mathbf{r}}{dt^2}$.

Therefore we have $\int \left(2 \frac{d\mathbf{r}}{dt} \bullet \frac{d^2\mathbf{r}}{dt^2} \right) dt = \left(\frac{d\mathbf{r}}{dt} \right)^2 + c.$

Here the constant of integration c is a scalar quantity.

Also $\left(\frac{d\mathbf{r}}{dt} \right)^2 = \frac{d\mathbf{r}}{dt} \bullet \frac{d\mathbf{r}}{dt}.$

4. We have $\frac{d}{dt} \left(\mathbf{r} \times \frac{d\mathbf{r}}{dt} \right) = \frac{d\mathbf{r}}{dt} \times \frac{d\mathbf{r}}{dt} + \mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2} = \mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2}.$

$\therefore \int \left(\mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2} \right) dt = \mathbf{r} \times \frac{d\mathbf{r}}{dt} + \mathbf{c}.$

Here the constant of integration $\mathbf{c}$ is a vector quantity since the integrand $\mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2}$ is also a vector quantity.

5. If $\mathbf{a}$ is a constant vector, we have

$$\frac{d}{dt} (\mathbf{a} \times \mathbf{r}) = \frac{d\mathbf{a}}{dt} \times \mathbf{r} + \mathbf{a} \times \frac{d\mathbf{r}}{dt} = \mathbf{a} \times \frac{d\mathbf{r}}{dt}.$$

Therefore $\int \left(\mathbf{a} \times \frac{d\mathbf{r}}{dt} \right) dt = \mathbf{a} \times \mathbf{r} + \mathbf{c}.$

Here the constant of integration $\mathbf{c}$ is a vector quantity.

6. If $r = |\mathbf{r}|$ and $\hat{\mathbf{r}}$ is a unit vector in the direction of $\mathbf{r}$, then

$$\frac{d}{dt} (\hat{\mathbf{r}}) = \frac{d}{dt} \left(\frac{1}{r} \mathbf{r} \right) = \frac{1}{r} \frac{d\mathbf{r}}{dt} - \frac{1}{r^2} \frac{dr}{dt} \mathbf{r}.$$

Therefore $\int \left(\frac{1}{r} \frac{d\mathbf{r}}{dt} - \frac{1}{r^2} \frac{dr}{dt} \mathbf{r} \right) dt = \hat{\mathbf{r}} + \mathbf{c}.$

7. If c is a constant scalar and $\mathbf{r}$ a vector function of a scalar t , then obviously

$$\int c \mathbf{r} dt = c \int \mathbf{r} dt.$$

8. If $\mathbf{r}$ and $\mathbf{s}$ are two vector functions of the scalar t , then obviously

$$\int (\mathbf{r} + \mathbf{s}) dt = \int \mathbf{r} dt + \int \mathbf{s} dt.$$

Illustrative Examples

Example 1: Evaluate $\int_1^2 \mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} dt$, where $\mathbf{r} = 2t^2 \mathbf{i} + t \mathbf{j} - 3t^3 \mathbf{k}$.

(Bundelkhand 2004; Kumaun 08; Kanpur 13)

Solution: Given $\mathbf{r} = 2t^2 \mathbf{i} + t \mathbf{j} - 3t^3 \mathbf{k}$.

$$\therefore \frac{d\mathbf{r}}{dt} = 4t \mathbf{i} + \mathbf{j} - 9t^2 \mathbf{k} \text{ and } \frac{d^2 \mathbf{r}}{dt^2} = 4\mathbf{i} + 0\mathbf{j} - 18t\mathbf{k}.$$

$$\therefore \mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} = (2t^2 \mathbf{i} + t\mathbf{j} - 3t^3 \mathbf{k}) \times (4\mathbf{i} + 0\mathbf{j} - 18t\mathbf{k})$$

$$\begin{aligned} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2t^2 & t & -3t^3 \\ 4 & 0 & -18t \end{vmatrix} \\ &= -18t^2 \mathbf{i} - (-36t^3 + 12t^3) \mathbf{j} - 4t \mathbf{k} \\ &= -18t^2 \mathbf{i} + 24t^3 \mathbf{j} - 4t \mathbf{k}. \end{aligned}$$

$$\begin{aligned} \therefore \int_1^2 \mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} dt &= \int_1^2 (-18t^2 \mathbf{i} + 24t^3 \mathbf{j} - 4t \mathbf{k}) dt \\ &= -18\mathbf{i} \int_1^2 t^2 dt + 24\mathbf{j} \int_1^2 t^3 dt - 4\mathbf{k} \int_1^2 t dt \\ &= -18\mathbf{i} \left[\frac{t^3}{3} \right]_1^2 + 24\mathbf{j} \left[\frac{t^4}{4} \right]_1^2 - 4\mathbf{k} \left[\frac{t^2}{2} \right]_1^2 \\ &= -6(8-1)\mathbf{i} + 6(16-1)\mathbf{j} - 2(4-1)\mathbf{k} \\ &= -42\mathbf{i} + 90\mathbf{j} - 6\mathbf{k}. \end{aligned}$$

Example 2: Find the value of $\mathbf{r}$ satisfying the equation $\frac{d^2 \mathbf{r}}{dt^2} = t\mathbf{a} + \mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors.

Solution: Integrating the equation $\frac{d^2 \mathbf{r}}{dt^2} = t\mathbf{a} + \mathbf{b}$, we get $\frac{d\mathbf{r}}{dt} = \frac{1}{2} t^2 \mathbf{a} + t\mathbf{b} + \mathbf{c}$, where $\mathbf{c}$ is constant.

Again integrating, we get

$$\mathbf{r} = \frac{1}{6} t^3 \mathbf{a} + \frac{1}{2} t^2 \mathbf{b} + t\mathbf{c} + \mathbf{d}, \text{ where } \mathbf{d} \text{ is constant.}$$

Example 3: If $\mathbf{r}(t) = 5t^2 \mathbf{i} + t\mathbf{j} - t^3 \mathbf{k}$, prove that

$$\int_1^2 \left(\mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} \right) dt = -14\mathbf{i} + 75\mathbf{j} - 15\mathbf{k}. \quad (\text{Kumaun 2000; Meerut 01, 04, 05, 07, 10, 10B; Kanpur 10; Rohilkhand 13})$$

Solution: We have $\int \left(\mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} \right) dt = \mathbf{r} \times \frac{d\mathbf{r}}{dt} + \mathbf{c}$.

$$\therefore \int_1^2 \left(\mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} \right) dt = \left[\mathbf{r} \times \frac{d\mathbf{r}}{dt} \right]_1^2.$$

Let us now find $\mathbf{r} \times \frac{d\mathbf{r}}{dt}$. We have $\frac{d\mathbf{r}}{dt} = 10t \mathbf{i} + \mathbf{j} - 3t^2 \mathbf{k}$.

$$\begin{aligned} \therefore \mathbf{r} \times \frac{d\mathbf{r}}{dt} &= (5t^2 \mathbf{i} + t \mathbf{j} - t^3 \mathbf{k}) \times (10t \mathbf{i} + \mathbf{j} - 3t^2 \mathbf{k}) \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5t^2 & t & -t^3 \\ 10t & 1 & -3t^2 \end{vmatrix} = -2t^3 \mathbf{i} + 5t^4 \mathbf{j} - 5t^2 \mathbf{k}. \end{aligned}$$

$$\begin{aligned} \therefore \int_1^2 \left(\mathbf{r} \times \frac{d^2 \mathbf{r}}{dt^2} \right) dt &= \left[-2t^3 \mathbf{i} + 5t^4 \mathbf{j} - 5t^2 \mathbf{k} \right]_1^2 \\ &= \left[-2t^3 \right]_1^2 \mathbf{i} + \left[5t^4 \right]_1^2 \mathbf{j} - \left[5t^2 \right]_1^2 \mathbf{k} \\ &= -14\mathbf{i} + 75\mathbf{j} - 15\mathbf{k}. \end{aligned}$$

Example 4: Given that $\mathbf{r}(t) = \begin{cases} 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}, & \text{when } t = 2 \\ 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}, & \text{when } t = 3, \end{cases}$

show that $\int_2^3 \left(\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} \right) dt = 10$.

(Meerut 2003, 13B; Bundelkhand 08;
Kanpur 09, 11; Agra 06; Avadh 10; Purvanchal 13)

Solution: We have $\int \left(\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} \right) dt = \left[\frac{1}{2} \mathbf{r}^2 \right]_2^3$.

When $t = 3$, $\mathbf{r} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$.

$\therefore$ when $t = 3$, $\mathbf{r}^2 = (4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) \cdot (4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) = 16 + 4 + 9 = 29$.

When $t = 2$, $\mathbf{r} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

$\therefore$ when $t = 2$, $\mathbf{r}^2 = 4 + 1 + 4 = 9$.

$$\therefore \int_2^3 \left(\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} \right) dt = \frac{1}{2} [29 - 9] = 10.$$

Example 5: The acceleration of a particle at any time $t \geq 0$ is given by

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 12 \cos 2t \mathbf{i} - 8 \sin 2t \mathbf{j} + 16t \mathbf{k}.$$

If the velocity $\mathbf{v}$ and displacement $\mathbf{r}$ are zero at $t = 0$, find $\mathbf{v}$ and $\mathbf{r}$ at any time.

(Agra 2007)

Solution: We have $\frac{d\mathbf{v}}{dt} = 12 \cos 2t \mathbf{i} - 8 \sin 2t \mathbf{j} + 16t \mathbf{k}$.

Integrating, we get

$$\mathbf{v} = \mathbf{i} \int 12 \cos 2t \, dt + \mathbf{j} \int -8 \sin 2t \, dt + \mathbf{k} \int 16t \, dt$$

or $\mathbf{v} = 6 \sin 2t \, \mathbf{i} + 4 \cos 2t \, \mathbf{j} + 8t^2 \, \mathbf{k} + \mathbf{c}.$

When $t = 0$, $\mathbf{v} = \mathbf{0}.$

$$\therefore \mathbf{0} = 0\mathbf{i} + 4\mathbf{j} + 0\mathbf{k} + \mathbf{c} \quad \text{or} \quad \mathbf{c} = -4\mathbf{j}.$$

$$\therefore \mathbf{v} = \frac{d\mathbf{r}}{dt} = 6 \sin 2t \, \mathbf{i} + (4 \cos 2t - 4) \, \mathbf{j} + 8t^2 \, \mathbf{k}.$$

Integrating, we get

$$\begin{aligned} \mathbf{r} &= \mathbf{i} \int 6 \sin 2t \, dt + \mathbf{j} \int (4 \cos 2t - 4) \, dt + \mathbf{k} \int 8t^2 \, dt \\ &= -3 \cos 2t \, \mathbf{i} + (2 \sin 2t - 4t) \, \mathbf{j} + \frac{8}{3} t^3 \, \mathbf{k} + \mathbf{d}, \end{aligned}$$

where $\mathbf{d}$ is constant.

When $t = 0$, $\mathbf{r} = \mathbf{0}.$

$$\therefore \mathbf{0} = -3\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} + \mathbf{d}. \quad \therefore \mathbf{d} = 3\mathbf{i}.$$

$$\begin{aligned} \therefore \mathbf{r} &= -3 \cos 2t \, \mathbf{i} + (2 \sin 2t - 4t) \, \mathbf{j} + \frac{8}{3} t^3 \, \mathbf{k} + 3\mathbf{i} \\ &= (3 - 3 \cos 2t) \, \mathbf{i} + (2 \sin 2t - 4t) \, \mathbf{j} + \frac{8}{3} t^3 \, \mathbf{k}. \end{aligned}$$

Comprehensive Exercise 1

1. If $\mathbf{f}(t) = (t - t^2) \, \mathbf{i} + 2t^3 \, \mathbf{j} - 3\mathbf{k}$, find (i) $\int \mathbf{f}(t) \, dt$ (ii) $\int_1^2 \mathbf{f}(t) \, dt$.
2. Evaluate $\int_0^1 e^t \, \mathbf{i} + e^{-2t} \, \mathbf{j} + t \, \mathbf{k} \, dt$. (Garhwal 2001, 02)
3. If $\mathbf{f}(t) = t \, \mathbf{i} + (t^2 - 2t) \, \mathbf{j} + (3t^2 + 3t^3) \, \mathbf{k}$, find $\int_0^1 \mathbf{f}(t) \, dt$.
(Garhwal 2003; Bundelkhand 07)
4. If $\mathbf{r} = t \, \mathbf{i} - t^2 \, \mathbf{j} + (t - 1) \, \mathbf{k}$ and $\mathbf{s} = 2t^2 \, \mathbf{i} + 6t \, \mathbf{k}$, evaluate
(i) $\int_0^2 \mathbf{r} \cdot \mathbf{s} \, dt$, (ii) $\int_0^2 \mathbf{r} \times \mathbf{s} \, dt$
(Rohilkhand 2008)
5. (i) Find the value of $\mathbf{r}$ satisfying the equation $\frac{d^2 \mathbf{r}}{dt^2} = \mathbf{a}$, where $\mathbf{a}$ is a constant vector. Also it is given that when $t = 0$, $\mathbf{r} = \mathbf{0}$ and $\frac{d\mathbf{r}}{dt} = \mathbf{u}$.
(ii) Solve the equation $\frac{d^2 \mathbf{r}}{dt^2} = \mathbf{a}$ where $\mathbf{a}$ is a constant vector; given that $\mathbf{r} = \mathbf{0}$ and $\frac{d\mathbf{r}}{dt} = \mathbf{0}$ when $t = 0$.
(Bundelkhand 2008)

6. Find the value of $\mathbf{r}$ satisfying the equation $\frac{d^2 \mathbf{r}}{dt^2} = 6t\mathbf{i} - 24t^2\mathbf{j} + 4\sin t\mathbf{k}$,
given that $\mathbf{r} = 2\mathbf{i} + \mathbf{j}$ and $d\mathbf{r}/dt = -\mathbf{i} - 3\mathbf{k}$ at $t = 0$.
(Agra 2001; Meerut 11)
7. The acceleration of a particle at any time t is $e^t\mathbf{i} + e^{2t}\mathbf{j} + \mathbf{k}$.
Find $\mathbf{v}$, given that $\mathbf{v} = \mathbf{i} + \mathbf{j}$ at $t = 0$.
8. Evaluate $\int_1^2 (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}) dt$, where
 $\mathbf{a} = t\mathbf{i} - 3\mathbf{j} + 2t\mathbf{k}$, $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$, $\mathbf{c} = 3\mathbf{i} + t\mathbf{j} - \mathbf{k}$. (Meerut 2013)
9. Integrate $\frac{d^2 \mathbf{r}}{dt^2} = -n\mathbf{r}^2$.
(Kumaun 2009)

Answers 1

1. (i) $\left(\frac{t^2}{2} + \frac{t^3}{3}\right)\mathbf{i} + \frac{t^4}{2}\mathbf{j} - 3t\mathbf{k} + \mathbf{c}$ (ii) $-\frac{5}{6}\mathbf{i} + \frac{15}{2}\mathbf{j} - 3\mathbf{k}$
2. $(e-1)\mathbf{i} - \frac{1}{2}(e^{-2}-1)\mathbf{j} + \frac{1}{2}\mathbf{k}$ 3. $\frac{1}{2}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{7}{4}\mathbf{k}$
4. (i) 12 (ii) $-24\mathbf{i} - \frac{40}{3}\mathbf{j} + \frac{64}{5}\mathbf{k}$
5. (i) $\frac{1}{2}t^2\mathbf{a} + t\mathbf{u}$ (ii) $\frac{1}{2}t^2\mathbf{a}$
6. $\mathbf{r} = (t^3 - t + 2)\mathbf{i} + (1 - 2t^4)\mathbf{j} + (t - 4\sin t)\mathbf{k}$
7. $e^t\mathbf{i} + \frac{1}{2}(e^{2t} + 1)\mathbf{j} + t\mathbf{k}$ 8. 0
9. $-n^2\mathbf{r}^2 + \mathbf{c}$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If $\mathbf{F}(t) = t\mathbf{i} + (t^2 - 2t)\mathbf{j} + (3t^2 + 3t^3)\mathbf{k}$, then the value of $\int_0^1 \mathbf{F}(t) dt$ is
- (a) $\frac{1}{2}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{7}{4}\mathbf{k}$ (b) $\frac{1}{2}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{7}{4}\mathbf{k}$
- (c) $-\frac{1}{2}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{7}{4}\mathbf{k}$ (d) None of these

(Bundelkhand 2001)

2. If $\mathbf{r} = t\mathbf{i} - t^2\mathbf{j} + (t-1)\mathbf{k}$, and $\mathbf{s} = 2t^2\mathbf{i} + 6t\mathbf{k}$, then the value of $\int_0^1 \mathbf{r} \cdot \mathbf{s} \, dt$ is
- (a) 10 (b) 12
(c) 15 (d) None of these

(Kumaun 2007, 10)

Fill in the Blank(s)*Fill in the blanks “.....”, so that the following statements are complete and correct.*

1. $\int_0^1 [t\mathbf{i} + (t^2 - 2t)\mathbf{j}] \, dt = \dots\dots\dots$
2. If $\mathbf{F}(t) = 3t^2\mathbf{i} + t\mathbf{j} + 2\mathbf{k}$ and $\mathbf{G}(t) = 6t^2\mathbf{i} + (t-1)\mathbf{j} + 3t\mathbf{k}$, then
- $$\int_0^1 \left(\frac{d\mathbf{F}}{dt} \cdot \mathbf{G} + \mathbf{F} \cdot \frac{d\mathbf{G}}{dt} \right) dt = \dots\dots\dots$$
- (Meerut 2011)
3. If $\mathbf{r}(t) = 5t^2\mathbf{i} + t\mathbf{j} - t^3\mathbf{k}$, then $\int_1^2 \mathbf{r} \times \frac{d^2\mathbf{r}}{dt^2} dt = \dots\dots\dots$
- (Kumaun 2013)
4. If $\mathbf{r} = t^2\mathbf{i} + \mathbf{j} - t\mathbf{k}$, then $\int_0^1 \mathbf{r} \cdot \frac{d\mathbf{r}}{dt} dt = \dots\dots\dots$
- (Kumaun 2014)

True or False*Write ‘T’ for true and ‘F’ for false statement.*

1. $\int \left(2 \frac{d\mathbf{r}}{dt} \cdot \frac{d^2\mathbf{r}}{dt^2} \right) dt = \left(\frac{d\mathbf{r}}{dt} \right)^2 + c.$
2. The value of $\int_0^1 (e^t\mathbf{i} + e^{-2t}\mathbf{j} + t\mathbf{k}) dt$ is $(e-1)\mathbf{i} - (e^{-2}-1)\mathbf{j} + \mathbf{k}.$

Answers**Multiple Choice Questions**

1. (b) 2. (d)

Fill in the Blank(s)

1. $\frac{1}{2}\mathbf{i} - \frac{2}{3}\mathbf{j}$ 2. 24
3. $-14\mathbf{i} + 75\mathbf{j} - 15\mathbf{k}$ 4. 1

True or False

1. T 2. F



Chapter

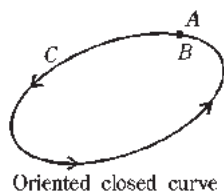
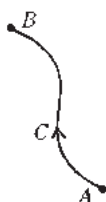
5



Line Integrals

1 Some Preliminary Concepts

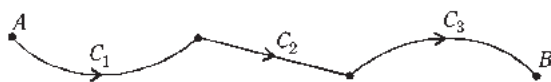
Oriented curve. Suppose C is a curve in space. Let us orient C by taking one of the two directions along C as the *positive direction*; the opposite direction along C is then called the *negative direction*. Suppose A is the initial point and B the terminal point of C under the chosen orientation. In case these two points coincide, the curve C is called a *closed curve*.



Smooth curve. Let $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, where $\mathbf{r}(t)$ is the position vector of (x, y, z) , be the parametric representation of a curve C joining the points A and B , where $t = t_1$ and $t = t_2$ respectively. We know that $d\mathbf{r}/dt$ is a tangent vector

to this curve at the point $\mathbf{r}$. Suppose the function $\mathbf{r}(t)$ is continuous and has a continuous first derivative not equal to zero vector for all values of t under consideration. Then the curve C possesses a unique tangent at each of its points. A curve satisfying these assumptions is called a **smooth curve**.

A curve C is said to be **piecewise smooth** if it is composed of a finite number of smooth curves. The curve C in the adjoining figure is piecewise smooth as it is



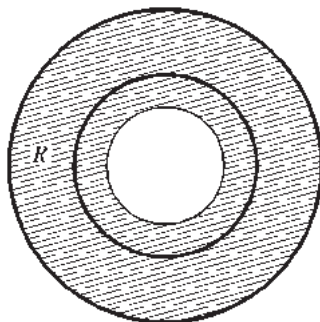
Piecewise Smooth Curve

composed of three smooth curves C_1 , C_2 and C_3 . The circle is a smooth closed curve while the curve consisting of the four sides of a rectangle is a piecewise smooth closed curve.

Smooth surface. Suppose S is a surface which has a unique normal at each of its points and the direction of this normal depends continuously on the points of S . Then S is called a **smooth surface**.

If a surface S is not smooth but can be subdivided into a finite number of smooth surfaces, then it is called a **piecewise smooth surface**. The surface of a sphere is smooth while the surface of a cube is piecewise smooth.

Classification of regions. A region R in which every closed curve can be contracted to a point without passing out of the region is called a **simply connected region**. Otherwise the region R is **multiply-connected**. The region interior to a circle is a simply-connected plane region. The region interior to a sphere is a simply-connected region in space. The region between two concentric circles lying in the same plane is a multiply-connected plane region.



If we take a closed curve in this region surrounding the inner circle, then it cannot be contracted to a point without passing out of the region. Therefore the region is not simply-connected. However the region between two concentric spheres is a simply-connected region in space. The region between two infinitely long coaxial cylinders is a multiply-connected region in space.

2 Line Integrals

(Avadh 2014)

Any integral which is to be evaluated along a curve is called a line integral.

Suppose $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, where $\mathbf{r}(t)$ is the position vector of (x, y, z) i.e., $\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, defines a piecewise smooth curve joining two points A and B . Let $t = t_1$ at A and $t = t_2$ at B . Suppose $\mathbf{F}(x, y, z) = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is a vector point function defined and continuous along C . If s denotes the arc length of the curve C , then $\frac{d\mathbf{r}}{ds} = \mathbf{t}$ is a unit vector along the tangent to the curve C at the point $\mathbf{r}$.

The component of the vector $\mathbf{F}$ along this tangent is $\mathbf{F} \cdot \frac{d\mathbf{r}}{ds}$. The integral of $\mathbf{F} \cdot \frac{d\mathbf{r}}{ds}$ along C from A to B written as

$$\int_A^B \left[\mathbf{F} \cdot \frac{d\mathbf{r}}{ds} \right] ds = \int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

is an example of a *line integral*. It is called the *tangent line integral* of $\mathbf{F}$ along C .

Since $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, therefore, $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$.

$$\begin{aligned} \therefore \quad \mathbf{F} \cdot d\mathbf{r} &= (F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\ &= F_1 dx + F_2 dy + F_3 dz. \end{aligned}$$

Therefore in components form the above line integral is written as

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (F_1 dx + F_2 dy + F_3 dz).$$

The parametric equations of the curve C are $x = x(t)$, $y = y(t)$ and $z = z(t)$.

Therefore we may write

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{t_1}^{t_2} \left[F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right] dt.$$

Circulation: If C is a simple closed curve (i.e. a curve which does not intersect itself anywhere), then the tangent line integral of $\mathbf{F}$ around C is called the *circulation* of $\mathbf{F}$ about C . It is often denoted by

$$\oint \mathbf{F} \cdot d\mathbf{r} = \oint (F_1 dx + F_2 dy + F_3 dz).$$

Work done by a Force. Suppose a force $\mathbf{F}$ acts upon a particle. Let the particle be displaced along a given path C in space. If $\mathbf{r}$ denotes the position vector of a point on C , then $\frac{d\mathbf{r}}{ds}$ is a unit vector along the tangent to C at the point $\mathbf{r}$ in the direction of s

increasing. The component of force $\mathbf{F}$ along tangent to C is $\mathbf{F} \cdot \frac{d\mathbf{r}}{ds}$. Therefore the

work done by $\mathbf{F}$ during a small displacement ds of the particle along C is $\left[\mathbf{F} \cdot \frac{d\mathbf{r}}{ds} \right] ds$

i.e., $\mathbf{F} \cdot d\mathbf{r}$. The total work W done by $\mathbf{F}$ in this displacement along C , is given by the line integral

$$W = \int_C \mathbf{F} \cdot d\mathbf{r},$$

the integration being taken in the sense of the displacement.

Illustrative Examples

Example 1: Evaluate $\int_C \mathbf{F} \bullet d\mathbf{r}$, where $\mathbf{F} = x^2 \mathbf{i} + y^3 \mathbf{j}$ and curve C is the arc of the parabola $y = x^2$ in the x - y plane from $(0, 0)$ to $(1, 1)$.

Solution: We shall illustrate two methods for the solution of such a problem.

Method 1. The curve C is the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

Let $x = t$; then $y = t^2$. If $\mathbf{r}$ is the position vector of any point (x, y) on C , then

$$\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j} = t\mathbf{i} + t^2\mathbf{j}. \quad \therefore \quad \frac{d\mathbf{r}}{dt} = \mathbf{i} + 2t\mathbf{j}.$$

Also in terms of t , $\mathbf{F} = t^2 \mathbf{i} + t^6 \mathbf{j}$.

At the point $(0, 0)$, $t = x = 0$. At the point $(1, 1)$, $t = 1$.

$$\begin{aligned} \therefore \int_C \mathbf{F} \bullet d\mathbf{r} &= \int_C \left(\mathbf{F} \bullet \frac{d\mathbf{r}}{dt} \right) dt = \int_0^1 (t^2 \mathbf{i} + t^6 \mathbf{j}) \bullet (\mathbf{i} + 2t \mathbf{j}) dt \\ &= \int_0^1 (t^2 + 2t^7) dt = \left[\frac{t^3}{3} + \frac{2t^8}{8} \right]_0^1 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}. \end{aligned}$$

Method 2: In the xy -plane we have $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$.

$$\therefore \quad d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j}.$$

Therefore, $\mathbf{F} \bullet d\mathbf{r} = (x^2 \mathbf{i} + y^3 \mathbf{j}) \bullet (dx \mathbf{i} + dy \mathbf{j}) = x^2 dx + y^3 dy$.

$$\therefore \quad \int_C \mathbf{F} \bullet d\mathbf{r} = \int_C (x^2 dx + y^3 dy).$$

Now along the curve C , $y = x^2$. Therefore $dy = 2x dx$.

$$\begin{aligned} \therefore \int_C \mathbf{F} \bullet d\mathbf{r} &= \int_{x=0}^1 [x^2 dx + x^6 (2x) dx] \\ &= \int_0^1 (x^2 + 2x^7) dx = \left[\frac{x^3}{3} + \frac{2x^8}{8} \right]_0^1 = \frac{7}{12}. \end{aligned}$$

Example 2: If $\mathbf{F} = 3xy \mathbf{i} - y^2 \mathbf{j}$, evaluate $\int_C \mathbf{F} \bullet d\mathbf{r}$, where C is the curve in the xy -plane, $y = 2x^2$, from $(0, 0)$ to $(1, 2)$. (Garhwal 2001, 02; Kumaun 07; Rohilkhand 12)

Solution: The parametric equations of the parabola $y = 2x^2$ can be taken as

$$x = t, y = 2t^2.$$

At the point $(0, 0)$, $x = 0$ and so $t = 0$. Again at the point $(1, 2)$, $x = 1$ and so $t = 1$.

$$\text{Now} \quad \int_C \mathbf{F} \bullet d\mathbf{r} = \int_C (3xy\mathbf{i} - y^2\mathbf{j}) \bullet (dx \mathbf{i} + dy \mathbf{j})$$

$$[\because \mathbf{r} = x\mathbf{i} + y\mathbf{j}, \text{ so that } d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j}]$$

$$\begin{aligned}
 &= \int_C (3xy \, dx - y^2 \, dy) = \int_{t=0}^1 \left(3xy \frac{dx}{dt} - y^2 \frac{dy}{dt} \right) dt \\
 &= \int_0^1 (3 \cdot t \cdot 2t^2 \cdot 1 - 4t^4 \cdot 4t) dt \\
 &\quad [\because x = t, y = 2t^2 \text{ so that } dx/dt = 1 \text{ and } dy/dt = 4t] \\
 &= \int_0^1 (6t^3 - 16t^5) dt = \left[6 \cdot \frac{t^4}{4} - 16 \cdot \frac{t^6}{6} \right]_0^1 \\
 &= \frac{6}{4} - \frac{16}{6} = \frac{3}{2} - \frac{8}{3} = -\frac{7}{6}.
 \end{aligned}$$

Example 3: Evaluate $\int \mathbf{F} \cdot d\mathbf{r}$ along the curve $x^2 + y^2 = 1$, $z = 1$ in the positive direction from $(0, 1, 1)$ to $(1, 0, 1)$ where $\mathbf{F} = (2x + yz) \mathbf{i} + xz \mathbf{j} + (xy + 2z) \mathbf{k}$.

Solution: Let the given curve be denoted by C and let A and B be points $(0, 1, 1)$ and $(1, 0, 1)$ respectively.

Along the given curve C , we have $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

$$\therefore d\mathbf{r} = dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}.$$

$$\begin{aligned}
 \therefore \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_A^B [(2x + yz) \mathbf{i} + xz \mathbf{j} + (xy + 2z) \mathbf{k}] \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\
 &= \int_A^B [(2x + yz) dx + xz dy + (xy + 2z) dz]. \quad \dots(1)
 \end{aligned}$$

In moving from A to B , x varies from 0 to 1, y varies from 1 to 0 and z remains constant. We have $z = 1$ and so $dz = 0$.

Hence from (1)

$$\begin{aligned}
 \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 (2x + y) dx + \int_1^0 x dy + 0 \\
 &= \int_0^1 [2x + \sqrt{1 - x^2}] dx - \int_0^1 \sqrt{1 - y^2} dy = [x^2]_0^1 = 1,
 \end{aligned}$$

the last two integrals cancel by a property of definite integrals.

Example 4: Evaluate $\int (x \, dy - y \, dx)$ around the circle $x^2 + y^2 = 1$. (Meerut 2002)

Solution: Let C denote the circle $x^2 + y^2 = 1$. The parametric equations of this circle are $x = \cos t$, $y = \sin t$.

To integrate around the circle C we should vary t from 0 to 2π .

$$\begin{aligned}
 \therefore \oint_C (x \, dy - y \, dx) &= \int_0^{2\pi} \left(x \frac{dy}{dt} - y \frac{dx}{dt} \right) dt \\
 &= \int_0^{2\pi} (\cos^2 t + \sin^2 t) dt = \int_0^{2\pi} dt = 2\pi.
 \end{aligned}$$

Example 5: If $\mathbf{F} = (2x + y) \mathbf{i} + (3y - x) \mathbf{j}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the curve in the xy -plane consisting of the straight lines from $(0, 0)$ to $(2, 0)$ and then to $(3, 2)$.

(Agra 2007; Meerut 11; Purvanchal 14)

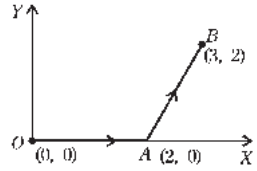
Solution: The path of integration C has been shown in the figure.

It consists of the straight lines OA and AB .

We have $\int_C \mathbf{F} \cdot d\mathbf{r}$

$$= \int_C [(2x + y) \mathbf{i} + (3y - x) \mathbf{j}] \cdot (dx \mathbf{i} + dy \mathbf{j})$$

$$= \int_C [(2x + y) dx + (3y - x) dy].$$



Now along the straight line OA , $y = 0$, $dy = 0$ and x varies from 0 to 2. The equation of the straight line AB is

$$y - 0 = \frac{2 - 0}{3 - 2} (x - 2) \text{ i.e., } y = 2x - 4.$$

$\therefore$ along AB , $y = 2x - 4$, $dy = 2 dx$ and x varies from 2 to 3.

$$\begin{aligned} \therefore \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^2 [(2x + 0) dx + 0] + \int_2^3 [(2x + 2x - 4) dx \\ &\quad + (6x - 12 - x) 2 dx] \end{aligned}$$

$$= \left[x^2 \right]_0^2 + \int_2^3 (14x - 28) dx$$

$$= 4 + 14 \int_2^3 (x - 2) dx = 4 + 14 \left[\frac{(x - 2)^2}{2} \right]_2^3 = 4 + 7 = 11.$$

Example 6: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = (x^2 + y^2) \mathbf{i} - 2xy \mathbf{j}$, curve C is the rectangle in the xy -plane bounded by $y = 0$, $x = a$, $y = b$, $x = 0$.

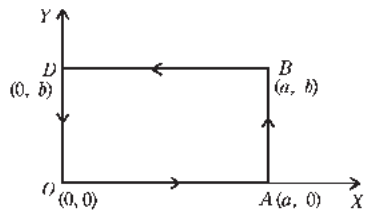
(Meerut 2000, 06B, 07B, 09B, 12, 13; Kanpur 10; Bundelkhand 09; Purvanchal 09)

Solution: In the xy -plane $z = 0$.

Therefore

$$\mathbf{r} = x \mathbf{i} + y \mathbf{j} \text{ and } d\mathbf{r} = dx \mathbf{i} + dy \mathbf{j}.$$

The path of integration C has been shown in the figure. It consists of the straight lines OA , AB , BD and DO .



We have

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C [(x^2 + y^2) \mathbf{i} - 2xy \mathbf{j}] \cdot (dx \mathbf{i} + dy \mathbf{j}) \\ &= \int_C [(x^2 + y^2) dx - 2xy dy]. \end{aligned}$$

Now on OA , $y = 0$, $dy = 0$ and x varies from 0 to a ; on AB , $x = a$, $dx = 0$ and y varies from 0 to b ; on BD , $y = b$, $dy = 0$ and x varies from a to 0; on DO , $x = 0$, $dx = 0$ and y varies from b to 0.

$$\therefore \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^a x^2 dx - \int_0^b 2ay dy + \int_a^0 (x^2 + b^2) dx + \int_b^0 0 dy$$

$$= \left[\frac{x^3}{3} \right]_0^a - 2a \left[\frac{y^2}{2} \right]_0^b + \left[\frac{x^3}{3} + b^2 x \right]_a^0 + 0 = -2ab^2.$$

Example 7: Find the total work done in moving a particle in a force field given by

$$\mathbf{F} = 3xy\mathbf{i} - 5z\mathbf{j} + 10x\mathbf{k}$$

along the curve $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$.

Solution: Let C denote the arc of the given curve from $t = 1$ to $t = 2$. Then the total work done

$$\begin{aligned} &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (3xy\mathbf{i} - 5z\mathbf{j} + 10x\mathbf{k}) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\ &= \int_C (3xy\,dx - 5z\,dy + 10x\,dz) \\ &= \int_1^2 \left(3xy \frac{dx}{dt} - 5z \frac{dy}{dt} + 10x \frac{dz}{dt} \right) dt \\ &= \int_1^2 [3(t^2 + 1)(2t)^2(2t) - (5t^3)(4t) + 10(t^2 + 1)(3t^2)] dt \\ &= \int_1^2 (12t^5 + 12t^3 - 20t^4 + 30t^4 + 30t^2) dt \\ &= \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2) dt = 303. \end{aligned}$$

Example 8: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}$ and C is the portion of the curve $\mathbf{r} = a \cos t\mathbf{i} + b \sin t\mathbf{j} + ct\mathbf{k}$, from $t = 0$ to $t = \pi/2$. (Avadh 2010)

Solution: Along the curve C ,

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = a \cos t\mathbf{i} + b \sin t\mathbf{j} + ct\mathbf{k}.$$

$$\therefore x = a \cos t, \quad y = b \sin t, \quad z = ct.$$

$$\begin{aligned} \text{Now } \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\ &= \int_C (yz\,dx + zx\,dy + xy\,dz) = \int_C d(xyz) \\ &= [xyz]_{t=0}^{t=\pi/2} = [(a \cos t) \cdot (b \sin t) \cdot (ct)]_0^{\pi/2} \\ &= abc [t \cos t \sin t]_0^{\pi/2} = abc (0 - 0) = 0. \end{aligned}$$

Comprehensive Exercise 1

- Find $\int_C \mathbf{t} \cdot d\mathbf{r}$ where $\mathbf{t}$ is the unit tangent vector and C is the unit circle, in xy -plane, with centre at the origin. (Bundelkhand 2008)
- (i) Integrate the function $\mathbf{F} = x^2\mathbf{i} - xy\mathbf{j}$ from the point $(0, 0)$ to $(1, 1)$ along parabola $y^2 = x$.

- (ii) Evaluate $\int_C xy^3 ds$, where C is the segment of the line $y = 2x$ in the xy -plane from $(-1, -2)$ to $(1, 2)$.
3. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \mathbf{i} \cos y - \mathbf{j} x \sin y$ and C is the curve $y = \sqrt{1-x^2}$ in the xy -plane from $(1, 0)$ to $(0, 1)$. (Agra 2001)
4. (i) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}$ is $x^2 y^2 \mathbf{i} + y \mathbf{j}$ and C is $y^2 = 4x$ in the xy -plane from $(0, 0)$ to $(4, 4)$.
(Garhwal 2003; Meerut 04, 06; Kanpur 09, 11, 12; Rohilkhand 11)
- (ii) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = xy \mathbf{i} + yz \mathbf{j} + zx \mathbf{k}$ and C is the curve $\mathbf{r} = t\mathbf{i} + t^2 \mathbf{j} + t^3 \mathbf{k}$, t varying from -1 to $+1$.
- (iii) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = (2x+y) \mathbf{i} + (3y-x) \mathbf{j} + yzx \mathbf{k}$ and C is the curve $x = 2t^2$, $y = t$, $z = t^3$ from $t = 0$ to $t = 1$. (Kumaun 2011)
5. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = c [-3a \sin^2 t \cos t \mathbf{i} + a (2 \sin t - 3 \sin^3 t) \mathbf{j} + b \sin 2t \mathbf{k}]$ and C is given by $\mathbf{r} = a \cos t \mathbf{i} + a \sin t \mathbf{j} + bt \mathbf{k}$ from $t = \pi/4$ to $\pi/2$.
6. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = z \mathbf{i} + x \mathbf{j} + y \mathbf{k}$ and C is the arc of the curve $\mathbf{r} = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$ from $t = 0$ to $t = 2\pi$.
7. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = xy \mathbf{i} + (x^2 + y^2) \mathbf{j}$ and C is the x -axis from $x = 2$ to $x = 4$ and the straight line $x = 4$ from $y = 0$ to $y = 12$.
8. Find the work done in moving a particle in a force field $\mathbf{F} = 3x^2 \mathbf{i} + (2xz - y) \mathbf{j} + z \mathbf{k}$ along the line joining $(0, 0, 0)$ to $(2, 1, 3)$.
9. Calculate $\int_C [(x^2 + y^2) \mathbf{i} + (x^2 - y^2) \mathbf{j}] \cdot d\mathbf{r}$ where C is the curve :
(i) $y^2 = x$ joining $(0, 0)$ to $(1, 1)$. (Meerut 2005B)
(ii) $x^7 = y$ joining $(0, 0)$ to $(1, 1)$.
(iii) consisting of two straight lines joining $(0, 0)$ to $(1, 0)$ and $(1, 0)$ to $(1, 1)$.
(iv) consisting of three straight lines joining $(0, 0)$ to $(2, -2)$, $(2, -2)$ to $(0, -1)$ and $(0, -1)$ to $(1, 1)$.
10. Find the circulation of $\mathbf{F}$ round the curve C , where $\mathbf{F} = e^x \sin y \mathbf{i} + e^x \cos y \mathbf{j}$ and C is the rectangle whose vertices are $(0, 0)$, $(1, 0)$, $(1, \frac{1}{2}\pi)$, $(0, \frac{1}{2}\pi)$.
11. Find the circulation of $\mathbf{F}$ round the curve C , where $\mathbf{F} = (x - y) \mathbf{i} + (x + y) \mathbf{j}$ and C is the circle $x^2 + y^2 = 4$, $z = 0$.

12. (i) If $\mathbf{F} = (2x^2 + y^2) \mathbf{i} + (3y - 4x) \mathbf{j}$, evaluate $\int \mathbf{F} \cdot d\mathbf{r}$ around the triangle ABC whose vertices are $A(0, 0)$, $B(2, 0)$ and $C(2, 1)$.

(Kumaun 2008)

(ii) If $\mathbf{F} = (3x^2 + 6y) \mathbf{i} - 14yz \mathbf{j} + 20xz^2 \mathbf{k}$, then evaluate $\int \mathbf{F} \cdot d\mathbf{r}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the curve $x = t$, $y = t^2$, $z = t^3$.

(Kumaun 2012)

Answers 1

- | | | |
|--|--|------------------------|
| 1. 2π | 2. (i) $\frac{1}{12}$ (ii) $\frac{16}{\sqrt{5}}$ | 3. -1 |
| 4. (i) 264 (ii) $\frac{10}{7}$ | 5. $\frac{1}{2}c(a^2 + b^2)$ | 6. $2\pi + \pi = 3\pi$ |
| 7. 768 | 8. 16 | |
| 9. (i) $\frac{7}{10}$ (ii) $\frac{38}{45}$ | (iii) 1 (iv) $-\frac{7}{3}$ | 10. 0 |
| 11. 8π | 12. (i) $-14/3$ (ii) 5 | |

3 Surface Integrals

(Avadh 2014)

Any integral which is to be evaluated over a surface is called a surface integral.

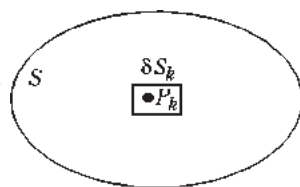
Suppose S is a surface of finite area. Suppose $f(x, y, z)$ is a single valued function of position defined over S . Subdivide the area S into n elements of areas $\delta S_1, \delta S_2, \dots, \delta S_n$. In each part δS_k we choose an arbitrary point P_k whose coordinates are (x_k, y_k, z_k) .

We define $f(P_k) = f(x_k, y_k, z_k)$.

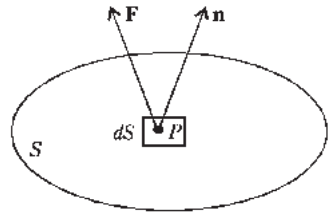
Form the sum $\sum_{k=1}^n f(P_k) \delta S_k$.

Now take the limit of this sum as $n \rightarrow \infty$ in such a way that the largest of the areas δS_k approaches zero. This limit if it exists, is called the *surface integral* of $f(x, y, z)$ over S and is denoted by $\iint_S f(x, y, z) dS$.

It can be shown that if the surface S is piecewise smooth and the function $f(x, y, z)$ is continuous over S , then the above limit exists *i.e.*, is independent of the choice of sub-divisions and points P_k .



Flux: Suppose S is a piecewise smooth surface and $\mathbf{F}(x, y, z)$ is a vector function of position defined and continuous over S . Let P be any point on the surface S and let $\mathbf{n}$ be the unit vector at P in the direction of **outward drawn normal** to the surface S at P . Then $\mathbf{F} \cdot \mathbf{n}$ is the normal component of $\mathbf{F}$ at P . The integral of $\mathbf{F} \cdot \mathbf{n}$ over S is $\iint_S \mathbf{F} \cdot \mathbf{n} dS$.



It is called the *flux* of $\mathbf{F}$ over S .

Let us associate with the differential of surface area dS a vector $d\mathbf{S}$ (called *vector area*) whose magnitude is dS and whose direction is that of $\mathbf{n}$. Then $d\mathbf{S} = \mathbf{n} dS$. Therefore we can write $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_S \mathbf{F} \cdot d\mathbf{S}$.

Suppose the outward drawn normal to the surface S at P makes angles α, β, γ with the positive directions of x, y and z axes respectively. If l, m, n are the direction cosines of the outward drawn normal, then

$$l = \cos \alpha, m = \cos \beta, n = \cos \gamma.$$

Also $\mathbf{n} = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k} = l \mathbf{i} + m \mathbf{j} + n \mathbf{k}$.

Let $\mathbf{F}(x, y, z) = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}$. Then

$$\mathbf{F} \cdot \mathbf{n} = F_1 \cos \alpha + F_2 \cos \beta + F_3 \cos \gamma = F_1 l + F_2 m + F_3 n.$$

Therefore, we can write

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iint_S (F_1 \cos \alpha + F_2 \cos \beta + F_3 \cos \gamma) dS \\ &= \iint_S (F_1 dy dz + F_2 dz dx + F_3 dx dy), \end{aligned}$$

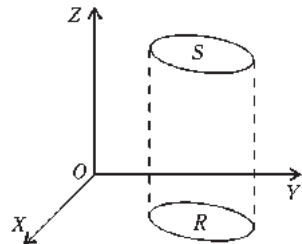
$$\text{if we define } \iint_S F_1 \cos \alpha dS = \iint_S F_1 dy dz,$$

$$\iint_S F_2 \cos \beta dS = \iint_S F_2 dz dx, \quad \iint_S F_3 \cos \gamma dS = \iint_S F_3 dx dy.$$

Note 1: Other examples of surface integrals are $\iint_S f \mathbf{n} dS, \iint_S \mathbf{F} \times d\mathbf{S}$

where $f(x, y, z)$ is a scalar function of position.

Note 2: Important. In order to evaluate surface integrals it is convenient to express them as double integrals taken over the orthogonal projection of the surface S on one of the coordinate planes. But this is possible only if any line perpendicular to the co-ordinate plane chosen meets the surface S in no more than one point. If the surface S does not satisfy this condition, then it can be sub-divided into surfaces which do satisfy this condition.



Suppose the surface S is such that any line perpendicular to the xy -plane meets S in no more than one point. Then the equation of the surface S can be written in the form $z = h(x, y)$.

Let R be the orthogonal projection of S on the xy -plane. If γ is the acute angle which the undirected normal $\mathbf{n}$ at $P(x, y, z)$ to the surface S makes with

z -axis, then it can be shown that $\cos \gamma \, dS = dx \, dy$,

where dS is the small element of area of surface S at the point P .

Therefore $dS = \frac{dx \, dy}{\cos \gamma} = \frac{dx \, dy}{|\mathbf{n} \cdot \mathbf{k}|}$, where $\mathbf{k}$ is the unit vector along z -axis.

Hence
$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \mathbf{F} \cdot \mathbf{n} \frac{dx \, dy}{|\mathbf{n} \cdot \mathbf{k}|}.$$

Thus the surface integral on S can be evaluated with the help of a double integral integrated over R .

Illustrative Examples

Example 9: Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, where $\mathbf{F} = yz \mathbf{i} + zx \mathbf{j} + xy \mathbf{k}$ and S is that part of the surface of the sphere $x^2 + y^2 + z^2 = 1$ which lies in the first octant.

(Garhwal 2003; kumaun 14))

Solution: A vector normal to the surface S is given by

$$\nabla (x^2 + y^2 + z^2) = 2x \mathbf{i} + 2y \mathbf{j} + 2z \mathbf{k}.$$

Therefore $\mathbf{n}$ = a unit normal to any point (x, y, z) of S

$$= \frac{2x \mathbf{i} + 2y \mathbf{j} + 2z \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k},$$

since $x^2 + y^2 + z^2 = 1$ on the surface S .

We have
$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \mathbf{F} \cdot \mathbf{n} \frac{dx \, dy}{|\mathbf{n} \cdot \mathbf{k}|},$$

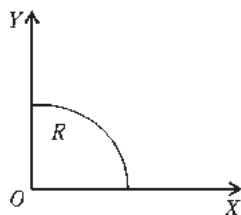
where R is the projection of S on the xy -plane.

The region R is bounded by x -axis, y -axis and the circle $x^2 + y^2 = 1$, $z = 0$.

We have $\mathbf{F} \cdot \mathbf{n} = (yz \mathbf{i} + zx \mathbf{j} + xy \mathbf{k}) \cdot (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) = 3xyz$.

Also $\mathbf{n} \cdot \mathbf{k} = (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) \cdot \mathbf{k} = z$. $\therefore |\mathbf{n} \cdot \mathbf{k}| = z$.

Hence
$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_R \frac{3xyz}{z} \, dx \, dy = 3 \iint_R xy \, dx \, dy \\ &= 3 \int_{\theta=0}^{\pi/2} \int_{r=0}^1 (r \cos \theta) (r \sin \theta) r \, d\theta \, dr, \text{ on changing to polars} \\ &= 3 \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_0^1 \cos \theta \sin \theta \, d\theta = \frac{3}{4} \left(\frac{1}{2} \right) = \frac{3}{8}. \end{aligned}$$



Example 10: Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, where $\mathbf{F} = z \mathbf{i} + x \mathbf{j} - 3y^2 z \mathbf{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.

Solution: A vector normal to the surface S is given by

$$\nabla (x^2 + y^2) = 2x \mathbf{i} + 2y \mathbf{j}.$$

Therefore $\mathbf{n}$ = a unit normal to any point of S

$$= \frac{2x \mathbf{i} + 2y \mathbf{j}}{\sqrt{4x^2 + 4y^2}} = \frac{x \mathbf{i} + y \mathbf{j}}{4},$$

since $x^2 + y^2 = 16$, on the surface S .

We have $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \mathbf{F} \cdot \mathbf{n} \frac{dx \, dz}{|\mathbf{n} \cdot \mathbf{j}|}$, where R is the projection of S on the x - z

plane. It should be noted that in this case we cannot take the projection of S on the x - y plane as the surface S is perpendicular to the x - y plane.

$$\text{Now} \quad \mathbf{F} \cdot \mathbf{n} = (z \mathbf{i} + x \mathbf{j} - 3y^2 z \mathbf{k}) \cdot \left(\frac{x \mathbf{i} + y \mathbf{j}}{4} \right) = \frac{1}{4} (xz + xy),$$

$$\mathbf{n} \cdot \mathbf{j} = \left(\frac{x \mathbf{i} + y \mathbf{j}}{4} \right) \cdot \mathbf{j} = \frac{y}{4}.$$

Therefore the required surface integral is

$$\begin{aligned} &= \iint_R \frac{xz + xy}{4} \frac{dx \, dz}{y/4} \\ &= \int_{z=0}^5 \int_{x=0}^4 \left(\frac{xz}{\sqrt{16-x^2}} + x \right) dx \, dz, \text{ since } y = \sqrt{16-x^2} \text{ on } S \\ &= \int_0^5 (4z + 8) dz = 90. \end{aligned}$$

4 Volume Integrals

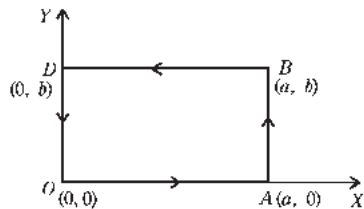
(Avadh 2014)

Suppose V is a volume bounded by a surface S . Suppose $f(x, y, z)$ is a single valued function of position defined over V . Subdivide the volume V into n elements of volumes $\delta V_1, \delta V_2, \dots, \delta V_n$. In each part δV_k we choose an arbitrary point P_k whose co-ordinates are (x_k, y_k, z_k) . We define

$$f(P_k) = f(x_k, y_k, z_k).$$

Form the sum

$$\sum_{k=1}^n f(P_k) \delta V_k.$$



Now take the limit of this sum as $n \rightarrow \infty$ in such a way that the largest of the volumes δV_k approaches zero. This limit, if it exists, is called the volume integral of $f(x, y, z)$ over V and is denoted by $\iiint_V f(x, y, z) dV$.

It can be shown that if the surface is piecewise smooth and the function $f(x, y, z)$ is continuous over V , then the above limit exists i.e., is independent of the choice of sub-divisions and points P_k .

If we subdivide the volume V into small cuboids by drawing lines parallel to the three co-ordinates axes, then $dV = dx dy dz$ and the above volume integral becomes $\iiint_V f(x, y, z) dx dy dz$.

If $\mathbf{F}(x, y, z)$ is a vector function, then $\iiint_V \mathbf{F} dV$ is also an example of a volume integral.

Example 11: If $\mathbf{F} = (2x^2 - 3z) \mathbf{i} - 2xy \mathbf{j} - 4x \mathbf{k}$, then evaluate $\iiint_V \nabla \cdot \mathbf{F} dV$ where V is the closed region bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$. Also Evaluate $\iiint_V \nabla \times \mathbf{F} dV$.

Solution: We have $\mathbf{F} = (2x^2 - 3z) \mathbf{i} - 2xy \mathbf{j} - 4x \mathbf{k}$.

$$\begin{aligned} \therefore \nabla \cdot \mathbf{F} &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot [(2x^2 - 3z) \mathbf{i} - 2xy \mathbf{j} - 4x \mathbf{k}] \\ &= \frac{\partial}{\partial x} (2x^2 - 3z) + \frac{\partial}{\partial y} (-2xy) + \frac{\partial}{\partial z} (-4x) = 4x - 2x = 2x. \end{aligned}$$

$$\begin{aligned} \therefore \iiint_V \nabla \cdot \mathbf{F} dV &= \iiint_V 2x dx dy dz \quad [\because dV = dx dy dz] \\ &= 2 \int_{x=0}^2 \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} x dx dy dz. \end{aligned}$$

[Note that we have taken a thin column parallel to z -axis as the elementary volume. It cuts the boundary at $z = 0$ and $z = 4 - 2x - 2y$. Also the projection of the plane $2x + 2y + z = 4$ on the xy -plane is bounded by the axes $y = 0$, $x = 0$ and the line $x + y = 2$. Hence the limits for y are from 0 to $2 - x$ and those for x are from 0 to 2]

$$\begin{aligned} \therefore \iiint_V \nabla \cdot \mathbf{F} dV &= 2 \int_{x=0}^2 \int_{y=0}^{2-x} x \left[z \right]_{z=0}^{4-2x-2y} dx dy \\ &= 2 \int_{x=0}^2 \int_{y=0}^{2-x} x (4 - 2x - 2y) dx dy \\ &= 2 \int_{x=0}^2 \left[4xy - 2x^2 y - xy^2 \right]_{y=0}^{2-x} dx \\ &= 2 \int_0^2 [4x(2-x) - 2x^2(2-x) - x(2-x)^2] dx \\ &= 2 \int_0^2 [x^3 - 4x^2 + 4x] dx, \text{ on simplifying} \\ &= 2 \left[\frac{1}{4} x^4 - \frac{4}{3} x^3 + 2x^2 \right]_0^2 = 2 \left[4 - \frac{32}{3} + 8 \right] = \frac{8}{3}. \end{aligned}$$

Second part: We have

$$\begin{aligned}\nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^2 - 3z & -2xy & -4x \end{vmatrix} \\ &= \left[\frac{\partial}{\partial y} (-4x) - \frac{\partial}{\partial z} (-2xy) \right] \mathbf{i} - \left[\frac{\partial}{\partial x} (-4x) - \frac{\partial}{\partial z} (2x^2 - 3z) \right] \mathbf{j} \\ &\quad + \left[\frac{\partial}{\partial x} (-2xy) - \frac{\partial}{\partial y} (2x^2 - 3z) \right] \mathbf{k} \\ &= 0 \mathbf{i} - (-4 + 3) \mathbf{j} + (-2y) \mathbf{k} = \mathbf{j} - 2y \mathbf{k} .\end{aligned}$$

$$\begin{aligned}\therefore \iiint_V \nabla \times \mathbf{F} dV &= \iiint_V (\mathbf{j} - 2y \mathbf{k}) dx dy dz \\ &= \int_{x=0}^2 \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} (\mathbf{j} - 2y \mathbf{k}) dx dy dz \\ &= \int_{x=0}^2 \int_{y=0}^{2-x} (\mathbf{j} - 2y \mathbf{k}) (4 - 2x - 2y) dx dy \\ &= \int_{x=0}^2 \left[\mathbf{j} (4y - 2xy - y^2) - 2 \mathbf{k} (2y^2 - xy^2 - \frac{2}{3} y^3) \right]_{y=0}^{2-x} dx \\ &= \int_{x=0}^2 \left[\mathbf{j} (2-x) (4-2x-2+x) \right. \\ &\quad \left. - 2 \mathbf{k} (2-x)^2 \left\{ 2-x - \frac{2}{3} (2-x) \right\} \right] dx \\ &= \int_0^2 \left[(2-x)^2 \mathbf{j} - \frac{2}{3} (2-x)^3 \mathbf{k} \right] dx \\ &= \int_0^2 \left[(x-2)^2 \mathbf{j} + \frac{2}{3} (x-2)^3 \mathbf{k} \right] dx \\ &= \left[\frac{(x-2)^3}{3} \right]_0^2 \mathbf{j} + \left[\frac{2}{3} \frac{(x-2)^4}{4} \right]_0^2 \mathbf{k} = \frac{8}{3} \mathbf{j} - \frac{8}{3} \mathbf{k} = \frac{8}{3} (\mathbf{j} - \mathbf{k}).\end{aligned}$$

Comprehensive Exercise 2

1. (i) Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} dS$, where $\mathbf{F} = 18z \mathbf{i} - 12 \mathbf{j} + 3y \mathbf{k}$ and S is the surface of the plane $2x + 3y + 6z = 12$ in the first octant.
(Bundelkhand 2005)
- (ii) Evaluate $\iiint_V \phi dV$, where $\phi = 45 x^2 y$ and V is the closed region bounded by the planes $4x + 2y + z = 8$, $x = 0$, $y = 0$, $z = 0$.

2. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, where $\mathbf{F} = (x + y^2) \mathbf{i} - 2x \mathbf{j} + 2yz \mathbf{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant.
3. Evaluate $\iint_S \mathbf{A} \cdot \mathbf{n} \, dS$, where $\mathbf{A} = xy \mathbf{i} - x^2 \mathbf{j} + (x + z) \mathbf{k}$, S is the portion of the plane $2x + 2y + z = 6$ included in the first octant and $\mathbf{n}$ is a unit normal to S .
4. If $\mathbf{F} = 2y\mathbf{i} - 3\mathbf{j} + x^2 \mathbf{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, then evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$.
5. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, where $\mathbf{F} = y\mathbf{i} + 2x\mathbf{j} - z \mathbf{k}$ and S is the surface of the plane $2x + y = 6$ in the first octant cut off by the plane $z = 4$.
6. Evaluate $\iiint_V \mathbf{F} \, dV$ where $\mathbf{F} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$ and V is the region bounded by the surfaces $x = 0$, $x = 2$, $y = 0$, $y = 6$, $z = 4$ and $z = x^2$.

Answers 2

- | | | |
|-----------|---|--------|
| 1. (i) 24 | (ii) 128 | |
| 2. 81 | 3. $\frac{27}{4}$ | 4. 132 |
| 5. 108 | 6. $24 \mathbf{i} + 96 \mathbf{j} + \frac{384}{5} \mathbf{k}$ | |

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If $\mathbf{F} = x^2 \mathbf{i} + y^3 \mathbf{j}$ and curve C is the arc of the parabola $y = x^2$ in the xy -plane from $(0, 0)$ to $(1, 1)$, then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is

(a) $7/12$	(b) $5/12$	(c) $11/12$	(d) None of these
------------	------------	-------------	-------------------
2. The work done in moving a particle in a force field

$$\mathbf{F} = 3x^2 \mathbf{i} + (2xz - y) \mathbf{j} + 3z \mathbf{k}$$
 along the line joining $(0, 0, 0)$ to $(2, 1, 3)$ is

(a) 12	(b) 16	(c) 0	(d) 20
--------	--------	-------	--------

3. For a closed surface S , the value of $\iint_S \mathbf{r} \cdot \mathbf{n} \, dS$ is

- (a) V (b) $2V$ (c) $3V$ (d) 0

where V is the volume enclosed by S .

(Kumaun 2013)

Fill in the Blank(s)

Fill in the blanks “.....”, so that the following statements are complete and correct.

- Any integral which is to be evaluated along a curve is called a
- Any integral which is to be evaluated over a surface is called a
- If $\mathbf{r}$ denotes the position vector of a point on a curve C , then $\frac{d\mathbf{r}}{ds}$ is a unit vector along the to C at the point $\mathbf{r}$ in the direction of s increasing.
- $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ is called the of $\mathbf{F}$ over S .
- If $\mathbf{t}$ is the unit tangent vector and C is the unit circle in $x-y$ -plane, with centre at the origin, then $\int_C \mathbf{t} \cdot d\mathbf{r} = \dots\dots$.
- The value of $\int (x \, dy - y \, dx)$ around the circle $x^2 + y^2 = 1$ is
- If $\mathbf{F} = (3x^2 + 6y) \mathbf{i} - 14yz \mathbf{j} + 20xz^2 \mathbf{k}$ and C is a straight line joining $(0, 0, 0)$ to $(1, 1, 1)$, then $\int_C \mathbf{F} \cdot d\mathbf{r} = \dots\dots$.

True or False

Write ‘T’ for true and ‘F’ for false statement.

- If C is a simple closed curve, then $\oint_C \mathbf{F} \cdot d\mathbf{r}$ is called the circulation of $\mathbf{F}$ about C .
- If $\mathbf{F} = ax \mathbf{i} + by \mathbf{j} + cz \mathbf{k}$, a, b, c are constants, then $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \frac{2}{3} \pi (a + b + c)$, where S is the surface of a unit sphere.

Answers

Multiple Choice Questions

1. (a) 2. (b) 3. (c)

Fill in the Blank(s)

- | | |
|------------------|---------------------|
| 1. line integral | 2. surface integral |
| 3. tangent | 4. flux |
| 5. 2π | 6. 2π 7. $13/3$ |

True or False

1. T 2. F



Chapter

6



Green's, Gauss's and Stoke's Theorems

1 Green's Theorem in the Plane

Let R be a closed bounded region in the x - y plane whose boundary C consists of finitely many smooth curves. Let M and N be continuous functions of x and y having continuous partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ in R . Then

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy = \oint_C (M \, dx + N \, dy),$$

the line integral being taken along the entire boundary C of R such that R is on the left as one advances in the direction of integration.

(Meerut 2004, 05, 07, 10; Avadh 10; Purvanchal 08; Rohilkhand 09B, 12; Kashi 13)

Proof: We shall first prove the theorem for a special region R bounded by a closed curve C and having the property that any straight line parallel to any one of the coordinate axes and intersecting R has only one segment (or a single point) in common with R . This means that R can be represented in both of the forms

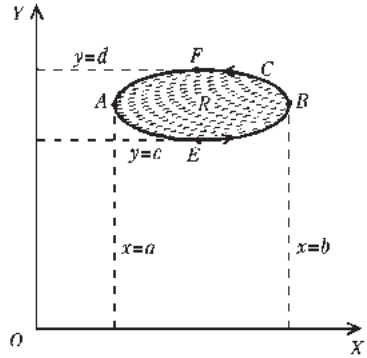
$$a \leq x \leq b, \quad f(x) \leq y \leq g(x)$$

and
$$c \leq y \leq d, \quad p(y) \leq x \leq q(y).$$

In the adjoining figure, the equations of the curves AEB and BFA are $y = f(x)$ and $y = g(x)$ respectively. Similarly the equations of the curves FAE and EBF are $x = p(y)$ and $x = q(y)$ respectively.

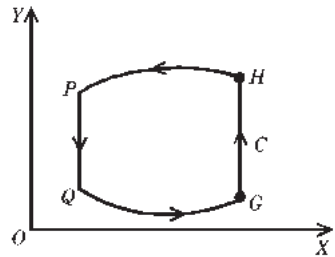
We have

$$\begin{aligned}
 & \iint_R \frac{\partial M}{\partial y} dx dy \\
 &= \int_{x=a}^b \left[\int_{y=f(x)}^{g(x)} \frac{\partial M}{\partial y} dy \right] dx \\
 &= \int_{x=a}^b \left[M(x, y) \right]_{y=f(x)}^{y=g(x)} dx \\
 &= \int_{x=a}^b [M(x, g(x)) - M(x, f(x))] dx \\
 &= - \int_a^b M(x, f(x)) dx - \int_b^a M(x, g(x)) dx \\
 &= - \oint_C M(x, y) dx, \text{ since } y = f(x) \text{ represents the curve } AEB
 \end{aligned}$$



and $y = g(x)$ represents the curve BFA .

If portions of C are segments parallel to y -axis such as GH and PQ in the adjoining figure, then the above result is not affected. The line integral $\int M dx$ over GH is zero because on GH , we have $x = \text{constant}$ implies $dx = 0$. Similarly the line integral over PQ is zero. The equations of QG and HP are $y = f(x)$ and $y = g(x)$ respectively. Hence we have



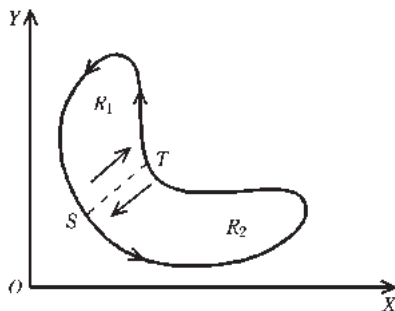
$$- \iint_R \frac{\partial M}{\partial y} dx dy = \oint_C M(x, y) dx. \quad \dots (1)$$

Similarly,

$$\begin{aligned}
 \iint_R \frac{\partial N}{\partial x} dx dy &= \int_{y=c}^d \left[\int_{x=p(y)}^{q(y)} \frac{\partial N}{\partial x} dx \right] dy \\
 &= \int_{y=c}^d \left[N(x, y) \right]_{x=p(y)}^{x=q(y)} dy \\
 &= \int_{y=c}^d [N(q(y), y) - N(p(y), y)] dy \\
 &= \int_c^d N(q(y), y) dy + \int_d^c N(p(y), y) dy \\
 &= \oint_C N(x, y) dy. \quad \dots (2)
 \end{aligned}$$

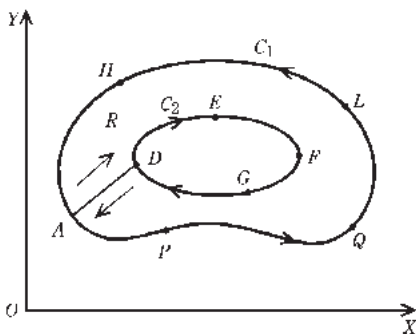
From (1) and (2), we get on adding $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \oint_C (M dx + N dy)$.

The proof of the theorem can now be extended to a region R which can be subdivided into finitely many special regions of the above type by drawing lines (TS in the adjoining figure). In this case we apply the theorem to each subregion (R_1 and R_2 in the figure) and then add the results. The sum of the left hand members will be equal to the integral over R . The sum of the right hand members will be equal to the integral over C plus the line integrals over the curves introduced for subdividing R . Each of the latter integrals comes twice, taken once in each direction (as ST and TS in the figure). Therefore these two integrals cancel each other and thus the sum of the right hand members will be equal to the line integral over C .



Note: Extension of Green's theorem in plane to multiply connected regions.

Green's theorem in the plane is also valid for a multiply-connected region R such as shown in the adjoining figure. Here the boundary C of R consists of two parts; the exterior boundary C_1 is traversed in the anticlockwise sense so that R is on the left, while the interior boundary C_2 is traversed in the clockwise sense so that R is on the left.



In order to establish the theorem, we construct a line such as AD (called a *cross cut*) connecting the exterior and interior boundaries. The region bounded by $ADEFGDAPQLHA$ is simply-connected and so Green's theorem is valid for it. Therefore

$$\oint_{ADEFGDAPQLHA} M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy.$$

The integral on the left hand side leaving out the integrand is equal to

$$\int_{AD} + \int_{C_2} + \int_{DA} + \int_{C_1} = \int_{C_2} + \int_{C_1}, \text{ since } \int_{AD} = -\int_{DA} \\ = \oint_C (M dx + N dy).$$

Hence the theorem.

2 Green's Theorem in the Plane in Vector Notation

We have $\mathbf{r} = x \mathbf{i} + y \mathbf{j}$ so that $d\mathbf{r} = dx \mathbf{i} + dy \mathbf{j}$.

Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$.

Then $M dx + N dy = (M \mathbf{i} + N \mathbf{j}) \cdot (dx \mathbf{i} + dy \mathbf{j}) = \mathbf{F} \cdot d\mathbf{r}$.

Also $\text{curl} \mathbf{F} = \nabla \times \mathbf{F}$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & 0 \end{vmatrix} = -\frac{\partial N}{\partial z} \mathbf{i} + \frac{\partial M}{\partial z} \mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mathbf{k}.$$

$$\therefore (\nabla \times \mathbf{F}) \cdot \mathbf{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}.$$

Hence Green's theorem in plane can be written as

$$\iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} dR = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

where $dR = dx dy$ and $\mathbf{k}$ is unit vector perpendicular to the xy -plane.

If s denotes the arc length of C and $\mathbf{t}$ is the unit tangent vector to C , then

$d\mathbf{r} = \frac{d\mathbf{r}}{ds} ds = \mathbf{t} ds$. Therefore the above result can also be written as

$$\iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} dR = \oint_C \mathbf{F} \cdot \mathbf{t} ds.$$

Illustrative Examples

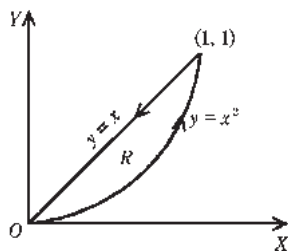
Example 1: Verify Green's theorem in the plane for $\oint_C (xy + y^2) dx + x^2 dy$, where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. (Rohilkhand 2011)

Solution: By Green's theorem in plane, we have

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \oint_C (M dx + N dy).$$

Here $M = xy + y^2$, $N = x^2$.

The curves $y = x$ and $y = x^2$ intersect at $(0, 0)$ and $(1, 1)$. The positive direction in traversing C is as shown in the figure.



$$\text{We have } \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \iint_R \left[\frac{\partial}{\partial x} (x^2) - \frac{\partial}{\partial y} (xy + y^2) \right] dx dy$$

$$= \iint_R (2x - x - 2y) dx dy = \iint_R (x - 2y) dx dy$$

$$\begin{aligned}
 &= \int_{x=0}^1 \int_{y=x^2}^x (x-2y) dy dx = \int_{x=0}^1 \left[xy - y^2 \right]_{y=x^2}^x dx \\
 &= \int_0^1 [x^2 - x^2 - x^3 + x^4] dx = \int_0^1 (x^4 - x^3) dx \\
 &= \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}.
 \end{aligned}$$

Now let us evaluate the line integral along C . Along $y = x^2$, $dy = 2x dx$. Therefore along $y = x^2$, the line integral equals

$$\int_0^1 [\{(x)(x^2) + x^4\} dx + x^2(2x) dx] = \int_0^1 (3x^3 + x^4) dx = \frac{19}{20}.$$

Along $y = x$, $dy = dx$. Therefore along $y = x$, the line integral equals

$$\int_1^0 [\{(x)(x) + x^2\} dx + x^2 dx] = \int_1^0 3x^4 dx = -1.$$

Therefore the required line integral $= \frac{19}{20} - 1 = -\frac{1}{20}$. Hence the theorem is verified.

Example 2: Evaluate by Green's theorem $\oint_C (x^2 - \cosh y) dx + (y + \sin x) dy$, where C is the rectangle with vertices $(0, 0)$, $(\pi, 0)$, $(\pi, 1)$, $(0, 1)$.

(Meerut 2002, 05B, 06, 13B; Rohilkhand 14)

Solution: By Green's theorem in plane, we have

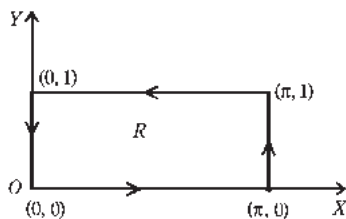
$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \oint_C (M dx + N dy).$$

Here $M = x^2 - \cosh y$, $N = y + \sin x$.

$$\therefore \frac{\partial N}{\partial x} = \cos x, \quad \frac{\partial M}{\partial y} = -\sinh y.$$

Hence the given line integral is equal to

$$\begin{aligned}
 &\iint_R (\cos x + \sinh y) dx dy \\
 &= \int_{x=0}^{\pi} \int_{y=0}^1 (\cos x + \sinh y) dy dx \\
 &= \int_{x=0}^{\pi} \left[y \cos x + \cosh y \right]_{y=0}^1 dx \\
 &= \int_{x=0}^{\pi} [\cos x + \cosh 1 - 1] dx = \left[\sin x + x \cosh 1 - x \right]_0^{\pi} = (\cosh 1 - 1).
 \end{aligned}$$



Example 3: Show that the area bounded by a simple closed curve C is given by $\frac{1}{2} \oint_C (x dy - y dx)$. Hence find the area of the ellipse $x = a \cos \theta$, $y = b \sin \theta$.

Solution: By Green's theorem in plane, if R is a plane region bounded by a simple closed curve C , then

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \oint_C M dx + N dy.$$

Putting $M = -y$, $N = x$, we get

$$\begin{aligned} \oint_C (x dy - y dx) &= \iint_R \left[\frac{\partial}{\partial x} (x) - \frac{\partial}{\partial y} (-y) \right] dx dy \\ &= 2 \iint_R dx dy \end{aligned}$$

$= 2 A$, where A is the area bounded by C .

Hence
$$A = \frac{1}{2} \oint_C (x dy - y dx).$$

The area of the ellipse

$$\begin{aligned} &= \frac{1}{2} \oint_C (x dy - y dx) = \frac{1}{2} \int_{\theta=0}^{2\pi} \left(a \cos \theta \frac{dy}{d\theta} - b \sin \theta \frac{dx}{d\theta} \right) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} (ab \cos^2 \theta + ab \sin^2 \theta) d\theta = \frac{1}{2} ab \int_0^{2\pi} d\theta = \pi ab. \end{aligned}$$

Comprehensive Exercise 1

1. Verify Green's theorem in the plane for

$$\int_C [(2xy - x^2) dx + (x^2 + y^2) dy],$$

where C is the boundary of the region enclosed by $y = x^2$ and $y^2 = x$ described in the positive sense.

2. Verify Green's theorem in the plane for

$$\oint_C [(3x^2 - 8y^2) dx + (4y - 6xy) dy],$$

where C is the boundary of the region defined by $y = \sqrt{x}$, $y = x^2$.

3. Apply Green's theorem in the plane to evaluate

$$\int_C \{(y - \sin x) dx + \cos x dy\},$$

where C is the triangle enclosed by the lines $y = 0$, $x = 2\pi$, $\pi y = 2x$.

(Avadh 2010)

4. Evaluate by Green's theorem in plane $\int_C (e^{-x} \sin y dx + e^{-x} \cos y dy)$,

where C is the rectangle with vertices $(0, 0)$, $(\pi, 0)$, $\left(\pi, \frac{1}{2}\pi\right)$, $\left(0, \frac{1}{2}\pi\right)$.

5. Evaluate by Green's theorem $\oint_C (\cos x \sin y - xy) dx + \sin x \cos y dy$,

where C is the circle $x^2 + y^2 = 1$.

(Kumaun 2012)

6. If $\mathbf{F} = (x^2 - y^2) \mathbf{i} + 2xy \mathbf{j}$ and $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$, find the value of $\int \mathbf{F} \cdot d\mathbf{r}$ around the rectangular boundary $x = 0$, $x = a$, $y = 0$, $y = b$.

7. Verify Green's theorem in the plane for

$$\int_C (x^2 - xy^3) dx + (y^2 - 2xy) dy,$$

where C is the square with vertices $(0, 0)$, $(2, 0)$, $(2, 2)$, $(0, 2)$.

(Meerut 2001)

8. Apply Green's theorem in the plane to evaluate

$$\int_C [(2x^2 - y^2) dx + (x^2 + y^2) dy], \text{ where } C \text{ is the boundary of the surface enclosed by the } x\text{-axis and the semi-circle } y = (1 - x^2)^{1/2}.$$

9. If C is the simple closed curve in the xy -plane not enclosing the origin,

$$\text{show that } \int_C \mathbf{F} \cdot d\mathbf{r} = 0, \text{ where } \mathbf{F} = \frac{-\mathbf{i} y + \mathbf{j} x}{x^2 + y^2}.$$

Answers 1

3. $-\frac{\pi}{4} - \frac{2}{\pi}$

4. $2(e^{-\pi} - 1)$

5. 0

6. $2ab^2$

3 The Divergence Theorem of Gauss

Suppose V is the volume bounded by a closed piecewise smooth surface S . Suppose $\mathbf{F}(x, y, z)$ is a vector function of position which is continuous and has continuous first partial derivatives in V . Then
$$\iiint_V \nabla \cdot \mathbf{F} dV = \iint_S \mathbf{F} \cdot \mathbf{n} dS,$$

where $\mathbf{n}$ is the outward drawn unit normal vector to S .

(Meerut 2000, 01, 06, 10B, 12, 12B;
Bundelkhand 09, 11; Avadh 14; Kashi 14)

Since $\mathbf{F} \cdot \mathbf{n}$ is the normal component of vector $\mathbf{F}$, therefore divergence theorem may also be stated as follows :

The surface integral of the normal component of a vector $\mathbf{F}$ taken over a closed surface is equal to the integral of the divergence of $\mathbf{F}$ taken over the volume enclosed by the surface.

Cartesian equivalent of Divergence Theorem:

Let $\mathbf{F} = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}.$

Then $\nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$

If α, β, γ are the angles which outward drawn unit normal $\mathbf{n}$ makes with positive directions of x, y, z -axes, then $\cos \alpha, \cos \beta, \cos \gamma$ are direction cosines of $\mathbf{n}$ and we have
$$\mathbf{n} = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}.$$

$$\begin{aligned} \therefore \mathbf{F} \cdot \mathbf{n} &= (F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}) \cdot (\cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}) \\ &= F_1 \cos \alpha + F_2 \cos \beta + F_3 \cos \gamma. \end{aligned}$$

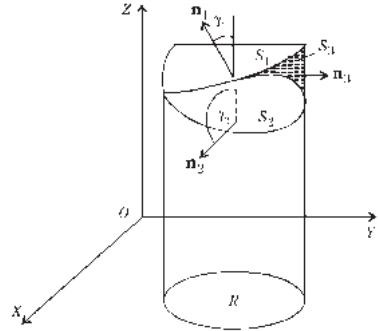
Therefore the divergence theorem can be written as

$$\begin{aligned} & \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dx \, dy \, dz \\ &= \iint_S (F_1 \cos \alpha + F_2 \cos \beta + F_3 \cos \gamma) dS \\ &= \iint F_1 \, dy \, dz + F_2 \, dz \, dx + F_3 \, dx \, dy). \end{aligned}$$

The significance of divergence theorem lies in the fact that a surface integral may be expressed as a volume integral and vice versa.

Proof of the divergence theorem:

We shall first prove the theorem for a special region V which is bounded by a piecewise smooth closed surface S and has the property that any straight line parallel to any one of the coordinate axes and intersecting V has only one segment (or a single point) in common with V . If R is the orthogonal projection of S on the xy -plane, then V can be represented in the form $f(x, y) \leq z \leq g(x, y)$ where (x, y) varies in R .



Obviously $z = g(x, y)$ represents the upper portion S_1 of S , $z = f(x, y)$ represents the lower portion S_2 of S and there may be a remaining vertical portion S_3 of S .

We have

$$\begin{aligned} & \iiint_V \frac{\partial F_3}{\partial z} dV = \iiint_V \frac{\partial F_3}{\partial z} dx \, dy \, dz \\ &= \iint_R \left[\int_{z=f(x,y)}^{z=g(x,y)} \frac{\partial F_3}{\partial z} dz \right] dx \, dy \\ &= \iint_R [F_3(x, y, z)]_{z=f(x,y)}^{z=g(x,y)} dx \, dy \\ &= \iint_R [F_3[x, y, g(x, y)] - F_3[x, y, f(x, y)]] dx \, dy \\ &= \iint_R F_3[x, y, g(x, y)] dx \, dy \\ &\quad - \iint_R F_3[x, y, f(x, y)] dx \, dy \quad \dots(1) \end{aligned}$$

Now for the vertical portion S_3 of S , the normal $\mathbf{n}_3$ to S_3 makes a right angle γ with $\mathbf{k}$. Therefore $\iint_{S_3} F_3 \mathbf{k} \cdot \mathbf{n}_3 \, dS_3 = 0$, since $\mathbf{k} \cdot \mathbf{n}_3 = 0$.

For the upper portion S_1 of S , the normal $\mathbf{n}_1$ to S_1 makes an acute angle γ_1 with $\mathbf{k}$. Therefore $\mathbf{k} \cdot \mathbf{n}_1 \, dS_1 = \cos \gamma_1 \, dS_1 = dx \, dy$.

Hence $\iint_{S_1} F_3 \mathbf{k} \cdot \mathbf{n}_1 \, dS_1 = \iint_R F_3[x, y, g(x, y)] dx \, dy$.

For the lower portion S_2 of S , the normal $\mathbf{n}_2$ to S_2 makes an obtuse angle γ_2 with $\mathbf{k}$.

Therefore $\mathbf{k} \cdot \mathbf{n}_2 dS_2 = \cos \gamma_2 dS_2 = -dx dy$.

Hence $\iint_{S_1} F_3 \mathbf{k} \cdot \mathbf{n}_2 dS_2 = - \iint_R F_3 [x, y, f(x, y)] dx dy$.

$$\begin{aligned} \therefore \iint_S F_3 \mathbf{k} \cdot \mathbf{n}_3 dS_3 + \iint_{S_1} F_3 \mathbf{k} \cdot \mathbf{n}_1 dS_1 + \iint_{S_2} F_3 \mathbf{k} \cdot \mathbf{n}_2 dS_2 \\ = 0 + \iint_R F_3 [x, y, g(x, y)] dx dy \\ - \iint_R F_3 [x, y, f(x, y)] dx dy \end{aligned}$$

or with the help of (1), we get

$$\iint_S F_3 \mathbf{k} \cdot \mathbf{n} dS = \iiint_V \frac{\partial F_3}{\partial z} dV. \quad \dots(2)$$

Similarly, by projecting S on the other co-ordinate planes, we get

$$\iint_S F_2 \mathbf{j} \cdot \mathbf{n} dS = \iiint_V \frac{\partial F_2}{\partial y} dV \quad \dots(3)$$

$$\text{and} \quad \iint_S F_1 \mathbf{i} \cdot \mathbf{n} dS = \iiint_V \frac{\partial F_1}{\partial x} dV \quad \dots(4)$$

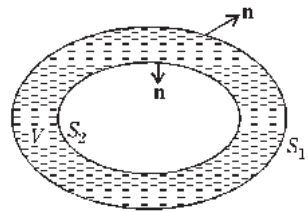
Adding (2), (3) and (4), we get

$$\iint_S (F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}) \cdot \mathbf{n} dS = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dV$$

$$\text{or} \quad \iiint_V \nabla \cdot \mathbf{F} dV = \iint_S \mathbf{F} \cdot \mathbf{n} dS.$$

The proof of the theorem can now be extended to a region V which can be subdivided into finitely many special regions of the above type by drawing auxiliary surfaces. In this case we apply the theorem to each sub-region and then add the results. The sum of the volume integrals over parts of V will be equal to the volume integral over V . The surface integrals over auxiliary surfaces cancel in pairs, while the sum of the remaining surface integrals is equal to the surface integral over the whole boundary S of V .

Note: The divergence theorem is applicable for a region V if it is bounded by two closed surfaces S_1 and S_2 one of which lies within the other. Here outward drawn normals will have the directions as shown in the figure.



4 Some Deductions from Divergence Theorem

1. Green's theorem: Let ϕ and ψ be scalar point functions which together with their derivatives in any direction are uniform and continuous within the region V bounded by a closed surface S , then

$$\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} dS.$$

Proof: By divergence theorem, we have $\iiint_V \nabla \cdot \mathbf{F} dV = \iint_S \mathbf{F} \cdot \mathbf{n} dS$.

Putting $\mathbf{F} = \phi \nabla \psi$, we get $\nabla \cdot \mathbf{F} = \nabla \cdot (\phi \nabla \psi)$
 $= \phi (\nabla \cdot \nabla \psi) + (\nabla \phi) \cdot (\nabla \psi) = \phi \nabla^2 \psi + (\nabla \phi) \cdot (\nabla \psi).$

Also $\mathbf{F} \cdot \mathbf{n} = (\phi \nabla \psi) \cdot \mathbf{n}.$

$\therefore$ divergence theorem gives

$$\iiint_V [\phi \nabla^2 \psi + (\nabla \phi) \cdot (\nabla \psi)] dV = \iint_S (\phi \nabla \psi) \cdot \mathbf{n} dS \quad \dots(1)$$

This is called *Green's first identity* or *theorem*.

Interchanging ϕ and ψ in (1), we get

$$\iiint_V [\psi \nabla^2 \phi + (\nabla \psi) \cdot (\nabla \phi)] dV = \iint_S [\psi \nabla \phi] \cdot \mathbf{n} dS \quad \dots(2)$$

Subtracting (2) from (1), we get

$$\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} dS \quad \dots(3)$$

This is called *Green's second identity* or *Green's theorem in symmetrical form*.

Since $\nabla \psi = \frac{\partial \psi}{\partial n} \mathbf{n}$ and $\nabla \phi = \frac{\partial \phi}{\partial n} \mathbf{n}$, therefore

$$\begin{aligned} (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} &= \left(\phi \frac{\partial \psi}{\partial n} \mathbf{n} - \psi \frac{\partial \phi}{\partial n} \mathbf{n} \right) \cdot \mathbf{n} \\ &= \phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n}. \end{aligned}$$

Hence (3) can also be written as

$$\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = \iint_S \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) dS.$$

Note: Harmonic function: If a scalar point function ϕ satisfies Laplace's equation $\nabla^2 \phi = 0$, then ϕ is called *harmonic function*. If ϕ and ψ are both harmonic functions, then $\nabla^2 \phi = 0$, $\nabla^2 \psi = 0$.

Hence from Green's second identity, we get $\iint_S \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) dS = 0$.

2. Prove that $\iiint_V \nabla \phi dV = \iint_S \phi \mathbf{n} dS$.

Proof: By divergence theorem, we have $\iiint_V \nabla \cdot \mathbf{F} dV = \iint_S \mathbf{F} \cdot \mathbf{n} dS$.

Taking $\mathbf{F} = \phi \mathbf{C}$ where $\mathbf{C}$ is an arbitrary constant non-zero vector, we get

$$\iiint_V \nabla \cdot (\phi \mathbf{C}) dV = \iint_S (\phi \mathbf{C}) \cdot \mathbf{n} dS. \quad \dots(1)$$

Now $\nabla \cdot (\phi \mathbf{C}) = (\nabla \phi) \cdot \mathbf{C} + \phi(\nabla \cdot \mathbf{C}) = (\nabla \phi) \cdot \mathbf{C}$, since $\nabla \cdot \mathbf{C} = 0$.

Also $(\phi \mathbf{C}) \cdot \mathbf{n} = \mathbf{C} \cdot (\phi \mathbf{n})$.

$\therefore$ (1) becomes

$$\iiint_V \mathbf{C} \cdot (\nabla \phi) dV = \iint_S \mathbf{C} \cdot (\phi \mathbf{n}) dS$$

$$\text{or} \quad \mathbf{C} \cdot \iiint_V \nabla \phi dV = \mathbf{C} \cdot \iint_S (\phi \mathbf{n}) dS$$

$$\text{or} \quad \mathbf{C} \cdot \left[\iiint_V \nabla \phi dV - \iint_S \phi \mathbf{n} dS \right] = 0.$$

Since $\mathbf{C}$ is an arbitrary vector, therefore we must have

$$\iiint_V \nabla \phi dV = \iint_S \phi \mathbf{n} dS.$$

$$3. \text{ Prove that } \iiint_V \nabla \times \mathbf{B} dV = \iint_S \mathbf{n} \times \mathbf{B} dS.$$

Proof: In divergence theorem taking $\mathbf{F} = \mathbf{B} \times \mathbf{C}$, where $\mathbf{C}$ is an arbitrary constant vector, we get

$$\iiint_V \nabla \cdot (\mathbf{B} \times \mathbf{C}) dV = \iint_S (\mathbf{B} \times \mathbf{C}) \cdot \mathbf{n} dS. \quad \dots(1)$$

Now $\nabla \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{C} \cdot \text{curl } \mathbf{B} - \mathbf{B} \cdot \text{curl } \mathbf{C} = \mathbf{C} \cdot \text{curl } \mathbf{B}$, since $\text{curl } \mathbf{C} = \mathbf{0}$.

Also $(\mathbf{B} \times \mathbf{C}) \cdot \mathbf{n} = [\mathbf{B}, \mathbf{C}, \mathbf{n}] = [\mathbf{C}, \mathbf{n}, \mathbf{B}] = \mathbf{C} \cdot (\mathbf{n} \times \mathbf{B})$.

$\therefore$ (1) becomes

$$\iiint_V (\mathbf{C} \cdot \text{curl } \mathbf{B}) dV = \iint_S \mathbf{C} \cdot (\mathbf{n} \times \mathbf{B}) dS$$

$$\text{or} \quad \mathbf{C} \cdot \iiint_V (\nabla \times \mathbf{B}) dV = \mathbf{C} \cdot \iint_S (\mathbf{n} \times \mathbf{B}) dS$$

$$\text{or} \quad \mathbf{C} \cdot \left[\iiint_V (\nabla \times \mathbf{B}) dV - \iint_S (\mathbf{n} \times \mathbf{B}) dS \right] = 0.$$

Since $\mathbf{C}$ is an arbitrary vector therefore we can take $\mathbf{C}$ as a non-zero vector which is not perpendicular to the vector

$$\iiint_V (\nabla \times \mathbf{B}) dV - \iint_S (\mathbf{n} \times \mathbf{B}) dS.$$

Hence we must have

$$\iiint_V (\nabla \times \mathbf{B}) dV - \iint_S (\mathbf{n} \times \mathbf{B}) dS = \mathbf{0}$$

$$\text{or} \quad \iiint_V (\nabla \times \mathbf{B}) dV = \iint_S (\mathbf{n} \times \mathbf{B}) dS.$$

Illustrative Examples

Example 4: (i) For any closed surface S , prove that $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = 0$.

(Meerut 2009B; Purvanchal 14; Avadh 14)

(ii) Evaluate $\iint_S \mathbf{r} \cdot \mathbf{n} \, dS$, where S is a closed surface.

(Agra 2006; Kumaun 07; Bundelkhand 09; Purvanchal 14)

(iii) If $\mathbf{F} = ax \mathbf{i} + by \mathbf{j} + cz \mathbf{k}$, a, b, c are constants show that

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \frac{4}{3} \pi (a + b + c),$$

where S is the surface of a unit sphere.

(Bundelkhand 2001, 07, 08; Rohilkhand 07; Kashi 13)

Solution: (i) By divergence theorem, we have

$$\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V (\text{div curl } \mathbf{F}) \, dV,$$

where V is the volume enclosed by S

$$= 0, \text{ since } \text{div curl } \mathbf{F} = 0.$$

(ii) By the divergence theorem, we have

$$\iint_S \mathbf{r} \cdot \mathbf{n} \, dS = \iiint_V \nabla \cdot \mathbf{r} \, dV = \iiint_V 3 \, dV,$$

$$\text{since } \nabla \cdot \mathbf{r} = \text{div } \mathbf{r} = 3$$

$$= 3V, \text{ where } V \text{ is the volume enclosed by } S.$$

(iii) By the divergence theorem, we have

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V (\nabla \cdot \mathbf{F}) \, dV,$$

where V is the volume enclosed by S

$$= \iiint_V [\nabla \cdot (ax \mathbf{i} + by \mathbf{j} + cz \mathbf{k})] \, dV$$

$$= \iiint_V \left[\frac{\partial}{\partial x} (ax) + \frac{\partial}{\partial y} (by) + \frac{\partial}{\partial z} (cz) \right] \, dV$$

$$= \iiint_V (a + b + c) \, dV$$

$$= (a + b + c) V = (a + b + c) \frac{4}{3} \pi,$$

since the volume V enclosed by a sphere of unit radius is equal to $\frac{4}{3} \pi (1)^3$ i.e., $\frac{4}{3} \pi$.

Example 5(i): Show that $\iint_S \mathbf{n} \, dS = \mathbf{0}$ for any closed surface S .

(Purvanchal 14)

(ii) Prove that $\iint_S \mathbf{r} \times \mathbf{n} \, dS = \mathbf{0}$ for any closed surface S .

(Agra 2007)

(iii) Prove that $\iint_S \mathbf{n} \times (\mathbf{a} \times \mathbf{r}) \, dS = 2V\mathbf{a}$, where $\mathbf{a}$ is a constant vector and V is the volume

enclosed by the closed surface S .

(Avadh 2011, 12)

Solution: (i) Let $\mathbf{C}$ be any arbitrary constant vector.

Then

$$\mathbf{C} \cdot \iint_S \mathbf{n} \, dS = \iint_S \mathbf{C} \cdot \mathbf{n} \, dS$$

$$= \iiint_V (\nabla \cdot \mathbf{C}) \, dV, \text{ by divergence theorem}$$

$$= 0, \text{ since } \text{div } \mathbf{C} = 0.$$

Thus $\mathbf{C} \cdot \iint_S \mathbf{n} \, dS = 0$, where $\mathbf{C}$ is an arbitrary vector.

Therefore we must have $\iint_S \mathbf{n} \, dS = \mathbf{0}$.

(ii) Let $\mathbf{C}$ be any arbitrary constant vector. Then

$$\begin{aligned} \mathbf{C} \cdot \iint_S \mathbf{r} \times \mathbf{n} \, dS &= \iint_S \mathbf{C} \cdot [(\mathbf{r} \times \mathbf{n})] \, dS = \iint_S (\mathbf{C} \times \mathbf{r}) \cdot \mathbf{n} \, dS \\ &= \iiint_V [\nabla \cdot (\mathbf{C} \times \mathbf{r})] \, dV, \text{ by divergence theorem} \\ &= \iiint_V [\mathbf{r} \cdot \text{curl } \mathbf{C} - \mathbf{C} \cdot \text{curl } \mathbf{r}] \, dV = 0, \end{aligned}$$

since $\text{curl } \mathbf{C} = \mathbf{0}$ and $\text{curl } \mathbf{r} = \mathbf{0}$.

Thus $\mathbf{C} \cdot \iint_S \mathbf{r} \times \mathbf{n} \, dS = 0$, where $\mathbf{C}$ is an arbitrary vector.

Therefore, we must have $\iint_S \mathbf{r} \times \mathbf{n} \, dS = \mathbf{0}$.

(iii) We know that

$$\iiint_V \nabla \times \mathbf{B} \, dV = \iint_S \mathbf{n} \times \mathbf{B} \, dS. \quad [\text{See article 4, part 3}]$$

Putting $\mathbf{B} = \mathbf{a} \times \mathbf{r}$, we get

$$\begin{aligned} \iint_S \mathbf{n} \times (\mathbf{a} \times \mathbf{r}) \, dS &= \iiint_V \nabla \times (\mathbf{a} \times \mathbf{r}) \, dV \\ &= \iiint_V \text{curl } (\mathbf{a} \times \mathbf{r}) \, dV \\ &= \iiint_V 2\mathbf{a} \, dV, \text{ since } \text{curl } (\mathbf{a} \times \mathbf{r}) = 2\mathbf{a} \\ &= 2\mathbf{a} \iiint_V dV = 2V\mathbf{a}. \end{aligned}$$

Example 6: Using the divergence theorem, show that the volume V of a region T bounded by a surface S is

$$\begin{aligned} V &= \iint_S x \, dy \, dz = \iint_S y \, dz \, dx = \iint_S z \, dx \, dy \\ &= \frac{1}{3} \iint_S (x \, dy \, dz + y \, dz \, dx + z \, dx \, dy). \end{aligned}$$

(Meerut 2010)

Solution: By divergence theorem, we have

$$\begin{aligned} \iint_S x \, dy \, dz &= \iiint_V \left(\frac{\partial}{\partial x} (x) \right) dV = \iiint_V dV = V \\ \iint_S y \, dz \, dx &= \iiint_V \left(\frac{\partial}{\partial y} (y) \right) dV = \iiint_V dV = V \\ \iint_S z \, dx \, dy &= \iiint_V \left(\frac{\partial}{\partial z} (z) \right) dV = \iiint_V dV = V. \end{aligned}$$

Adding these results, we get

$$3V = \iint_S (x \, dy \, dz + y \, dz \, dx + z \, dx \, dy)$$

or
$$V = \frac{1}{3} \iint_S (x \, dy \, dz + y \, dz \, dx + z \, dx \, dy).$$

Example 7: Verify divergence theorem for

$$\mathbf{F} = (x^2 - yz) \mathbf{i} + (y^2 - zx) \mathbf{j} + (z^2 - xy) \mathbf{k}$$

taken over the rectangular parallelepiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$.

(Meerut 2006B; Avadh 09; Rohilkhand 13)

Solution: We have

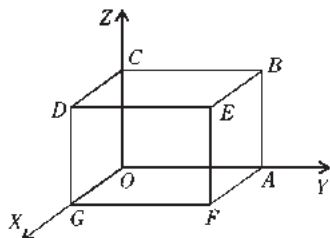
$$\begin{aligned} \operatorname{div} \mathbf{F} &= \nabla \cdot \mathbf{F} \\ &= \frac{\partial}{\partial x} (x^2 - yz) + \frac{\partial}{\partial y} (y^2 - zx) + \frac{\partial}{\partial z} (z^2 - xy) \\ &= 2x + 2y + 2z. \end{aligned}$$

$$\begin{aligned} \therefore \text{volume integral} &= \iiint_V \nabla \cdot \mathbf{F} dV = \iiint_V 2(x + y + z) dV \\ &= 2 \int_{z=0}^c \int_{y=0}^b \int_{x=0}^a (x + y + z) dx dy dz \\ &= 2 \int_{z=0}^c \int_{y=0}^b \left[\frac{x^2}{2} + yx + zx \right]_{x=0}^a dy dz \\ &= 2 \int_{z=0}^c \int_{y=0}^b \left[\frac{a^2}{2} + ay + az \right] dy dz \\ &= 2 \int_{z=0}^c \left[\frac{a^2}{2} y + a \frac{y^2}{2} + azy \right]_{y=0}^b dz \\ &= 2 \int_{z=0}^c \left[\frac{a^2 b}{2} + \frac{ab^2}{2} + abz \right] dz \\ &= 2 \left[\frac{a^2 b}{2} z + \frac{ab^2}{2} z + ab \frac{z^2}{2} \right]_0^c \\ &= [a^2 bc + ab^2 c + abc^2] = abc (a + b + c). \end{aligned}$$

Surface Integral: We shall now calculate $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ over the six faces of the rectangular parallelepiped.

Over the face $DEFG$, $\mathbf{n} = \mathbf{i}$, $x = a$.

$$\begin{aligned} \text{Therefore, } \iint_{DEFG} \mathbf{F} \cdot \mathbf{n} dS &= \int_{z=0}^c \int_{y=0}^b [(a^2 - yz) \mathbf{i} \\ &\quad + (y^2 - za) \mathbf{j} + (z^2 - ay) \mathbf{k}] \cdot \mathbf{i} dy dz \\ &= \int_{z=0}^c \int_{y=0}^b (a^2 - yz) dy dz \\ &= \int_{z=0}^c \left[a^2 y - z \frac{y^2}{2} \right]_{y=0}^b dz \end{aligned}$$



$$= \int_{z=0}^c \left[a^2 b - \frac{zb^2}{2} \right] dz = \left[a^2 bz - \frac{z^2}{4} b^2 \right]_0^c = a^2 bc - \frac{c^2 b^2}{4}.$$

Over the face $ABCO$, $\mathbf{n} = -\mathbf{i}$, $x = 0$. Therefore

$$\begin{aligned} \iint_{ABCO} \mathbf{F} \bullet \mathbf{n} \, dS &= \iint [(0 - yz) \mathbf{i} + \dots + \dots] \bullet (-\mathbf{i}) \, dy \, dz \\ &= \int_{z=0}^c \int_{y=0}^b yz \, dy \, dz = \int_{z=0}^c \left[\frac{y^2}{2} z \right]_{y=0}^b dz \\ &= \int_{z=0}^c \frac{b^2}{2} z \, dz = \frac{b^2 c^2}{4}. \end{aligned}$$

Over the face $ABEF$, $\mathbf{n} = \mathbf{j}$, $y = b$. Therefore

$$\begin{aligned} \iint_{ABEF} \mathbf{F} \bullet \mathbf{n} \, dS &= \int_{z=0}^c \int_{x=0}^a [(x^2 - bz) \mathbf{i} + (b^2 - zx) \mathbf{j} \\ &\quad + (z^2 - bx) \mathbf{k}] \bullet \mathbf{j} \, dx \, dz \\ &= \int_{z=0}^c \int_{x=0}^a (b^2 - zx) \, dx \, dz = b^2 ca - \frac{a^2 c^2}{4}. \end{aligned}$$

Over the face $OGDC$, $\mathbf{n} = -\mathbf{j}$, $y = 0$. Therefore

$$\iint_{OGDC} \mathbf{F} \bullet \mathbf{n} \, dS = \int_{z=0}^c \int_{x=0}^a zx \, dx \, dz = \frac{c^2 a^2}{4}.$$

Over the face $BCDE$, $\mathbf{n} = \mathbf{k}$, $z = c$. Therefore

$$\iint_{BCDE} \mathbf{F} \bullet \mathbf{n} \, dS = \int_{y=0}^b \int_{x=0}^a (c^2 - xy) \, dx \, dy = c^2 ab - \frac{a^2 b^2}{4}.$$

Over the face $AFGO$, $\mathbf{n} = -\mathbf{k}$, $z = 0$. Therefore

$$\iint_{AFGO} \mathbf{F} \bullet \mathbf{n} \, dS = \int_{y=0}^b \int_{x=0}^a x y \, dx \, dy = \frac{a^2 b^2}{4}.$$

Adding the six surface integrals, we get

$$\begin{aligned} \iint_S \mathbf{F} \bullet \mathbf{n} \, dS &= \left(a^2 bc - \frac{c^2 b^2}{4} + \frac{c^2 b^2}{4} \right) + \left(b^2 ca - \frac{a^2 c^2}{4} + \frac{a^2 c^2}{4} \right) \\ &\quad + \left(c^2 ab - \frac{a^2 b^2}{4} + \frac{a^2 b^2}{4} \right) \\ &= abc (a + b + c). \end{aligned}$$

Hence the theorem is verified.

Example 8: If $\mathbf{F} = x \mathbf{i} - y \mathbf{j} + (z^2 - 1) \mathbf{k}$, find the value of $\iint_S \mathbf{F} \bullet \mathbf{n} \, dS$ where S is the closed surface bounded by the planes $z = 0$, $z = 1$ and the cylinder $x^2 + y^2 = 4$.

(Garhwal 2000; Kanpur 05; Avadh 13; Kumaun 15)

Solution: By divergence theorem, we have $\iint_S \mathbf{F} \bullet \mathbf{n} \, dS = \iiint_V \operatorname{div} \mathbf{F} \, dV$.

Here $\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(-y) + \frac{\partial}{\partial z}(z^2 - 1) = 1 - 1 + 2z = 2z.$

$$\begin{aligned}
 \therefore \iiint_V \operatorname{div} \mathbf{F} dV &= \int_{z=0}^1 \int_{y=-2}^2 \int_{x=-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 2z dx dy dz \\
 &= \int_{z=0}^1 \int_{y=-2}^2 [2zx]_{x=-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dy dz \\
 &= \int_{z=0}^1 \int_{y=-2}^2 4z \sqrt{4-y^2} dy dz \\
 &= \int_{y=-2}^2 \left[4 \frac{z^2}{2} \sqrt{4-y^2} \right]_{z=0}^1 dy \\
 &= 2 \int_{y=-2}^2 \sqrt{4-y^2} dy = 4 \int_0^2 \sqrt{4-y^2} dy \\
 &= 4 \left[\frac{y}{2} \sqrt{4-y^2} + 2 \sin^{-1} \frac{y}{2} \right]_0^2 \\
 &= 4 [2 \sin^{-1} 1] = 4 (2) \frac{\pi}{2} = 4\pi.
 \end{aligned}$$

Comprehensive Exercise 2

1. (i) Verify divergence theorem for $\mathbf{F} = (2x - z)\mathbf{i} + x^2 y \mathbf{j} - xz^2 \mathbf{k}$ taken over the region bounded by $x = 0, x = 2, y = 0, y = 2, z = 0, z = 2$.
 (ii) Verify divergence theorem for $\mathbf{F} = (2x - z)\mathbf{i} + x^2 y \mathbf{j} - xz^2 \mathbf{k}$ taken over the region bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.
 (Garhwal 2001; Kumaun 14)
2. (i) If $\mathbf{F} = 4xz \mathbf{i} - y^2 \mathbf{j} + yz \mathbf{k}$ and S is the surface bounded by $x = 0, y = 0, z = 0, x = 1, y = 1, z = 1$, evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} dS$.
 (ii) Evaluate $\iint_S x^2 dy dz + y^2 dz dx + 2z (xy - x - y) dx dy$ where S is the surface of the cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.
 (Kumaun 2015)
3. (i) Evaluate $\iint_S [4xz dy dz - y^2 dz dx + yz dx dy]$ where S is the surface of the cube bounded by the planes $x = 0, y = 0, z = 0, x = 1, y = 1$ and $z = 1$.
 (Meerut 2005, 06B, 10B, 11)

- (ii) Apply Gauss's divergence theorem to evaluate

$$\iint_S [(x^3 - yz) dy dz - 2x^2 y dz dx + z dx dy]$$

over the surface of a cube bounded by the coordinate planes and the planes $x = y = z = a$.

(Rohilkhand 2011)

4. (i) State divergence theorem of Gauss.

- (ii) Use Gauss divergence theorem to show that

$$\iint_S \{ (x^3 - yz) \mathbf{i} - 2x^2 y \mathbf{j} + 2z \mathbf{k} \} \cdot \mathbf{n} dS = \frac{1}{3} a^5,$$

where S denotes the surface of the cube bounded by the planes

$$x = 0, x = a, y = 0, y = a, z = 0, z = a. \quad (\text{Bundelkhand 2005, 06})$$

5. Evaluate $\iint_S (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) \cdot \mathbf{n} dS$ where S denotes the surface of the cube

bounded by the planes $x = 0, y = 0, z = 0, x = a, y = a, z = a$ by the application of Gauss divergence theorem. Verify your answer by evaluating the integral directly. (Garhwal 2003)

6. (i) Evaluate by divergence theorem the integral

$$\iint_S xz^2 dy dz + (x^2 y - z^3) dz dx + (2xy + y^2 z) dx dy,$$

where S is the entire surface of the hemispherical region bounded by $z = \sqrt{(a^2 - x^2 - y^2)}$ and $z = 0$.

- (ii) Evaluate $\iint_S (y^2 z^2 \mathbf{i} + z^2 x^2 \mathbf{j} + z^2 y^2 \mathbf{k}) \cdot \mathbf{n} dS$

where S is the part of the sphere $x^2 + y^2 + z^2 = 1$ above the xy -plane and bounded by this plane. (Bundelkhand 2006)

7. (i) If $\mathbf{F} = ax \mathbf{i} + by \mathbf{j} + cz \mathbf{k}$, where a, b, c are constants, show that

$$\iint_S (\mathbf{n} \cdot \mathbf{F}) dS = \frac{4\pi}{3} (a + b + c),$$

S being the surface of the sphere $(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 1$.

- (ii) If S is any closed surface enclosing a volume V and

$$\mathbf{F} = x \mathbf{i} + 2y \mathbf{j} + 3z \mathbf{k}, \text{ prove that } \iint_S \mathbf{F} \cdot \mathbf{n} dS = 6V.$$

(Rohilkhand 2009B)

8. Verify the divergence theorem for $\mathbf{F} = 4x \mathbf{i} - 2y^2 \mathbf{j} + z^2 \mathbf{k}$

taken over the region bounded by the surfaces $x^2 + y^2 = 4, z = 0, z = 3$.

(Garhwal 2002; Bundelkhand 08)

9. Use Gauss divergence theorem to find $\iint_S \mathbf{F} \cdot \mathbf{n} dS$, where

$\mathbf{F} = 2x^2 y \mathbf{i} - y^2 \mathbf{j} + 4xz^2 \mathbf{k}$ and S is the closed surface in the first octant bounded by $y^2 + z^2 = 9$ and $x = 2$.

10. If $\mathbf{F} = y \mathbf{i} + (x - 2xz) \mathbf{j} - xy \mathbf{k}$, evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$ where S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$ above the xy -plane.

11. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$,
 where $\mathbf{F} = (x^2 + y - 4) \mathbf{i} + 3xy \mathbf{j} + (2xz + z^2) \mathbf{k}$ and S is the surface of the paraboloid $z = 4 - (x^2 + y^2)$ above the xy -plane.
12. Compute
 (i) $\iint_S (a^2 x^2 + b^2 y^2 + c^2 z^2)^{1/2} \, dS$, and
 (ii) $\iint_S (a^2 x^2 + b^2 y^2 + c^2 z^2)^{-1/2} \, dS$
 over the ellipsoid $ax^2 + by^2 + cz^2 = 1$.
13. Evaluate $\iint_S (x^2 + y^2) \, dS$, where S is the surface of the cone $z^2 = 3(x^2 + y^2)$ bounded by $z = 0$ and $z = 3$.
14. Show that $\iint_S (x^2 \mathbf{i} + y^2 \mathbf{j} + z^2 \mathbf{k}) \cdot \mathbf{n} \, dS$ vanishes where S denotes the surface of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
 (Meerut 2005, 07; Kumaun 11, 13)
15. If $\mathbf{n}$ is the unit outward drawn normal to any closed surface S , show that

$$\iiint_V \operatorname{div} \mathbf{n} \, dV = S.$$

Answers 2

2. (i) $\frac{3}{2}$ (ii) $\frac{1}{2}$ 3. (i) $\frac{3}{2}$ (ii) $a^2 \left(\frac{a^3}{3} + a \right)$
5. $3a^3$ 6. (i) $\frac{2\pi a^5}{5}$ (ii) $\frac{\pi}{12}$
9. 180 10. 0
11. -4π 12. (i) $\frac{4}{3} \pi abc$. (ii) $\frac{4\pi}{\sqrt{abc}}$
13. 9π

5 Stoke's Theorem

Let S be a piecewise smooth open surface bounded by a piecewise smooth simple closed curve C . Let $\mathbf{F}(x, y, z)$ be a continuous vector function which has continuous first partial derivatives in a region of space which contains S in its interior. Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S (\operatorname{curl} \mathbf{F}) \cdot d\mathbf{S}$$

where C is traversed in the positive direction. The direction of C is called positive if an observer, walking on the boundary of S in this direction, with his head pointing in the direction of outward drawn normal $\mathbf{n}$ to S , has the surface on the left.

(Meerut 2009; Bundelkhand 10)

Note: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \left(\mathbf{F} \cdot \frac{d\mathbf{r}}{ds} \right) ds = \oint_C (\mathbf{F} \cdot \mathbf{t}) ds$, where $\mathbf{t}$ is unit tangent vector

to C . Therefore $\mathbf{F} \cdot \mathbf{t}$ is the component of $\mathbf{F}$ in the direction of the tangent vector of C . Also $(\nabla \times \mathbf{F}) \cdot \mathbf{n}$ is the component of curl $\mathbf{F}$ in the direction of outward drawn normal vector $\mathbf{n}$ of S . Therefore in words Stoke's theorem may be stated as follows:

The line integral of the tangential component of vector $\mathbf{F}$ taken around a simple closed curve C is equal to the surface integral of the normal component of the curl of $\mathbf{F}$ taken over any surface S having C as its boundary.

Cartesian equivalent of Stoke's theorem:

Let $\mathbf{F} = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}$. Let outward drawn normal vector $\mathbf{n}$ of S make angles α, β, γ with positive directions of x, y, z axes.

Then $\mathbf{n} = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}$.

$$\begin{aligned} \text{Also } \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\ &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}. \end{aligned}$$

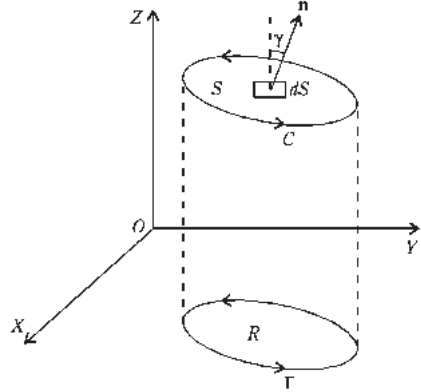
$$\begin{aligned} \therefore (\nabla \times \mathbf{F}) \cdot \mathbf{n} &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \cos \alpha + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \cos \beta \\ &\quad + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \cos \gamma. \end{aligned}$$

$$\begin{aligned} \text{Also } \mathbf{F} \cdot d\mathbf{r} &= (F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\ &= F_1 dx + F_2 dy + F_3 dz. \end{aligned}$$

$\therefore$ Stoke's theorem can be written as

$$\begin{aligned} \oint_C F_1 dx + F_2 dy + F_3 dz \\ &= \iint_S \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \cos \alpha + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \cos \beta \right. \\ &\quad \left. + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \cos \gamma \right] dS. \end{aligned}$$

Proof of Stoke's theorem: Let S be a surface which is such that its projections on the xy , yz and zx planes are regions bounded by simple closed curves. Suppose S can be represented simultaneously in the forms $z = f(x, y)$, $y = g(x, z)$, $x = h(z, y)$ where f, g, h are continuous functions and have continuous first partial derivatives.



Consider the integral

$$\iint_S [\nabla \times (F_1 \mathbf{i})] \cdot \mathbf{n} \, dS.$$

We have
$$\nabla \times (F_1 \mathbf{i}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & 0 & 0 \end{vmatrix} = \frac{\partial F_1}{\partial z} \mathbf{j} - \frac{\partial F_1}{\partial y} \mathbf{k}.$$

$$\therefore [\nabla \times (F_1 \mathbf{i})] \cdot \mathbf{n} = \left(\frac{\partial F_1}{\partial z} \mathbf{j} \cdot \mathbf{n} - \frac{\partial F_1}{\partial y} \mathbf{k} \cdot \mathbf{n} \right) = \frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma.$$

$$\therefore \iint_S [\nabla \times (F_1 \mathbf{i})] \cdot \mathbf{n} \, dS = \iint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dS.$$

We shall prove that

$$\iint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dS = \oint_C F_1 \, dx.$$

Let R be the orthogonal projection of S on the xy -plane and let Γ be its boundary which is oriented as shown in the figure. Using the representation $z = f(x, y)$ of S , we may write the line integral over C as a line integral over Γ . Thus

$$\begin{aligned} \oint_C F_1(x, y, z) \, dx &= \oint_\Gamma F_1[x, y, f(x, y)] \, dx \\ &= \oint_\Gamma \{F_1[x, y, f(x, y)] \, dx + 0 \, dy\} \\ &= - \iint_R \frac{\partial F_1}{\partial y} \, dx \, dy, \end{aligned}$$

by Green's theorem in plane for the region R .

But
$$\frac{\partial F_1[x, y, f(x, y)]}{\partial y} = \frac{\partial F_1(x, y, z)}{\partial y} + \frac{\partial F_1(x, y, z)}{\partial z} \frac{\partial f}{\partial y}.$$
 [$\because z = f(x, y)$]

$$\therefore \oint_C F_1(x, y, z) \, dx = - \iint_R \left(\frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z} \frac{\partial f}{\partial y} \right) dx \, dy \quad \dots(1)$$

Now the equation $z = f(x, y)$ of the surface S can be written as

$$\phi(x, y, z) \equiv z - f(x, y) = 0.$$

We have $\text{grad } \phi = -\frac{\partial f}{\partial x} \mathbf{i} - \frac{\partial f}{\partial y} \mathbf{j} + \mathbf{k}$.

Let $|\text{grad } \phi| = a$.

Since $\text{grad } \phi$ is normal to S , therefore, we get $\mathbf{n} = \pm \frac{\text{grad } \phi}{a}$.

But the components of both $\mathbf{n}$ and $\text{grad } \phi$ in positive direction of z -axis are positive. Therefore

$$\mathbf{n} = + \frac{\text{grad } \phi}{a}$$

$$\text{or} \quad \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k} = -\frac{1}{a} \frac{\partial f}{\partial x} \mathbf{i} - \frac{1}{a} \frac{\partial f}{\partial y} \mathbf{j} + \frac{1}{a} \mathbf{k}.$$

$$\therefore \quad \cos \alpha = -\frac{1}{a} \frac{\partial f}{\partial x}, \cos \beta = -\frac{1}{a} \frac{\partial f}{\partial y}, \cos \gamma = \frac{1}{a}.$$

$$\text{Now} \quad dS = \frac{dx \, dy}{\cos \gamma} = a \, dx \, dy.$$

$$\begin{aligned} \therefore \quad \iint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dS \\ = \iint_R \left[\frac{\partial F_1}{\partial z} \left(-\frac{1}{a} \frac{\partial f}{\partial y} \right) - \frac{\partial F_1}{\partial y} \frac{1}{a} \right] a \, dx \, dy \\ = - \iint_R \left(\frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z} \frac{\partial f}{\partial y} \right) dx \, dy. \end{aligned} \quad \dots(2)$$

From (1) and (2), we get

$$\begin{aligned} \oint_C F_1 \, dx &= \iint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dS \\ &= \iint_S [\nabla \times (F_1 \mathbf{i})] \cdot \mathbf{n} \, dS \end{aligned} \quad \dots(3)$$

Similarly, by projections on the other coordinate planes, we get

$$\oint_C F_2 \, dy = \iint_S [\nabla \times (F_2 \mathbf{j})] \cdot \mathbf{n} \, dS \quad \dots(4)$$

$$\oint_C F_3 \, dz = \iint_S [\nabla \times (F_3 \mathbf{k})] \cdot \mathbf{n} \, dS \quad \dots(5)$$

Adding (3), (4), (5), we get

$$\oint_C (F_1 \, dx + F_2 \, dy + F_3 \, dz) = \iint_S [\nabla \times (F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k})] \cdot \mathbf{n} \, dS$$

$$\text{or} \quad \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS.$$

If the surface S does not satisfy the restrictions imposed above, even then Stoke's theorem will be true provided S can be subdivided into surfaces $S_1, S_2, \dots, S_k$ with boundaries $C_1, C_2, \dots, C_k$ which do satisfy the restrictions. Stoke's theorem

holds for each such surface. The sum of surface integrals over $S_1, S_2, \dots, S_k$ will give us surface integral over S while the sum of the integrals over $C_1, C_2, \dots, C_k$ will give us line integral over C .

Note: Green's theorem in plane is a special case of Stoke's theorem. If R is a region in the xy -plane bounded by a closed curve C , then in vector form Green's theorem in plane can be written as

$$\iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} \, dR = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

This is nothing but a special case of Stoke's theorem because here $\mathbf{k} = \mathbf{n}$ = outward drawn unit normal to the surface of region R .

Illustrative Examples

Example 9: Prove that $\oint_C \mathbf{r} \cdot d\mathbf{r} = 0$.

(Meerut 2010)

Solution: By Stoke's theorem $\oint_C \mathbf{r} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{r}) \cdot \mathbf{n} \, dS = 0$, since $\text{curl } \mathbf{r} = \mathbf{0}$.

Example 10: By Stoke's theorem prove that $\text{div curl } \mathbf{F} = 0$.

Solution: Let V be any volume enclosed by a closed surface.

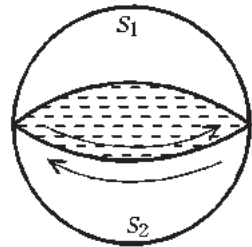
Then by divergence theorem

$$\iiint_V \nabla \cdot (\text{curl } \mathbf{F}) \, dV = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS.$$

Divide the surface S into two portions S_1 and S_2

by a closed curve C . Then

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \iint_{S_1} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS_1 \\ &+ \iint_{S_2} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS_2. \quad \dots (1) \end{aligned}$$



By Stoke's theorem right hand side of (1) is $= \oint_C \mathbf{F} \cdot d\mathbf{r} - \oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.

Negative sign has been taken in the second integral because the positive directions about the boundaries of the two surfaces are opposite.

$$\therefore \iiint_V \nabla \cdot (\text{curl } \mathbf{F}) \, dV = 0.$$

Now this equation is true for all volume elements V . Therefore we have

$$\nabla \cdot (\text{curl } \mathbf{F}) = 0 \quad \text{or} \quad \text{div curl } \mathbf{F} = 0.$$

Example 11: Verify Stoke's theorem for $\mathbf{F} = y \mathbf{i} + z \mathbf{j} + x \mathbf{k}$ where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

(Agra 2000, 06; Kanpur 09; Kumaun 07, 10, 13)

Solution: The boundary C of S is a circle in the xy -plane of radius unity and centre origin. The equations of the curve C are $x^2 + y^2 = 1, z = 0$. Suppose $x = \cos t, y = \sin t, z = 0, 0 \leq t < 2\pi$ are parametric equation of C . Then

$$\begin{aligned}
\oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C (y \mathbf{i} + z \mathbf{j} + x \mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\
&= \oint_C (y dx + z dy + x dz) \\
&= \oint_C y dx, \text{ since on } C, z = 0 \text{ and } dz = 0 \\
&= \int_0^{2\pi} \sin t \frac{dx}{dt} dt = \int_0^{2\pi} -\sin^2 t dt \\
&= -\frac{1}{2} \int_0^{2\pi} (1 - \cos 2t) dt = -\frac{1}{2} \left[t - \frac{\sin 2t}{2} \right]_0^{2\pi} \\
&= -\pi.
\end{aligned} \tag{1}$$

Now let us evaluate $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$. We have

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} = -\mathbf{i} - \mathbf{j} - \mathbf{k}.$$

If S_1 is the plane region bounded by the circle C , then by an application of divergence theorem, we have

$$\begin{aligned}
\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS &= \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{k} dS \\
&\quad [\text{See example 4 after article 4}] \\
&= \iint_{S_1} (-\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot \mathbf{k} dS \\
&= \iint_{S_1} (-1) dS = -\iint_{S_1} dS = -S_1.
\end{aligned}$$

But $S_1 = \text{area of a circle of radius } 1 = \pi (1)^2 = \pi$.

$$\therefore \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = -\pi. \tag{2}$$

Hence from (1) and (2), the theorem is verified.

Example 12: Verify Stoke's theorem for $\mathbf{F} = (2x - y) \mathbf{i} - yz^2 \mathbf{j} - y^2 z \mathbf{k}$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

(Kumaun 2003; Kanpur 10, 14; Avadh 09)

Solution: The boundary C of S is a circle in the xy -plane of radius unity and centre origin. Suppose $x = \cos t$, $y = \sin t$, $z = 0, 0 \leq t < 2\pi$ are parametric equations of C . Then

$$\begin{aligned}
\oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C [(2x - y) \mathbf{i} - yz^2 \mathbf{j} - y^2 z \mathbf{k}] \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\
&= \oint_C [(2x - y) dx - yz^2 dy - y^2 z dz] \\
&= \oint_C (2x - y) dx, \text{ since } z = 0 \text{ and } dz = 0 \\
&= \int_0^{2\pi} (2 \cos t - \sin t) \frac{dx}{dt} dt
\end{aligned}$$

$$\begin{aligned}
&= - \int_0^{2\pi} (2 \cos t - \sin t) \sin t \, dt \\
&= - \int_0^{2\pi} \left[\sin 2t - \frac{1}{2} (1 - \cos 2t) \right] dt \\
&= - \left[-\frac{\cos 2t}{2} - \frac{1}{2} t + \frac{1}{2} \frac{\sin 2t}{2} \right]_0^{2\pi} \\
&= - \left[\left(-\frac{1}{2} + \frac{1}{2} \right) - \frac{1}{2} (\pi - 0) + \frac{1}{4} (0 - 0) \right] = \pi. \quad \dots(1)
\end{aligned}$$

And $(\nabla \times \mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix}$

$$= (-2yz + 2yz) \mathbf{i} - (0 - 0) \mathbf{j} + (0 + 1) \mathbf{k} = \mathbf{k}.$$

Let S_1 be the plane region bounded by the circle C . If S' is the surface consisting of the surfaces S and S_1 , then S' is a closed surface.

$\therefore$ by an application of Gauss divergence theorem, we have

$$\iint_{S'} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = 0 \quad [\text{See example 4(i) after article 4}]$$

or $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS + \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = 0$

$[\because S' \text{ consists of } S \text{ and } S_1]$

or $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS - \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{k} \, dS = 0 \quad [\because \text{ on } S_1, \mathbf{n} = -\mathbf{k}]$

or $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{k} \, dS.$

$\therefore \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{k} \, dS$

$$= \iint_{S_1} \mathbf{k} \cdot \mathbf{k} \, dS = \iint_{S_1} dS = S_1 = \pi. \quad \dots(2)$$

Note that $S_1 = \text{area of a circle of radius } 1 = \pi (1)^2 = \pi.$

Hence from (1) and (2) Stoke's theorem is verified.

Example 13: Verify Stoke's theorem for $\mathbf{F} = (x^2 + y^2) \mathbf{i} - 2xy \mathbf{j}$ taken round the rectangle bounded by $x = \pm a, y = 0, y = b.$ (Bundelkhand 2007; Agra 08; Kumaun 15)

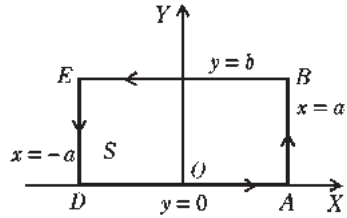
Solution: We have $\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix}$

$$= (-2y - 2y) \mathbf{k} = -4y \mathbf{k}.$$

Also $\mathbf{n} = \mathbf{k}.$

$\therefore \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \int_{y=0}^b \int_{x=-a}^a (-4y \mathbf{k}) \cdot \mathbf{k} \, dx \, dy$

$$\begin{aligned}
 &= -4 \int_{y=0}^b \int_{x=-a}^a y \, dx \\
 &= -4 \int_{y=0}^b [xy]_{x=-a}^a \, dy \\
 &= -4 \int_{y=0}^b 2ay \, dy \\
 &= -4 [ay^2]_0^b = -4ab^2.
 \end{aligned}$$



Also

$$\begin{aligned}
 \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C [(x^2 + y^2) \mathbf{i} - 2xy \mathbf{j}] \cdot (dx \mathbf{i} + dy \mathbf{j}) \\
 &= \oint_C [(x^2 + y^2) dx - 2xy \, dy] \\
 &= \int_{DA} [(x^2 + y^2) dx - 2xy \, dy] + \int_{AB} + \int_{BE} + \int_{ED}.
 \end{aligned}$$

Along DA, $y = 0$ and $dy = 0$. Along AB, $x = a$ and $dx = 0$.

Along BE, $y = b$ and $dy = 0$. Along ED, $x = -a$ and $dx = 0$.

$$\begin{aligned}
 \therefore \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_{x=-a}^a x^2 \, dx + \int_{y=0}^b -2ay \, dy \\
 &\quad + \int_{x=a}^{-a} (x^2 + b^2) \, dx + \int_{y=b}^0 2ay \, dy \\
 &= \int_{-a}^a x^2 \, dx - \int_{-a}^a (x^2 + b^2) \, dx - 4a \int_0^b y \, dy \\
 &= - \int_{-a}^a x^2 \, dx - 4a \int_0^b y \, dy = -2ab^2 - 4a \left[\frac{y^2}{2} \right]_0^b = -4ab^2.
 \end{aligned}$$

Thus $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$.

Hence the theorem is verified.

Example 14: Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ by Stoke's theorem where

$$\mathbf{F} = y^2 \mathbf{i} + x^2 \mathbf{j} - (x + z) \mathbf{k}$$

and C is the boundary of the triangle with vertices at $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$.

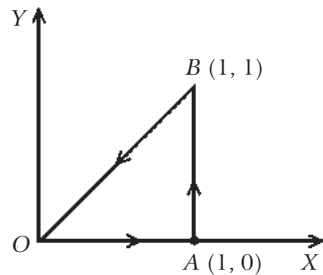
(Avadh 2013)

Solution: We have

$$\begin{aligned}
 \text{Curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & (x + z) \end{vmatrix} \\
 &= 0 \mathbf{i} + \mathbf{j} + 2(x - y) \mathbf{k}.
 \end{aligned}$$

Also we note that z co-ordinate of each vertex of the triangle is zero. Therefore the triangle lies in the xy -plane. So $\mathbf{n} = \mathbf{k}$.

$$\therefore \text{Curl } \mathbf{F} \cdot \mathbf{n} = [\mathbf{j} + 2(x - y)\mathbf{k}] \cdot \mathbf{k} = 2(x - y).$$



In the figure, we have only considered the xy plane.

The equation of the line OB is $y = x$.

By Stoke's theorem

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS \\ &= \int_{x=0}^1 \int_{y=0}^x 2(x-y) \, dx \, dy \\ &= 2 \int_{x=0}^1 \left[xy - \frac{y^2}{2} \right]_{y=0}^x dx \\ &= 2 \int_0^1 \left[x^2 - \frac{x^2}{2} \right] dx = 2 \int_0^1 \frac{x^2}{2} dx \\ &= \int_0^1 x^2 \, dx = \frac{1}{3}.\end{aligned}$$

Comprehensive Exercise 3

- State Stoke's theorem.
 - By Stoke's theorem prove that $\text{curl grad } \phi = \mathbf{0}$. (Kumaun 2014)
 - Verify Stoke's theorem for the function $\mathbf{F} = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$ where curve is the unit circle in the xy -plane bounding the hemisphere $z = \sqrt{(1-x^2 - y^2)}$. (Garhwal 2003; Agra 07)
- Verify Stoke's theorem for the vector $\mathbf{F} = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$ taken over the half of the sphere $x^2 + y^2 + z^2 = a^2$ lying above the xy -plane.
- Verify Stoke's theorem for the function $\mathbf{F} = x^2\mathbf{i} + xy\mathbf{j}$ integrated along the rectangle, in the plane $z = 0$, whose sides are along the lines $x = 0$, $y = 0$, $x = a$ and $y = b$.
 - Verify Stoke's theorem for the function $\mathbf{F} = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j}$ in the rectangular region in the xy -plane bounded by the lines $x = 0$, $x = a$, $y = 0$ and $y = b$. (Kanpur 2008)
 - Verify Stoke's theorem for the function $\mathbf{F} = x^2\mathbf{i} + xy\mathbf{j}$, integrated round the square, in the plane $z = 0$, whose sides are along the lines $x = 0$, $y = 0$, $x = a$, $y = a$. (Agra 2002)
- Verify Stoke's theorem for the vector $\mathbf{A} = 3y\mathbf{i} - xz\mathbf{j} + yz^2\mathbf{k}$, where S is the surface of the paraboloid $2z = x^2 + y^2$ bounded by $z = 2$ and C is its boundary.

5. (i) By converting into a line integral evaluate

$$\iint_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} \, dS, \text{ where } \mathbf{A} = (x - z) \mathbf{i} + (x^3 + yz) \mathbf{j} - 3xy^2 \mathbf{k}$$

and S is the surface of the cone $z = 2 - \sqrt{(x^2 + y^2)}$ above the xy -plane.

- (ii) By converting into a line integral evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$

where $\mathbf{F} = (x^2 + y - 4) \mathbf{i} + 3xy \mathbf{j} + (2xy + z^2) \mathbf{k}$ and S is the surface of the paraboloid $z = 4 - (x^2 + y^2)$ above the xy -plane.

6. (i) Evaluate by Stoke's theorem $\oint_C (e^x \, dx + 2y \, dy - dz)$

where C is the curve $x^2 + y^2 = 4, z = 2$. (Garhwal 2001; Meerut 09B)

- (ii) Evaluate by Stoke's theorem $\oint_C (yz \, dx + xz \, dy + xy \, dz)$

where C is the curve $x^2 + y^2 = 1, z = y^2$. (Meerut 2006B)

7. (i) Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$, where $\mathbf{F} = (y - z + 2) \mathbf{i} +$

$(yz + 4) \mathbf{j} - xz \mathbf{k}$ and S is the surface of the cube $x = y = z = 0$, $x = y = z = 2$ above the xy -plane.

- (ii) Evaluate by Stoke's theorem $\oint_C (\sin z \, dx - \cos x \, dy + \sin y \, dz)$ where

C is the boundary of the rectangle $0 \leq x \leq \pi, 0 \leq y \leq 1, z = 3$.

(Meerut 2009)

8. If $\mathbf{f} = \nabla \phi$ and $\mathbf{g} = \nabla \psi$ are two vector point functions, such that $\nabla^2 \phi = 0$, $\nabla^2 \psi = 0$, show that

$$\iint_S (\mathbf{g} \cdot \nabla) \mathbf{f} \cdot d\mathbf{S} = \int_C (\mathbf{f} \times \mathbf{g}) \cdot d\mathbf{r} + \iint_S (\mathbf{f} \cdot \nabla) \mathbf{g} \cdot d\mathbf{S}.$$

9. Prove that a necessary and sufficient condition that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C lying in a simply connected region R is that $\nabla \times \mathbf{F} = \mathbf{0}$ identically.

10. Apply Stoke's theorem to prove that

$$\int_C (y \, dx + z \, dy + x \, dz) = -2\sqrt{2} \pi a^2$$

where C is the curve given by $x^2 + y^2 + z^2 - 2ax - 2ay = 0, x + y = 2a$ and begins at the point $(2a, 0, 0)$ and goes at first below the z -plane.

(Meerut 2005, 06B)

11. Use Stoke's theorem to evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$,

where $\mathbf{F} = y \mathbf{i} + (x - 2xz) \mathbf{j} - xy \mathbf{k}$ and S is the surface of sphere $x^2 + y^2 + z^2 = a^2$, above the xy -plane.

(Kumaun 2012)

Answers 3

5. (i) 12π (ii) -4π 7. (i) -4 (ii) 2
 11. 0

6 Line Integrals Independent of Path

Let $\mathbf{F}(x, y, z) = f(x, y, z) \mathbf{i} + g(x, y, z) \mathbf{j} + h(x, y, z) \mathbf{k}$ be a vector point function defined and continuous in a region R of space. Let P and Q be two points in R and let C be a path joining P to Q . Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int (f dx + g dy + h dz) \quad \dots (1)$$

is called the line integral of $\mathbf{F}$ along C . In general the value of this line integral depends not only on the end points P and Q of the path C but also on C .

In other words, if we integrate from P to Q along different paths, we shall, in general, get different values of the integral. *The line integral (1) is said to be independent of path in R , if for every pair of end points P and Q in R the value of the integral is the same for all paths C in R starting from P and ending at Q .*

In this case the value of this line integral will depend on the choice of P and Q and not on the choice of the path joining P to Q .

Definition: *The expression $f dx + g dy + h dz$ is said to be an exact differential if there exists a single valued scalar point function $\phi(x, y, z)$, having continuous first partial derivatives such that $d\phi = f dx + g dy + h dz$.*

It can be easily seen that $f dx + g dy + h dz$ is an exact differential if and only if the vector function $\mathbf{F} = f \mathbf{i} + g \mathbf{j} + h \mathbf{k}$ is the gradient of a single valued scalar function $\phi(x, y, z)$.

Because $\mathbf{F} = \text{grad } \phi$

$$\text{if, and only if } f \mathbf{i} + g \mathbf{j} + h \mathbf{k} = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}$$

$$\text{if, and only if } f = \frac{\partial \phi}{\partial x}, \quad g = \frac{\partial \phi}{\partial y}, \quad h = \frac{\partial \phi}{\partial z}$$

$$\text{if, and only if } f dx + g dy + h dz = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$\text{if, and only if } f dx + g dy + h dz = d\phi.$$

Thus $\mathbf{F} = \text{grad } \phi$ if, and only if $f dx + g dy + h dz$ is an exact differential $d\phi$.

Theorem 1: Let $f(x, y, z)$, $g(x, y, z)$ and $h(x, y, z)$ be continuous in a region R of space. Then the line integral

$$\int (f dx + g dy + h dz)$$

is independent of path in R if and only if the differential form under the integral sign is exact in R .

Or

Let $\mathbf{F}(x, y, z)$ be continuous in region R of space. Then the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$

is independent of the path C in R joining P and Q if and only if $\mathbf{F} = \text{grad } \phi$ where $\phi(x, y, z)$ is a single-valued scalar function having continuous first partial derivatives in R .

Proof: Suppose $\mathbf{F} = \text{grad } \phi$ in R . Let P and Q be any two points in R and let C be any path from P to Q in R .

$$\begin{aligned} \text{Then } \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \nabla \phi \cdot d\mathbf{r} \\ &= \int_C \left(\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\ &= \int_C \left(\frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \right) \\ &= \int_C d\phi \\ &= \int_P^Q d\phi = [\phi]_P^Q \\ &= \phi(Q) - \phi(P). \end{aligned}$$

Thus the line integral depends only on points P and Q and not on the path joining them. This is true, of course, only if $\phi(x, y, z)$ is single valued at all points P and Q .

Conversely, suppose the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C joining any two points P and Q in R . Let P be a fixed point (x_0, y_0, z_0) in R and let Q be any point (x, y, z) in R .

$$\begin{aligned} \text{Let } \phi(x, y, z) &= \int_{(x_0, y_0, z_0)}^{(x, y, z)} \mathbf{F} \cdot d\mathbf{r} \\ &= \int_{x_0, y_0, z_0}^{(x, y, z)} \left(\mathbf{F} \cdot \frac{d\mathbf{r}}{ds} \right) ds. \end{aligned}$$

Differentiating both sides with respect to s , we get

$$\frac{d\phi}{ds} = \mathbf{F} \cdot \frac{d\mathbf{r}}{ds}.$$

$$\begin{aligned} \text{But } \frac{d\phi}{ds} &= \frac{\partial \phi}{\partial x} \frac{dx}{ds} + \frac{\partial \phi}{\partial y} \frac{dy}{ds} + \frac{\partial \phi}{\partial z} \frac{dz}{ds} \\ &= \left(\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \cdot \left(\frac{dx}{ds} \mathbf{i} + \frac{dy}{ds} \mathbf{j} + \frac{dz}{ds} \mathbf{k} \right) \end{aligned}$$

$$= \nabla \phi \bullet \frac{d\mathbf{r}}{ds}.$$

$$\therefore \mathbf{F} \bullet \frac{d\mathbf{r}}{ds} = \nabla \phi \bullet \frac{d\mathbf{r}}{ds}$$

$$\text{or} \quad (\nabla \phi - \mathbf{F}) \bullet \frac{d\mathbf{r}}{ds} = 0.$$

Now this result is true irrespective of the path joining P to Q i.e. this result is true irrespective of the direction of $\frac{d\mathbf{r}}{ds}$ which is tangent vector to C . Therefore we must

have $\nabla \phi - \mathbf{F} = \mathbf{0}$ i.e., $\nabla \phi = \mathbf{F}$.

This completes the proof of the theorem.

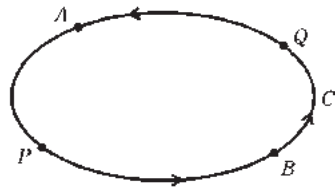
Definition: A vector field $\mathbf{F}(x, y, z)$ defined and continuous in a region R of space is said to be a conservative vector field if the line integral $\int_C \mathbf{F} \bullet d\mathbf{r}$ is independent of the path C in R joining P and Q where P and Q are any two points in R .

By theorem 1, vector field $\mathbf{F}(x, y, z)$ is conservative if and only if $\mathbf{F} = \nabla \phi$ where $\phi(x, y, z)$ is a single valued scalar function having continuous first partial derivatives in R . The function $\phi(x, y, z)$ is called the **scalar potential** of the vector field $\mathbf{F}$.

Theorem 2: Let $\mathbf{F}(x, y, z)$ be a vector function defined and continuous in a region R of space. Then the line integral $\int_P^Q \mathbf{F} \bullet d\mathbf{r}$ is independent of the path joining any two points P and Q in R if and only if $\oint_C \mathbf{F} \bullet d\mathbf{r} = 0$ for every simple closed path in R .

Proof: Let C be any simple closed path in R and let the line integral be independent of path in R . Take two points P and Q on C and subdivide C into two arcs PBQ and QAP . Then

$$\begin{aligned} \oint_C \mathbf{F} \bullet d\mathbf{r} &= \int_{PBQAP} \mathbf{F} \bullet d\mathbf{r} \\ &= \int_{PBQ} \mathbf{F} \bullet d\mathbf{r} + \int_{QAP} \mathbf{F} \bullet d\mathbf{r} \\ &= \int_{PBQ} \mathbf{F} \bullet d\mathbf{r} - \int_{PAQ} \mathbf{F} \bullet d\mathbf{r} \\ &= 0, \end{aligned}$$



since the integral from P to Q along a path through B is equal to the integral from P to Q along a path through A .

Conversely, suppose that the integral under consideration is zero on every simple closed path in R . Let P and Q be any two points in R which join P to Q and do not cross. Then

$$\begin{aligned}\oint_{PBQAP} \mathbf{F} \cdot d\mathbf{r} &= \int_{PBQ} \mathbf{F} \cdot d\mathbf{r} + \int_{QAP} \mathbf{F} \cdot d\mathbf{r} \\ &= \int_{PBQ} \mathbf{F} \cdot d\mathbf{r} - \int_{PAQ} \mathbf{F} \cdot d\mathbf{r}.\end{aligned}$$

But as given, we have

$$\oint_{PBQAP} \mathbf{F} \cdot d\mathbf{r} = 0.$$

$$\therefore \int_{PBQ} \mathbf{F} \cdot d\mathbf{r} - \int_{PAQ} \mathbf{F} \cdot d\mathbf{r} = 0$$

$$\text{or} \quad \int_{PBQ} \mathbf{F} \cdot d\mathbf{r} = \int_{PAQ} \mathbf{F} \cdot d\mathbf{r}.$$

This completes the proof of the theorem.

Theorem 3: Let $\mathbf{F}(x, y, z) = f\mathbf{i} + g\mathbf{j} + h\mathbf{k}$ be a continuous vector function having continuous first partial derivatives in a region R of space. If $\int f dx + g dy + h dz$ is independent of path in R and consequently $f dx + g dy + h dz$ is an exact differential in R , then $\text{curl } \mathbf{F} = \mathbf{0}$ everywhere in R . Conversely, if R is simply connected and $\text{curl } \mathbf{F} = \mathbf{0}$ everywhere in R , then $f dx + g dy + h dz$ is an exact differential in R or $\int f dx + g dy + h dz$ is independent of path in R .

Proof: Suppose $\int (f dx + g dy + h dz)$ is independent of path in R . Then $f dx + g dy + h dz$ is an exact differential in R . Therefore

$$\mathbf{F} = f\mathbf{i} + g\mathbf{j} + h\mathbf{k} = \text{grad } \phi.$$

$$\therefore \text{curl } \mathbf{F} = \text{curl } (\text{grad } \phi) = \mathbf{0}.$$

Converse. Suppose R is simply connected and $\text{curl } \mathbf{F} = \mathbf{0}$ everywhere in R . Let C be any simple closed path in R . Since R is simply connected, therefore we can find a surface S in R having C as its boundary. Therefore by Stoke's theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = 0.$$

Thus $\oint_C \mathbf{F} \cdot d\mathbf{r}$ is zero for every simple closed path C in R .

Therefore $\int \mathbf{F} \cdot d\mathbf{r}$ is independent of path in R .

Therefore $\mathbf{F} = \nabla \phi$ and consequently $f dx + g dy + h dz$ is an exact differential $d\phi$.

Note: The assumption that R be simply connected is essential and cannot be omitted. It is obvious from the following illustration.

Illustration: Let $\mathbf{F} = -\frac{y}{x^2 + y^2} \mathbf{i} + \frac{x}{x^2 + y^2} \mathbf{j}.$

Here $\mathbf{F}$ is not defined at origin. In every region R of the xy -plane not containing the origin, we have

$$\begin{aligned}
 \text{Curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\frac{y}{x^2 + y^2} & \frac{x}{x^2 + y^2} & 0 \end{vmatrix} \\
 &= 0\mathbf{i} + 0\mathbf{j} + \left\{ \frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2} \right) + \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right) \right\} \mathbf{k} \\
 &= \left\{ \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} + \frac{x^2 + y^2 - 2y^2}{(x^2 + y^2)^2} \right\} \mathbf{k} \\
 &= 0 \mathbf{k} \\
 &= \mathbf{0}.
 \end{aligned}$$

Suppose R is simply connected. For example let R be the region enclosed by a simple closed curve C not enclosing the origin. Then

$$\begin{aligned}
 \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C \left(-\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy \right) \\
 &= \int \int_R \left[\frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2} \right) - \frac{\partial}{\partial y} \left(-\frac{y}{x^2 + y^2} \right) \right] dx dy, \\
 &\quad \text{by Green's theorem in plane} \\
 &= 0.
 \end{aligned}$$

Suppose R is not simply connected. Let R be the region of the xy -plane contained between concentric circles of radii $\frac{1}{2}$ and $\frac{3}{2}$ and having centre at origin. Obviously

R is not simply connected. We have $z = 0$, everywhere in R . Let C be a closed curve in R . The parametric equations of C can be taken as $x = \cos t$, $y = \sin t$, $z = 0$, $0 \leq t < 2\pi$.

$$\begin{aligned}
 \text{We have } \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \left(-\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy \right) \\
 &= \int_{t=0}^{2\pi} \left[-\frac{\sin t}{\cos^2 t + \sin^2 t} \frac{dx}{dt} + \frac{\cos t}{\cos^2 t + \sin^2 t} \frac{dy}{dt} \right] dt \\
 &= \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt \\
 &= 2\pi.
 \end{aligned}$$

Thus we see that $\oint_C \mathbf{F} \cdot d\mathbf{r} \neq 0$.

Definition: Irrotational vector field: A vector field $\mathbf{F}$ is said to be irrotational if $\text{curl } \mathbf{F} = \mathbf{0}$.

We see that an irrotational field $\mathbf{F}$ is characterized by any one of the three conditions :

- (i) $\mathbf{F} = \nabla \phi$,
- (ii) $\nabla \times \mathbf{F} = \mathbf{0}$,
- (iii) $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path.

Any one of these conditions implies the other two.

Illustrative Examples

Example 15: Are the following forms exact ?

- (i) $e^y dx + e^x dy + e^z dz$.
- (ii) $yz dx + xz dy + xy dz$.

Solution:

- (i) Here $\mathbf{F} = e^y \mathbf{i} + e^x \mathbf{j} + e^z \mathbf{k}$.

We have

$$\begin{aligned} \text{Curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^y & e^x & e^z \end{vmatrix} \\ &= 0 \mathbf{i} + 0 \mathbf{j} + (e^x - e^y) \mathbf{k}. \end{aligned}$$

Since $\text{curl } \mathbf{F} \neq \mathbf{0}$, therefore the given form is not exact.

- (ii) Here $\mathbf{F} = yz \mathbf{i} + xz \mathbf{j} + xy \mathbf{k}$.

We have

$$\begin{aligned} \text{Curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix} \\ &= (x - x)\mathbf{i} - (y - y)\mathbf{j} + (z - z)\mathbf{k} \\ &= \mathbf{0}. \end{aligned}$$

Since $\text{curl } \mathbf{F} = \mathbf{0}$, therefore the given form is exact.

Example 16: In each of following cases show that the given differential form is exact and find a function ϕ such that the form equals $d\phi$:

- (i) $\cos x dx - 2yz dy - y^2 dz$.
- (ii) $(z^2 - 2xy) dx - x^2 dy + 2xz dz$.

Solution: (i) Here $\mathbf{F} = \cos x \mathbf{i} - 2yz \mathbf{j} - y^2 \mathbf{k}$.

We have

$$\begin{aligned}\text{Curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos x & -2yz & -y^2 \end{vmatrix} \\ &= (-2y + 2y) \mathbf{i} + 0 \mathbf{j} + 0 \mathbf{k} = \mathbf{0}.\end{aligned}$$

$\therefore$ the given form is exact.

Let $\mathbf{F} = \nabla \phi$,

$$\text{or} \quad \cos x \mathbf{i} - 2yz \mathbf{j} - y^2 \mathbf{k} = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}.$$

Then

$$\frac{\partial \phi}{\partial x} = \cos x \quad \text{whence} \quad \phi = \sin x + f_1(y, z) \quad \dots (1)$$

$$\frac{\partial \phi}{\partial y} = -2yz \quad \text{whence} \quad \phi = -y^2 z + f_2(x, z) \quad \dots (2)$$

$$\frac{\partial \phi}{\partial z} = -y^2 \quad \text{whence} \quad \phi = -y^2 z + f_3(x, y). \quad \dots (3)$$

(1), (2), (3) each represents ϕ . These agree if we choose

$$f_1(y, z) = -y^2 z, \quad f_2(x, z) = \sin x, \quad f_3(x, y) = \sin x.$$

$\therefore \quad \phi = \sin x - y^2 z$ to which may be added any constant.

$\therefore \quad \phi = \sin x - y^2 z + C.$

(ii) Here $\mathbf{F} = (z^2 - 2xy) \mathbf{i} - x^2 \mathbf{j} + 2xz \mathbf{k}$. We have

$$\begin{aligned}\text{Curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 - 2xy & -x^2 & 2xz \end{vmatrix} \\ &= 0 \mathbf{i} + 0 \mathbf{j} + 0 \mathbf{k} = \mathbf{0}.\end{aligned}$$

$\therefore$ the given form is exact.

Let $\mathbf{F} = \nabla \phi$

$$\text{or} \quad (z^2 - 2xy) \mathbf{i} - x^2 \mathbf{j} + 2xz \mathbf{k} = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}.$$

$$\text{Then} \quad \frac{\partial \phi}{\partial x} = z^2 - 2xy \quad \text{whence} \quad \phi = z^2 x - x^2 y + f_1(y, z) \quad \dots (1)$$

$$\frac{\partial \phi}{\partial y} = -x^2 \quad \text{whence} \quad \phi = -x^2 y + f_2(x, z) \quad \dots (2)$$

$$\frac{\partial \phi}{\partial z} = 2xz \quad \text{whence} \quad \phi = xz^2 + f_3(x, y). \quad \dots (3)$$

(1), (2), (3) each represents ϕ . These agree if we choose

$$f_1(y, z) = 0, \quad f_2(x, z) = xz^2, \quad f_3(x, y) = -x^2 y.$$

$\therefore \phi = z^2 x - x^2 y$ to which may be added any constant.

$\therefore \phi = z^2 x - x^2 y + C.$

Example 17: Show that the vector field $\mathbf{F}$ given by

$$\mathbf{F} = (x^2 - yz) \mathbf{i} + (y^2 - zx) \mathbf{j} + (z^2 - xy) \mathbf{k}$$

is irrotational. Find a scalar ϕ such that $\mathbf{F} = \nabla\phi$.

Solution: We have

$$\begin{aligned} \text{Curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - yz & y^2 - zx & z^2 - xy \end{vmatrix} \\ &= (-x + x) \mathbf{i} - (-y + y) \mathbf{j} + (-z + z) \mathbf{k} = \mathbf{0}. \end{aligned}$$

$\therefore$ The vector field $\mathbf{F}$ is irrotational.

Let $\mathbf{F} = \nabla\phi$

or $(x^2 - yz) \mathbf{i} + (y^2 - zx) \mathbf{j} + (z^2 - xy) \mathbf{k} = \frac{\partial\phi}{\partial x} \mathbf{i} + \frac{\partial\phi}{\partial y} \mathbf{j} + \frac{\partial\phi}{\partial z} \mathbf{k}.$

Then $\frac{\partial\phi}{\partial x} = x^2 - yz$ whence $\phi = \frac{x^3}{3} - xyz + f_1(y, z)$... (1)

$\frac{\partial\phi}{\partial y} = y^2 - zx$ whence $\phi = \frac{y^3}{3} - xyz + f_2(x, z)$... (2)

$\frac{\partial\phi}{\partial z} = z^2 - xy$ whence $\phi = \frac{z^3}{3} - xyz + f_3(x, y).$... (3)

(1), (2), (3) each represents ϕ . These agree if we choose

$$f_1(y, z) = \frac{y^3}{3} + \frac{z^3}{3}, \quad f_2(x, z) = \frac{x^3 + z^3}{3}, \quad f_3(x, y) = \frac{x^3 + y^3}{3}.$$

Therefore $\phi = \frac{x^3 + y^3 + z^3}{3} - xyz + C.$

7 Physical Interpretation of Divergence and Curl

Physical interpretation of divergence: Suppose that there is a fluid motion whose velocity at any point is $\mathbf{v}(x, y, z)$. Then the loss of fluid per unit volume per unit time in a small parallelepiped having centre at $P(x, y, z)$ and edges parallel to the co-ordinate axes and having lengths $\delta x, \delta y, \delta z$ respectively, is given approximately by

$$\text{div } \mathbf{v} = \nabla \cdot \mathbf{v}.$$

Let $\mathbf{v} = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k}$.

x -component of velocity $\mathbf{v}$ at $P = v_1(x, y, z)$.

$\therefore$ x -component of $\mathbf{v}$ at centre of face $AFED$

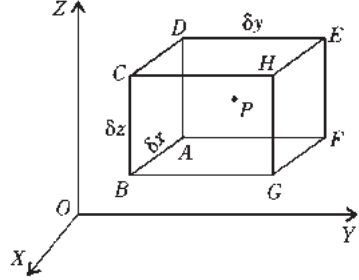
which is perpendicular to x -axis and is nearer to origin

$$= v_1 \left(x - \frac{\delta x}{2}, y, z \right)$$

$$= v_1(x, y, z) - \frac{\delta x}{2} \frac{\partial v_1}{\partial x} + \dots$$

by Taylor's theorem

$$= v_1(x, y, z) - \frac{\delta x}{2} \frac{\partial v_1}{\partial x} \text{ approximately.}$$



Similarly x -component of $\mathbf{v}$ at centre of opposite face $GHCB$

$$= v_1 + \frac{\delta x}{2} \frac{\partial v_1}{\partial x} \text{ approximately.}$$

$\therefore$ volume of fluid entering the parallelopiped across $AFED$ per unit time

$$= \left(v_1 - \frac{\delta x}{2} \frac{\partial v_1}{\partial x} \right) \delta y \delta z.$$

Also volume of fluid going out the parallelopiped across $GHCB$ per unit time

$$= \left(v_1 + \frac{\delta x}{2} \frac{\partial v_1}{\partial x} \right) \delta y \delta z.$$

$\therefore$ loss in volume per unit time in the direction of x -axis

$$\begin{aligned} &= \left(v_1 + \frac{\delta x}{2} \frac{\partial v_1}{\partial x} \right) \delta y \delta z - \left(v_1 - \frac{\delta x}{2} \frac{\partial v_1}{\partial x} \right) \delta y \delta z \\ &= \frac{\partial v_1}{\partial x} \delta x \delta y \delta z. \end{aligned}$$

Similarly, loss in volume per unit time in y direction

$$= \frac{\partial v_2}{\partial y} \delta x \delta y \delta z,$$

and loss in volume per unit time in z direction

$$= \frac{\partial v_3}{\partial z} \delta x \delta y \delta z.$$

$\therefore$ total loss of the fluid per unit volume per unit time symbol

$$\begin{aligned} &= \frac{\left(\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \right) \delta x \delta y \delta z}{\delta x \delta y \delta z} \\ &= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} = \nabla \cdot \mathbf{v} = \text{div } \mathbf{v}. \end{aligned}$$

Physical interpretation of curl: Let S be a circular disc of small radius r and centre P bounded by the circle C . Let $\mathbf{F}(x, y, z)$ be a continuously differentiable vector function in S . Then by Stoke's theorem

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \overline{(\text{curl } \mathbf{F}) \cdot \mathbf{n}} \iint_S dS,$$

by mean value theorem of integral calculus where $\overline{(\text{curl } \mathbf{F}) \cdot \mathbf{n}}$ is some value intermediate between the maximum and minimum values of $(\text{curl } \mathbf{F}) \cdot \mathbf{n}$ over S .

$$\therefore \int_C \mathbf{F} \cdot d\mathbf{r} = \overline{(\text{curl } \mathbf{F}) \cdot \mathbf{n}} \, S.$$

$$\therefore \overline{(\text{curl } \mathbf{F}) \cdot \mathbf{n}} = \frac{\left[\oint_C \mathbf{F} \cdot d\mathbf{r} \right]}{S}.$$

Taking limit as $r \rightarrow 0$, we get at P ,

$$(\text{curl } \mathbf{F}) \cdot \mathbf{n} = \lim_{r \rightarrow 0} \frac{\oint_C \mathbf{F} \cdot d\mathbf{r}}{S}.$$

Now $(\text{curl } \mathbf{F}) \cdot \mathbf{n}$ is normal component of $\text{curl } \mathbf{F}$ at P and $\oint_C \mathbf{F} \cdot d\mathbf{r}$ is circulation of $\mathbf{F}$ about C . Therefore the normal component of the curl can be interpreted physically as the limit of the circulation per unit area.

Comprehensive Exercise 4

- Are the following forms exact ?
 - $x \, dx - y \, dy + z \, dz$.
 - $y^2 z^3 \, dx + 2xyz^3 \, dy + 3xy^2 z^2 \, dz$.
- In each of following cases show that the given differential form is exact and find a function ϕ such that the form equals $d\phi$:
 - $x \, dx - y \, dy - z \, dz$.
 - $dx + z \, dy + y \, dz$.
- Show that $(y^2 z^3 \cos x - 4x^3 z) \, dx + 2z^3 y \sin x \, dy + (3y^2 z^2 \sin x - x^4) \, dz$ is an exact differential of some function ϕ and find this function.
 - Show that $\mathbf{F} = (2xy + z^3) \mathbf{i} + x^2 \mathbf{j} + 3xz^2 \mathbf{k}$ is a conservative force field. Find the scalar potential. Find also the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4)$.

4. (i) Show that the vector field

$$\mathbf{F} = (2xy^2 + yz) \mathbf{i} + (2x^2y + xz + 2yz^2) \mathbf{j} + (2y^2z + xy) \mathbf{k}$$

is conservative.

(ii) Show that $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is conservative and find ϕ such that $\mathbf{F} = \nabla\phi$.

5. (i) Show that $\mathbf{F} = (\sin y + z) \mathbf{i} + (x \cos y - z) \mathbf{j} + (x - y) \mathbf{k}$ is a

conservative vector field and find a function ϕ such that $\mathbf{F} = \nabla\phi$.

(ii) Evaluate $\int_C 2xyz^2 dx + (x^2z^2 + z \cos yz) dy + (2x^2yz + y \cos yz) dz$

where C is any path from $(0, 0, 1)$ to $(1, \pi/4, 2)$.

6. Show that the following vector functions $\mathbf{F}$ are irrotational and find the corresponding scalar ϕ such that $\mathbf{F} = \nabla\phi$.

(i) $\mathbf{F} = (\sin y + z \cos x) \mathbf{i} + (x \cos y + \sin z) \mathbf{j} + (y \cos z + \sin x) \mathbf{k}$.

(ii) $\mathbf{F} = (y \sin z - \sin x) \mathbf{i} + (x \sin z + 2yz) \mathbf{j} + (xy \cos z + y^2) \mathbf{k}$.

(iii) $\mathbf{F} = x^3 \mathbf{i} + y^3 \mathbf{j} + z^3 \mathbf{k}$.

7. Find a, b, c if $\mathbf{F} = (3x - 3y + az) \mathbf{i} + (bx + 2y - 4z) \mathbf{j} + (2x + cy + z) \mathbf{k}$ is irrotational.

8. Evaluate $\int_C yz dx + (xz + 1) dy + xy dz$, where C is any path from $(1, 0, 0)$ to $(2, 1, 4)$.

9. Show that the form under the integral sign is exact and evaluate

$$\int_{(0,2,1)}^{(2,0,1)} [ze^x dx + 3yz dy + (e^x + y^2) dz].$$

Answers 4

1. (i) Exact (ii) Exact
2. (i) Exact; $\phi = \frac{x^2 - y^2 - z^2}{2} + C$
- (ii) Exact; $\phi = x + yz + C$
3. (i) $\phi = y^2 z^3 \sin x - x^4 z + C$
- (ii) $\phi = x^2 y - xz^3 + C$; 202
4. (ii) $\phi = \frac{1}{2} (x^2 + y^2 + z^2) + C$.
5. (i) $\phi = x \sin y + xz - yz + C$.
- (ii) $\pi + 1$

6. (i) $\phi = x \sin y + z \sin x + y \sin z + C$
 (ii) $\phi = xy \sin z + \cos x + y^2 z + C$
 (iii) $\phi = \frac{1}{4} (x^4 + y^4 + z^4) + C$
7. $a = 2, b = -3, c = -4$.
8. 9
9. $e^2 - 5$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If C is the curve $x^2 + y^2 = 1, z = y^2$, then by Stoke's theorem $\oint_C (yz \, dx + zx \, dy + xy \, dz)$ is
- (a) 0 (b) 3
 (c) 5 (d) None of these
2. If S denotes the surface of the cube bounded by the planes $x = 0, x = a, y = 0, y = a, z = 0, z = a$ then by the application of Gauss divergence theorem the value of $\iint_S (x \, \mathbf{i} + y \, \mathbf{j} + z \, \mathbf{k}) \cdot \mathbf{n} \, dS$ is
- (a) a^3 (b) $2a^3$
 (c) $3a^3$ (d) 0

Fill in the Blank(s)

Fill in the blanks ".....", so that the following statements are complete and correct.

1. For any closed surface $S, \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \dots\dots\dots$
2. If $\mathbf{n}$ is the unit outward drawn normal to any closed surface S , then $\iiint_V \text{div } \mathbf{n} \, dV = \dots\dots\dots$ (Bundelkhand 2008)
3. The value of $\oint_C \mathbf{r} \cdot d\mathbf{r} = \dots\dots\dots$ (Agra 2008)
4. A necessary and sufficient condition that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C lying in a simply connected region R is that $\nabla \times \mathbf{F} = \dots\dots\dots$ identically.

5. By Stoke's theorem, $\oint_C \mathbf{F} \cdot d\mathbf{r} = \dots\dots\dots$
6. $\iiint_S \mathbf{r} \cdot \mathbf{n} \, dS = \dots\dots\dots$

(Kumaun 2009)

True or False*Write 'T' for true and 'F' for false statement.*

- $\iint_S \mathbf{n} \, dS = \mathbf{0}$ for any surface S .
- Green's theorem in plane is a special case of Stoke's theorem.
- Green's theorem states that "the surface integral of the normal component of a vector $\mathbf{F}$ taken over a closed surface is equal to the integral of the divergence of $\mathbf{F}$ taken over the volume enclosed by the surface".

Answers

Multiple Choice Questions

1. (a) 2. (c)

Fill in the Blank(s)

- | | | | |
|---|---------|------|------|
| 1. 0 | 2. S | 3. 0 | 4. 0 |
| 5. $\iiint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$ | 6. $3V$ | | |

True or False

1. F 2. T 3. F

